

High Energy Physics And Accelerators

Quantum Field Theory

$$(i\gamma^\mu\partial_\mu - m)\psi = 0$$

Guillem Arias

*Class notes of the course High Energy Physics and Accelerators, taught by Professor Ricardo Vazquez
Gomez and Professor Lukas Calefice*

Universitat de Barcelona
Curs 2025–2026

Contents

1	Introduction	3
2	Klein Gordon Equation	5
2.1	Solutions of Klein-Gordon Equation	6
2.2	Continuity equations	6
2.3	Klein Paradox	7
2.4	Problems	9
3	Dirac Equation	12
3.1	Conserved Charges and Current	13
3.2	Solutions of the Dirac Equation	14
3.3	Normalization	16
3.4	Spin of Dirac Particle	17
3.5	Commutation Relations	18
3.6	Non-Relativistic Limit of Dirac Equation	19
3.7	Non-Relativistic Limit in an External Field	20
3.8	Clifford Algebra	21
3.9	Dirac's Hole Theory	23
3.10	Lorentz covariance of Dirac equation	24
3.11	Rotations and Boosts	27
3.12	Spin and Helicity Projection Operators	30
3.13	Parity, time reversal and Charge Conjugation	31
3.14	Problems	33
4	From Classical Field Theory to Quantum Field theory	43
4.1	Quantization of non relativistic string	43
4.2	Noether's Theorem	45
4.3	Spacetime Translation	47
4.4	Quantization of String	48
4.5	Number Operator	51
5	Quantization of the scalar field	53
5.1	Hamiltonian Operator and Eigenstates	54
5.2	Localized Particle	56
5.3	Causality	57
5.4	The Klein-Gordon Propagator	58
5.5	Quantization of Charged scalar fields	62
5.6	Charge Operators	65
5.7	Problems	67
6	Quantization of the Dirac Field	73
6.1	Field Decomposition	73
6.2	Hamiltonian of the Dirac Field	74
6.3	Charge Operator and Conserved Charges	77
6.4	Green Function and Dirac Propagator	78
6.5	Recapitulation on Propagators	80
6.6	Majorana Fermions	81
6.7	Problems	83
7	Quantization of the Photon Field	89
7.1	Quantization of the Maxwell Field	90
7.2	Spin of the Photons	93
7.3	Photon Propagator	94
7.4	Problems	95

8	Interactions	101
8.1	Interaction Representation	101
8.2	S-Matrix	103
8.3	Feynman Rules for Yukawa Coupling	108
8.4	Problems	109
9	Quantum Electrodynamics	120
9.1	Feynman Rules for QED	122
9.2	Problems	126
10	Experimental Particle Physics	129
10.1	Particle Accelerators	129
10.2	Particle Detectors	133
10.2.1	Photoelectric effect	134
10.2.2	Compton scattering	134
10.2.3	Pair production	134
10.3	Particle Identification	138
10.4	Problems	140

1 Introduction

At short distances we need quantum mechanics. Because of the uncertainty principle short distances imply large momentum, we will need accelerators. We need to work at high $v \sim c$ so we also need special relativity. We need to combine quantum mechanics and special relativity. There are two main approaches, relativistic quantum mechanics that fixes the number of particles before hand and turns out to be inconsistent and quantum field theory which is the formalism we will develop in this notes. We will also be working in natural units i.e $\hbar = c = G = 1$

Let's begin by considering the Schrödinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi \quad (1.1)$$

with $\psi(x, t)$ a wave function and the Hamiltonian $H = \frac{\vec{p}^2}{2m} + V(\vec{x})$. For a free particle

$$i\hbar \frac{\partial \psi}{\partial t} = \frac{p^2}{2m} \psi \quad (1.2)$$

using $\hat{p} = -i\hbar \vec{\nabla}$ we obtain

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi \quad (1.3)$$

which is linear in t and quadratic in $\vec{x}$ so it is clearly non-covariant.

We will denote vectors as follows $\vec{x} = (x_1, x_2, x_3)$, $\vec{J} = (J_1, J_2, J_3)$ then we can write the scalar product of two vectors

$$\vec{A} \cdot \vec{B} = A_i B_i = \delta_{ij} A_i B_j \quad (1.4)$$

where Einstein notation for summation has been used and δ_{ij} is the Kronecker delta (metric in euclidean space).

The length of a vector is then

$$\vec{A} \cdot \vec{A} = \vec{A}^2 = A_i \cdot A_i = \delta_{ij} A_i A_j = \delta^{ij} A_i A_j \quad (1.5)$$

and must satisfy $\vec{A}^2 \geq 0$ and $\vec{A}^2 = 0 \rightarrow \vec{A} = 0$.

In 4D we can represent contravariant $X^\mu = (ct, \vec{x})$ and covariant $X_\mu = (ct, -\vec{x})$ vectors. They are related by the metric $X_\mu = \eta_{\mu\nu} X^\nu$, $X^\mu = \eta^{\mu\nu} X_\nu$.

$\eta^{\mu\nu}$ is the metric tensor of 4D Minkowski space

$$\eta^{\mu\lambda} \eta_{\lambda\nu} = \delta_\nu^\mu, \quad \eta^{\mu\nu} = \eta^{\nu\mu} \quad (1.6)$$

The signature of the metric is $(+, -, -, -)$:

$$\eta^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \eta_{\mu\nu}. \quad (1.7)$$

Given two 4-vectors $A^\mu = (A^0, \vec{A})$ and $B^\mu = (B^0, \vec{B})$, we can define a scalar product invariant under Lorentz transformations:

$$A \cdot B = A^\mu B_\mu = A_\mu B^\mu = \eta^{\mu\nu} A_\mu B_\nu = \eta_{\mu\nu} A^\mu B^\nu = A^0 B^0 - \vec{A} \cdot \vec{B}. \quad (1.8)$$

We can now define the length of a 4-vector:

$$A^2 = A \cdot A = A^\mu A_\mu = \eta^{\mu\nu} A_\mu A_\nu = \eta_{\mu\nu} A^\mu A^\nu = (A^0)^2 - \vec{A}^2, \quad (1.9)$$

which can be negative.

The invariant interval of a spacetime point from the origin is

$$x^2 = x^\mu x_\mu = \eta^{\mu\nu} x_\mu x_\nu = c^2 t^2 - \vec{x}^2 = \eta_{\mu\nu} x^\mu x^\nu. \quad (1.10)$$

There are three cases:

1. If $c^2t^2 - \vec{x}^2 > 0 \rightarrow$ timelike region of spacetime
2. If $c^2t^2 - \vec{x}^2 < 0 \rightarrow$ spacelike region of spacetime
3. If $c^2t^2 - \vec{x}^2 = 0 \rightarrow$ lightlike region of spacetime

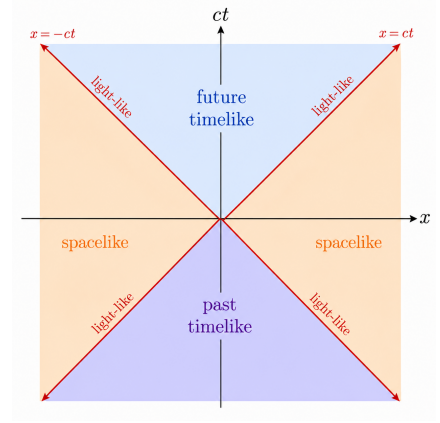


Figure 1: Light cone and the timelike, spacelike, and lightlike regions of Minkowski spacetime.

Let's now define the cogradient $\partial^\mu = \frac{\partial}{\partial x_\mu} = \left(\frac{\partial}{\partial t}, -\vec{\nabla} \right)$ and the contragradient $\partial_\mu = \frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial t}, \vec{\nabla} \right)$ with this two we can construct a Lorentz invariant quadratic operator

$$\square \equiv \partial^2 = \partial^\mu \partial_\mu \quad (1.11)$$

the so called d'Alembertian. We define the Energy-Momentum 4-vector

$$P^\mu = (E, \vec{p}), \quad P_\mu = (E, -\vec{p}) \quad (1.12)$$

Squaring it we find another lorentz invariant quantity

$$P^2 = P^\mu P_\mu = E^2 - \vec{p}^2 = m^2 \quad (1.13)$$

and we can write Einstein relation

$$E^2 = \vec{p}^2 + m^2 \quad (1.14)$$

Let's now write the coordinate representation of E and $\vec{P}$, $E \rightarrow i\hbar \frac{\partial}{\partial t}$, $\vec{p} = -i\hbar \vec{\nabla}$ then

$$P^\mu = i\hbar \partial^\mu = i\hbar \frac{\partial}{\partial x_\mu} = \left(i\hbar \frac{\partial}{\partial t}, -i\hbar \vec{\nabla} \right) \quad (1.15)$$

$$P_\mu = i\hbar \partial_\mu = i\hbar \frac{\partial}{\partial x^\mu} = \left(i\hbar \frac{\partial}{\partial t}, i\hbar \vec{\nabla} \right) \quad (1.16)$$

In 4D we have two totally antisymmetric 4-rank tensors: $\varepsilon^{\mu\nu\lambda\delta}$ and $\varepsilon_{\mu\nu\lambda\delta}$, the Levi-Civita tensors.

We choose $\varepsilon^{0123} = 1 = -\varepsilon_{0123} \rightarrow \varepsilon^{0ijk} = \varepsilon_{ijk} = -\varepsilon_{0ijk}$ with ε_{ijk} being the 3D Levi-Civita.

We can lower and upper its indices with the metric $\varepsilon^i_{jk} = \eta^{il} \varepsilon_{ljk}$

2 Klein Gordon Equation

Klein Gordon equation describes the motion of spin-0 particles. It is the relativistic version of the Schrödinger equation.

Let's begin by considering the non relativistic hamiltonian of a free particle

$$H = \frac{\vec{p}^2}{2m} \quad (2.1)$$

In coordinate representation $H \rightarrow \hat{E} \equiv i\hbar \frac{\partial}{\partial t}$ and $\vec{p} \rightarrow \hat{p} = i\hbar \vec{\nabla}$. Applying operators to the wave-function $\psi(\vec{x}, t)$

$$\hat{E}\psi(\vec{x}, t) = \frac{\hat{p}^2}{2m}\psi(\vec{x}, t) \rightarrow i\hbar \frac{\partial}{\partial t}\psi(\vec{x}, t) = -\frac{\hbar^2}{2m}\nabla^2\psi \quad (2.2)$$

and our result is clearly non relativistic.

We need to start from Einstein relation $E^2 = \vec{p}^2 + m^2$ and promote $E, \vec{p}$ to operators

$$-\hbar^2 \frac{\partial^2}{\partial t^2}\psi(\vec{x}, t) = (-\hbar^2\nabla^2 + m^2)\psi(\vec{x}, t) \quad (2.3)$$

rearranging

$$\frac{\partial^2\psi(\vec{x}, t)}{\partial t^2} - \nabla^2\psi(\vec{x}, t) + \frac{m^2}{\hbar^2}\psi(\vec{x}, t) = 0 \quad (2.4)$$

using 4-vectors

$$\frac{\partial^2\psi(\vec{x}, t)}{(\partial x^0)^2} - \frac{\partial^2\psi(\vec{x}, t)}{(\partial x^i)^2} + \frac{m^2}{\hbar^2}\psi(\vec{x}, t) = 0 \quad (2.5)$$

putting $\hbar = 1$ we finally obtain the so called Klein-Gordon equation

$$\boxed{(\square + m^2)\psi(\vec{x}, t) = 0} \quad (2.6)$$

which is clearly relativistic. A few comments on Klein Gordon's equation, $(\square + m^2)$ is an scalar operator.

If we set $m = 0$ then $\square\psi(\vec{x}, t) = 0 \rightarrow \frac{\partial^2\psi}{\partial t^2} = \frac{\partial^2\psi}{\partial x^2}$ which is a wave equation.

Let's imagine that we perform the following Lorentz transformation:

$$x^\mu \longrightarrow x'^\mu.$$

The Klein-Gordon equation in the transformed and original coordinates is

$$\left(\frac{\partial}{\partial x'^\mu} \frac{\partial}{\partial x'_\mu} + m^2\right)\psi'(x') = 0, \quad (2.7)$$

$$\left(\frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x_\mu} + m^2\right)\psi(x) = 0. \quad (2.8)$$

For both equations to have the same form, the field must satisfy

$$\psi'(x') = \psi(x), \quad (2.9)$$

which is the transformation law of a Lorentz scalar.

From now on, we will denote a scalar field by $\phi(\vec{x}, t)$.

A scalar field transforms as

$$\phi(x) \longrightarrow \phi'(x') = \phi(x). \quad (2.10)$$

so we define two states under space inversion $x'^i = -x^i$ $x'^0 = x^0$ and $x'^i = x^i$ $x'^0 = x^0$. if $\lambda = +1$ then $\phi(\vec{x}, t)$ is a scalar and if $\lambda = -1$ then $\phi(\vec{x}, t)$ is a pseudoscalar. The value of λ is what is called intrinsic parity.

2.1 Solutions of Klein-Gordon Equation

We have already seen that if $m = 0$ the Klein-Gordon equation becomes a wave equation. This suggests that in order to find solutions of the Klein-Gordon equation when $m \neq 0$ we can take an ansatz in the form of plane waves

$$\phi(\vec{x}, t) = e^{\mp i k \cdot x} = e^{\mp i k_\mu x^\mu} = e^{\mp i k^\mu x_\mu} = e^{\mp i(k_0 t - \vec{k} \cdot \vec{x})} \quad (2.11)$$

where $k^\mu = (k^0, \vec{k})$ are eigenfunctions of the energy momentum operator.

Plugging $\phi(\vec{x}, t)$ into the Klein-Gordon equation

$$(\square + m^2)\phi(x) = (-p_0^2 + \vec{p}^2 + m^2)\phi(x) = 0 \quad (2.12)$$

$$(\square + m^2)\phi(x) = (\partial^\mu \partial_\mu + m^2)\phi(x) = [(-ip^\mu)(-ip_\mu) + m^2]\phi(x) = 0 \quad (2.13)$$

$$i\partial^\mu e^{-ip \cdot x} = i \frac{\partial}{\partial x_\mu} e^{-ip \cdot x} = p^\mu e^{-ip \cdot x} = p^\mu \phi(x) \rightarrow (-p^\mu p_\mu + m^2)\phi(x) = 0 = (-p_0^2 + \vec{p}^2 + m^2)\phi(x) \quad (2.14)$$

for the equation to be 0

$$-p_0^2 + \vec{p}^2 + m^2 = 0 \rightarrow p_0^2 = \vec{p}^2 + m^2 \rightarrow p_0 = \pm \sqrt{\vec{p}^2 + m^2} = \pm E \quad (2.15)$$

we cannot discard the negative solutions. In quantum mechanics they form a base.

We identify two solutions

$$\phi_p^+(x) = e^{-i(Et - \vec{p} \cdot \vec{x})} \quad p_0 = E \quad (2.16)$$

$$\phi_p^-(x) = e^{i(Et - \vec{p} \cdot \vec{x})} \quad p_0 = -E \quad (2.17)$$

in compact notation

$$\phi_p^\pm(x) = e^{\mp i p \cdot x} \quad (2.18)$$

2.2 Continuity equations

From quantum mechanics, the Schrödinger equation leads to

$$\frac{\partial \rho}{\partial t} = \frac{i}{2m} \vec{\nabla}(\psi^*(\vec{\nabla}\psi) - (\vec{\nabla}\psi^*)\psi) = -\nabla \cdot \vec{J} \quad (2.19)$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0 \quad (2.20)$$

this equation is interpreted as probability conservation with $\rho = \psi^* \psi = |\psi|^2$. We want to find conserved charge and current for Klein-Gordon

$$(\square + m^2)\phi = 0 \rightarrow (\square + m^2)\phi^* = 0 \quad (2.21)$$

Multiplying Klein-Gordon equation for ϕ by ϕ^* on the left

$$\phi^*(\partial_\mu \partial^\mu + m^2)\phi = 0 \rightarrow \partial_\mu(\phi^* \partial^\mu \phi) - (\partial_\mu \phi^*)(\partial^\mu \phi) + m^2 \phi^* \phi = 0 \quad (2.22)$$

using the chain rule

$$\partial_\mu(\phi^* \partial^\mu \phi) = (\partial_\mu \phi^*)(\partial^\mu \phi) + \phi^*(\partial_\mu \partial^\mu \phi) \quad (2.23)$$

$$\phi^*(\partial_\mu \partial^\mu \phi) = \partial_\mu(\phi^* \partial^\mu \phi) - (\partial_\mu \phi^*)(\partial^\mu \phi) \quad (2.24)$$

now multiplying Klein-Gordon equation for ϕ^* by ϕ on the left

$$\phi(\square + m^2)\phi^* = 0 \rightarrow \partial_\mu(\phi \partial^\mu \phi^*) - (\partial_\mu \phi)(\partial^\mu \phi^*) + m^2 \phi \phi^* = 0 \quad (2.25)$$

subtracting both equations

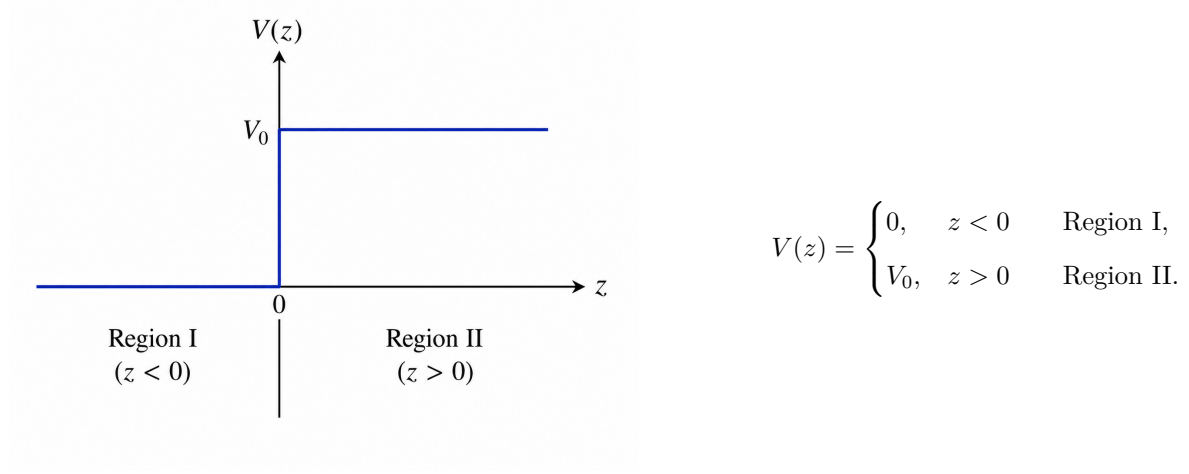
$$\partial_\mu(\phi^* \partial^\mu \phi) - \partial_\mu(\phi \partial^\mu \phi^*) = 0 \quad (2.26)$$

$$\partial_\mu(\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*) = \partial_\mu j^\mu = 0 \quad (2.27)$$

which is the covariant version of (2.20). In Schrödinger $\rho = |\psi|^2$ can not be negative instead $\rho = j^0 = \phi^* \partial_t \phi - \phi \partial_t \phi^*$ can be negative. This hints at ϕ not describing single particles but fields.

2.3 Klein Paradox

Let's localize a particle in a strong potential i.e confinement.



the 4-potential is $A^\mu = (\frac{V(z)}{q}, \vec{0})$. We will consider the two regions independently. For reg I

$$(\square + m^2)\phi(x) = 0 \quad (2.28)$$

$$(\partial^\mu \partial_\mu + m^2)\phi_I = 0 \quad (2.29)$$

for interacting electromagnetic fields we consider the minimal coupling

$$\begin{cases} \partial_\mu \rightarrow \partial_\mu + ieA_\mu \\ p_\mu \rightarrow p_\mu - eA_\mu \end{cases} \quad (2.30)$$

some texts use the following notation $D_\mu \equiv \partial_\mu + ieA_\mu$. Applying the minimal coupling to the Klein Gordon equation for reg II

$$[(\partial_\mu + ieA_\mu)(\partial^\mu + ieA^\mu) + m^2]\phi = 0 \quad (2.31)$$

$$[\partial_\mu \partial^\mu + m^2 - e^2 V_0^2 + 2ieA^0 \partial_t]\phi = 0 \quad (2.32)$$

$$(\square + m^2 + 2ieV_0 \frac{\partial}{\partial t} - e^2 V_0^2)\phi_{II}. \quad (2.33)$$

We will need to apply continuity between reg I and reg II to obtain the boundary conditions. In reg I we will have a transmitted and a reflected wave

$$\phi_I(t, z) = e^{-iEt}(e^{ipz} + Re^{-ipz}) \quad z < 0 \quad (2.34)$$

in reg II we'll only have the transmitted wave

$$\phi_{II}(t, z) = Te^{-iEt+ip'z} \quad z > 0. \quad (2.35)$$

R, T are related to reflected and transmitted amplitudes. Continuity at $z = 0$ requires $E_I = E_{II} \rightarrow E = \sqrt{p^2 + m^2}$, $p' = \pm \sqrt{(E - V_0)^2 - m^2}$.

We have three cases depending on the value of V_0 for p'

1. weak potential $E - V_0 > m \rightarrow p'$ is real, we have standard transmission and reflection
2. intermediate potential $-m < E - V_0 < m \rightarrow p'$ is purely imaginary, we have total reflection, no transmission
3. strong potential $E - V_0 < -m \rightarrow p'$ is real, we have a new transmission $\mathcal{T} < 0$ and a new reflection $\mathcal{R} > 1$.

Imposing boundary conditions at $z = 0$

$$R = \frac{p - p'}{p + p'}, \quad T = \frac{2p'}{p + p'} \rightarrow R + T = 1 \quad (2.36)$$

determining the probability density current for different beams

$$J = \frac{1}{2im}(\phi^* \vec{\nabla} \phi - \phi \vec{\nabla} \phi^*) \quad (2.37)$$

and the current to the left and right of the barrier

$$j_{<} = \frac{1}{2im}[(e^{-ipz} + R^* e^{ipz})ip(e^{ipz} - R e^{-ipz}) - (e^{ipz} + R e^{-ipz})ip((-e^{-ipz} + R^* e^{ipz}))] = \frac{p}{m}(1 - |R|^2) \quad (2.38)$$

$$j_{>} = \frac{p' + p'^*}{2m}|T|^2 e^{i(p' - p'^*)z} = \begin{cases} \frac{p'}{m}|T|^2 & \text{for } p' \text{ real} \\ 0 & \text{for } p' \text{ imaginary} \end{cases} \quad (2.39)$$

and the new coefficients

$$\mathcal{T} = \frac{j_{>}}{j_I} = \frac{2p(p' + p'^*)}{(p' + p)(p'^* + p)} e^{i(p' - p^*)z} \quad (2.40)$$

$$\mathcal{R} = \frac{j_I + j_{<}}{j_I} = |R|^2 = \left(\frac{p - p'}{p + p'}\right)^2 \quad (2.41)$$

- For a weak potential (p' real):

$$\mathcal{T} = \frac{4pp'}{(p + p')^2}, \quad \mathcal{R} = \left(\frac{p - p'}{p + p'}\right)^2 \implies \mathcal{T} + \mathcal{R} = 1. \quad (2.42)$$

- For an intermediate potential (p' imaginary):

$$\mathcal{T} = 0, \quad \mathcal{R} = 1 \implies \mathcal{T} + \mathcal{R} = 1. \quad (2.43)$$

- For a strong potential (p' real and negative):

$$\mathcal{T} = \frac{-4p|p'|}{(p - |p'|)^2}, \quad \mathcal{R} = \left(\frac{p + |p'|}{p - |p'|}\right)^2 \implies \mathcal{T} + \mathcal{R} = 1. \quad (2.44)$$

This apparent paradox is solved by the consideration that ϕ , as a wave function, does not describe a single particle but infinitely many particles.

Above a certain potential threshold, particle pairs are created, which explains why

$$\mathcal{T} < 0, \quad \mathcal{R} > 1. \quad (2.45)$$

We have seen three problems with

$$(\square + m^2)\phi = 0 : \quad (2.46)$$

- Negative energies.
- A density of the form

$$\rho = \phi^* \frac{\partial \phi}{\partial t} - \phi \frac{\partial \phi^*}{\partial t}, \quad (2.47)$$

which is not necessarily positive.

- The Klein paradox.

These three problems can be solved by considering ϕ as a field.

However, historically this was not solved in this way; it was “solved” by the Dirac equation.

2.4 Problems

Problem 2.1 Obtain the Schrödinger equation for a spinless free particle from the Klein–Gordon equation.

Hint: Write the Klein–Gordon field as

$$\phi(x, t) = e^{-imt}\psi(x, t)$$

and use

$$m|\psi(x, t)| \gg |i\partial_i\psi(x, t)| \gg |i\partial_0\psi(x, t)|.$$

Solution.

$$(\square + m^2)\phi(x, t) = 0 \quad (2.48)$$

$$(\partial_\mu\partial^\mu + m^2)e^{-imt}\psi(x, t) = 0 \quad (2.49)$$

$$\partial_t^2 [e^{-imt}\psi(x, t)] - e^{-imt}\vec{\nabla}^2\psi(x, t) + m^2e^{-imt}\psi(x, t) = 0 \quad (2.50)$$

$$\partial_t [-ime^{-imt}\psi(x, t) + e^{-imt}\partial_t\psi(x, t)] - e^{-imt}\vec{\nabla}^2\psi(x, t) + m^2e^{-imt}\psi(x, t) = 0 \quad (2.51)$$

$$\begin{aligned} & -m^2e^{-imt}\psi(x, t) - ime^{-imt}\partial_t\psi(x, t) - ime^{-imt}\partial_t\psi(x, t) \\ & + e^{-imt}\partial_t^2\psi(x, t) - e^{-imt}\vec{\nabla}^2\psi(x, t) + m^2e^{-imt}\psi(x, t) = 0 \end{aligned} \quad (2.52)$$

$$\begin{aligned} & e^{-imt} \left[-m^2\psi(x, t) - 2mi\partial_t\psi(x, t) + \partial_t^2\psi(x, t) \right. \\ & \quad \left. - \vec{\nabla}^2\psi(x, t) + m^2\psi(x, t) \right] = 0 \end{aligned} \quad (2.53)$$

$$e^{-imt} \left[-2mi\partial_t\psi(x, t) + \partial_t^2\psi(x, t) - \vec{\nabla}^2\psi(x, t) \right] = 0 \quad (2.54)$$

Considering that

$$|\partial_t^2\psi(x, t)| \ll |-2im\partial_t\psi(x, t)|, \quad (2.55)$$

$$\begin{aligned} -2im\partial_t\psi(x, t) &= \vec{\nabla}^2\psi(x, t), \\ i\partial_t\psi(x, t) &= -\frac{\nabla^2}{2m}\psi(x, t). \end{aligned} \quad (2.56)$$

□

Problem 2.2 Obtain the Schrödinger equation for a spinless charged particle from the Klein–Gordon equation coupled to an electromagnetic field.

Hint: Use

$$m|\psi(x, t)| \gg |iD_i\psi(x, t)| \gg |iD_0\psi(x, t)|.$$

Solution.

For interacting electromagnetic fields, we consider the minimal coupling

$$\begin{cases} \partial_\mu \rightarrow \partial_\mu + ieA_\mu \equiv D_\mu, \\ p_\mu \rightarrow p_\mu - eA_\mu. \end{cases} \quad (2.57)$$

With $A_\mu = (V, \vec{A})$, therefore

$$[D_\mu D^\mu + m^2]\phi(x, t) = 0. \quad (2.58)$$

Writing the Klein–Gordon field as $\phi(x, t) = e^{-imt}\psi(\vec{x}, t)$,

$$[D_\mu D^\mu + m^2]e^{-imt}\psi(\vec{x}, t) = 0. \quad (2.59)$$

$$[(\partial_\mu + ieA_\mu)(\partial^\mu + ieA^\mu) + m^2]e^{-imt}\psi(\vec{x}, t) = 0. \quad (2.60)$$

$$\left[\partial_\mu \partial^\mu + ie A_\mu \partial^\mu + ie \partial_\mu A^\mu - e^2 A_\mu A^\mu + m^2 \right] e^{-imt} \psi(\vec{x}, t) = 0. \quad (2.61)$$

$$\begin{aligned} & \partial_t^2 [e^{-imt} \psi(\vec{x}, t)] - e^{-imt} \vec{\nabla}^2 \psi(\vec{x}, t) + ie V \partial_t [e^{-imt} \psi(\vec{x}, t)] \\ & + ie \vec{A} \cdot \vec{\nabla} [e^{-imt} \psi(\vec{x}, t)] + ie (\partial_t V) e^{-imt} \psi(\vec{x}, t) \\ & + ie (\vec{\nabla} \cdot \vec{A}) e^{-imt} \psi(\vec{x}, t) - e^2 V^2 e^{-imt} \psi(\vec{x}, t) \\ & + e^2 \vec{A}^2 e^{-imt} \psi(\vec{x}, t) + m^2 e^{-imt} \psi(\vec{x}, t) = 0. \end{aligned} \quad (2.62)$$

$$\begin{aligned} & \partial_t [-ime^{-imt} \psi + e^{-imt} \partial_t \psi] - e^{-imt} \nabla^2 \psi + 2meV e^{-imt} \psi \\ & + 2ieV e^{-imt} \partial_t \psi + ie \vec{A} \cdot \vec{\nabla} [e^{-imt} \psi] + ie (\vec{\nabla} \cdot \vec{A}) e^{-imt} \psi \\ & - e^2 V^2 e^{-imt} \psi + e^2 \vec{A}^2 e^{-imt} \psi + m^2 e^{-imt} \psi = 0. \end{aligned} \quad (2.63)$$

$$\begin{aligned} & -m^2 e^{-imt} \psi - ime^{-imt} \partial_t \psi - ime^{-imt} \partial_t \psi + e^{-imt} \partial_t^2 \psi \\ & - e^{-imt} \nabla^2 \psi + 2meV e^{-imt} \psi + 2ieV e^{-imt} \partial_t \psi \\ & + ie \vec{A} \cdot \vec{\nabla} [e^{-imt} \psi] + ie (\vec{\nabla} \cdot \vec{A}) e^{-imt} \psi \\ & - e^2 V^2 e^{-imt} \psi + e^2 \vec{A}^2 e^{-imt} \psi + m^2 e^{-imt} \psi = 0. \end{aligned} \quad (2.64)$$

$$\begin{aligned} & e^{-imt} \left[-m^2 \psi - 2im \partial_t \psi + \partial_t^2 \psi - \nabla^2 \psi + 2eV m \psi \right. \\ & \quad \left. + 2eiV \partial_t \psi + ie \vec{A} \cdot \vec{\nabla} \psi + ie (\vec{\nabla} \cdot \vec{A}) \psi \right. \\ & \quad \left. - e^2 V^2 \psi + e^2 \vec{A}^2 \psi + m^2 \psi \right] = 0. \end{aligned} \quad (2.65)$$

$$\begin{aligned} & -2im \partial_t \psi + \partial_t^2 \psi - \nabla^2 \psi + 2eV m \psi + 2eiV \partial_t \psi + ie \vec{A} \cdot \vec{\nabla} \psi \\ & + ie \vec{\nabla} \cdot (\vec{A} \psi) - e^2 V^2 \psi + e^2 \vec{A}^2 \psi = 0. \end{aligned} \quad (2.66)$$

Considering the hint

$$m|\psi(x, t)| \gg |iD_i \psi(x, t)| \gg |iD_0 \psi(x, t)|,$$

then

$$m|\psi(x, t)| \gg |i\vec{\nabla} \psi(x, t)| \gg |i\partial_t \psi(x, t) - eV \psi(x, t)|. \quad (2.67)$$

So we can neglect the terms $\partial_t^2 \psi \sim 0$ and $-e^2 V^2 \psi \sim 0$:

$$\begin{aligned} & -2im \partial_t \psi - \nabla^2 \psi + 2eV m \psi + 2eiV \partial_t \psi + 2ie \vec{A} \cdot \vec{\nabla} \psi \\ & + ie \psi (\vec{\nabla} \cdot \vec{A}) + e^2 \vec{A}^2 \psi = 0. \end{aligned} \quad (2.68)$$

Considering the Lorenz gauge,

$$\begin{aligned} & \partial_\mu A^\mu = 0 \quad \rightarrow \quad \partial_t V + \vec{\nabla} \cdot \vec{A} = 0, \\ & i\partial_t \psi = \left(-\frac{\nabla^2}{2m} + eV + \frac{ei}{m} \vec{A} \cdot \vec{\nabla} + e^2 \vec{A}^2 \right) \psi. \end{aligned} \quad (2.69)$$

□

Problem 2.3 Prove that the current

$$j^\mu = -\frac{i}{2} (\phi \partial^\mu \phi^\dagger - \phi^\dagger \partial^\mu \phi) - q A^\mu \phi^\dagger \phi$$

satisfies

$$\partial_\mu j^\mu = 0,$$

where ϕ is a solution of the Klein–Gordon equation in an external electromagnetic potential A^μ .

Hint: Use the Lorenz gauge

$$\partial_\mu A^\mu = 0.$$

Solution.

$$\begin{aligned} \partial_\mu j^\mu &= -\frac{i}{2} \left[(\partial_\mu \phi) \partial^\mu \phi^\dagger + \phi \partial_\mu \partial^\mu \phi^\dagger - (\partial_\mu \phi^\dagger) \partial^\mu \phi - \phi^\dagger \partial_\mu \partial^\mu \phi \right] \\ &\quad - q (\partial_\mu A^\mu) \phi^\dagger \phi - q A^\mu (\partial_\mu \phi^\dagger) \phi - q A^\mu \phi^\dagger \partial_\mu \phi \\ &= -\frac{i}{2} \left[\phi \partial_\mu \partial^\mu \phi^\dagger - \phi^\dagger \partial_\mu \partial^\mu \phi \right] - q A^\mu (\partial_\mu \phi^\dagger) \phi - q A^\mu \phi^\dagger \partial_\mu \phi. \end{aligned} \quad (2.70)$$

Let's now write the Klein–Gordon equation with minimal coupling

$$\partial_\mu \rightarrow \partial_\mu + ie A_\mu.$$

$$[(\partial_r + iq A_r)(\partial^r + iq A^r) + m^2] \phi = 0. \quad (2.71)$$

$$\partial_r \partial^r \phi + iq \partial_r (A^r \phi) + iq A_r \partial^r \phi - q^2 A_r A^r \phi + m^2 \phi = 0. \quad (2.72)$$

$$\underbrace{iq \partial_r (A^r \phi)}_{0 + iq A^r \partial_r \phi}$$

$$\partial_r \partial^r \phi + 2iq A^r \partial_r \phi - q^2 A_r A^r \phi + m^2 \phi = 0. \quad (2.73)$$

Multiplying by $\phi^\dagger$ on the left,

$$\phi^\dagger \partial_r \partial^r \phi + 2iq \phi^\dagger A^r \partial_r \phi - q^2 \phi^\dagger A_r A^r \phi + m^2 \phi^\dagger \phi = 0. \quad (2.74)$$

Taking its complex conjugate,

$$\phi \partial_r \partial^r \phi^\dagger - 2iq \phi A^r \partial_r \phi^\dagger - q^2 \phi A_r A^r \phi^\dagger + m^2 \phi \phi^\dagger = 0. \quad (2.75)$$

Subtracting both equations,

$$\phi \partial_r \partial^r \phi^\dagger - \phi^\dagger \partial_r \partial^r \phi - 2iq A^r (\partial_r \phi^\dagger) \phi - 2iq A^r \phi^\dagger (\partial_r \phi) = 0. \quad (2.76)$$

Multiplying by $-\frac{1}{2}$,

$$-\frac{1}{2} (\phi \partial_r \partial^r \phi^\dagger - \phi^\dagger \partial_r \partial^r \phi) - q A^r [(\partial_r \phi^\dagger) \phi - \phi^\dagger (\partial_r \phi)] = 0. \quad (2.77)$$

$$\implies \partial_r j^r = 0.$$

□

3 Dirac Equation

In non relativistic quantum mechanics $\rho = |\psi|^2 > 0$. In relativistic quantum mechanics (Klein-Gordon) $\rho = \alpha \phi^* \frac{\partial \phi}{\partial t} - \phi \frac{\partial \phi^*}{\partial t}$ which can be both positive or negative.

Dirac tried to solve this problem by finding something covariant, that treats space and time equally and only contains first order derivative in time.

Dirac proposed the following equation

$$i\hbar \frac{\partial \psi}{\partial t} = \frac{\hbar c}{i} (\alpha_1 \frac{\partial \psi}{\partial x^1} + \alpha_2 \frac{\partial \psi}{\partial x^2} + \alpha_3 \frac{\partial \psi}{\partial x^3}) + \beta mc^2 \psi \quad (3.1)$$

$$i\hbar \frac{\partial \psi}{\partial t} = (c\vec{\alpha} \cdot \vec{p} + \beta mc^2) \psi = \hat{H} \psi \quad (3.2)$$

this needs to satisfy $E = \pm \sqrt{pc^2 + m^2 c^4}$.

$$\begin{aligned} H^2 &= (-i\hbar c \alpha_i \nabla_i + \beta mc^2)(-i\hbar c \alpha_j \nabla_j + \beta mc^2) \\ &= \alpha_i \alpha_j (-i\hbar c \nabla_i)(-i\hbar c \nabla_j) + mc^2 \alpha_i \beta (-i\hbar c \nabla_i) \\ &\quad + mc^2 \beta \alpha_j (-i\hbar c \nabla_j) + \beta^2 m^2 c^4 \end{aligned} \quad (3.3)$$

comparing with

$$H^2 = (-i\hbar c \vec{\nabla})^2 + m^2 c^4 \quad (3.4)$$

we find

$$\begin{cases} \alpha_i^2 = \beta^2 = 1 \\ \alpha_i \beta + \beta \alpha_i = 0 \\ \alpha_i \alpha_j + \alpha_j \alpha_i = 2\delta_{ij} \end{cases} \quad (3.5)$$

there is no number combination that satisfies this, which means that α_i, β are not numbers but matrices. Since the hamiltonian is hermitian $\rightarrow \alpha_i, \beta$ are hermitian matrices $\alpha_i^\dagger = \alpha_i, \beta^\dagger = \beta$.

$$\begin{aligned} \alpha_i \beta + \beta \alpha_i = 0 &\rightarrow \beta \alpha_i = -\alpha_i \beta \rightarrow \beta^2 \alpha_i = -\beta \alpha_i \beta \\ &\rightarrow \alpha_i = -\beta \alpha_i \beta \rightarrow Tr[\alpha_i] = -Tr[\beta \alpha_i \beta] \\ &\rightarrow -Tr[\beta^2 \alpha_i] = -Tr[\alpha_i] \rightarrow Tr[\alpha_i] = 0 \end{aligned} \quad (3.6)$$

Where we have used the cyclic property of the trace. By an analogous argument $Tr[\beta] = 0$.

Since $\alpha_i^2 = \beta^2 = \mathbf{1}$ the eigenvalues of α_i, β are ± 1 and since the matrices are traceless they must be even. In two dimensions only three matrices can fulfill this relations but we need 4 $\alpha_1, \alpha_2, \alpha_3, \beta$ therefore the smallest dimension allowed is 4×4 . A representation of the matrices is

$$\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}, \quad \beta = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}, \quad (3.7)$$

where σ_i are the 2×2 Pauli matrices and $\mathbf{1}_2$ is the 2×2 identity matrix.

The Pauli matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3.8)$$

Back to the original equation, we multiply everything by β/c :

$$\gamma^0 = \beta = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}, \quad \beta \alpha_i = \gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}. \quad (3.9)$$

The γ matrices are the Dirac matrices, where γ^μ has μ running from 0 to 3.

The Dirac matrices satisfy the anticommutation relation

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu} \mathbf{1}_4, \quad (3.10)$$

which defines the Clifford algebra.

The matrix γ^0 is Hermitian:

$$(\gamma^0)^\dagger = \beta^\dagger = \beta = \gamma^0. \quad (3.11)$$

The matrices γ^i are anti-Hermitian:

$$\begin{aligned} (\gamma^i)^\dagger &= (\beta\alpha_i)^\dagger \\ &= \alpha_i\beta \\ &= -\beta\alpha_i \\ &= -\gamma^i. \end{aligned} \quad (3.12)$$

Therefore,

$$\boxed{(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0}. \quad (3.13)$$

Writing back the Dirac equation

$$i\hbar\partial_t\psi = \beta mc^2 - i\hbar c\vec{\alpha} \cdot \vec{\nabla}\psi \quad (3.14)$$

$$i\hbar\frac{\partial}{\partial t}\psi + i\hbar c\vec{\alpha} \cdot \vec{\nabla}\psi - \beta mc^2 = 0 \quad (3.15)$$

multiplying by $\beta = \gamma^2$ and setting $c = 1$

$$i\hbar\gamma^0\frac{\partial\psi}{\partial t} + i\hbar\gamma^i\frac{\partial\psi}{\partial x^i} - \beta^2 m\psi = 0 \quad (3.16)$$

$$i\hbar\gamma^\mu\frac{\partial\psi}{\partial x^\mu} - m\psi = 0 \quad (3.17)$$

$$(i\hbar\gamma^\mu\partial_\mu - m)\psi = 0 \quad (3.18)$$

defining the Feynman dagger $\not{\partial}$

$$\boxed{(i\hbar\not{\partial} - mc)\psi = 0} \quad (3.19)$$

which is the most compact way to write Dirac equation. In general slash notation $\gamma^\mu a_\mu = \not{a} = g_{\mu\sigma}\gamma^\mu a^\sigma = \gamma^0 a^0 - \gamma^i a^i$. Fully expanding Dirac equation

$$\begin{aligned} &\left[i\hbar \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \frac{\partial}{\partial t} + i\hbar \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \frac{\partial}{\partial x} \right. \\ &\quad \left. + i\hbar \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \frac{\partial}{\partial y} + i\hbar \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \frac{\partial}{\partial z} - m \right] \psi(t, \vec{x}) = 0. \end{aligned} \quad (3.20)$$

$\psi(t, \vec{x})$ is a four-component wave function:

$$\psi(t, \vec{x}) = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} \longrightarrow \text{spinor}. \quad (3.21)$$

3.1 Conserved Charges and Current

Now we want to construc conserved quantities from Dirac's equation $\partial_\mu j^\mu$, $j^0 = \rho > 0$, we will proceed in an analogous fashion to the Klein-Gordon conserved current.

For now let's consider the usual adjoint $\psi^\dagger = (\psi_1^*, \psi_2^*, \psi_3^*, \psi_4^*)$ and multiply on the left side of Dirac's equation by $\psi^\dagger$

$$0 = \psi^\dagger(i\hbar\gamma^\mu\partial_\mu\psi - m\psi) = i\hbar\psi^\dagger\gamma^\mu\partial_\mu\psi - m\psi^\dagger\psi \quad (3.22)$$

and multiplying the hermitian conjugate of Dirac's equation by ψ

$$0 = (i\hbar\gamma^\mu\partial_\mu\psi - m\psi)^\dagger\psi = (-i\hbar(\partial_\mu\psi^\dagger)\gamma^{\mu\dagger} - m\psi^\dagger) = -i\hbar(\partial_\mu\psi^\dagger)\gamma^{\mu\dagger} - m\psi^\dagger\psi \quad (3.23)$$

subtracting both equations

$$\psi^\dagger \gamma^\mu \partial_\mu \psi^\dagger (\partial_\mu \psi^\dagger) \gamma^{\mu\dagger} \psi = 0 \quad (3.24)$$

which can be written as $\partial_\mu j^\mu$ if and only if $\gamma^{\mu\dagger} = \gamma^\mu$ but we know that $\gamma^{\mu\dagger} = \gamma^0 \gamma^\mu \gamma^0 \rightarrow \gamma^{\mu\dagger} \neq \gamma^\mu$. In order to find the conserved current we need to define the adjoint spinor

$$\bar{\psi} = \psi^\dagger \gamma^0 = (\psi_1^*, \psi_2^*, -\psi_3^*, -\psi_4^*). \quad (3.25)$$

Repeating the process, but instead of multiplying by $\psi^\dagger$ we multiply by $\bar{\psi}$. Multiplying the hermitian conjugate by $\gamma^\mu \psi$ on the right

$$0 = -i\hbar(\partial_\mu \psi^\dagger) \gamma^0 \gamma^0 \gamma^{\mu\dagger} \gamma^0 \psi - m \psi^\dagger \gamma^0 \psi = -i\hbar(\partial_\mu \bar{\psi}) \gamma^0 \gamma^{\mu\dagger} \gamma^0 \psi - m \bar{\psi} \psi \quad (3.26)$$

and multiplying the Dirac equation by $\bar{\psi}$ on the left

$$i\hbar \bar{\psi} \gamma^\mu (\partial_\mu \psi) - \bar{\psi} \psi = 0 \quad (3.27)$$

subtracting both equations

$$0 = \bar{\psi} \gamma^\mu (\partial_\mu \psi) + (\partial_\mu \bar{\psi}) \gamma^\mu \psi = 0 \quad (3.28)$$

$$\partial_\mu (\bar{\psi} \gamma^\mu \psi) = \partial_\mu j^\mu = 0 \quad (3.29)$$

looking at the zeroth component of j^μ

$$j^0 = \rho = \bar{\psi} \gamma^0 \psi = \psi^\dagger \psi = |\psi_1|^2 + |\psi_2|^2 + |\psi_3|^2 + |\psi_4|^2 \quad (3.30)$$

which is the result we intended.

3.2 Solutions of the Dirac Equation

$$(i\gamma^\mu \partial_\mu - m) \psi = 0. \quad (3.31)$$

Dirac's equation comes from satisfying

$$E^2 = \vec{p}^2 + m^2. \quad (3.32)$$

We are going to assume that

$$\psi(x) = u(p) e^{-ipx}, \quad (3.33)$$

where $u(p)$ is a spinor and e^{-ipx} is a plane wave.

The quantity $u(p)$ is a four-component spinor, independent of x .

Taking the particle at rest, $\vec{p} = 0$,

$$Hu = (\vec{\alpha} \cdot \vec{p} + \beta m) u = Eu. \quad (3.34)$$

Therefore,

$$Hu = \beta m u(p) = \begin{pmatrix} m \mathbf{1}_2 & 0 \\ 0 & -m \mathbf{1}_2 \end{pmatrix} u(p). \quad (3.35)$$

The eigenvalues are

$$m, \quad m, \quad -m, \quad -m, \quad (3.36)$$

and the corresponding eigenvectors are

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (3.37)$$

For $\vec{p} \neq 0$,

$$Hu = \begin{pmatrix} m & \vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & -m \end{pmatrix} \begin{pmatrix} u_A \\ u_B \end{pmatrix} = E \begin{pmatrix} u_A \\ u_B \end{pmatrix}, \quad (3.38)$$

where u_A and u_B are two-component spinors.

Therefore,

$$(\vec{\sigma} \cdot \vec{p}) u_B = (E - m) u_A, \quad (3.39)$$

$$(\vec{\sigma} \cdot \vec{p}) u_A = (E + m) u_B. \quad (3.40)$$

For the $E > 0$ solutions,

$$u_A^{(s)} = \chi^{(s)}, \quad s = 1, 2, \quad (3.41)$$

where

$$\chi^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3.42)$$

Hence,

$$u_B^{(s)} = \frac{\vec{\sigma} \cdot \vec{p}}{E + m} \chi^{(s)}, \quad (3.43)$$

and

$$u^{(s)} = N \begin{pmatrix} \chi^{(s)} \\ \frac{\vec{\sigma} \cdot \vec{p}}{E + m} \chi^{(s)} \end{pmatrix}, \quad s = 1, 2, \quad E > 0. \quad (3.44)$$

We still need to find two more solutions for negative energy.

For the $E < 0$ solutions,

$$u_B^{(s)} = \chi^{(s)}, \quad s = 1, 2. \quad (3.45)$$

Therefore,

$$u_A^{(s)} = \frac{\vec{\sigma} \cdot \vec{p}}{E - m} u_B^{(s)} = -\frac{\vec{\sigma} \cdot \vec{p}}{|E| + m} \chi^{(s)}, \quad (3.46)$$

and

$$u^{(s+2)} = N \begin{pmatrix} \frac{\vec{\sigma} \cdot \vec{p}}{E - m} \chi^{(s)} \\ \chi^{(s)} \end{pmatrix}, \quad s = 1, 2, \quad E < 0. \quad (3.47)$$

Explicitly, the positive-energy solutions are

$$\psi^{(1)}(x) = N \begin{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \frac{\vec{\sigma} \cdot \vec{p}}{E + m} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} e^{-ipx}, \quad (3.48)$$

$$\psi^{(2)}(x) = N \begin{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \frac{\vec{\sigma} \cdot \vec{p}}{E + m} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} e^{-ipx}. \quad (3.49)$$

The negative-energy solutions are

$$\psi^{(3)}(x) = N \begin{pmatrix} \frac{\vec{\sigma} \cdot \vec{p}}{E - m} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} e^{-ipx}, \quad (3.50)$$

$$\psi^{(4)}(x) = N \begin{pmatrix} \frac{\vec{\sigma} \cdot \vec{p}}{E - m} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} e^{-ipx}. \quad (3.51)$$

There exists no solution of the form

$$\begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}. \quad (3.52)$$

We want to determine the normalization.

The first two solutions,

$$u^{(1,2)}(\vec{p})e^{-ipx}, \quad (3.53)$$

describe a free electron of energy E and momentum $\vec{p}$.

The spinors $u^{(3,4)}$ correspond to antiparticles (positrons).

A positron with energy E and momentum $\vec{p}$ is described by an electron solution with energy $-E$ and momentum $-\vec{p}$,

$$u^{(3,4)}(-p)e^{-i(-p)x} \equiv v^{(2,1)}(\vec{p})e^{-ipx}, \quad (3.54)$$

where

$$p^0 \equiv E > 0. \quad (3.55)$$

The Dirac equation for $u(p)$ is

$$(\not{p} - m)u(p) = 0, \quad p^0 \equiv E > 0. \quad (3.56)$$

For $v(p)$,

$$(-\not{p} - m)u(-p) = 0. \quad (3.57)$$

Therefore,

$$\boxed{(-\not{p} + m)v(p) = 0}, \quad p^0 \equiv E > 0. \quad (3.58)$$

3.3 Normalization

We begin by interpreting j^0 as a probability density.

In a unit volume,

$$\int \rho dV = \int \psi^\dagger \psi dV = u^\dagger u = 2E. \quad (3.59)$$

Since the normalization is arbitrary, in physics we are free to choose the most convenient one. We choose the normalization

$$u^\dagger u = 2E, \quad (3.60)$$

which represents $2E$ particles per unit volume.

With this choice of normalization, the orthogonality relations are

$$u^{(r)\dagger} u^{(s)} = 2E \delta_{rs}, \quad r, s = 1, 2, \quad (3.61)$$

and

$$v^{(r)\dagger} v^{(s)} = 2E \delta_{rs}, \quad r, s = 1, 2. \quad (3.62)$$

Using the positive-energy Dirac solutions,

$$u^{(s)} = N \begin{pmatrix} \chi^{(s)} \\ \frac{\vec{\sigma} \cdot \vec{p}}{E + m} \chi^{(s)} \end{pmatrix}, \quad (3.63)$$

we obtain

$$u^{(s)\dagger} u^{(s)} = |N|^2 \left[1 + \frac{(\vec{\sigma} \cdot \vec{p})^2}{(E + m)^2} \right] = |N|^2 \frac{2E}{E + m} = 2E. \quad (3.64)$$

Therefore,

$$\boxed{N = \sqrt{E + m}}. \quad (3.65)$$

The same normalization is obtained for $v(p)$.

In the calculation we have used

$$(\vec{\sigma} \cdot \vec{p})^2 = |\vec{p}|^2. \quad (3.66)$$

The normalized solutions are

$$\psi^{(1,2)}(x) = \sqrt{E + m} \begin{pmatrix} \chi^{(1,2)} \\ \frac{\vec{\sigma} \cdot \vec{p}}{E + m} \chi^{(1,2)} \end{pmatrix} e^{-ipx}, \quad (3.67)$$

corresponding to particle spinors $u(p)$,

and

$$\psi^{(3,4)}(x) = \sqrt{E+m} \begin{pmatrix} \vec{\sigma} \cdot \vec{p} \\ E+m \\ \chi^{(1,2)} \end{pmatrix} e^{ipx}, \quad (3.68)$$

corresponding to antiparticle spinors $v(p)$.

These form a basis in spinor space.

We define the Dirac adjoint as

$$\bar{u} = u^\dagger \gamma^0. \quad (3.69)$$

Starting from

$$(\not{p} - m)u(p) = 0, \quad (3.70)$$

taking the Hermitian conjugate gives

$$u^\dagger \gamma^{\mu\dagger} p_\mu - m u^\dagger = 0. \quad (3.71)$$

Multiplying on the right by γ^0 and using

$$\gamma^{\mu\dagger} = \gamma^0 \gamma^\mu \gamma^0, \quad (3.72)$$

we obtain

$$\boxed{\bar{u}(p)(\not{p} - m) = 0.} \quad (3.73)$$

Proceeding similarly,

$$\boxed{\bar{v}(p)(\not{p} + m) = 0.} \quad (3.74)$$

It is then easy to prove that

$$\bar{u}^{(s)}(p)u^{(s)}(p) = 2m, \quad \bar{v}^{(s)}(p)v^{(s)}(p) = -2m, \quad (3.75)$$

and the completeness relations

$$\sum_{s=1}^2 u^{(s)} \bar{u}^{(s)} = \not{p} + m, \quad (3.76)$$

$$\sum_{s=1}^2 v^{(s)} \bar{v}^{(s)} = \not{p} - m. \quad (3.77)$$

3.4 Spin of Dirac Particle

Particles that satisfy the Dirac equation carry intrinsic angular momentum.

Therefore,

$$\text{spin} = \frac{1}{2}. \quad (3.78)$$

This result appears naturally from the mathematics and does not need to be imposed.

Let us start by defining the generalized Pauli matrices

$$\Sigma_i = \begin{pmatrix} \sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix}, \quad i = 1, 2, 3. \quad (3.79)$$

They are related to the Dirac matrices α_i through

$$\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} = \rho \Sigma_i = \Sigma_i \rho, \quad (3.80)$$

where

$$\rho \equiv \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}. \quad (3.81)$$

Since

$$\rho^2 = \mathbf{1}_4, \quad (3.82)$$

the relation can be inverted:

$$\Sigma_i = \rho \alpha_i = \alpha_i \rho. \quad (3.83)$$

The generalized Pauli matrices satisfy the same commutation relations as the Pauli matrices:

$$[\Sigma_i, \Sigma_j] = \begin{pmatrix} [\sigma_i, \sigma_j] & 0 \\ 0 & [\sigma_i, \sigma_j] \end{pmatrix} \quad (3.84)$$

$$= \begin{pmatrix} 2i\epsilon_{ijk}\sigma_k & 0 \\ 0 & 2i\epsilon_{ijk}\sigma_k \end{pmatrix} \quad (3.85)$$

$$= 2i\epsilon_{ijk}\Sigma_k. \quad (3.86)$$

This implies that

$$\frac{1}{2}\Sigma_i \quad (3.87)$$

satisfies the angular momentum algebra.

3.5 Commutation Relations

The Dirac Hamiltonian is

$$H = \vec{\alpha} \cdot \vec{p} + \beta m = \alpha_i p_i + \beta m, \quad \beta = \gamma^0. \quad (3.88)$$

We will need the relations

$$[\Sigma_i, \alpha_j] = 2i\epsilon_{ijk}\alpha_k \quad (3.89)$$

and

$$[\Sigma_i, \gamma^0] = 0. \quad (3.90)$$

Let us check whether the Dirac Hamiltonian commutes with the orbital angular momentum

$$\vec{L} = \vec{r} \times \vec{p}. \quad (3.91)$$

Its components are

$$L_i = \epsilon_{ijk} r_j p_k, \quad i, j, k = 1, 2, 3. \quad (3.92)$$

Then,

$$\begin{aligned} [L_i, H] &= [\epsilon_{ijk} r_j p_k, \alpha_\ell p_\ell + \beta m] \\ &= i\epsilon_{ijk} \alpha_j p_k. \end{aligned} \quad (3.93)$$

Therefore,

$$\vec{L} \text{ is not a conserved quantity.} \quad (3.94)$$

Let us now check whether the generalized Pauli matrices commute with the Hamiltonian:

$$\begin{aligned} [\Sigma_i, H] &= [\Sigma_i, \alpha_j p_j + \beta m] \\ &= [\Sigma_i, \alpha_j] p_j + [\Sigma_i, \beta] m \\ &= 2i\epsilon_{ijk} \alpha_k p_j \\ &= -2i\epsilon_{ijk} \alpha_j p_k. \end{aligned} \quad (3.95)$$

Thus, $\vec{\Sigma}$ does not commute with the Hamiltonian either.

The total angular momentum that commutes with H is

$$\vec{J} = \vec{L} + \frac{1}{2}\vec{\Sigma}. \quad (3.96)$$

This allows us to identify the spin angular momentum operator as

$$S_i = \frac{1}{2}\Sigma_i. \quad (3.97)$$

In particular,

$$S_3 = \frac{1}{2}\Sigma_3 = \frac{1}{2} \begin{pmatrix} \sigma_3 & 0 \\ 0 & \sigma_3 \end{pmatrix}. \quad (3.98)$$

This operator has doubly degenerate eigenvalues

$$\pm \frac{1}{2}. \quad (3.99)$$

Therefore, Dirac particles carry intrinsic spin

$$s = \frac{1}{2}. \quad (3.100)$$

Moreover,

$$\begin{aligned} \left| \frac{1}{2} \vec{\Sigma} \right|^2 &= \frac{1}{4} (\Sigma_1^2 + \Sigma_2^2 + \Sigma_3^2) \\ &= \frac{3}{4} = s(s+1), \quad s = \frac{1}{2}. \end{aligned} \quad (3.101)$$

3.6 Non-Relativistic Limit of Dirac Equation

For the positive-energy solutions,

$$\psi^{(+)}(x) = \sqrt{E+m} \begin{pmatrix} \chi^{(s)} \\ \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \chi^{(s)} \end{pmatrix} e^{-ipx} = \begin{pmatrix} u_L(p) \\ u_S(p) \end{pmatrix} e^{-ipx}, \quad (3.102)$$

where L denotes the large component and S the small component.

For the negative-energy solutions,

$$\psi^{(-)}(x) = \sqrt{E+m} \begin{pmatrix} \frac{\vec{\sigma} \cdot \vec{p}}{E-m} \chi^{(s')} \\ \chi^{(s')} \end{pmatrix} e^{ipx} = \begin{pmatrix} v_S(p) \\ v_L(p) \end{pmatrix} e^{ipx}. \quad (3.103)$$

The two-component spinors are

$$\chi^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (3.104)$$

and

$$\chi^{(1)'} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \chi^{(2)'} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (3.105)$$

For example,

$$u_L(p) = \sqrt{E+m} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad v_L(p) = \sqrt{E+m} \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3.106)$$

The small and large components are related by

$$u_S(p) = \frac{\vec{\sigma} \cdot \vec{p}}{E+m} u_L(p), \quad (3.107)$$

and

$$v_S(p) = \frac{\vec{\sigma} \cdot \vec{p}}{E-m} v_L(p). \quad (3.108)$$

In the non-relativistic limit,

$$|\vec{p}| \ll m, \quad (3.109)$$

and therefore

$$u_S(p) \ll u_L(p), \quad v_S(p) \ll v_L(p). \quad (3.110)$$

For a positive-energy solution,

$$(\vec{\alpha} \cdot \vec{p} + \beta m) \psi^{(+)}(p) = E \psi^{(+)}(p). \quad (3.111)$$

In terms of the large and small components,

$$\begin{pmatrix} m\mathbf{1}_2 & \vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & -m\mathbf{1}_2 \end{pmatrix} \begin{pmatrix} u_L(p) \\ u_S(p) \end{pmatrix} = E \begin{pmatrix} u_L(p) \\ u_S(p) \end{pmatrix}. \quad (3.112)$$

Thus,

$$\vec{\sigma} \cdot \vec{p} u_S(p) = (E-m) u_L(p), \quad (3.113)$$

$$\vec{\sigma} \cdot \vec{p} u_L(p) = (E+m) u_S(p). \quad (3.114)$$

Therefore,

$$u_S(p) = \frac{\vec{\sigma} \cdot \vec{p}}{E + m} u_L(p). \quad (3.115)$$

Substituting this expression into the first equation,

$$(\vec{\sigma} \cdot \vec{p}) \frac{\vec{\sigma} \cdot \vec{p}}{E + m} u_L(p) = (E - m) u_L(p). \quad (3.116)$$

In the non-relativistic limit,

$$E + m \simeq 2m, \quad (3.117)$$

so that

$$\frac{\vec{p}^2}{2m} u_L(p) \simeq (E - m) u_L(p). \quad (3.118)$$

Defining the non-relativistic energy as

$$E_{\text{NR}} = E - m, \quad (3.119)$$

we obtain

$$\boxed{\frac{\vec{p}^2}{2m} u_L(p) = E_{\text{NR}} u_L(p)}. \quad (3.120)$$

This is the Schrödinger equation for a two-component spinor.

3.7 Non-Relativistic Limit in an External Field

We introduce the electromagnetic interaction through minimal coupling:

$$p_\mu \longrightarrow p_\mu - eA_\mu, \quad A_\mu = (A_0, \vec{A}), \quad (3.121)$$

or equivalently,

$$\partial_\mu \longrightarrow \partial_\mu + ieA_\mu. \quad (3.122)$$

Setting

$$A_0 = 0, \quad \vec{B} = \vec{\nabla} \times \vec{A}, \quad \vec{A} = \vec{A}(x), \quad (3.123)$$

the Dirac equation for a positive-energy solution becomes

$$\left[\vec{\alpha} \cdot (\vec{p} - e\vec{A}) + \beta m \right] u(p) = E u(p). \quad (3.124)$$

In matrix form,

$$\begin{pmatrix} m\mathbf{1}_2 & \vec{\sigma} \cdot (\vec{p} - e\vec{A}) \\ \vec{\sigma} \cdot (\vec{p} - e\vec{A}) & -m\mathbf{1}_2 \end{pmatrix} \begin{pmatrix} u_L(p) \\ u_S(p) \end{pmatrix} = E \begin{pmatrix} u_L(p) \\ u_S(p) \end{pmatrix}. \quad (3.125)$$

Therefore,

$$\vec{\sigma} \cdot (\vec{p} - e\vec{A}) u_S(p) = (E - m) u_L(p), \quad (3.126)$$

$$\vec{\sigma} \cdot (\vec{p} - e\vec{A}) u_L(p) = (E + m) u_S(p). \quad (3.127)$$

Hence,

$$u_S(p) = \frac{\vec{\sigma} \cdot (\vec{p} - e\vec{A})}{E + m} u_L(p) \simeq \frac{\vec{\sigma} \cdot (\vec{p} - e\vec{A})}{2m} u_L(p), \quad (3.128)$$

where we have used

$$|\vec{p}| \ll m \quad \implies \quad E \simeq m. \quad (3.129)$$

Substituting into the equation for the large component,

$$\left[\vec{\sigma} \cdot (\vec{p} - e\vec{A}) \right] \frac{\vec{\sigma} \cdot (\vec{p} - e\vec{A})}{2m} u_L(p) \simeq (E - m) u_L(p). \quad (3.130)$$

We use the identity

$$(\vec{\sigma} \cdot \vec{a})(\vec{\sigma} \cdot \vec{b}) = \vec{a} \cdot \vec{b} \mathbf{1}_2 + i\vec{\sigma} \cdot (\vec{a} \times \vec{b}). \quad (3.131)$$

Therefore,

$$\begin{aligned} & \left[\vec{\sigma} \cdot (\vec{p} - e\vec{A}) \right] \left[\vec{\sigma} \cdot (\vec{p} - e\vec{A}) \right] \\ &= (\vec{p} - e\vec{A})^2 - ie\vec{\sigma} \cdot (\vec{p} \times \vec{A} + \vec{A} \times \vec{p}). \end{aligned} \quad (3.132)$$

Considering the last term,

$$(\vec{p} \times \vec{A} + \vec{A} \times \vec{p})_i = \epsilon_{ijk} (p_j A_k + A_j p_k) \quad (3.133)$$

$$= \epsilon_{ijk} (p_j A_k - A_k p_j) \quad (3.134)$$

$$= \epsilon_{ijk} [p_j, A_k] \quad (3.135)$$

$$= -i\epsilon_{ijk} [\nabla_j, A_k] \quad (3.136)$$

$$= -i(\vec{\nabla} \times \vec{A})_i \quad (3.137)$$

$$= -iB_i. \quad (3.138)$$

Thus,

$$\left[\vec{\sigma} \cdot (\vec{p} - e\vec{A}) \right]^2 = (\vec{p} - e\vec{A})^2 - e\vec{\sigma} \cdot \vec{B}. \quad (3.139)$$

Substituting this result back into the equation,

$$\frac{1}{2m} \left[(\vec{p} - e\vec{A})^2 - e\vec{\sigma} \cdot \vec{B} \right] u_L(p) = (E - m)u_L(p). \quad (3.140)$$

Therefore,

$$Hu_L(p) = E_{\text{NR}}u_L(p). \quad (3.141)$$

The magnetic moment operator can be identified as

$$\vec{\mu} = \frac{e}{2m} \vec{\sigma}. \quad (3.142)$$

For any particle, the magnetic moment is defined by

$$\vec{\mu} = \frac{eg}{2m} \vec{S}, \quad (3.143)$$

where g is the gyromagnetic ratio.

Since

$$\vec{S} = \frac{1}{2} \vec{\sigma}, \quad (3.144)$$

comparison of both expressions gives

$$\boxed{g = 2}. \quad (3.145)$$

Thus, the electron gyromagnetic ratio $g = 2$ is obtained directly from the Dirac equation.

3.8 Clifford Algebra

The gamma matrices satisfy the anticommutation relation

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \quad \mu, \nu = 0, 1, 2, 3. \quad (3.146)$$

The spatial matrices are anti-Hermitian,

$$(\gamma^i)^\dagger = -\gamma^i, \quad (3.147)$$

whereas

$$(\gamma^0)^\dagger = \gamma^0. \quad (3.148)$$

Therefore,

$$(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0. \quad (3.149)$$

We define

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3 = \frac{i}{4!}\epsilon_{\mu\nu\rho\sigma}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma. \quad (3.150)$$

The most common (Dirac) representation of the gamma matrices is

$$\gamma^0 = \begin{pmatrix} \mathbf{1}_2 & 0 \\ 0 & -\mathbf{1}_2 \end{pmatrix}, \quad (3.151)$$

$$\gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}, \quad (3.152)$$

and

$$\gamma^5 = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}. \quad (3.153)$$

In four dimensions there are sixteen independent combinations of gamma matrices that form a basis:

- Scalar:

$$\mathbf{1}.$$

- Vector:

$$\gamma^\mu = \{\gamma^0, \gamma^1, \gamma^2, \gamma^3\}.$$

- Tensor:

$$\sigma^{\mu\nu} = \gamma^0\gamma^1, \gamma^0\gamma^2, \gamma^0\gamma^3, \gamma^1\gamma^2, \gamma^1\gamma^3, \gamma^2\gamma^3.$$

- Pseudovector:

$$\gamma^\mu\gamma^5.$$

- Pseudoscalar:

$$\gamma^5.$$

These matrices satisfy the relations

$$(\Gamma^A)^2 = \mathbf{1}, \quad (3.154)$$

$$\text{Tr}(\Gamma^A) = 0, \quad (3.155)$$

$$\Gamma^A\Gamma^B = \lambda\Gamma^C, \quad (3.156)$$

for suitable λ and Γ^C .

Moreover, every 4×4 matrix can be expanded as

$$X = \frac{1}{4} \sum_A \text{Tr}(X\Gamma^A)\Gamma^A. \quad (3.157)$$

The matrix γ^5 anticommutes with every gamma matrix,

$$\{\gamma^\mu, \gamma^5\} = 0. \quad (3.158)$$

Other useful properties are

$$(\gamma^5)^\dagger = \gamma^5, \quad (\gamma^5)^2 = \mathbf{1}. \quad (3.159)$$

Now let us consider some useful traces involving Dirac matrices.

Using

$$AA = A^2, \quad AB = 2A \cdot B - BA, \quad (3.160)$$

one finds

$$\text{Tr}(A) = 0, \quad (3.161)$$

$$\text{Tr}(ABA_{2n+1}) = 0, \quad (3.162)$$

$$\text{Tr}(AB) = 4 A \cdot B, \quad (3.163)$$

which implies

$$\text{Tr}(\gamma^\mu \gamma^\nu) = 4 \eta^{\mu\nu}. \quad (3.164)$$

Similarly,

$$\text{Tr}(ABCD) = 4 (A \cdot B C \cdot D - A \cdot C B \cdot D + A \cdot D B \cdot C), \quad (3.165)$$

or, in terms of gamma matrices,

$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4 (\eta^{\mu\nu} \eta^{\rho\sigma} - \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho}). \quad (3.166)$$

The remaining useful traces are

$$\text{Tr}(\gamma^5) = 0, \quad (3.167)$$

$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^5) = 0, \quad (3.168)$$

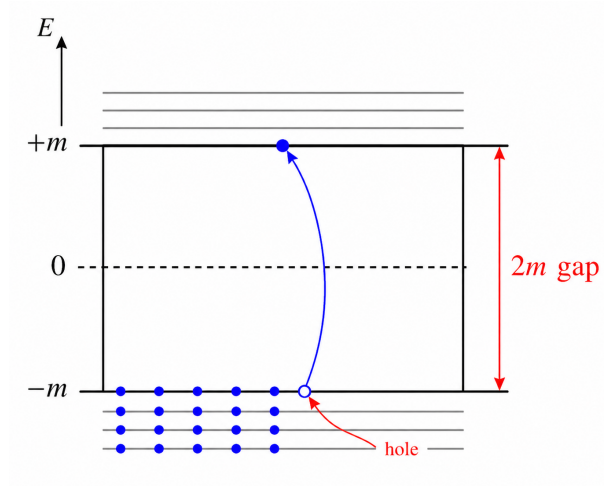
$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^5) = 0, \quad (3.169)$$

and

$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma^5) = 4i \epsilon^{\mu\nu\rho\sigma}. \quad (3.170)$$

3.9 Dirac's Hole Theory

Let's find an interpretation for the negative energy solutions.



$$p^0 = E_{\pm} = \pm \sqrt{|\vec{p}|^2 + m^2} \quad (3.171)$$

Negative E electrons would imply that atoms are not stable.

The solution is to reinterpret the vacuum. All negative energy states are filled with electrons $\rightarrow$ ground state contains infinitely many particles. Electrons in the ground state are passive, no charge, no mass.

This allows processes such as

$$\gamma \rightarrow e^+ e^- \quad (3.172)$$

with $E_\gamma \geq 2m$

3.10 Lorentz covariance of Dirac equation

The Dirac equation is

$$(\not{p} - m)\psi = 0. \quad (3.173)$$

How does the Dirac equation transform under Lorentz transformations, such as boosts and rotations?
In quantum mechanics,

$$U = e^{i\vec{\theta} \cdot \vec{J}}, \quad (3.174)$$

where $\vec{\theta}$ represents a rotation and $\vec{J}$ is the generator of rotations.

In relativistic quantum mechanics, we introduce

$$S(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\sigma^{\mu\nu}\right), \quad (3.175)$$

where Λ represents a Lorentz transformation.

For example, under a three-dimensional rotation around the z -axis,

$$\vec{x} \longrightarrow \vec{x}' = R\vec{x}, \quad (3.176)$$

with

$$\begin{pmatrix} x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^1 \\ x^2 \\ x^3 \end{pmatrix}. \quad (3.177)$$

Under a four-dimensional boost along the x -axis,

$$\begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}. \quad (3.178)$$

Equivalently,

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu, \quad (3.179)$$

where $\Lambda^\mu{}_\nu$ represents a Lorentz transformation and

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}. \quad (3.180)$$

We know that lowering and raising the indices of the Lorentz transformation gives

$$\Lambda_\mu{}^\nu = \eta_{\mu\rho}\eta^{\rho\sigma}\Lambda^\sigma{}_\mu. \quad (3.181)$$

Lorentz transformations preserve the Minkowski metric:

$$\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \eta_{\mu\nu} = \eta_{\rho\sigma}. \quad (3.182)$$

Equivalently,

$$(\Lambda^{-1})^\mu{}_\nu \Lambda^\nu{}_\rho = \delta^\mu{}_\rho. \quad (3.183)$$

We also need to know how the derivatives transform:

$$\begin{aligned} \partial'_\mu &= \frac{\partial}{\partial x'^\mu} = \frac{\partial x^\nu}{\partial x'^\mu} \frac{\partial}{\partial x^\nu} \\ &= (\Lambda^{-1})^\nu{}_\mu \partial_\nu. \end{aligned} \quad (3.184)$$

What about spinors?

Under a Lorentz transformation,

$$\psi(x) \longrightarrow \psi'(x') = S(\Lambda)\psi(x). \quad (3.185)$$

The matrix $S(\Lambda)$ represents the action of the Lorentz transformation in spinor space. It is a 4×4 matrix.

Lorentz covariance means that

$$(i\gamma^\mu \partial'_\mu - m)\psi'(x') = 0 \quad (3.186)$$

must imply

$$(i\gamma^\mu \partial_\mu - m) \psi(x) = 0. \quad (3.187)$$

Since

$$\partial'_\mu = (\Lambda^{-1})^\nu{}_\mu \partial_\nu, \quad (3.188)$$

we have

$$\gamma^\mu \partial'_\mu = \gamma^\mu (\Lambda^{-1})^\nu{}_\mu \partial_\nu. \quad (3.189)$$

Multiplying the Dirac equation on the left by $S^{-1}(\Lambda)$,

$$S^{-1} (i\gamma^\mu \partial'_\mu - m) S \psi(x) = 0. \quad (3.190)$$

Since $S^{-1}S = \mathbf{1}$,

$$[iS^{-1}\gamma^\mu S(\Lambda^{-1})^\nu{}_\mu \partial_\nu - m] \psi(x) = 0. \quad (3.191)$$

Therefore, the Dirac equation is Lorentz covariant if

$$\boxed{S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu \gamma^\nu}. \quad (3.192)$$

If we prove that, for every Lorentz transformation $\Lambda^\mu{}_\nu$, there exists a matrix $S(\Lambda)$ satisfying this relation, then the Lorentz covariance of the Dirac equation is proven.

The Lorentz transformations satisfy

$$\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \eta_{\mu\nu} = \eta_{\rho\sigma}. \quad (3.193)$$

Taking determinants,

$$\det(\Lambda)^2 = 1, \quad (3.194)$$

and therefore

$$\det(\Lambda) = \pm 1. \quad (3.195)$$

If

$$\det(\Lambda) = 1, \quad (3.196)$$

the transformation is a proper Lorentz transformation, such as a rotation or a boost.

If

$$\det(\Lambda) = -1, \quad (3.197)$$

the transformation is an improper Lorentz transformation, such as time reversal or space inversion.

Consider an infinitesimal Lorentz transformation,

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \delta\omega^\mu{}_\nu, \quad (3.198)$$

where $\delta\omega^\mu{}_\nu$ is infinitesimal.

Using the invariance of the metric,

$$\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \eta_{\mu\nu} = \eta_{\rho\sigma}, \quad (3.199)$$

we find, to first order,

$$\begin{aligned} (\delta^\mu{}_\rho + \delta\omega^\mu{}_\rho) (\delta^\nu{}_\sigma + \delta\omega^\nu{}_\sigma) \eta_{\mu\nu} &= \eta_{\rho\sigma}, \\ \eta_{\rho\sigma} + \delta\omega_{\sigma\rho} + \delta\omega_{\rho\sigma} &= \eta_{\rho\sigma}. \end{aligned} \quad (3.200)$$

Therefore,

$$\boxed{\delta\omega_{\rho\sigma} = -\delta\omega_{\sigma\rho}}. \quad (3.201)$$

Thus, the infinitesimal Lorentz parameters are antisymmetric.

We now expand $S(\Lambda)$ in powers of $\delta\omega^{\mu\nu}$:

$$S(\Lambda) = S(\delta\omega^{\mu\nu}). \quad (3.202)$$

By hand-waving, we propose

$$S = \mathbf{1} - \frac{i}{4} \sigma_{\mu\nu} \delta\omega^{\mu\nu}, \quad (3.203)$$

where both $\sigma_{\mu\nu}$ and $\delta\omega^{\mu\nu}$ are antisymmetric.

Its inverse, to first order, is

$$S^{-1} = \mathbf{1} + \frac{i}{4} \sigma_{\mu\nu} \delta\omega^{\mu\nu}. \quad (3.204)$$

Substituting the infinitesimal transformations into

$$S^{-1} \gamma^\rho S = \Lambda^\rho{}_\nu \gamma^\nu, \quad (3.205)$$

we obtain

$$(\delta^\rho{}_\nu + \delta\omega^\rho{}_\nu) \gamma^\nu = \left(\mathbf{1} + \frac{i}{4} \sigma_{\alpha\beta} \delta\omega^{\alpha\beta} \right) \gamma^\rho \left(\mathbf{1} - \frac{i}{4} \sigma_{\alpha\beta} \delta\omega^{\alpha\beta} \right). \quad (3.206)$$

Keeping only first-order terms,

$$\delta\omega^\rho{}_\nu \gamma^\nu = -\frac{i}{4} \delta\omega^{\alpha\beta} [\gamma^\rho, \sigma_{\alpha\beta}]. \quad (3.207)$$

Using the antisymmetry of $\delta\omega^{\alpha\beta}$,

$$\begin{aligned} \delta\omega^\rho{}_\nu \gamma^\nu &= \frac{1}{2} (\delta\omega^\rho{}_\alpha \gamma^\alpha + \delta\omega^\rho{}_\beta \gamma^\beta) \\ &= -\frac{i}{4} \delta\omega^{\alpha\beta} 2i (\eta^\rho{}_\alpha \gamma_\beta - \eta^\rho{}_\beta \gamma_\alpha). \end{aligned} \quad (3.208)$$

Therefore, $\sigma_{\alpha\beta}$ must satisfy

$$\boxed{[\gamma^\rho, \sigma_{\alpha\beta}] = 2i (\eta^\rho{}_\alpha \gamma_\beta - \eta^\rho{}_\beta \gamma_\alpha)}. \quad (3.209)$$

From the contraction involving $\sigma_{\alpha\beta}$, we define

$$\sigma_{\mu\nu} = \frac{i}{2} [\gamma_\mu, \gamma_\nu]. \quad (3.210)$$

Then,

$$\begin{aligned} [\gamma^\rho, \sigma_{\alpha\beta}] &= \frac{i}{2} [\gamma^\rho, [\gamma_\alpha, \gamma_\beta]] \\ &= 2i (\eta^\rho{}_\alpha \gamma_\beta - \eta^\rho{}_\beta \gamma_\alpha). \end{aligned} \quad (3.211)$$

Thus,

$$\boxed{S(\delta\omega^{\mu\nu}) = \mathbf{1} + \frac{1}{8} [\gamma_\mu, \gamma_\nu] \delta\omega^{\mu\nu}}. \quad (3.212)$$

Successive infinitesimal transformations define

$$\delta\omega^\mu{}_\nu = \delta\omega I^\mu{}_\nu, \quad (3.213)$$

where $\delta\omega$ is an infinitesimal parameter and $I^\mu{}_\nu$ are 4×4 matrices.

Taking

$$\delta\omega = \frac{\omega}{N} \quad (3.214)$$

and the limit $N \rightarrow \infty$, we obtain

$$\begin{aligned} S(\Lambda) &= \lim_{N \rightarrow \infty} \left(\mathbf{1} - \frac{i}{4} \sigma_{\mu\nu} \frac{\omega^{\mu\nu}}{N} \right)^N \\ &= \exp \left(-\frac{i}{4} \omega^{\mu\nu} \sigma_{\mu\nu} \right). \end{aligned} \quad (3.215)$$

Therefore,

$$\boxed{S(\Lambda) = \exp \left(-\frac{i}{4} \omega^{\mu\nu} \sigma_{\mu\nu} \right)}. \quad (3.216)$$

Each Lorentz transformation has its corresponding generator $I^{\mu\nu}$. Since $\omega^{\mu\nu}$ is antisymmetric, there are six independent generators: three rotations and three boosts.

3.11 Rotations and Boosts

Consider a rotation around the z -axis by an angle ϕ . The corresponding generator is

$$(I_3)^\mu{}_\nu = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (3.217)$$

Using

$$\sigma_{\mu\nu} I_3^{\mu\nu} = 2\sigma_{12} = -2i\gamma^1\gamma^2 = -2\Sigma_3, \quad (3.218)$$

the corresponding Lorentz transformation in spinor space is

$$S = \exp\left(-\frac{i}{4}\phi\sigma_{\mu\nu}I_3^{\mu\nu}\right) = \exp\left(i\frac{\phi}{2}\Sigma_3\right). \quad (3.219)$$

For a rotation around an arbitrary axis $\hat{n}$,

$$S = \exp\left(i\frac{\phi}{2}\hat{n} \cdot \vec{\Sigma}\right). \quad (3.220)$$

Since

$$(\hat{n} \cdot \vec{\Sigma})^2 = \mathbf{1}, \quad (3.221)$$

we obtain

$$S = \cos\left(\frac{\phi}{2}\right)\mathbf{1} + i\sin\left(\frac{\phi}{2}\right)\hat{n} \cdot \vec{\Sigma}. \quad (3.222)$$

Therefore,

$$S(\phi + 2\pi) = -S(\phi), \quad (3.223)$$

whereas

$$S(\phi + 4\pi) = S(\phi). \quad (3.224)$$

Thus, only a rotation of 4π recovers the original spinor state.

Observables in Dirac theory are bilinears. Therefore, a rotation of 2π is sufficient to recover the original observable.

For completeness, the six generators $I^{\mu\nu}$ include the three rotation generators

$$(I_x)^\mu{}_\nu = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad (3.225)$$

$$(I_y)^\mu{}_\nu = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad (3.226)$$

and

$$(I_z)^\mu{}_\nu = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (3.227)$$

In compact notation, for

$$\vec{n} = (n_1, n_2, n_3), \quad (3.228)$$

the rotation generator can be written as

$$(I_{\vec{n}})^\mu{}_\nu = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -n_3 & n_2 \\ 0 & n_3 & 0 & -n_1 \\ 0 & -n_2 & n_1 & 0 \end{pmatrix}. \quad (3.229)$$

The general spinor transformation is

$$S(\Lambda) = \exp\left(-\frac{i}{4}\omega_{\mu\nu}\sigma^{\mu\nu}\right). \quad (3.230)$$

For spatial indices,

$$\sigma^{ij} = \frac{i}{2}[\gamma^i, \gamma^j], \quad (3.231)$$

which are the generators of rotations.

Since

$$(\gamma^i)^\dagger = -\gamma^i, \quad (3.232)$$

we have

$$(\sigma^{ij})^\dagger = \sigma^{ij}. \quad (3.233)$$

Therefore,

$$S = \exp\left(-\frac{i}{4}\omega_{ij}\sigma^{ij}\right) \quad (3.234)$$

is a unitary transformation.

For boosts,

$$\sigma^{0i} = \frac{i}{2}[\gamma^0, \gamma^i]. \quad (3.235)$$

Using

$$(\gamma^0)^\dagger = \gamma^0, \quad (3.236)$$

we obtain

$$(\sigma^{0i})^\dagger = -\sigma^{0i}. \quad (3.237)$$

Therefore, boost transformations are not unitary.

In general,

$$S^\dagger S \neq \mathbf{1}, \quad (3.238)$$

and this equality is only true for rotations.

The spinor transforms as

$$\psi'(x') = S(\Lambda)\psi(x). \quad (3.239)$$

Consider the bilinear

$$\psi^\dagger\psi. \quad (3.240)$$

Under a Lorentz transformation,

$$\psi'^\dagger(x') = (S\psi(x))^\dagger = \psi^\dagger(x)S^\dagger. \quad (3.241)$$

Therefore,

$$\psi'^\dagger(x')\psi'(x') = \psi^\dagger(x)S^\dagger S\psi(x). \quad (3.242)$$

Since, in general,

$$S^\dagger S \neq \mathbf{1}, \quad (3.243)$$

the quantity $\psi^\dagger\psi$ is not a Lorentz scalar.

We need to find another quantity that transforms as a scalar.

Using

$$(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0, \quad (3.244)$$

we find

$$(\sigma^{\mu\nu})^\dagger = \gamma^0\sigma^{\mu\nu}\gamma^0. \quad (3.245)$$

Thus,

$$S^\dagger = \exp\left[\frac{i}{4}\omega_{\mu\nu}(\sigma^{\mu\nu})^\dagger\right] \quad (3.246)$$

$$= \exp\left[\frac{i}{4}\omega_{\mu\nu}\gamma^0\sigma^{\mu\nu}\gamma^0\right] \quad (3.247)$$

$$= \gamma^0 \exp\left[\frac{i}{4}\omega_{\mu\nu}\sigma^{\mu\nu}\right] \gamma^0 \quad (3.248)$$

$$= \gamma^0 S^{-1} \gamma^0 = S^\dagger. \quad (3.249)$$

To prove the intermediate identity, consider an exponential operator

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}. \quad (3.250)$$

Then,

$$\gamma^0 e^A \gamma^0 = \sum_{n=0}^{\infty} \frac{\gamma^0 A^n \gamma^0}{n!} \quad (3.251)$$

$$= \sum_{n=0}^{\infty} \frac{(\gamma^0 A \gamma^0)^n}{n!} \quad (3.252)$$

$$= e^{\gamma^0 A \gamma^0}. \quad (3.253)$$

Now consider

$$\psi'^{\dagger}(x') \gamma^0 \psi'(x') = \psi^{\dagger}(x) S^{\dagger} \gamma^0 S \psi(x) \quad (3.254)$$

$$= \psi^{\dagger}(x) \gamma^0 S^{-1} \gamma^0 S \psi(x) \quad (3.255)$$

$$= \psi^{\dagger}(x) \gamma^0 \psi(x). \quad (3.256)$$

Therefore, this quantity is Lorentz invariant.

For notational convenience, we define the Dirac adjoint

$$\bar{\psi} \equiv \psi^{\dagger} \gamma^0. \quad (3.257)$$

The scalar bilinear is therefore

$$\bar{\psi} \psi. \quad (3.258)$$

The transformation laws are

$$\psi'(x') = S(\Lambda) \psi(x), \quad (3.259)$$

and

$$\bar{\psi}'(x') = \bar{\psi}(x) S^{-1}(\Lambda). \quad (3.260)$$

Since the gamma matrices form a basis of the space of 4×4 matrices, different bilinears can be constructed as

$$\bar{\psi} \mathcal{O} \psi, \quad (3.261)$$

where

$$\mathcal{O} = \mathbf{1}, \quad \gamma^{\mu}, \quad \sigma^{\mu\nu}, \quad \gamma^{\mu} \gamma^5, \quad \gamma^5. \quad (3.262)$$

Recall that

$$\gamma^5 = i \gamma^0 \gamma^1 \gamma^2 \gamma^3, \quad (3.263)$$

with

$$(\gamma^5)^{\dagger} = \gamma^5 \quad (3.264)$$

and

$$\gamma^{\mu} \gamma^5 = -\gamma^5 \gamma^{\mu}. \quad (3.265)$$

Moreover,

$$[\gamma^5, \sigma^{\mu\nu}] = 0. \quad (3.266)$$

Thus, γ^5 commutes with $S(\Lambda)$ for proper Lorentz transformations. For improper Lorentz transformations,

$$S(\Lambda) \gamma^5 = \det(\Lambda) \gamma^5 S(\Lambda). \quad (3.267)$$

The transformation properties of the Dirac bilinears are therefore:

- Scalar:

$$\bar{\psi}'(x') \psi'(x') = \bar{\psi}(x) \psi(x). \quad (3.268)$$

- Pseudoscalar:

$$\bar{\psi}'(x') \gamma^5 \psi'(x') = \det(\Lambda) \bar{\psi}(x) \gamma^5 \psi(x). \quad (3.269)$$

- Vector:

$$\bar{\psi}'(x')\gamma^\mu\psi'(x') = \Lambda^\mu{}_\nu\bar{\psi}(x)\gamma^\nu\psi(x). \quad (3.270)$$

- Pseudovector:

$$\bar{\psi}'(x')\gamma^5\gamma^\mu\psi'(x') = \det(\Lambda)\Lambda^\mu{}_\nu\bar{\psi}(x)\gamma^5\gamma^\nu\psi(x). \quad (3.271)$$

- Tensor:

$$\bar{\psi}'(x')\sigma^{\mu\nu}\psi'(x') = \Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma\bar{\psi}(x)\sigma^{\rho\sigma}\psi(x). \quad (3.272)$$

3.12 Spin and Helicity Projection Operators

We have defined the spin operator

$$S_i = \frac{\Sigma_i}{2}, \quad \Sigma_i = \begin{pmatrix} \sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix}. \quad (3.273)$$

It does not commute with H_{Dirac} :

$$[H, S_i] = -i\epsilon_{ijk}P^j\alpha^k. \quad (3.274)$$

We need an operator that commutes with H , acting on spinors $u(p, \pm s)$ and $v(p, \pm s)$, and returns the correct eigenvalue of spin $\pm 1/2$.

$$S_\pm = \frac{1 \pm \gamma^5 \not{s}}{2}. \quad (3.275)$$

s_μ is the polarization vector. In the rest frame it is normalized as

$$s_\mu = (0, \hat{n}). \quad (3.276)$$

$$\not{s} = \gamma^\mu s_\mu, \quad s^2 = -1. \quad (3.277)$$

$S_\pm$ satisfies the projection operator properties:

$$S_\pm S_\pm = \frac{1 \pm \gamma^5 \not{s}}{2} \frac{1 \pm \gamma^5 \not{s}}{2} \quad (3.278)$$

$$= \frac{1 \pm 2\gamma^5 \not{s} + (\gamma^5)^2 s^2}{2 \cdot 2} \quad (3.279)$$

$$= \frac{1 \pm \gamma^5 \not{s}}{2} = S_\pm, \quad (3.280)$$

$$S_\pm S_\mp = \frac{1 \pm \gamma^5 \not{s}}{2} \frac{1 \mp \gamma^5 \not{s}}{2} \quad (3.281)$$

$$= \frac{1 - \gamma^5 \not{s} \gamma^5 \not{s}}{2} \quad (3.282)$$

$$= \frac{1 + (\gamma^5)^2 s^2}{2} = 0, \quad (3.283)$$

$$S_\pm + S_\mp = \frac{1 \pm \gamma^5 \not{s}}{2} + \frac{1 \mp \gamma^5 \not{s}}{2} \quad (3.284)$$

$$= 1. \quad (3.285)$$

We define the helicity operator that commutes with H , which is the spin projected onto the momentum direction:

$$\hat{h} = \frac{\vec{\Sigma} \cdot \vec{p}}{2|\vec{p}|} = \frac{\vec{S} \cdot \vec{p}}{|\vec{p}|}. \quad (3.286)$$

$$\hat{h} = \frac{1}{2} (\hat{p} \cdot \vec{\Sigma}) \gamma^0 = \frac{1}{2} \gamma^5 \not{s}. \quad (3.287)$$

The helicity operator projects two physical states:

- spin along the momentum direction: right helicity;
- spin along the opposite momentum direction: left helicity.

This is valid for massless particles also.

For massless particles it is Lorentz invariant.

3.13 Parity, time reversal and Charge Conjugation

We have to find $S(\Lambda)$ for non-infinitesimal transformations.

Space reflection (Parity)

For a space reflection,

$$t' = t, \quad \vec{x}' = -\vec{x}, \quad (3.288)$$

$$\Lambda^\nu{}_\mu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \eta^{\mu\nu}. \quad (3.289)$$

The condition for the Dirac equation to remain covariant is

$$S^{-1}\gamma^\mu S = \Lambda^\mu{}_\nu \gamma^\nu. \quad (3.290)$$

The Lorentz transformation for spatial reflection is denoted by

$$S(\eta^{\mu\nu}) = P. \quad (3.291)$$

It satisfies

$$P^{-1}\gamma^0 P = \gamma^0, \quad P^{-1}\gamma^i P = -\gamma^i. \quad (3.292)$$

A solution is

$$P = \gamma^0, \quad (3.293)$$

or, more generally,

$$P = \gamma^0 e^{i\phi}. \quad (3.294)$$

Furthermore,

$$P^{-1} = P^\dagger, \quad (3.295)$$

so P is unitary.

Time reversal

For time reversal,

$$t' = -t, \quad \vec{x}' = \vec{x}, \quad (3.296)$$

$$\Lambda^\mu{}_\nu = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (3.297)$$

Again we impose

$$S^{-1}\gamma^\mu S = \Lambda^\mu{}_\nu \gamma^\nu. \quad (3.298)$$

A solution is

$$S(\Lambda^\mu{}_\nu) = T = i\gamma^1\gamma^3. \quad (3.299)$$

If we apply T to a spinor,

$$\psi'(x') = T\psi(x). \quad (3.300)$$

This transforms an electron with spin up and momentum $\vec{p}$ into a positron with spin down and momentum $-\vec{p}$.

The change in spin and momentum is expected, but not the particle $\rightarrow$ antiparticle transformation.

To avoid this, we apply T to the complex conjugate:

$$\psi'(x') = T\psi^*(x). \quad (3.301)$$

Charge conjugation

The existence of antiparticles implies that we need a transformation that connects particles and antiparticles alone.

This symmetry is called C .

Consider the Dirac equation in an electromagnetic field,

$$(i\cancel{\partial} - e\cancel{A} - m) \psi = 0, \quad (\text{negative charge particle}) \quad (1)$$

and

$$(i\cancel{\partial} + e\cancel{A} - m) \psi_c = 0, \quad (\text{positive charge particle}) \quad (2)$$

Taking the complex conjugate of (1),

$$-i\gamma_r^* \partial^r \psi^* - e\gamma_r^* A^r \psi^* - m\psi^* = 0. \quad (3.302)$$

Comparing with (2),

$$i\gamma_r \partial^r \psi_c + e\gamma_r A^r \psi_c - m\psi_c = 0. \quad (3.303)$$

We need to accomplish

$$C\gamma_r^* C^{-1} = \gamma_r, \quad C\psi^* = \psi_c. \quad (3.304)$$

Hence,

$$C = i\gamma^0 \gamma^2. \quad (3.305)$$

For a Dirac field,

$$(CPT) \psi(x) (CPT)^{-1} = \gamma^5 C \psi^\dagger(-x). \quad (3.306)$$

The three discrete symmetries act as follows:

- C turns a particle into its antiparticle.
- P flips the spatial momentum.
- T flips spin, momentum and performs complex conjugation.

For the spinors,

$$u(p, s) \longleftrightarrow v(-p, -s), \quad (3.307)$$

$$v(p, s) \longleftrightarrow u(-p, -s). \quad (3.308)$$

A fundamental result is the CPT theorem:

There cannot exist a Lorentz invariant quantum field theory with a Hermitian Hamiltonian that violates CPT, regardless of whether C , P or T are individually violated.

Therefore, particles and antiparticles have the same mass and lifetime.

3.14 Problems

Problem 3.1 Consider the Dirac Hamiltonian

$$H = \vec{\alpha} \cdot \vec{p} + \beta m = \gamma^0(\gamma^i p_i + m),$$

where the spatial indices run from 1 to 3.

1. Show that

$$[L_i, H] = i\gamma^0 \epsilon_{ijk} \gamma^j p^k,$$

where

$$L_i = \epsilon_{ijk} r^j p^k.$$

2. Show that

$$\left[\frac{1}{2} \Sigma_i, H \right] = -i\gamma^0 \epsilon_{ijk} \gamma^j p^k.$$

Hence prove that

$$\vec{J} = \vec{L} + \frac{1}{2} \vec{\Sigma}$$

is a conserved quantity, namely

$$[\vec{J}, H] = 0.$$

3. Prove that the helicity operator

$$h = \frac{1}{2} \frac{\vec{\Sigma} \cdot \vec{p}}{|\vec{p}|}$$

commutes with the Dirac Hamiltonian.

4. By direct diagonalization show that the eigenvalues of h are $\pm \frac{1}{2}$.

Solution.

$$\gamma^0 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \Sigma^i = \begin{pmatrix} \sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix}. \quad (3.309)$$

$$[p^i, p^j] = [r^i, r^j] = 0, \quad [r^i, p^j] = i\delta^{ij}. \quad (3.310)$$

$$[\sigma^i, \sigma^j] = 2i\epsilon^{ijk} \sigma_k, \quad \sigma^i \sigma^j = \delta^{ij} + i\epsilon^{ijk} \sigma_k. \quad (3.311)$$

(i)

$$L^i = \epsilon^{ijk} r_j p_k, \quad H = \gamma^0(\gamma^i p^i + m). \quad (3.312)$$

$$[L^i, H] = [\epsilon^{ijk} r_j p_k, \gamma^0(\gamma^l p^l + m)] \quad (3.313)$$

$$= [\epsilon^{ijk} r_j p_k, \gamma^0 \gamma^l p^l] + [\epsilon^{ijk} r_j p_k, \gamma^0 m] \quad (3.314)$$

$$= \epsilon^{ijk} p_k \gamma^0 \gamma^l [r_j, p_l] \quad (3.315)$$

$$= i\gamma^0 \epsilon^{ijk} p_k \gamma^l \delta_{jl} \quad (3.316)$$

$$= i\gamma^0 \epsilon^{ijk} p_k \gamma_j. \quad (3.317)$$

(ii)

$$\left[\frac{\Sigma^i}{2}, H \right] = \left[\frac{\Sigma^i}{2}, \gamma^0(\gamma^j p^j + m) \right] \quad (3.318)$$

$$= \frac{1}{2} \gamma^0 p^j [\Sigma^i, \gamma^j]. \quad (3.319)$$

Here,

$$[\Sigma^i, \gamma^0] = 0, \quad [\Sigma^i, p^k] = 0. \quad (3.320)$$

$$[\Sigma^i, \gamma^j] = \Sigma^i \gamma^j - \gamma^j \Sigma^i \quad (3.321)$$

$$= \begin{pmatrix} 0 & \sigma^i \sigma^j \\ -\sigma^i \sigma^j & 0 \end{pmatrix} - \begin{pmatrix} 0 & \sigma^j \sigma^i \\ -\sigma^j \sigma^i & 0 \end{pmatrix} \quad (3.322)$$

$$= \begin{pmatrix} 0 & \sigma^i \sigma^j - \sigma^j \sigma^i \\ -\sigma^i \sigma^j + \sigma^j \sigma^i & 0 \end{pmatrix} \quad (3.323)$$

$$= \begin{pmatrix} 0 & 2i\epsilon^{ijk}\sigma_k \\ -2i\epsilon^{ijk}\sigma_k & 0 \end{pmatrix} \quad (3.324)$$

$$= 2i\epsilon^{ijk}\gamma_k. \quad (3.325)$$

Therefore,

$$\left[\frac{\Sigma^i}{2}, H \right] = \frac{1}{2} \gamma^0 p^j 2i\epsilon^{ijk} \gamma_k \quad (3.326)$$

$$= i\gamma^0 p^j \epsilon^{ijk} \gamma_k \quad (3.327)$$

$$= i\gamma^0 \epsilon^{ijk} p_j \gamma_k \quad (3.328)$$

$$= -i\gamma^0 \epsilon^{ijk} p_k \gamma_j. \quad (3.329)$$

Hence,

$$[\vec{J}, H] = \left[\vec{L} + \frac{\vec{\Sigma}}{2}, H \right] = 0. \quad (3.330)$$

(iii)

$$h = \frac{1}{2} \frac{\vec{\Sigma} \cdot \vec{p}}{|\vec{p}|}. \quad (3.331)$$

$$\left[\frac{1}{2} \Sigma^i p^i, \gamma^0 \gamma^j p^j + \gamma^0 m \right] = \left[\frac{1}{2} \Sigma^i p^i, \gamma^0 \gamma^j p^j \right] \quad (3.332)$$

$$= \frac{1}{2} p^i \gamma^0 p^j [\Sigma^i, \gamma^j] \quad (3.333)$$

$$= \frac{1}{2} p^i \gamma^0 p^j 2i\epsilon^{ijk} \gamma_k \quad (3.334)$$

$$= i\epsilon^{ijk} p^i p^j \gamma^0 \gamma_k \quad (3.335)$$

$$= 0. \quad (3.336)$$

Here, $p^i p^j$ is symmetric and ϵ^{ijk} is antisymmetric.

(iv)

$$h = \frac{1}{2} \frac{\vec{\Sigma} \cdot \vec{p}}{|\vec{p}|} = \frac{1}{2|\vec{p}|} \begin{pmatrix} \vec{\sigma} \cdot \vec{p} & 0 \\ 0 & \vec{\sigma} \cdot \vec{p} \end{pmatrix}. \quad (3.337)$$

$$\vec{\sigma} \cdot \vec{p} = \begin{pmatrix} 0 & P_1 \\ p_1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -ip_2 \\ ip_2 & 0 \end{pmatrix} + \begin{pmatrix} p_3 & 0 \\ 0 & -p_3 \end{pmatrix} \quad (3.338)$$

$$= \begin{pmatrix} p_3 & p_1 - ip_2 \\ p_1 + ip_2 & -p_3 \end{pmatrix}. \quad (3.339)$$

The characteristic equation is

$$\begin{vmatrix} \frac{p_3}{2|\vec{p}|} - \lambda & \frac{p_1 - ip_2}{2|\vec{p}|} \\ \frac{p_1 + ip_2}{2|\vec{p}|} & -\frac{p_3}{2|\vec{p}|} - \lambda \end{vmatrix} = 0. \quad (3.340)$$

Therefore,

$$-\left(\frac{p_3}{2|\vec{p}|} - \lambda\right) \left(\frac{p_3}{2|\vec{p}|} + \lambda\right) - \frac{1}{4|\vec{p}|^2} (p_1 - ip_2)(p_1 + ip_2) = 0 \quad (3.341)$$

$$-\frac{p_3^2}{4|\vec{p}|^2} - \frac{p_3\lambda}{2|\vec{p}|} + \frac{p_3\lambda}{2|\vec{p}|} + \lambda^2 - \frac{p_1^2 + p_2^2}{4|\vec{p}|^2} = 0 \quad (3.342)$$

$$\lambda^2 - \frac{p_1^2 + p_2^2 + p_3^2}{4|\vec{p}|^2} = 0 \quad (3.343)$$

$$\lambda^2 - \frac{1}{4} = 0. \quad (3.344)$$

Hence,

$$\lambda^2 = \frac{1}{4} \quad \implies \quad \lambda = \pm \frac{1}{2}. \quad (3.345)$$

□

Problem 3.2 prove that

$$[\gamma^\mu, B^{\alpha\beta}] = (M^{\alpha\beta})^\mu{}_\nu \gamma^\nu,$$

where

$$B^{\alpha\beta} = \frac{1}{4}[\gamma^\alpha, \gamma^\beta],$$

and

$$(M^{\alpha\beta})^\mu{}_\nu = \eta^{\alpha\mu}\eta^\beta{}_\nu - \eta^{\beta\mu}\eta^\alpha{}_\nu.$$

Solution.

Developing the right-hand side,

$$(M^{\alpha\beta})^\gamma{}_\nu \gamma^\nu = (\eta^{\alpha\gamma}\delta_\nu^\beta - \eta^{\beta\gamma}\delta_\nu^\alpha) \gamma^\nu \quad (3.346)$$

$$= \eta^{\alpha\gamma}\gamma^\beta - \eta^{\beta\gamma}\gamma^\alpha. \quad (3.347)$$

Developing the left-hand side,

$$[\gamma^\gamma, \sigma^{\alpha\beta}] = \left[\gamma^\gamma, \frac{1}{4} [\gamma^\alpha, \gamma^\beta] \right] \quad (3.348)$$

$$= \frac{1}{4} \gamma^\gamma [\gamma^\alpha, \gamma^\beta] - \frac{1}{4} [\gamma^\alpha, \gamma^\beta] \gamma^\gamma \quad (3.349)$$

$$= \frac{1}{4} \left(\underbrace{\gamma^\gamma \gamma^\alpha}_{2\eta^{\gamma\alpha} - \gamma^\alpha \gamma^\gamma} \gamma^\beta - \underbrace{\gamma^\gamma \gamma^\beta}_{2\eta^{\gamma\beta} - \gamma^\beta \gamma^\gamma} \gamma^\alpha - \gamma^\alpha \underbrace{\gamma^\beta \gamma^\gamma}_{2\eta^{\beta\gamma} - \gamma^\gamma \gamma^\beta} + \gamma^\beta \underbrace{\gamma^\alpha \gamma^\gamma}_{2\eta^{\alpha\gamma} - \gamma^\gamma \gamma^\alpha} \right). \quad (3.350)$$

Using

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \quad (3.351)$$

we have

$$\gamma^\mu \gamma^\nu = 2\eta^{\mu\nu} - \gamma^\nu \gamma^\mu. \quad (3.352)$$

Therefore,

$$[\gamma^\gamma, \sigma^{\alpha\beta}] = \frac{1}{4} \left(2\eta^{\gamma\alpha}\gamma^\beta - \gamma^\alpha\gamma^\gamma\gamma^\beta - 2\eta^{\gamma\beta}\gamma^\alpha + \gamma^\beta\gamma^\gamma\gamma^\alpha - \gamma^\alpha(2\eta^{\beta\gamma} - \gamma^\gamma\gamma^\beta) + \gamma^\beta(2\eta^{\alpha\gamma} - \gamma^\gamma\gamma^\alpha) \right) \quad (3.353)$$

$$= \frac{1}{2} (\eta^{\gamma\alpha}\gamma^\beta - \eta^{\gamma\beta}\gamma^\alpha - \eta^{\beta\gamma}\gamma^\alpha + \eta^{\alpha\gamma}\gamma^\beta) \quad (3.354)$$

$$= \eta^{\alpha\gamma}\gamma^\beta - \eta^{\beta\gamma}\gamma^\alpha. \quad (3.355)$$

□

Problem 3.3 Let

$$\chi_+ = \sqrt{\frac{1}{2(1+p^3)}} \begin{pmatrix} 1+p^3 \\ p^+ \end{pmatrix}, \quad \chi_- = \sqrt{\frac{1}{2(1+p^3)}} \begin{pmatrix} -p^- \\ 1+p^3 \end{pmatrix},$$

where

$$p_{\pm} = p^1 \pm ip^2, \quad \bar{p}^2 = 1.$$

Show that

1. $\chi_{\pm}$ are eigenvectors of

$$\frac{1}{2} p_i \sigma^i$$

with eigenvalues $\pm \frac{1}{2}$.

- 2.

$$\chi_{\lambda}^{\dagger} \chi_{\lambda'} = \delta_{\lambda\lambda'}.$$

- 3.

$$\sum_{\lambda=\pm} \chi_{\lambda} \chi_{\lambda}^{\dagger} = \mathbf{1}.$$

Solution.

This describes the Weyl basis for helicity spinors.

Spinors describe massless particles and are uniquely normalized eigenvectors of $\frac{1}{2} p^i \sigma^i$.

$$\frac{1}{2} p^i \sigma^i = \frac{1}{2} \left[p^1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + p^2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + p^3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \quad (3.356)$$

$$= \frac{1}{2} \begin{pmatrix} p^3 & p^1 - ip^2 \\ p^1 + ip^2 & -p^3 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} p^3 & p^- \\ p^+ & -p^3 \end{pmatrix}. \quad (3.357)$$

$$\left(\frac{1}{2} p^i \sigma^i \right) \chi_+ = \frac{1}{2} \frac{1}{\sqrt{2(1+p^3)}} \begin{pmatrix} p^3 & p^- \\ p^+ & -p^3 \end{pmatrix} \begin{pmatrix} 1+p^3 \\ p^+ \end{pmatrix} \quad (3.358)$$

$$= \frac{1}{2} \frac{1}{\sqrt{2(1+p^3)}} \begin{pmatrix} p^3 + (p^3)^2 + p^- p^+ \\ p^+ + p^+ p^3 - p^3 p^+ \end{pmatrix} \quad (3.359)$$

$$= \frac{1}{2} \frac{1}{\sqrt{2(1+p^3)}} \begin{pmatrix} p^3 + (p^3)^2 + (p^1)^2 + (p^2)^2 \\ p^+ \end{pmatrix} \quad (3.360)$$

$$= \frac{1}{2} \frac{1}{\sqrt{2(1+p^3)}} \begin{pmatrix} 1+p^3 \\ p^+ \end{pmatrix} \quad (3.361)$$

$$= \frac{1}{2} \chi_+. \quad (3.362)$$

Here we have used

$$|\vec{p}|^2 = 1. \quad (3.363)$$

Doing the same calculation for χ_- , we obtain the eigenvalue $-\frac{1}{2}$.

$$\chi_+^{\dagger} \chi_+ = \chi_-^{\dagger} \chi_- = 1, \quad (3.364)$$

$$\chi_+^{\dagger} \chi_- = \chi_-^{\dagger} \chi_+ = 0. \quad (3.365)$$

$$\chi_+^{\dagger} \chi_+ = \frac{1}{2(1+p^3)} \begin{pmatrix} 1+p^3 & (p^+)^* \end{pmatrix} \begin{pmatrix} 1+p^3 \\ p^+ \end{pmatrix} \quad (3.366)$$

$$= \frac{1}{2(1+p^3)} (1 + 2p^3 + (p^3)^2 + (p^1)^2 + (p^2)^2) \quad (3.367)$$

$$= \frac{1}{2(1+p^3)} (2 + 2p^3) \quad (3.368)$$

$$= 1. \quad (3.369)$$

$$\chi_+^\dagger \chi_- = \frac{1}{2(1+p^3)} \begin{pmatrix} 1+p^3 & (p^+)^* \\ -p^- & 1+p^3 \end{pmatrix} \begin{pmatrix} -p^- \\ 1+p^3 \end{pmatrix} \quad (3.370)$$

$$= \frac{1}{2(1+p^3)} (-p^- - p^3 p^- + p^- + p^- p^3) \quad (3.371)$$

$$= 0. \quad (3.372)$$

$$\sum_\lambda \chi_\lambda \chi_\lambda^\dagger = \chi_+ \chi_+^\dagger + \chi_- \chi_-^\dagger \quad (3.373)$$

$$= \frac{1}{2(1+p^3)} \left[\begin{pmatrix} 1+p^3 & (p^+)^* \\ p^+ & 1+p^3 \end{pmatrix} + \begin{pmatrix} -p^- & \\ 1+p^3 & \end{pmatrix} \begin{pmatrix} -(p^-)^* & 1+p^3 \end{pmatrix} \right] \quad (3.374)$$

$$= \frac{1}{2(1+p^3)} \left[\begin{pmatrix} 1+2p^3+(p^3)^2 & p^-+p^-p^3 \\ p^++p^+p^3 & p^+p^- \end{pmatrix} + \begin{pmatrix} p^-p^+ & -p^- - p^-p^3 \\ -p^+ - p^+p^3 & 1+2p^3+(p^3)^2 \end{pmatrix} \right] \quad (3.375)$$

$$= \frac{1}{2(1+p^3)} \begin{pmatrix} 1+2p^3+(p^3)^2+(p^1)^2+(p^2)^2 & 0 \\ 0 & 1+2p^3+(p^3)^2+(p^1)^2+(p^2)^2 \end{pmatrix} \quad (3.376)$$

$$= \frac{2+2p^3}{2(1+p^3)} \mathbf{1} \quad (3.377)$$

$$= \mathbf{1}. \quad (3.378)$$

□

Problem 3.4 Let the Dirac spinors be

$$u_\lambda(p) = \sqrt{E+m} \begin{pmatrix} \chi_\lambda \\ \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \chi_\lambda \end{pmatrix},$$

and

$$v_\lambda(p) = \sqrt{E+m} \begin{pmatrix} \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_\lambda \\ \hat{\chi}_\lambda \end{pmatrix},$$

with

$$\bar{u} = u^\dagger \gamma^0, \quad \bar{v} = v^\dagger \gamma^0.$$

Show that

1.

$$u_\lambda^\dagger u_{\lambda'} = v_\lambda^\dagger v_{\lambda'} = 2E \delta_{\lambda\lambda'}.$$

2.

$$\bar{u}_\lambda u_{\lambda'} = -\bar{v}_\lambda v_{\lambda'} = 2m \delta_{\lambda\lambda'}.$$

3.

$$\bar{u}_\lambda v_{\lambda'} = \bar{v}_\lambda u_{\lambda'} = 0.$$

4.

$$u_\lambda^\dagger(-p) v_{\lambda'} = v_\lambda^\dagger(p) u_{\lambda'}(-p) = 0.$$

5.

$$\sum_\lambda u_\lambda(p) \bar{u}_\lambda(p) = \not{p} + m.$$

6.

$$\sum_\lambda v_\lambda(p) \bar{v}_\lambda(p) = \not{p} - m.$$

Solution.

$$u_{\lambda}^{\dagger}(\vec{p}) = \sqrt{E+m} \left(\chi_{\lambda}^{\dagger}, \chi_{\lambda}^{\dagger} \frac{(\vec{\sigma} \cdot \vec{p})^{\dagger}}{E+m} \right), \quad (3.379)$$

$$v_{\lambda}^{\dagger}(\vec{p}) = \sqrt{E+m} \left(\hat{\chi}_{\lambda}^{\dagger} \frac{(\vec{\sigma} \cdot \vec{p})^{\dagger}}{E+m}, \hat{\chi}_{\lambda}^{\dagger} \right). \quad (3.380)$$

$$\bar{u}_{\lambda}(\vec{p}) = u_{\lambda}^{\dagger}(\vec{p}) \gamma^0 \quad (3.381)$$

$$= \sqrt{E+m} \left(\chi_{\lambda}^{\dagger}, \chi_{\lambda}^{\dagger} \frac{(\vec{\sigma} \cdot \vec{p})^{\dagger}}{E+m} \right) \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix} \quad (3.382)$$

$$= \sqrt{E+m} \left(\chi_{\lambda}^{\dagger}, -\chi_{\lambda}^{\dagger} \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \right), \quad (3.383)$$

$$\bar{v}_{\lambda}(\vec{p}) = v_{\lambda}^{\dagger}(\vec{p}) \gamma^0 \quad (3.384)$$

$$= \sqrt{E+m} \left(\hat{\chi}_{\lambda}^{\dagger} \frac{\vec{\sigma} \cdot \vec{p}}{E+m}, -\hat{\chi}_{\lambda}^{\dagger} \right). \quad (3.385)$$

(i)

$$u_{\lambda}^{\dagger}(\vec{p}) u_{\lambda'}(\vec{p}) = (E+m) \left(\chi_{\lambda}^{\dagger}, \chi_{\lambda}^{\dagger} \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \right) \begin{pmatrix} \chi_{\lambda'} \\ \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \chi_{\lambda'} \end{pmatrix} \quad (3.386)$$

$$= (E+m) \left(\chi_{\lambda}^{\dagger} \chi_{\lambda'} + \chi_{\lambda}^{\dagger} \frac{(\vec{\sigma} \cdot \vec{p})^2}{(E+m)^2} \chi_{\lambda'} \right) \quad (3.387)$$

$$= (E+m) \left(\delta_{\lambda\lambda'} + \chi_{\lambda}^{\dagger} \frac{E-m}{E+m} \chi_{\lambda'} \right) \quad (3.388)$$

$$= (E+m) \left(1 + \frac{E-m}{E+m} \right) \delta_{\lambda\lambda'} \quad (3.389)$$

$$= \frac{E+m}{E+m} 2E \delta_{\lambda\lambda'} \quad (3.390)$$

$$= 2E \delta_{\lambda\lambda'}. \quad (3.391)$$

$$v_{\lambda}^{\dagger}(\vec{p}) v_{\lambda'}(\vec{p}) = 2E \delta_{\lambda\lambda'}. \quad (3.392)$$

(ii)

$$\bar{u}_{\lambda}(\vec{p}) u_{\lambda'}(\vec{p}) = (E+m) \left(\chi_{\lambda}^{\dagger} \chi_{\lambda'} - \chi_{\lambda}^{\dagger} \frac{(\vec{\sigma} \cdot \vec{p})^2}{(E+m)^2} \chi_{\lambda'} \right) \quad (3.393)$$

$$= (E+m) \left(\delta_{\lambda\lambda'} - \frac{E-m}{E+m} \delta_{\lambda\lambda'} \right) \quad (3.394)$$

$$= 2m \delta_{\lambda\lambda'}. \quad (3.395)$$

$$\bar{v}_{\lambda}(\vec{p}) v_{\lambda'}(\vec{p}) = -2m \delta_{\lambda\lambda'}. \quad (3.396)$$

(iii)

$$\bar{u}_{\lambda}(\vec{p}) v_{\lambda'}(\vec{p}) = (E+m) \left(\chi_{\lambda}^{\dagger}, -\chi_{\lambda}^{\dagger} \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \right) \begin{pmatrix} \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_{\lambda'} \\ \hat{\chi}_{\lambda'} \end{pmatrix} \quad (3.397)$$

$$= (E+m) \left[\chi_{\lambda}^{\dagger} \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_{\lambda'} - \chi_{\lambda}^{\dagger} \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_{\lambda'} \right] \quad (3.398)$$

$$= 0. \quad (3.399)$$

$$\bar{v}_{\lambda'}(\vec{p}) u_{\lambda}(\vec{p}) = 0. \quad (3.400)$$

(iv)

$$u_\lambda^\dagger(-\vec{p})v_{\lambda'}(\vec{p}) = (E+m) \begin{pmatrix} \chi_\lambda^\dagger & -\chi_\lambda^\dagger \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \end{pmatrix} \begin{pmatrix} \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_{\lambda'} \\ \hat{\chi}_{\lambda'} \end{pmatrix} \quad (3.401)$$

$$= (E+m) \left(\chi_\lambda^\dagger \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_{\lambda'} - \chi_\lambda^\dagger \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_{\lambda'} \right) \quad (3.402)$$

$$= 0. \quad (3.403)$$

$$v_{\lambda'}^\dagger(\vec{p})u_\lambda(-\vec{p}) = 0. \quad (3.404)$$

(v)

$$\sum_\lambda u_\lambda(\vec{p})\bar{u}_\lambda(\vec{p}) = \sum_\lambda (E+m) \begin{pmatrix} \chi_\lambda \\ \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \chi_\lambda \end{pmatrix} \begin{pmatrix} \chi_\lambda^\dagger & -\chi_\lambda^\dagger \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \end{pmatrix} \quad (3.405)$$

$$= \sum_\lambda (E+m) \begin{pmatrix} \chi_\lambda \chi_\lambda^\dagger & -\chi_\lambda \chi_\lambda^\dagger \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \\ \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \chi_\lambda \chi_\lambda^\dagger & -\frac{(\vec{\sigma} \cdot \vec{p})^2}{(E+m)^2} \chi_\lambda \chi_\lambda^\dagger \end{pmatrix} \quad (3.406)$$

$$= \begin{pmatrix} (E+m)\mathbf{1} & -\vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & (m-E)\mathbf{1} \end{pmatrix}. \quad (3.407)$$

Moreover,

$$m\mathbf{1} + \begin{pmatrix} E & -\vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & -E \end{pmatrix} = m\mathbf{1} + E\gamma^0 + \begin{pmatrix} 0 & -\vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & 0 \end{pmatrix} \quad (3.408)$$

$$= m\mathbf{1} + E\gamma^0 - \gamma^i p^i \quad (3.409)$$

$$= \gamma^\mu p_\mu + m. \quad (3.410)$$

Therefore,

$$\sum_\lambda u_\lambda(\vec{p})\bar{u}_\lambda(\vec{p}) = \gamma^\mu p_\mu + m. \quad (3.411)$$

(vi)

$$v_\lambda(\vec{p})\bar{v}_\lambda(\vec{p}) = v_\lambda(\vec{p})v_\lambda^\dagger(\vec{p})\gamma^0 \quad (3.412)$$

$$= (E+m) \begin{pmatrix} \frac{(\vec{\sigma} \cdot \vec{p})^2}{(E+m)^2} \hat{\chi}_\lambda \hat{\chi}_\lambda^\dagger & -\frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_\lambda \hat{\chi}_\lambda^\dagger \\ \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_\lambda \hat{\chi}_\lambda^\dagger & -\hat{\chi}_\lambda \hat{\chi}_\lambda^\dagger \end{pmatrix}. \quad (1)$$

$$v_{-\lambda}(\vec{p})\bar{v}_{-\lambda}(\vec{p}) = (E+m) \begin{pmatrix} \frac{(\vec{\sigma} \cdot \vec{p})^2}{(E+m)^2} \hat{\chi}_{-\lambda} \hat{\chi}_{-\lambda}^\dagger & -\frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_{-\lambda} \hat{\chi}_{-\lambda}^\dagger \\ \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_{-\lambda} \hat{\chi}_{-\lambda}^\dagger & -\hat{\chi}_{-\lambda} \hat{\chi}_{-\lambda}^\dagger \end{pmatrix}. \quad (2)$$

Adding (1) and (2),

$$\sum_\lambda v_\lambda(\vec{p})\bar{v}_\lambda(\vec{p}) = \begin{pmatrix} (E-m)\mathbf{1} & 0 \\ 0 & -(E+m)\mathbf{1} \end{pmatrix} + (-\gamma^i)p^i \quad (3.413)$$

$$= \gamma^0 E + \gamma^i p^i - m\mathbf{1}_{4 \times 4}. \quad (3.414)$$

Therefore,

$$\boxed{\sum_\lambda v_\lambda(\vec{p})\bar{v}_\lambda(\vec{p}) = \gamma^\mu p_\mu - m}. \quad (3.415)$$

□

Problem 3.5 prove the following identities.

1. The trace of the product of an odd number of Dirac matrices vanishes.

2.

$$\text{Tr}(\gamma^\mu \gamma^\nu) = 4\eta^{\mu\nu}.$$

3.

$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(\eta^{\mu\nu} \eta^{\rho\sigma} - \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho}).$$

Solution.

$$(\gamma^5)^2 = \mathbf{1}, \quad \gamma^5 = i\gamma^0 \gamma^1 \gamma^2 \gamma^3. \quad (3.416)$$

$$\{\gamma^5, \gamma^\mu\} = 0, \quad \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}. \quad (3.417)$$

(i) Consider

$$\text{Tr} \left(\underbrace{\gamma^\alpha \cdots \gamma^\beta}_{2n+1 \text{ } \gamma\text{'s}} \right). \quad (3.418)$$

$$\text{Tr}(\gamma^\alpha \cdots \gamma^\beta) = \text{Tr}(\gamma^\alpha \cdots \gamma^5 \gamma^5 \gamma^\beta) \quad (3.419)$$

$$= -\text{Tr}(\gamma^\alpha \cdots \gamma^5 \gamma^\beta \gamma^5) \quad (3.420)$$

$$= (-1)^{2n} (-1) \text{Tr}(\gamma^5 \gamma^\alpha \cdots \gamma^\beta \gamma^5) \quad (3.421)$$

$$= -\text{Tr}(\gamma^\alpha \cdots \gamma^\beta \gamma^5 \gamma^5) \quad (3.422)$$

$$= -\text{Tr}(\gamma^\alpha \cdots \gamma^\beta). \quad (3.423)$$

Therefore,

$$\text{Tr}(\gamma^\alpha \cdots \gamma^\beta) = 0. \quad (3.424)$$

(ii)

$$\text{Tr}(\gamma^\mu \gamma^\nu) = 4\eta^{\mu\nu}. \quad (3.425)$$

Using the cyclic property,

$$\text{Tr}(\gamma^\mu \gamma^\nu) = \text{Tr}(\gamma^\nu \gamma^\mu). \quad (3.426)$$

Since the trace is a linear operator,

$$\text{Tr}(\gamma^\mu \gamma^\nu) = \frac{2}{2} \text{Tr}(\gamma^\mu \gamma^\nu) \quad (3.427)$$

$$= \frac{\text{Tr}(\gamma^\mu \gamma^\nu) + \text{Tr}(\gamma^\nu \gamma^\mu)}{2} \quad (3.428)$$

$$= \frac{1}{2} \text{Tr}(\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu) \quad (3.429)$$

$$= \frac{1}{2} \text{Tr}(\{\gamma^\mu, \gamma^\nu\}) \quad (3.430)$$

$$= \frac{1}{2} \text{Tr}(2\eta^{\mu\nu} \mathbf{1}) \quad (3.431)$$

$$= 4\eta^{\mu\nu}. \quad (3.432)$$

(iii)

$$\text{Tr}[\gamma^\mu \gamma^\nu (2\eta^{\rho\sigma} - \gamma^\sigma \gamma^\rho)] = 2\eta^{\rho\sigma} \text{Tr}(\gamma^\mu \gamma^\nu) - \text{Tr}(\gamma^\mu \gamma^\nu \gamma^\sigma \gamma^\rho) \quad (3.433)$$

$$= 8\eta^{\rho\sigma} \eta^{\mu\nu} - \text{Tr}(\gamma^\mu \gamma^\nu \gamma^\sigma \gamma^\rho). \quad (3.434)$$

$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\sigma \gamma^\rho) = \text{Tr}[\gamma^\mu (2\eta^{\nu\sigma} - \gamma^\sigma \gamma^\nu) \gamma^\rho] \quad (3.435)$$

$$= 8\eta^{\nu\sigma} \eta^{\mu\rho} - \text{Tr}(\gamma^\mu \gamma^\sigma \gamma^\nu \gamma^\rho). \quad (3.436)$$

$$\text{Tr}(\gamma^\mu \gamma^\sigma \gamma^\nu \gamma^\rho) = \text{Tr}[(2\eta^{\mu\sigma} - \gamma^\sigma \gamma^\mu) \gamma^\nu \gamma^\rho] \quad (3.437)$$

$$= 2\eta^{\mu\sigma} \text{Tr}(\gamma^\nu \gamma^\rho) - \text{Tr}(\gamma^\sigma \gamma^\mu \gamma^\nu \gamma^\rho) \quad (3.438)$$

$$= 8\eta^{\mu\sigma} \eta^{\nu\rho} - \text{Tr}(\gamma^\sigma \gamma^\mu \gamma^\nu \gamma^\rho). \quad (3.439)$$

Using the cyclic property,

$$\text{Tr}(\gamma^\sigma \gamma^\mu \gamma^\nu \gamma^\rho) = \text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma). \quad (3.440)$$

Hence,

$$\begin{aligned} \text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) &= 8\eta^{\mu\nu} \eta^{\rho\sigma} - 8\eta^{\mu\rho} \eta^{\nu\sigma} + 8\eta^{\mu\sigma} \eta^{\nu\rho} \\ &\quad - \text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma). \end{aligned} \quad (3.441)$$

Therefore,

$$\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(\eta^{\mu\nu} \eta^{\rho\sigma} - \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho}). \quad (3.442)$$

□

Problem 3.6 Calculate the trace

$$\text{Tr}[(\not{p} - m)\gamma^\mu(1 - \gamma^5)(\not{q} + m)\gamma^\nu].$$

Solution.

$$\begin{aligned} &\text{Tr}[(\not{p} - m)\gamma_\mu(1 - \gamma^5)(\not{q} + m)\gamma_\nu] \\ &= \text{Tr}(\not{p}\gamma_\mu\not{q}\gamma_\nu) - \underbrace{\text{Tr}(m\gamma_\mu\not{q}\gamma_\nu)}_0 + \underbrace{\text{Tr}(\not{p}\gamma_\mu m\gamma_\nu)}_0 \\ &\quad - \text{Tr}(m^2\gamma_\mu\gamma_\nu) - \text{Tr}(\not{p}\gamma_\mu\gamma^5\not{q}\gamma_\nu) \\ &\quad + \underbrace{\text{Tr}(m\gamma_\mu\gamma^5\not{q}\gamma_\nu)}_0 - \underbrace{\text{Tr}(\not{p}\gamma_\mu\gamma^5 m\gamma_\nu)}_0 \\ &\quad + \underbrace{\text{Tr}(m^2\gamma_\mu\gamma^5\gamma_\nu)}_0. \end{aligned} \quad (3.443)$$

$$\text{Tr}(\not{p}\gamma_\mu\not{q}\gamma_\nu) = \text{Tr}(p^\alpha \gamma_\alpha \gamma_\mu q^\beta \gamma_\beta \gamma_\nu) \quad (3.444)$$

$$= p^\alpha q^\beta \text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu) \quad (3.445)$$

$$= 4p^\alpha q^\beta (\eta_{\alpha\mu} \eta_{\beta\nu} - \eta_{\alpha\beta} \eta_{\mu\nu} + \eta_{\alpha\nu} \eta_{\mu\beta}) \quad (3.446)$$

$$= 4(p_\mu q_\nu - \vec{p} \cdot \vec{q} \eta_{\mu\nu} + p_\nu q_\mu). \quad (3.447)$$

$$\text{Tr}(m^2\gamma_\mu\gamma_\nu) = 4m^2\eta_{\mu\nu}. \quad (3.448)$$

$$\text{Tr}(\not{p}\gamma_\mu\gamma^5\not{q}\gamma_\nu) = p^\alpha q^\beta \text{Tr}(\gamma_\alpha \gamma_\mu \gamma^5 \gamma_\beta \gamma_\nu) \quad (3.449)$$

$$= -p^\alpha q^\beta \underbrace{4i\epsilon_{\alpha\mu\beta\nu}}_{\text{Tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\nu \gamma^5)} \quad (3.450)$$

$$= 0, \quad (3.451)$$

since $p^\alpha q^\beta$ is symmetric in α, β , while $\epsilon_{\alpha\mu\beta\nu}$ is antisymmetric in α, β .

Therefore,

$$\text{Tr}[(\not{p} - m)\gamma_\mu(1 - \gamma^5)(\not{q} + m)\gamma_\nu] = 4(p_\mu q_\nu - \vec{p} \cdot \vec{q} \eta_{\mu\nu} + p_\nu q_\mu) - 4m^2\eta_{\mu\nu}. \quad (3.452)$$

□

Problem 3.7 prove the Gordon identity

$$\bar{u}(\vec{p}_1) [(p_1 + p_2)^\mu + i\sigma^{\mu\nu}(p_1 - p_2)_\nu] u(\vec{p}_2) = 2m\bar{u}(\vec{p}_1)\gamma^\mu u(\vec{p}_2),$$

where

$$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu].$$

Using the previous result, write the current

$$J^\mu = \bar{u}(\vec{p}_2)\gamma^\mu u(\vec{p}_1)$$

as

$$J^\mu = \bar{u}(\vec{p}_2) [F_1(m, q^2)\gamma^\mu + F_2(m, q^2)\sigma^{\mu\nu}q_\nu] u(\vec{p}_1),$$

where

$$q^\mu = p_2^\mu - p_1^\mu,$$

and determine the functions $F_1(m, q^2)$ and $F_2(m, q^2)$.

Solution. We know that $\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}$ therefore $= -\gamma_\nu\gamma_\mu + (2\eta_{\mu\nu} - \gamma_\nu\gamma_\mu)$. Then

$$[\gamma_\mu, \gamma_\nu] = \gamma_\mu\gamma_\nu - \gamma_\nu\gamma_\mu = \gamma_\mu\gamma_\nu - (2\eta_{\mu\nu} - \gamma_\mu\gamma_\nu) = 2\gamma_\mu\gamma_\nu - 2\eta_{\mu\nu} = 2\eta_{\mu\nu} - 2\gamma_\nu\gamma_\mu \quad (3.453)$$

$$i\sigma_{\mu\nu} = -(\gamma_\mu\gamma_\nu - \eta_{\mu\nu}) = \eta_{\mu\nu} - \gamma_\mu\gamma_\nu = \gamma_\nu\gamma_\mu - \eta_{\mu\nu} \quad (3.454)$$

now let's compute the term with $\sigma_{\mu\nu}$

$$\bar{u}(\vec{p}_1)i\sigma^{\mu\nu}(p_1^\nu - p_2^\nu)u(\vec{p}_2) \quad (3.455)$$

$$= \bar{u}(\vec{p}_1)[(\gamma^\nu\gamma^\mu - \eta^{\mu\nu})p_{1\nu} - (\eta^{\mu\nu} - \gamma^\mu\gamma^\nu)p_{2\nu}]u(\vec{p}_2) \quad (3.456)$$

$$= \bar{u}(\vec{p}_1)[\gamma^\nu p_{1\nu}\gamma^\mu - p_1^\mu - p_2^\mu + \gamma^\mu\gamma^\nu p_{2\nu}]u(\vec{p}_2) \quad (3.457)$$

$$= \bar{u}(\vec{p}_1)[\gamma \cdot p_1\gamma^\mu - (p_1 + p_2)^\mu + \gamma^\mu\gamma \cdot p_2]u(\vec{p}_2) \quad (3.458)$$

from dirac's equation

$$(\gamma \cdot p - m)u(p) = 0 \rightarrow \gamma \cdot pu(p) = mu(p) \quad (3.459)$$

$$\bar{u}(p')(\gamma \cdot p' - m) = 0 \rightarrow \bar{u}(p')\gamma \cdot p' = m\bar{u}(p') \quad (3.460)$$

$$\bar{u}(\vec{p}_1)[m\gamma^\mu - (p_1 + p_2)^\mu + m\gamma^\mu]u(p_2) \quad (3.461)$$

$$= \bar{u}(\vec{p}_1)[2m\gamma^\mu - (p_1 + p_2)^\mu]u(\vec{p}_2) \quad (3.462)$$

plugging it all into the original equation

$$\bar{u}(\vec{p}_1)[(p_1 + p_2)^\mu + i\sigma^{\mu\nu}(p_1 - p_2)_\nu]u(\vec{p}_2) \quad (3.463)$$

$$= \bar{u}(\vec{p}_1)[(p_1 + p_2)^\mu + 2m\gamma^\mu - (p_1 + p_2)^\mu]u(\vec{p}_2) \quad (3.464)$$

$$= 2m\bar{u}(\vec{p}_1)\gamma^\mu u(\vec{p}_2). \quad (3.465)$$

$$J_\mu = \bar{u}(\vec{p}_2)\not{p}_1\gamma_\mu\not{p}_2u(\vec{p}_1) \quad (3.466)$$

$$\not{p}_1\gamma_\mu\not{p}_2 = (2p_1^\nu\eta_{\mu\nu} - \gamma_\mu p_1)p_2 = 2p_{1\mu}\not{p}_2 - \gamma_\mu(2p_1 \cdot p_2 - \not{p}_2\not{p}_1) \quad (3.467)$$

$$= 2(p_{1\mu}\not{p}_2 + p_{2\mu}\not{p}_1 - p_1 \cdot p_2\gamma_\mu) - \not{p}_2\gamma_\mu\not{p}_1 \quad (3.468)$$

plugging it in the current

$$J_\mu = \bar{u}(\vec{p}_2)\not{p}_1\gamma_\mu\not{p}_2u(\vec{p}_1) \quad (3.469)$$

$$= \bar{u}(\vec{p}_2)[2(p_{1\mu}\not{p}_1 + p_{2\mu}\not{p}_1 - p_1 \cdot p_2\gamma_\mu) - \not{p}_2\gamma_\mu\not{p}_1]u(\vec{p}_1) \quad (3.470)$$

$$= \hat{u}(\vec{p}_2)[2m(p_{1\mu} + p_{2\mu}) - (2p_1 \cdot p_2 + m^2)\gamma_\mu]u(\vec{p}_1) \quad (3.471)$$

$$q^2 = p_1^2 + p_2^2 - 2p_1 \cdot p_2 = 2m^2 - 2p_1 \cdot p_2 \quad (3.472)$$

$$\bar{u}(\vec{p}_1)[(p_1 + p_2)_\mu + i\sigma_{\mu\nu}(p_1 - p_2)^\nu]u(p_2) = 2m\bar{u}(\vec{p}_1)\gamma_\mu u(\vec{p}_2) \quad (3.473)$$

applying gordon's identity

$$\rightarrow \bar{u}(\vec{p}_2)[-2mi\sigma_{\mu\nu}q^\nu + (q^2 + m^2)\gamma_\mu]u(\vec{p}_1) \quad (3.474)$$

and by comparison one finds

$$F_1 = m^2 + q^2, \quad F_2 = -2mi \quad (3.475)$$

□

4 From Classical Field Theory to Quantum Field theory

$\psi \rightarrow$ is not a wave function of a single particle.

Dirac hole theory failed to explain $n \rightarrow p + \bar{e} + \bar{\nu}_e$ which is a intrinsically many body process.

We want a theory that describes infinitely many degrees of freedom $\rightarrow$ field (a continuous function of space and time).

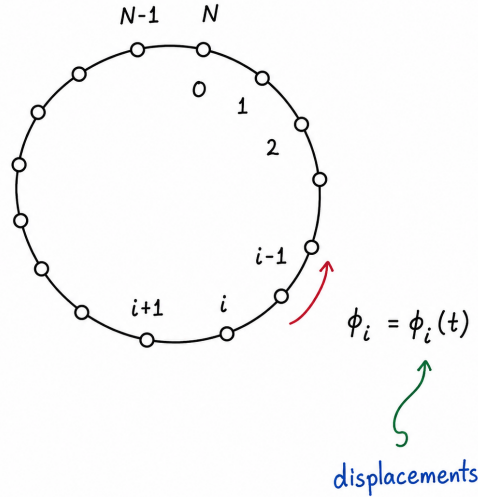
How do we choose the dynamical equations? How do we quantize them? we will use Lagrangian formulation (it is covariant)

4.1 Quantization of non relativistic string

$$L = T - V; \quad H = T + V \quad (4.1)$$

where L is the lagrangian, T the kinetic energy, V the potential energy and H is the hamiltonian.

We will describe the string as N oscillators, and, imposing periodicity $\phi_0 = \phi_N \rightarrow \frac{d\phi_0}{dt} = \frac{d\phi_N}{dt}$



the total kinetic energy is

$$T = \frac{1}{2} m \sum_{i=0}^{N-1} \left(\frac{d\phi_i}{dt} \right)^2 \quad (4.2)$$

and the total potential energy

$$V = \frac{1}{2} k \sum_{i=0}^{N-1} (\phi_{i+1} - \phi_i)^2 \quad (4.3)$$

where k is the spring constant.

Taking the limit as $N \rightarrow \infty$:

the distance between 2 oscillators is $a \rightarrow l = Na$. We also define the variables $\mu \equiv ma$,

$\tau = ka$ where l, μ, τ are fixed.

The spatial coordinates become continuous $\phi_i(t) = \phi_i(t, z_i) = \phi(t, z)$. The kinetic term

$$T = \frac{1}{2} m \sum_{i=0}^{N-1} \left(\frac{d\phi_i}{dt} \right)^2 \rightarrow \frac{1}{2} \mu \int_0^l \left[\frac{\partial \phi(t, z)}{\partial t} \right]^2 dz \quad (4.4)$$

and the potential

$$V = \frac{1}{2}k \sum_{i=0}^{N-1} (\phi_{i+1} - \phi_i)^2 \rightarrow \frac{1}{2}\tau \int_0^l \left[\frac{\partial \phi(t, z)}{\partial z} \right]^2 dz \quad (4.5)$$

L and H are

$$L = T - V = \int_0^l \left[\frac{1}{2}\mu \left(\frac{\partial \phi}{\partial t} \right)^2 - \frac{1}{2}\tau \left(\frac{\partial \phi}{\partial z} \right)^2 \right] dz = \int_0^l \mathcal{L} dz \quad (4.6)$$

$$H = T + V = \int_0^l \left[\frac{1}{2}\mu \left(\frac{\partial \phi}{\partial t} \right)^2 + \frac{1}{2}\tau \left(\frac{\partial \phi}{\partial z} \right)^2 \right] dz = \int_0^l \mathcal{H} dz \quad (4.7)$$

where $\mathcal{L}$ and $\mathcal{H}$ are the lagrangian and hamiltonian density respectively.

To find the equations of motion we invoke the principle of least action

$$\delta S = \delta \int L(\phi_i, \dot{\phi}_i, t) dt = 0 \quad (4.8)$$

for the discrete case, $\delta S = 0$, simply the Euler-Lagrange Equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}_i} \right) - \frac{\partial L}{\partial \phi_i} = 0 \text{ for } i = 0, 1, \dots, N-1 \quad (4.9)$$

in the continuum

$$-\frac{\partial \mathcal{L}}{\partial \phi} + \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial (\frac{\partial \phi}{\partial t})} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \mathcal{L}}{\partial (\frac{\partial \phi}{\partial z})} \right) \quad (4.10)$$

Injecting $\mathcal{L}/L$ into the Euler-Lagrange equations,

$$\frac{\rho}{T} \frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} = 0, \quad v = \sqrt{\frac{\tau}{\mu}}, \quad (4.11)$$

where v is the speed of the wave.

We have passed from an N -particle system to a continuous field, $N \rightarrow \infty$:

$$\frac{1}{v^2} \frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} = 0. \quad (4.12)$$

This is the continuous equation of motion of a string.

For $v = 1$,

$$\frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} = 0. \quad (4.13)$$

Extending to three dimensions,

$$\frac{\partial^2 \phi}{\partial t^2} - \sum_{i=1}^3 \frac{\partial^2 \phi}{\partial x_i^2} = 0. \quad (4.14)$$

Therefore,

$$\left(\frac{\partial^2}{\partial t^2} - \sum_{i=1}^3 \frac{\partial^2}{\partial x_i^2} \right) \phi = 0 \quad \implies \quad \square \phi = 0. \quad (4.15)$$

This is the massless Klein-Gordon equation, describing phonons.

As in classical mechanics, for a field

$$\phi_i(t, \vec{x}), \quad (4.16)$$

where i labels the i -th field, we can define its conjugate momentum:

$$\Pi_i(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_i}. \quad (4.17)$$

With this, we can construct the Hamiltonian density:

$$\mathcal{H} = \sum_i \Pi_i \dot{\phi}_i - \mathcal{L}. \quad (4.18)$$

The Hamiltonian is

$$H = \int d^3x \mathcal{H} = \int d^3x \left(\sum_i \Pi_i \dot{\phi}_i - \mathcal{L} \right). \quad (4.19)$$

In field theory, we distinguish between L , whose time integral is the action,

$$S = \int L dt, \quad (4.20)$$

and the Lagrangian density, which has to be integrated in all directions to obtain the action:

$$S = \int \mathcal{L} \left(\phi, \dot{\phi}, \frac{\partial \phi}{\partial \vec{x}}, \vec{x}, t \right) d^3x dt. \quad (4.21)$$

The spatial volume integral of $\mathcal{L}$ is L :

$$L = \int \mathcal{L} d^3\vec{x}. \quad (4.22)$$

In four dimensions,

$$S = \int_{t_i}^{t_f} d^4x \mathcal{L}, \quad \mathcal{L} = \mathcal{L}(\phi, \partial_\mu \phi). \quad (4.23)$$

The variation of the action is

$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta (\partial_\mu \phi) \right]. \quad (4.24)$$

Therefore, the Euler–Lagrange equation for a field in four dimensions is

$$\boxed{0 = \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right)}. \quad (4.25)$$

4.2 Noether's Theorem

In order to construct $\mathcal{L}$, we need guidance from symmetry principles.

Consider

$$S = \int \mathcal{L} d^4x, \quad (4.26)$$

with

$$\mathcal{L} = \mathcal{L}(\phi, \partial_\mu \phi). \quad (4.27)$$

Under a transformation,

$$x^\mu \longrightarrow x'^\mu, \quad (4.28)$$

$$\phi(x) \longrightarrow \phi'(x'), \quad (4.29)$$

and

$$\partial_\mu \phi(x) \longrightarrow \partial'_\mu \phi'(x'). \quad (4.30)$$

The dynamics, and therefore the action, have to be invariant:

$$S = \int d^4x \mathcal{L}(\phi(x), \partial_\mu \phi(x)) = \int d^4x' \mathcal{L}(\phi'(x'), \partial'_\mu \phi'(x')). \quad (4.31)$$

If, under a transformation, the action does not change, then the system has a symmetry.

Consider the transformation

$$x'^\mu = x^\mu. \quad (4.32)$$

For an infinitesimal transformation,

$$\left| \frac{\partial x'}{\partial x} \right| = 1, \quad (4.33)$$

which is sufficiently general for spacetime transformations.

Applying this transformation,

$$\delta S = \int d^4x' \mathcal{L}(\phi'(x'), \partial'_\mu \phi'(x')) - \int d^4x \mathcal{L}(\phi(x), \partial_\mu \phi(x)) \quad (4.34)$$

$$= 0. \quad (4.35)$$

Therefore,

$$\int d^4x [\mathcal{L}(\phi'(x), \partial_\mu \phi'(x)) - \mathcal{L}(\phi(x), \partial_\mu \phi(x))] = 0. \quad (4.36)$$

The difference of the Lagrangian densities can be written as a total divergence:

$$\mathcal{L}(\phi'(x), \partial_\mu \phi'(x)) - \mathcal{L}(\phi(x), \partial_\mu \phi(x)) = \partial_\mu k^\mu. \quad (4.37)$$

Here, k^μ can vanish or not.

Define the infinitesimal change

$$\phi'(x) - \phi(x) = \delta\phi(x). \quad (4.38)$$

The corresponding variation of the derivative is

$$\delta(\partial_\mu \phi(x)) = \partial_\mu(\delta\phi(x)). \quad (4.39)$$

Indeed,

$$\partial_\mu \phi(x) = \lim_{\epsilon \rightarrow 0} \frac{\phi(x + \epsilon) - \phi(x)}{\epsilon}, \quad (4.40)$$

$$\delta(\partial_\mu \phi(x)) = \lim_{\epsilon \rightarrow 0} \frac{\delta\phi(x + \epsilon) - \delta\phi(x)}{\epsilon} \quad (4.41)$$

$$= \partial_\mu(\delta\phi(x)). \quad (4.42)$$

The difference of the Lagrangian densities is

$$\begin{aligned} & \mathcal{L}(\phi'(x), \partial_\mu \phi'(x)) - \mathcal{L}(\phi(x), \partial_\mu \phi(x)) \\ &= \delta\phi(x) \frac{\partial \mathcal{L}}{\partial \phi(x)} + \delta(\partial_\mu \phi(x)) \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} \end{aligned} \quad (4.43)$$

$$= \delta\phi(x) \frac{\partial \mathcal{L}}{\partial \phi(x)} + [\partial_\mu \delta\phi(x)] \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))}. \quad (4.44)$$

Using the Euler–Lagrange equation,

$$\frac{\partial \mathcal{L}}{\partial \phi(x)} = \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} \right], \quad (4.45)$$

we obtain

$$\begin{aligned} & \delta\phi(x) \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} \right] + [\partial_\mu \delta\phi(x)] \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} \\ &= \partial_\mu \left[\delta\phi(x) \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} \right] \end{aligned} \quad (4.46)$$

$$= \partial_\mu k^\mu. \quad (4.47)$$

Therefore,

$$\partial_\mu \left[\delta\phi(x) \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} - k^\mu \right] = 0. \quad (4.48)$$

Whenever there is a continuous symmetry of the system, we can define a current

$$j^\mu(x) = \delta\phi(x) \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi(x))} - k^\mu, \quad (4.49)$$

which is conserved:

$$\partial_\mu j^\mu = 0. \quad (4.50)$$

The conserved current is not necessarily a vector.

Whenever there is a conserved current, we can define a charge:

$$Q = \int d^3x j^0(t, \vec{x}), \quad (4.51)$$

which is constant in time and is the generator of the symmetry.

Indeed,

$$\frac{dQ}{dt} = \int d^3x \partial_0 j^0 \quad (4.52)$$

$$= \int d^3x (\partial_0 j^0 + \partial_i j^i) \quad (4.53)$$

$$= \int d^3x \partial_\mu j^\mu \quad (4.54)$$

$$= 0. \quad (4.55)$$

Because of the time independence,

$$[Q, H] = 0. \quad (4.56)$$

If there is a continuous symmetry of the system, then there is a conserved charge that is the generator of the symmetry.

4.3 Spacetime Translation

Consider an infinitesimal translation

$$x^\mu \longrightarrow x'^\mu = x^\mu + \epsilon^\mu, \quad (4.57)$$

where ϵ^μ is an arbitrary infinitesimal parameter.

The variation of the field is

$$\delta\phi(x) = \phi'(x) - \phi(x) = \epsilon^\nu \partial_\nu \phi(x). \quad (4.58)$$

Therefore,

$$\delta\mathcal{L} = \delta\mathcal{L}(\phi, \partial_\mu \phi) = \epsilon^\nu \partial_\nu \mathcal{L}(\phi, \partial_\mu \phi). \quad (4.59)$$

On the other hand,

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi} \delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) \quad (4.60)$$

$$= \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi \right). \quad (4.61)$$

By Noether's theorem,

$$\partial_\mu j^\mu = 0. \quad (4.62)$$

Comparing both variations of the Lagrangian,

$$\partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi \right) - \epsilon^\nu \partial_\nu \mathcal{L} = 0. \quad (4.63)$$

Using

$$\delta\phi = \epsilon^\nu \partial_\nu \phi, \quad (4.64)$$

we obtain

$$\partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi)} \epsilon^\nu \partial_\nu \phi \right) - \epsilon^\nu \partial_\nu \mathcal{L} = 0 \quad (4.65)$$

$$\epsilon^\nu \left[\partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi \right) - \partial_\nu \mathcal{L} \right] = 0. \quad (4.66)$$

Since the current does not depend on ϵ^ν ,

$$\partial_\mu \left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \eta^\mu{}_\nu \mathcal{L} \right] = 0. \quad (4.67)$$

We identify the energy–momentum tensor as

$$T^\mu{}_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \eta^\mu{}_\nu \mathcal{L}. \quad (4.68)$$

The current j^μ is the source of the electromagnetic potential A^μ , whereas $T^{\mu\nu}$ is the source of the gravitational field $G^{\mu\nu}$.

The energy–momentum tensor is symmetric:

$$T^{\mu\nu} = T^{\nu\mu}. \quad (4.69)$$

It defines four conserved quantities,

$$P^\nu = \int d^3x T^{0\nu}. \quad (4.70)$$

For $\nu = 0$,

$$P^0 = \int d^3x T^{00} \quad (4.71)$$

$$= \int d^3x \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \dot{\phi} - \mathcal{L} \right) \quad (4.72)$$

$$= H. \quad (4.73)$$

For $\nu = 1, 2, 3$,

$$P^i = \int d^3x T^{0i} \quad (4.74)$$

$$= \int d^3x \frac{\partial \mathcal{L}}{\partial \dot{\phi}} \partial^i \phi \quad (4.75)$$

$$= - \int d^3x \Pi \partial_i \phi \quad (4.76)$$

$$= - \int d^3x \Pi \vec{\nabla} \phi. \quad (4.77)$$

This is interpreted as the linear momentum.

Earlier, the system was described by a finite number of coupled oscillators. Taking

$$N \longrightarrow \infty, \quad (4.78)$$

we obtained a continuous classical field, understood as an oscillator pattern with unshared energies.

We now want to quantize the field:

$$\text{classical field} \longrightarrow \text{quantized field}. \quad (4.79)$$

4.4 Quantization of String

We need to start by finding the normal modes:

$$\phi \sim e^{+i(k_n z - \omega_n t)}. \quad (4.80)$$

Periodicity implies

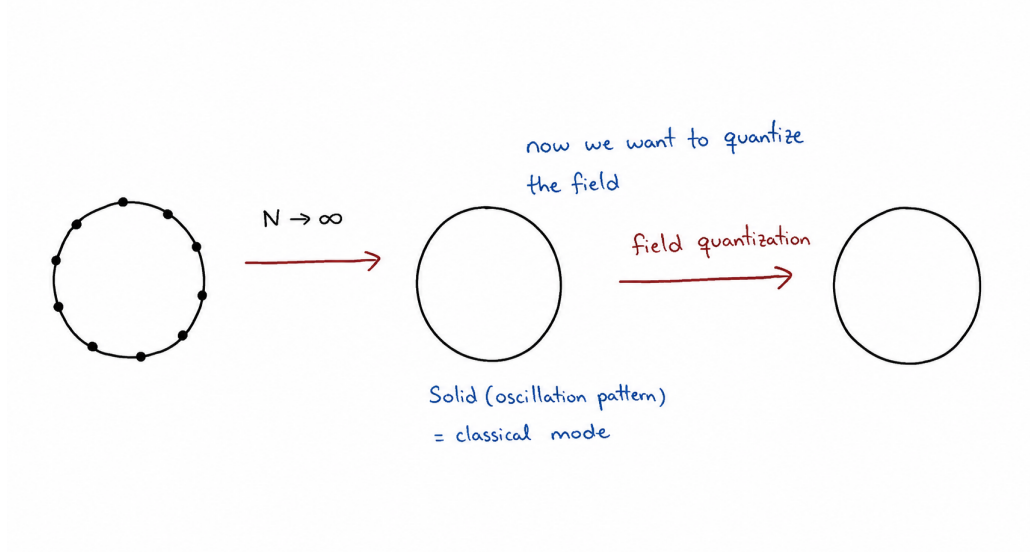
$$k_n = \frac{2\pi n}{L}, \quad n = 0, \pm 1, \pm 2, \dots \quad (4.81)$$

The wave equation gives

$$\omega_n^2 = v^2 k_n^2. \quad (4.82)$$

The positive-frequency solutions are

$$\phi_n(z, t) = \frac{1}{\sqrt{L}} e^{i(k_n z - \omega_n t)}, \quad \omega_n > 0. \quad (4.83)$$



The negative-frequency solutions are

$$\phi_n^*(z, t) = \frac{1}{\sqrt{L}} e^{i(-k_n z + \omega_n t)}, \quad \omega_n < 0. \quad (4.84)$$

Thus, $\phi_n^*(z, t)$ corresponds to the negative-frequency solution.

The normal modes satisfy

$$\int_0^L dz \phi_n^*(z, t) \phi_m(z, t) = \delta_{nm}. \quad (4.85)$$

The most general field is

$$\phi(z, t) = \sum_{n=-\infty}^{\infty} C_n [a_n(0) \phi_n(z, t) + a_n^*(0) \phi_n^*(z, t)] \quad (4.86)$$

$$= \sum_{n=-\infty}^{\infty} \frac{C_n}{\sqrt{L}} [a_n(t) e^{ik_n z} + a_n^*(t) e^{-ik_n z}]. \quad (4.87)$$

Here, $a_n(0)$ and $a_n^*(0)$ are the coefficients of each normal mode, and C_n is an overall normalization.

The time dependence is

$$a_n(t) = a_n(0) e^{-i\omega_n t}, \quad (4.88)$$

and

$$a_n^*(t) = a_n^*(0) e^{i\omega_n t}. \quad (4.89)$$

Each normal mode behaves as an independent harmonic oscillator:

$$\ddot{a}_n(t) + \omega_n^2 a_n(t) = 0. \quad (4.90)$$

Therefore, to quantize the field, we have to quantize the harmonic oscillators.

We can describe the Hamiltonian in terms of normal modes. Since

$$\dot{a}_n(t) = -i\omega_n a_n(t), \quad (4.91)$$

we obtain

$$H = T + V = \sum_{n=-\infty}^{\infty} C_n^2 \frac{2\omega_n^2}{v^2} a_n^*(t) a_n(t). \quad (4.92)$$

To make the coefficients dimensionless and H have dimensions of energy, we choose

$$C_n = \sqrt{\frac{v^2}{2\omega_n}}. \quad (4.93)$$

Therefore,

$$H = \sum_{n=-\infty}^{\infty} \omega_n a_n^*(t) a_n(t) \quad (4.94)$$

$$= \sum_{n=-\infty}^{\infty} \omega_n a_n^*(0) a_n(0). \quad (4.95)$$

Generalized position and momentum

An alternative description is

$$H = \sum_n \Pi_n \dot{\phi}_n - L. \quad (4.96)$$

We choose

$$\phi_n(t) = \frac{1}{\sqrt{2\omega_n}} [a_n(t) + a_n^*(t)]. \quad (4.97)$$

The conjugate momentum is

$$\Pi_n(t) = \frac{d\phi_n}{dt} \quad (4.98)$$

$$= -\frac{i\omega_n}{\sqrt{2\omega_n}} [a_n(t) - a_n^*(t)] \quad (4.99)$$

$$= -i\sqrt{\frac{\omega_n}{2}} [a_n(t) - a_n^*(t)]. \quad (4.100)$$

If we express $a_n(t)$ and $a_n^*(t)$ as functions of ϕ_n and Π_n , we obtain

$$H = \sum_{n=-\infty}^{\infty} \frac{1}{2} (\Pi_n^2 + \omega_n^2 \phi_n^2), \quad (4.101)$$

which is a sum of independent harmonic oscillators.

Quantization of the string

We will use canonical quantization.

The first type of field quantization applies to the string, the Klein–Gordon field, the Dirac field and the electromagnetic field.

In canonical quantization, the canonical variables are promoted to operators, transforming Poisson brackets into commutators.

For generalized position and momentum,

$$[\phi_i, \phi_j] = [\Pi_i, \Pi_j] = 0, \quad (4.102)$$

and

$$[\phi_i, \Pi_j] = i\delta_{ij}, \quad (4.103)$$

where we are setting $\hbar = 1$.

We can obtain the commutation relations for a_n and $a_n^\dagger$:

$$[a_i, a_j^\dagger] = \delta_{ij}, \quad (4.104)$$

and

$$[a_i, a_j] = [a_i^\dagger, a_j^\dagger] = 0. \quad (4.105)$$

We have replaced the coefficients a_n and a_n^* by the quantum operators a_n and $a_n^\dagger$.

The quantum field is

$$\phi(t, z) = v \sum_{n=-\infty}^{\infty} \frac{1}{\sqrt{2\omega_n L}} \left[a_n e^{i(k_n z - \omega_n t)} + a_n^\dagger e^{-i(k_n z - \omega_n t)} \right] \quad (4.106)$$

$$= \phi_+(t, z) + \phi_-(t, z). \quad (4.107)$$

Here, ϕ_+ contains the annihilation operator a_n , whereas ϕ_- contains the creation operator $a_n^\dagger$. This is called field quantization or second quantization. The uncertainty relation is

$$\Delta x \Delta p \geq \frac{1}{2}. \quad (4.108)$$

The relations

$$[\phi_i, \Pi_j] = i\delta_{ij}, \quad [\phi_i, \phi_j] = [\Pi_i, \Pi_j] = 0, \quad (4.109)$$

imply

$$[a_i, a_j^\dagger] = \delta_{ij}, \quad [a_i, a_j] = [a_i^\dagger, a_j^\dagger] = 0. \quad (4.110)$$

These are not universal quantization relations.

We have seen:

$$\text{Klein-Gordon: spin 0,} \quad \text{Dirac: spin } \frac{1}{2}, \quad \text{photon: spin 1.} \quad (4.111)$$

For the non-relativistic string, the normal modes have spin 0.

The relations derived here are the ones for the string.

4.5 Number Operator

The Hamiltonian of the string was

$$H = \sum_{n=-\infty}^{\infty} \omega_n a_n^\dagger a_n. \quad (4.112)$$

To find the eigenstates of H , we need to find the eigenstates of

$$a_n^\dagger a_n \equiv N, \quad (4.113)$$

where N is the number operator.

Let us study the relations between H , a_n and $a_n^\dagger$:

$$[H, a_m] = \omega_n [a_n^\dagger a_n, a_m] \quad (4.114)$$

$$= \omega_n [a_n^\dagger, a_m] a_n \quad (4.115)$$

$$= -\omega_n a_m, \quad (4.116)$$

$$[H, a_m^\dagger] = \omega_n a_m^\dagger. \quad (4.117)$$

Taking the basis of eigenvectors of H ,

$$H|E_n\rangle = E_n|E_n\rangle, \quad (4.118)$$

we can study the action of Ha_n and $Ha_n^\dagger$:

$$Ha_n|E_n\rangle = (a_n H - \omega_n a_n)|E_n\rangle \quad (4.119)$$

$$= (E_n - \omega_n)a_n|E_n\rangle, \quad (4.120)$$

$$Ha_n^\dagger|E_n\rangle = (a_n^\dagger H + \omega_n a_n^\dagger)|E_n\rangle \quad (4.121)$$

$$= (E_n + \omega_n)a_n^\dagger|E_n\rangle. \quad (4.122)$$

Therefore,

$$a_n|E_n\rangle \quad (4.123)$$

is an eigenvector of H with eigenvalue $E_n - \omega_n$, so a_n is the annihilation operator.

Similarly,

$$a_n^\dagger|E_n\rangle \quad (4.124)$$

is an eigenvector of H with eigenvalue $E_n + \omega_n$, so $a_n^\dagger$ is the creation operator.

The normalization is

$$a_n^\dagger |E_n\rangle = \sqrt{n+1} |E_{n+1}\rangle, \quad (4.125)$$

$$a_n |E_n\rangle = \sqrt{n} |E_{n-1}\rangle. \quad (4.126)$$

The number operator acts as

$$N |E_n\rangle = a^\dagger a |E_n\rangle \quad (4.127)$$

$$= a^\dagger \sqrt{n} |E_{n-1}\rangle \quad (4.128)$$

$$= \sqrt{n} \sqrt{n} |E_n\rangle \quad (4.129)$$

$$= n |E_n\rangle. \quad (4.130)$$

The energy levels are

$$E_{k,n} = \hbar \omega_k \left(n + \frac{1}{2} \right), \quad (4.131)$$

with

$$\omega_k = \sqrt{k^2 + m^2}. \quad (4.132)$$

The allowed momenta are

$$k \in \mathbb{Z}, \quad k = 0, \pm 1, \pm 2, \dots \quad (4.133)$$

and

$$k = \frac{2\pi n}{L}. \quad (4.134)$$

The number operator gives the number of phonons with energy ω_n .

There is no direct connection between phonons and the original mass points of the string.

Phonons are associated with the frequency and describe the motion of the string as a whole.

Phonons are localized in momentum space, whereas the particles of the string are localized in position space.

In the continuum limit,

$$L \longrightarrow \infty, \quad (4.135)$$

while keeping the tension and density fixed, the canonical commutation relations become

$$[\phi(t, \vec{x}), \Pi(t, \vec{y})] = i\delta^3(\vec{x} - \vec{y}), \quad (4.136)$$

$$[\phi(t, \vec{x}), \phi(t, \vec{y})] = [\Pi(t, \vec{x}), \Pi(t, \vec{y})] = 0. \quad (4.137)$$

For several fields,

$$[\phi_\alpha(t, \vec{x}), \Pi_\beta(t, \vec{y})] = i\delta_{\alpha\beta}\delta^3(\vec{x} - \vec{y}), \quad (4.138)$$

and

$$[\phi_\alpha(t, \vec{x}), \phi_\beta(t, \vec{y})] = [\Pi_\alpha(t, \vec{x}), \Pi_\beta(t, \vec{y})] = 0. \quad (4.139)$$

5 Quantization of the scalar field

We will quantize the Klein–Gordon field, both real and complex.

The Lagrangian giving rise to the Klein–Gordon equation is

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2). \quad (5.1)$$

Applying the Euler–Lagrange equations,

$$\frac{\partial \mathcal{L}}{\partial \phi} = -m^2 \phi, \quad (5.2)$$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi. \quad (5.3)$$

Therefore,

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = 0, \quad (5.4)$$

$$-m^2 \phi - \partial_\mu (\partial^\mu \phi) = 0 \quad (5.5)$$

$$\implies (\partial_\mu \partial^\mu + m^2) \phi = 0. \quad (5.6)$$

Given a complete set of solutions of the Klein–Gordon equation, we can expand the field operator $\phi(x)$ in that basis.

Our complete basis is formed by plane waves:

$$\phi(x) = \int \frac{d^3 p}{\sqrt{(2\pi)^3 2E}} [e^{-ipx} a(\vec{p}) + e^{ipx} a^\dagger(\vec{p})]. \quad (5.7)$$

The Hermitian part makes $\phi(x)$ real.

Decomposing into positive- and negative-frequency parts,

$$\phi^{(+)}(x) = \int \frac{d^3 p}{\sqrt{(2\pi)^3 2E}} e^{-ipx} a(\vec{p}), \quad (5.8)$$

$$\phi^{(-)}(x) = \int \frac{d^3 p}{\sqrt{(2\pi)^3 2E}} e^{ipx} a^\dagger(\vec{p}), \quad (5.9)$$

$$\phi(x) = \phi^{(+)}(x) + \phi^{(-)}(x). \quad (5.10)$$

The conjugate momentum is

$$\Pi(x) = \dot{\phi}(x) = -i \int \frac{d^3 p}{\sqrt{(2\pi)^3 2E}} E [e^{-ipx} a(\vec{p}) - e^{ipx} a^\dagger(\vec{p})]. \quad (5.11)$$

Considering $\phi(x)$ and $\Pi(x)$, we impose the equal-time commutation relations

$$[\phi(x), \phi(y)]_{x^0=y^0} = 0, \quad (5.12)$$

$$[\Pi(x), \Pi(y)]_{x^0=y^0} = 0, \quad (5.13)$$

$$[\phi(x), \Pi(y)]_{x^0=y^0} = i\delta^3(\vec{x} - \vec{y}). \quad (5.14)$$

We compute the field commutators in order to obtain the commutation relations between the coefficients $a(\vec{p})$ and $a^\dagger(\vec{p})$.

$$\begin{aligned}
[\phi(x), \phi(y)]_{x^0=y^0} &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} \left[e^{-ipx-ip'y} [a(\vec{p}), a(\vec{p}')] \right. \\
&\quad + e^{-ipx+ip'y} [a(\vec{p}), a^\dagger(\vec{p}')] \\
&\quad + e^{ipx-ip'y} [a^\dagger(\vec{p}), a(\vec{p}')] \\
&\quad \left. + e^{ipx+ip'y} [a^\dagger(\vec{p}), a^\dagger(\vec{p}')] \right] = 0.
\end{aligned} \tag{5.15}$$

Similarly,

$$\begin{aligned}
[\Pi(x), \Pi(y)]_{x^0=y^0} &= - \int d^3p \int d^3p' \frac{EE'}{(2\pi)^3 2\sqrt{EE'}} \left[e^{-ipx-ip'y} [a(\vec{p}), a(\vec{p}')] \right. \\
&\quad - e^{-ipx+ip'y} [a(\vec{p}), a^\dagger(\vec{p}')] \\
&\quad - e^{ipx-ip'y} [a^\dagger(\vec{p}), a(\vec{p}')] \\
&\quad \left. + e^{ipx+ip'y} [a^\dagger(\vec{p}), a^\dagger(\vec{p}')] \right] = 0.
\end{aligned} \tag{5.16}$$

For the mixed commutator,

$$\begin{aligned}
[\phi(x), \Pi(y)]_{x^0=y^0} &= -\frac{i}{2(2\pi)^3} \int d^3p \int d^3p' \sqrt{\frac{E'}{E}} \left[e^{-ipx-ip'y} [a(\vec{p}), a(\vec{p}')] \right. \\
&\quad - e^{-ipx+ip'y} [a(\vec{p}), a^\dagger(\vec{p}')] \\
&\quad + e^{ipx-ip'y} [a^\dagger(\vec{p}), a(\vec{p}')] \\
&\quad \left. - e^{ipx+ip'y} [a^\dagger(\vec{p}), a^\dagger(\vec{p}')] \right]
\end{aligned} \tag{5.17}$$

$$= i\delta^3(\vec{x} - \vec{y}), \tag{5.18}$$

where we have used

$$\int \frac{d^3p}{(2\pi)^3} e^{i\vec{p} \cdot (\vec{x} - \vec{y})} = \delta^3(\vec{x} - \vec{y}). \tag{5.19}$$

From the commutators between $\phi(x)$ and $\Pi(y)$, we deduce

$$[a(\vec{p}), a(\vec{p}')] = [a^\dagger(\vec{p}), a^\dagger(\vec{p}')] = 0, \tag{5.20}$$

$$[a(\vec{p}), a^\dagger(\vec{p}')] = \delta^3(\vec{p} - \vec{p}'). \tag{5.21}$$

These are the same commutation relations as for the harmonic oscillator.

5.1 Hamiltonian Operator and Eigenstates

$$H = \int d^3x \mathcal{H} \tag{5.22}$$

$$= \frac{1}{2} \int d^3x \left(\Pi^2(x) + \vec{\nabla}\phi \cdot \vec{\nabla}\phi + m^2\phi^2 \right) \tag{5.23}$$

$$= \frac{1}{2} \int d^3p E \left(a(\vec{p})a^\dagger(\vec{p}) + a^\dagger(\vec{p})a(\vec{p}) \right), \tag{5.24}$$

where in the last step Π , ϕ and $\vec{\nabla}\phi$ have been substituted as functions of a and $a^\dagger$.

Therefore, the Hamiltonian of the Klein–Gordon field is the sum of the Hamiltonians of an infinite number of harmonic oscillators of frequency E :

$$H = \frac{1}{2} \int d^3p E \left(a^\dagger(\vec{p})a(\vec{p}) + a(\vec{p})a^\dagger(\vec{p}) \right) \tag{5.25}$$

$$= \frac{1}{2} \int d^3p E \left(a^\dagger(\vec{p})a(\vec{p}) + a^\dagger(\vec{p})a(\vec{p}) + 1 \right) \tag{5.26}$$

$$= \int d^3p E \left(a^\dagger(\vec{p})a(\vec{p}) + \frac{1}{2} \right). \tag{5.27}$$

The term

$$\int d^3p \frac{1}{2} \quad (5.28)$$

is infinite but constant.

We can always redefine, or shift, our energy measurement from zero. This can always be done because energies are measured with respect to a ground state; absolute energies are never measured.

We eliminate the infinity using *normal ordering* of operators.

We define normal ordering as the ordering in which every creation operator stands to the left of every annihilation operator:

$$a(\vec{p})a^\dagger(\vec{p}') \longrightarrow a(\vec{p})a^\dagger(\vec{p}') := a^\dagger(\vec{p}')a(\vec{p}). \quad (5.29)$$

Thus,

$$: H : = \int d^3p E : a^\dagger(\vec{p})a(\vec{p}) : \quad (5.30)$$

$$= \int d^3p E N(\vec{p}), \quad (5.31)$$

where the number operator is

$$N(\vec{p}) = a^\dagger(\vec{p})a(\vec{p}). \quad (5.32)$$

When normal ordering is applied, there is no reordering using commutation relations.

Some examples of normal ordering are

$$: a^\dagger a a^\dagger : = a^\dagger a^\dagger a, \quad (5.33)$$

and

$$: a^\dagger a a^\dagger a^\dagger : = a^\dagger a^\dagger a^\dagger a. \quad (5.34)$$

From the number operator, we obtain the total number operator of the system:

$$N = \int d^3p N(\vec{p}) = \int d^3p a^\dagger(\vec{p})a(\vec{p}). \quad (5.35)$$

With these definitions, we can compute the energy of the state

$$|\vec{p}\rangle = a^\dagger(\vec{p})|0\rangle. \quad (5.36)$$

Then,

$$\langle \vec{p} | H | \vec{p} \rangle = \left\langle \vec{p} \left| \int d^3p' E_{\vec{p}'} N(\vec{p}') \right| \vec{p} \right\rangle \quad (5.37)$$

$$= \left\langle \vec{p} \left| \int d^3p' E_{\vec{p}'} \delta^3(\vec{p} - \vec{p}') \right| \vec{p} \right\rangle \quad (5.38)$$

$$= E_{\vec{p}}. \quad (5.39)$$

The symmetry of the Lagrangian under displacements implies four conserved quantities:

$$P^0 = \int d^3x T^{00} \quad (5.40)$$

$$= \int d^3x \left(\frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)} \partial^0 \phi - \mathcal{L} \right) \quad (5.41)$$

$$= \int d^3x \mathcal{H} \quad (5.42)$$

$$= H, \quad (5.43)$$

and

$$P^i = \int d^3x T^{0i} \quad (5.44)$$

$$= \int d^3x \left(\frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)} \partial^i \phi \right) \quad (5.45)$$

$$= \int d^3x \Pi \partial^i \phi. \quad (5.46)$$

Therefore,

$$\vec{P} = - \int d^3x \Pi \vec{\nabla} \phi. \quad (5.47)$$

Substituting the field expansions,

$$\vec{P} = \int d^3p \frac{\vec{p}}{2} (a^\dagger(\vec{p})a(\vec{p}) + a(\vec{p})a^\dagger(\vec{p})) \quad (5.48)$$

$$= \int d^3p \vec{p} \left(N(\vec{p}) + \frac{1}{2} \right). \quad (5.49)$$

Using normal ordering,

$$: \vec{P} : = \frac{1}{2} \int d^3p \vec{p} : (a^\dagger(\vec{p})a(\vec{p}) + a(\vec{p})a^\dagger(\vec{p})) : \quad (5.50)$$

$$= \int d^3p \vec{p} a^\dagger(\vec{p})a(\vec{p}) \quad (5.51)$$

$$= \int d^3p \vec{p} N(\vec{p}). \quad (5.52)$$

For a state containing n_i particles with momentum $\vec{p}_i$,

$$|n_1, \vec{p}_1; n_2, \vec{p}_2; \dots; n_i, \vec{p}_i\rangle, \quad (5.53)$$

the Hamiltonian acts as

$$H|n_1, \vec{p}_1; n_2, \vec{p}_2; \dots; n_i, \vec{p}_i\rangle = \int d^3p E N(\vec{p})|n_1, \vec{p}_1; n_2, \vec{p}_2; \dots; n_i, \vec{p}_i\rangle \quad (5.54)$$

$$= (n_1 E_1 + n_2 E_2 + \dots + n_i E_i) |n_1, \vec{p}_1; n_2, \vec{p}_2; \dots; n_i, \vec{p}_i\rangle. \quad (5.55)$$

Similarly,

$$\vec{P}|n_1, \vec{p}_1; n_2, \vec{p}_2; \dots; n_i, \vec{p}_i\rangle = \int d^3p \vec{p} N(\vec{p})|n_1, \vec{p}_1; n_2, \vec{p}_2; \dots; n_i, \vec{p}_i\rangle \quad (5.56)$$

$$= (n_1 \vec{p}_1 + n_2 \vec{p}_2 + \dots + n_i \vec{p}_i) |n_1, \vec{p}_1; n_2, \vec{p}_2; \dots; n_i, \vec{p}_i\rangle. \quad (5.57)$$

5.2 Localized Particle

$$\phi(x) = \phi^{(+)}(x) + \phi^{(-)}(x) = \int \frac{d^3\vec{p}}{\sqrt{(2\pi)^3 2E}} (e^{-ipx} a(\vec{p}) + e^{ipx} a^\dagger(\vec{p})). \quad (5.58)$$

Let us consider the state produced by the action of the field on the vacuum:

$$|\phi(x)\rangle = \phi(x)|0\rangle = \phi^{(-)}(x)|0\rangle. \quad (5.59)$$

Since $\phi(x)$ is linear in creation and annihilation operators,

$$\langle 0|\phi(x)|0\rangle = 0. \quad (5.60)$$

The only non-zero element is

$$\langle p|\phi(x)|0\rangle = \langle p|\phi(x)\rangle, \quad (5.61)$$

which is a function representing the projection of the state $|\phi(x)\rangle$ onto the one-particle state $|p\rangle$.

The function $\langle p|\phi(x)\rangle$ is a solution of the Klein-Gordon equation:

$$(\partial_\mu \partial^\mu + m^2) \langle p|\phi(x)\rangle = \langle p|(\partial_\mu \partial^\mu + m^2) \phi(x)|0\rangle \quad (5.62)$$

$$= 0. \quad (5.63)$$

Let us project the particle state into a space point:

$$\langle \vec{x}'|\phi(0, \vec{x})|0\rangle = \left\langle \vec{x}' \left| \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (a(\vec{p})e^{-ipx} + a^\dagger(\vec{p})e^{ipx}) \right| 0 \right\rangle \quad (5.64)$$

$$= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} e^{i\vec{p}\cdot\vec{x}} \langle \vec{x}'|a^\dagger(\vec{p})|0\rangle, \quad (5.65)$$

where the field has been considered at one time, for instance $t = 0$.

$$\langle \vec{x}' | a^\dagger(\vec{p}) | 0 \rangle = \langle \vec{x}' | p \rangle = \frac{1}{\sqrt{(2\pi)^3 2E}} e^{-i\vec{p} \cdot \vec{x}'} \quad (5.66)$$

Therefore,

$$\langle \vec{x}' | \phi(0, \vec{x}) | 0 \rangle = \int \frac{d^3 p}{\sqrt{(2\pi)^3 2E}} e^{i\vec{p} \cdot \vec{x}} \frac{1}{\sqrt{(2\pi)^3 2E}} e^{-i\vec{p} \cdot \vec{x}'} \quad (5.67)$$

$$= \int \frac{d^3 p}{(2\pi)^3 2E} e^{i\vec{p} \cdot (\vec{x} - \vec{x}')} \quad (5.68)$$

$$= \frac{1}{2E} \delta^3(\vec{x} - \vec{x}'). \quad (5.69)$$

Here we have used

$$\int \frac{d^3 p}{(2\pi)^3} e^{i\vec{p} \cdot (\vec{x} - \vec{x}')} = \delta^3(\vec{x} - \vec{x}'). \quad (5.70)$$

For small $\vec{p}$,

$$E \sim m, \quad (5.71)$$

whereas for large $\vec{p}$,

$$E \sim |\vec{p}|. \quad (5.72)$$

Large momenta are therefore more suppressed.

The quantity

$$\langle \vec{x}' | \phi(0, \vec{x}) | 0 \rangle \quad (5.73)$$

represents a sharply localized state at $\vec{x}$. Therefore, $\phi(x)$ acting on the vacuum creates a particle.

We can construct a multiparticle state:

$$|\phi(x_1), \phi(x_2), \dots, \phi(x_n)\rangle = \phi^{(-)}(x_1) \phi^{(-)}(x_2) \cdots \phi^{(-)}(x_n) | 0 \rangle. \quad (5.74)$$

This state consists of n particles at $x_1, x_2, \dots, x_n$.

Such states are meaningful only if

$$x_1^0 = x_2^0 = x_3^0 = \cdots = x_n^0, \quad (5.75)$$

that is, if all particles are considered at equal time.

The wavefunction is symmetric, as expected for Bose particles:

$$\langle p, p' | \phi(x_1), \phi(x_2) \rangle = \langle p, p' | \phi^{(-)}(x_1) \phi^{(-)}(x_2) | 0 \rangle. \quad (5.76)$$

The absolute square

$$\left| \langle p, p' | \phi^{(-)}(x_1) \phi^{(-)}(x_2) | 0 \rangle \right|^2 \quad (5.77)$$

gives the probability of finding two Klein–Gordon particles with four-momenta

$$p^\mu = (E, \vec{p}), \quad p'^\mu = (E', \vec{p}'), \quad (5.78)$$

at $\vec{x}_1$ and $\vec{x}_2$, respectively, at a given time.

Second quantization theory does not have problems with negative probabilities as first quantization theory did.

5.3 Causality

Quantum field theory is based on the same postulates of quantum mechanics, with fields and canonical variables, replacing position and momentum.

A key ingredient absent in quantum mechanics but present in special relativity is causality.

Information cannot propagate faster than light.

The field $\phi(x)$ creates a particle at $\vec{x}$ when acting on the vacuum. We say that the value of the field at $\vec{x}$ cannot influence another point $\vec{x}'$ if they cannot be connected by $v < c$.

We encode this within commutators. We expect

$$[\phi(x), \phi(x')] = 0 \quad \text{if} \quad (x - x')^2 < 0, \quad (5.79)$$

for spacelike trajectories.

Now we compute the commutator at any time, taking $x' = y$:

$$[\phi(x), \phi(y)] = [\phi^{(+)}(x) + \phi^{(-)}(x), \phi^{(+)}(y) + \phi^{(-)}(y)] \quad (5.80)$$

$$= [\phi^{(+)}(x), \phi^{(-)}(y)] + [\phi^{(-)}(x), \phi^{(+)}(y)], \quad (5.81)$$

since

$$[\phi^{(+)}(x), \phi^{(+)}(y)] = [\phi^{(-)}(x), \phi^{(-)}(y)] = 0. \quad (5.82)$$

For the first term,

$$[\phi^{(+)}(x), \phi^{(-)}(y)] = \left[\int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} a(\vec{p}) e^{-ipx}, \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} a^\dagger(\vec{p}') e^{ip'y} \right] \quad (5.83)$$

$$= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} e^{-i(p x - p' y)} [a(\vec{p}), a^\dagger(\vec{p}')] \quad (5.84)$$

$$= \int \frac{d^3p}{(2\pi)^3 2E} e^{-ip(x-y)} \quad (5.85)$$

$$\equiv i\Delta_+(x - y). \quad (5.86)$$

Analogously,

$$[\phi^{(-)}(x), \phi^{(+)}(y)] = - \int \frac{d^3p}{(2\pi)^3 2E} e^{-ip(y-x)} \equiv i\Delta_-(x - y). \quad (5.87)$$

Therefore,

$$[\phi(x), \phi(y)] = i\Delta_+(x - y) + i\Delta_-(x - y) \quad (5.88)$$

$$= i\Delta(x - y), \quad (5.89)$$

where $\Delta(x - y)$ is the Klein-Gordon function.

Explicitly,

$$[\phi(x), \phi(y)] = \int \frac{d^3p}{(2\pi)^3 2E} (e^{-ip(x-y)} - e^{ip(x-y)}). \quad (5.90)$$

The Klein-Gordon function is Lorentz invariant.

It is a correlation function and only depends on $x - y$.

A Lorentz-invariant integral of a function that depends only on the scalar product $p(x - y)$ is Lorentz invariant.

We know that

$$[\phi(x), \phi(y)]_{x^0=y^0} = 0. \quad (5.91)$$

For every spacelike distance,

$$(x - y)^2 < 0, \quad (5.92)$$

we can always boost to a frame in which the two events occur at equal times.

Therefore,

$$[\phi(x), \phi(y)] = 0 \quad \text{for} \quad (x - y)^2 < 0. \quad (5.93)$$

Hence, the theory is causal.

5.4 The Klein-Gordon Propagator

Eventually we will add interactions. When computing interactions, it is useful to compute propagators, which is to say the amplitude (probability) of having a particle created at x and annihilated at y .

Propagators appear in Feynman diagrams.

Propagators encapsulate mathematical information on causality.

They are fundamental Green functions of QFT.

The Green function of a system is the solution to the inhomogeneous equation with a delta source. For the Klein–Gordon equation,

$$(\partial_\mu \partial^\mu + m^2) G(x - y) = -\delta^4(x - y). \quad (5.94)$$

We define the Fourier transforms

$$\delta^4(x - y) = \frac{1}{(2\pi)^4} \int d^4p e^{-ip(x-y)}, \quad (5.95)$$

and

$$G(x - y) = \int \frac{d^4p}{(2\pi)^4} e^{-ip(x-y)} \tilde{G}(p). \quad (5.96)$$

Substituting,

$$(\partial_\mu \partial^\mu + m^2) \int \frac{d^4p}{(2\pi)^4} e^{-ip(x-y)} \tilde{G}(p) = -\frac{1}{(2\pi)^4} \int d^4p e^{-ip(x-y)}, \quad (5.97)$$

and hence

$$\frac{1}{(2\pi)^4} (-p^2 + m^2) \tilde{G}(p) = -\frac{1}{(2\pi)^4}. \quad (5.98)$$

Therefore,

$$\boxed{\tilde{G}(p) = \frac{1}{p^2 - m^2}} \quad (5.99)$$

in momentum space.

The Green function for the Klein–Gordon equation is

$$G(x - y) = \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip(x-y)}}{p^2 - m^2}. \quad (5.100)$$

The solutions of

$$(\square + m^2) \phi(x) = 0 \quad (5.101)$$

are proportional to the Green function, so we have to perform the integral.

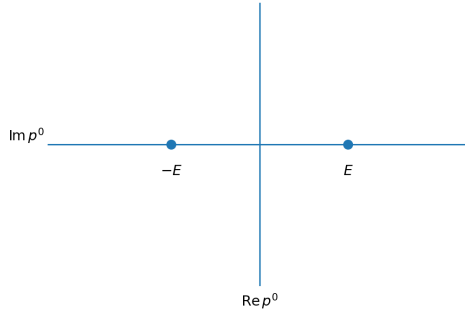
The integral of the Green function is complex and has poles when

$$p^2 - m^2 = 0. \quad (5.102)$$

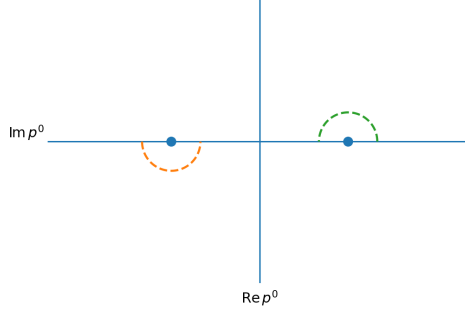
Thus,

$$E^2 = \vec{p}^2 + m^2, \quad E = \pm \sqrt{\vec{p}^2 + m^2}. \quad (5.103)$$

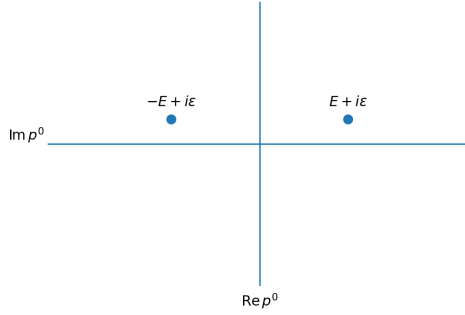
The poles occur when a free particle propagates on shell. We therefore use Cauchy integration methods.



We can choose different contours around the poles.



For the retarded prescription, both poles are displaced into the upper half-plane.



Choosing the lower circuit gives $I = 0$, whereas choosing the upper circuit encloses the poles. Using the residue theorem,

$$I = -2\pi i \frac{e^{-ip^0(x^0-y^0)}}{2p^0}. \quad (5.104)$$

Therefore,

$$G_R(x-y) = -\frac{i}{(2\pi)^3} \int \frac{d^3p}{2p^0} e^{-ip(x-y)} \quad (5.105)$$

$$= \Delta_+(x-y). \quad (5.106)$$

For

$$x^0 - y^0 < 0, \quad (5.107)$$

we obtain

$$G_R(x-y)|_{x^0-y^0<0} = -\frac{i}{(2\pi)^3} \int \frac{d^3p}{2p^0} e^{ip(x-y)} \quad (5.108)$$

$$= -\Delta_-(x-y). \quad (5.109)$$

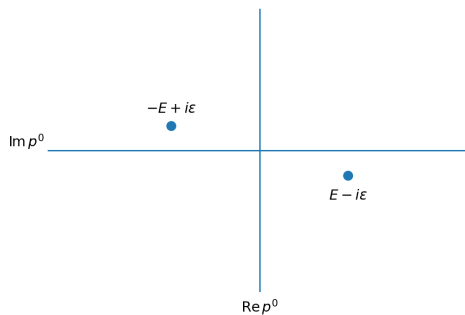
In total,

$$\Delta_R(x-y) = i\theta(x^0-y^0)\Delta_+(x-y) - i\theta(y^0-x^0)\Delta_-(x-y). \quad (5.110)$$

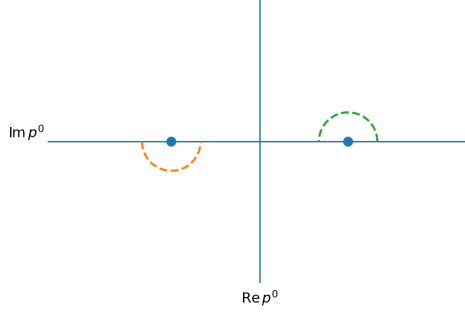
For bosonic fields, the time-ordering operator is defined by

$$T(\phi(x_1)\phi(x_2)) = \theta(t_1-t_2)\phi(x_1)\phi(x_2) + \theta(t_2-t_1)\phi(x_2)\phi(x_1). \quad (5.111)$$

The prescription useful for interactions is the Feynman prescription.



Equivalently, the integration contour can be represented as follows.



We define

$$G_F(x-y) = \lim_{\eta \rightarrow 0^+} \int \frac{d^4 p}{(2\pi)^4} \frac{e^{-ip(x-y)}}{p^2 - m^2 + i\eta}. \quad (5.112)$$

The poles are located at

$$p^0 = E - i\epsilon, \quad p^0 = -E + i\epsilon. \quad (5.113)$$

For

$$x^0 - y^0 > 0, \quad (5.114)$$

we close the contour in the lower half-plane and pick the positive-energy pole

$$p^0 = E - i\epsilon. \quad (5.115)$$

For

$$x^0 - y^0 < 0, \quad (5.116)$$

we close the contour in the upper half-plane and pick the negative-energy pole

$$p^0 = -E + i\epsilon. \quad (5.117)$$

For $x^0 - y^0 > 0$,

$$G_F(x-y)|_{x^0-y^0>0} = \frac{1}{(2\pi)^4} \int e^{i\vec{p} \cdot (\vec{x}-\vec{y})} d^3 p \int_{-\infty}^{+\infty} \frac{e^{-ip^0(x^0-y^0)}}{(p^0)^2 - (E - i\epsilon)^2} dp^0. \quad (5.118)$$

Defining

$$I = \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{+\infty} \frac{e^{-ip^0(x^0-y^0)}}{(p^0)^2 - (E - i\epsilon)^2} dp^0, \quad (5.119)$$

the residue theorem gives

$$I = -2\pi i \frac{e^{-ip^0(x^0-y^0)}}{2p^0}. \quad (5.120)$$

Consequently,

$$G_F(x-y) = -\frac{i}{(2\pi)^3} \int \frac{d^3 p}{2p^0} e^{-ip(x-y)} \quad (5.121)$$

$$= \Delta_+(x-y). \quad (5.122)$$

For $x^0 - y^0 < 0$,

$$G_F(x-y)|_{x^0-y^0<0} = -\frac{i}{(2\pi)^3} \int \frac{d^3 p}{2p^0} e^{ip(x-y)} \quad (5.123)$$

$$= -\Delta_-(x-y). \quad (5.124)$$

In total,

$$\Delta_F(x-y) = i\theta(x^0 - y^0)\Delta_+(x-y) - i\theta(y^0 - x^0)\Delta_-(x-y). \quad (5.125)$$

Thus,

$$i\Delta_F(x-y) = \langle 0 | T(\phi(x)\phi(y)) | 0 \rangle. \quad (5.126)$$

From causality,

$$[\phi^{(+)}(x), \phi^{(-)}(y)] = i\Delta_+(x-y) = \int \frac{d^3p}{(2\pi)^3 2E} e^{-ip(x-y)}, \quad (5.127)$$

and

$$[\phi^{(-)}(x), \phi^{(+)}(y)] = i\Delta_-(x-y). \quad (5.128)$$

Therefore,

$$i\Delta_F(x-y) = i[\theta(x^0 - y^0)\Delta_+(x-y) - \theta(y^0 - x^0)\Delta_-(x-y)]. \quad (5.129)$$

This is the vacuum expectation value of the ordered product of bosonic fields:

$$i\Delta_F(x-y) = \langle 0 | T(\phi(x)\phi(y)) | 0 \rangle. \quad (5.130)$$

The commutator

$$[\phi, \phi] \quad (5.131)$$

is a number, namely a c -number or classical number.

Although $\hat{O}$ is a quantum-mechanical operator,

$$\langle 0 | \hat{O} | 0 \rangle = C, \quad C \in \mathbb{C}, \quad (5.132)$$

is a number.

For example, if

$$\hat{O} = f(x, y)\mathbf{1}, \quad (5.133)$$

then

$$\langle 0 | f(x, y)\mathbf{1} | 0 \rangle = f(x, y)\langle 0 | \mathbf{1} | 0 \rangle = f(x, y). \quad (5.134)$$

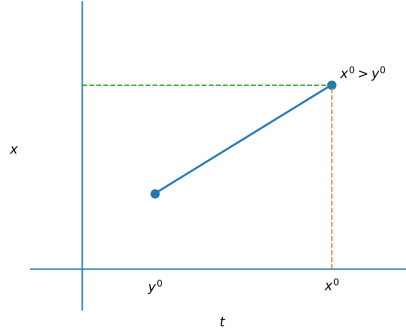
Hence,

$$\langle 0 | T(\phi(x)\phi(y)) | 0 \rangle \quad (5.135)$$

is the Feynman propagator in coordinate space.

Each field can create or destroy particles, but for the vacuum expectation value to be non-zero, anything created in the initial state must be destroyed in the final state.

Which of x^0 and y^0 occurs first is determined by the time-ordering operator T .



Finally,

$$\langle 0 | T(\phi(x)\phi(y)) | 0 \rangle = \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip(x-y)}}{p^2 - m^2 + i\epsilon} \equiv iD_F(x-y). \quad (5.136)$$

5.5 Quantization of Charged scalar fields

For a non-Hermitian field,

$$\phi \neq \phi^\dagger.$$

For non-Hermitian fields there are two different equations:

$$(\partial_\mu \partial^\mu + m^2) \phi(x) = 0,$$

$$(\partial_\mu \partial^\mu + m^2) \phi^\dagger(x) = 0.$$

A complex field operator $(\phi(x), \phi^\dagger(x))$ can be expressed as a linear combination of two Hermitian field operators (ϕ_1, ϕ_2) :

$$\begin{aligned}\phi(x) &= \frac{1}{\sqrt{2}} (\phi_1(x) + i\phi_2(x)), \\ \phi^\dagger(x) &= \frac{1}{\sqrt{2}} (\phi_1(x) - i\phi_2(x)),\end{aligned}$$

where ϕ_1 and ϕ_2 are two distinct spin-0 Hermitian fields.

We can easily invert these relations:

$$\begin{aligned}\phi_1(x) &= \frac{1}{\sqrt{2}} (\phi(x) + \phi^\dagger(x)), \\ \phi_2(x) &= -\frac{i}{\sqrt{2}} (\phi(x) - \phi^\dagger(x)).\end{aligned}$$

The Lagrangian that gives rise to their equations of motion is

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi_1 \partial^\mu \phi_1 - \frac{m^2}{2} \phi_1^2 + \frac{1}{2} \partial_\mu \phi_2 \partial^\mu \phi_2 - \frac{m^2}{2} \phi_2^2 \quad (5.137)$$

$$= \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi. \quad (5.138)$$

Defining the conjugate momenta,

$$\Pi_i(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_i(x)} = \dot{\phi}_i(x), \quad i = 1, 2.$$

Imposing the canonical commutation relations,

$$[\phi_i(x), \phi_j(y)]_{x^0=y^0} = 0 = [\Pi_i(x), \Pi_j(y)]_{x^0=y^0},$$

$$[\phi_i(x), \Pi_j(y)]_{x^0=y^0} = i\delta^3(\vec{x} - \vec{y})\delta_{ij}.$$

The Hamiltonian density is

$$\mathcal{H} = \sum_{i=1}^2 \left(\frac{1}{2} \Pi_i^2 + \frac{1}{2} \vec{\nabla} \phi_i \cdot \vec{\nabla} \phi_i + \frac{m^2}{2} \phi_i^2 \right).$$

The Hamiltonian for two complex scalar fields is the sum of the Hamiltonians for two non-interacting real Klein–Gordon fields with degenerate mass:

$$H = \int d^3x \mathcal{H}.$$

Alternatively,

$$\Pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^\dagger(x)} = \dot{\phi}(x) = \frac{1}{\sqrt{2}} (\dot{\phi}_1(x) + i\dot{\phi}_2(x)) = \frac{1}{\sqrt{2}} (\Pi_1(x) + i\Pi_2(x)),$$

$$\Pi^\dagger(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}(x)} = \dot{\phi}^\dagger(x) = \frac{1}{\sqrt{2}} (\dot{\phi}_1(x) - i\dot{\phi}_2(x)) = \frac{1}{\sqrt{2}} (\Pi_1(x) - i\Pi_2(x)).$$

The canonical commutation relations are

$$[\phi(x), \phi(y)] = [\phi(x), \phi^\dagger(y)] = [\phi^\dagger(x), \phi^\dagger(y)] = 0,$$

$$[\Pi(x), \Pi(y)] = [\Pi(x), \Pi^\dagger(y)] = [\Pi^\dagger(x), \Pi^\dagger(y)] = 0,$$

$$[\phi(x), \Pi^\dagger(y)] = [\phi^\dagger(x), \Pi(y)] = i\delta^3(\vec{x} - \vec{y}).$$

The Hamiltonian density $(\phi, \phi^\dagger)$ is

$$\mathcal{H} = \Pi \dot{\phi}^\dagger + \Pi^\dagger \dot{\phi} - \mathcal{L} \quad (5.139)$$

$$= \Pi^\dagger \Pi + \vec{\nabla} \phi^\dagger \cdot \vec{\nabla} \phi + m^2 \phi^\dagger \phi. \quad (5.140)$$

For a single field, we can expand ϕ_1 and ϕ_2 in the basis of plane waves:

$$\phi_i(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \left(e^{-ipx} a_i(\vec{p}) + e^{ipx} a_i^\dagger(\vec{p}) \right), \quad i = 1, 2.$$

The commutation relations are

$$\begin{aligned} [a_i(\vec{p}), a_j(\vec{p}')] &= 0 = [a_i^\dagger(\vec{p}), a_j^\dagger(\vec{p}')], \\ [a_i(\vec{p}), a_j^\dagger(\vec{p}')] &= \delta^3(\vec{p} - \vec{p}') \delta_{ij}. \end{aligned}$$

For charged spin-0 particles,

$$\begin{aligned} \Pi(x) &= \frac{\partial \mathcal{L}}{\partial \dot{\phi}^\dagger(x)} = \dot{\phi}(x), \\ \Pi^\dagger(x) &= \frac{\partial \mathcal{L}}{\partial \dot{\phi}(x)} = \dot{\phi}^\dagger(x). \end{aligned}$$

Therefore,

$$[\phi(x), \Pi^\dagger(y)] = [\phi^\dagger(x), \Pi(y)] = i\delta^3(\vec{x} - \vec{y}),$$

and

$$\mathcal{H} = \Pi \dot{\phi}^\dagger + \Pi^\dagger \dot{\phi} - \mathcal{L} \quad (5.141)$$

$$= \Pi^\dagger \Pi + \vec{\nabla} \phi^\dagger \cdot \vec{\nabla} \phi + m^2 \phi^\dagger \phi. \quad (5.142)$$

Alternatively,

$$\Pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}(x)} = \dot{\phi}^\dagger,$$

$$\Pi^\dagger(x) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^\dagger} = \dot{\phi}.$$

Thus,

$$[\phi(x), \Pi(y)] = [\phi^\dagger(x), \Pi^\dagger(y)] = i\delta^3(\vec{x} - \vec{y}),$$

and

$$\mathcal{H} = \Pi \dot{\phi} + \Pi^\dagger \dot{\phi}^\dagger - \mathcal{L} \quad (5.143)$$

$$= \Pi^\dagger \Pi + \vec{\nabla} \phi^\dagger \cdot \vec{\nabla} \phi + m^2 \phi^\dagger \phi. \quad (5.144)$$

We had found

$$\phi_i(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \left(e^{-ipx} a_i(\vec{p}) + e^{ipx} a_i^\dagger(\vec{p}) \right), \quad i = 1, 2,$$

and the commutation relations

$$\begin{aligned} [a_i(\vec{p}), a_j(\vec{p}')] &= 0 = [a_i^\dagger(\vec{p}), a_j^\dagger(\vec{p}')], \\ [a_i(\vec{p}), a_j^\dagger(\vec{p}')] &= \delta_{ij} \delta^3(\vec{p} - \vec{p}'). \end{aligned}$$

Then

$$\phi(x) = \frac{1}{\sqrt{2}} (\phi_1(x) + i\phi_2(x)) \quad (5.145)$$

$$= \frac{1}{\sqrt{2}} \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \left[e^{-ipx} (a_1(\vec{p}) + ia_2(\vec{p})) + e^{ipx} (a_1^\dagger(\vec{p}) + ia_2^\dagger(\vec{p})) \right]. \quad (5.146)$$

Defining

$$\begin{aligned} a(\vec{p}) &= \frac{1}{\sqrt{2}} (a_1(\vec{p}) + ia_2(\vec{p})), \\ b^\dagger(\vec{p}) &= \frac{1}{\sqrt{2}} (a_1^\dagger(\vec{p}) + ia_2^\dagger(\vec{p})), \end{aligned}$$

we obtain

$$\phi(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (e^{-ipx} a(\vec{p}) + e^{ipx} b^\dagger(\vec{p})) .$$

Similarly,

$$\phi^\dagger(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (e^{-ipx} b(\vec{p}) + e^{ipx} a^\dagger(\vec{p})) .$$

The operators

$$\begin{aligned} a(\vec{p}) &= \frac{1}{\sqrt{2}} (a_1(\vec{p}) + i a_2(\vec{p})) , \\ b^\dagger(\vec{p}) &= \frac{1}{\sqrt{2}} (a_1^\dagger(\vec{p}) + i a_2^\dagger(\vec{p})) \end{aligned}$$

are the annihilation and creation operators of $\phi(x)$ and $\phi^\dagger(x)$.

Their commutation relations are

$$[a(\vec{p}), a^\dagger(\vec{p}')] = [b(\vec{p}), b^\dagger(\vec{p}')] = \delta^3(\vec{p} - \vec{p}'),$$

and the rest vanish:

$$[a, a] = [b, b] = \dots = 0.$$

The Hamiltonian of the system is

$$H = \int d^3p \frac{E}{2} (a_1^\dagger(\vec{p}) a_1(\vec{p}) + a_1(\vec{p}) a_1^\dagger(\vec{p})) \quad (5.147)$$

$$+ a_2^\dagger(\vec{p}) a_2(\vec{p}) + a_2(\vec{p}) a_2^\dagger(\vec{p})) \quad (5.148)$$

$$= \int d^3p \frac{E}{2} (a^\dagger(\vec{p}) a(\vec{p}) + a(\vec{p}) a^\dagger(\vec{p}) + b^\dagger(\vec{p}) b(\vec{p}) + b(\vec{p}) b^\dagger(\vec{p})) . \quad (5.149)$$

Doing normal ordering with $a(\vec{p})$ and $b(\vec{p})$,

$$: H := \int d^3p E (a^\dagger(\vec{p}) a(\vec{p}) + b^\dagger(\vec{p}) b(\vec{p})) .$$

The vacuum is defined as the lowest-energy state, annihilated both by $a(\vec{p})$ and $b(\vec{p})$:

$$a(\vec{p})|0\rangle = 0, \quad b(\vec{p})|0\rangle = 0, \quad H|0\rangle = 0.$$

Equivalently,

$$\langle 0|a^\dagger(\vec{p}) = 0, \quad \langle 0|b^\dagger(\vec{p}) = 0.$$

There are two distinct one-particle states degenerate in momentum and energy:

$$|\vec{p}\rangle = a^\dagger(\vec{p})|0\rangle,$$

$$|\tilde{\vec{p}}\rangle = b^\dagger(\vec{p})|0\rangle.$$

We understand the degeneracy by looking at conserved quantities.

5.6 Charge Operators

There is now an extra symmetry beyond Lorentz invariance, translation, ...

$$\phi(x) \longrightarrow \phi'(x) = e^{-i\theta} \phi(x),$$

$$\phi^\dagger(x) \longrightarrow \phi'^\dagger(x) = e^{i\theta} \phi^\dagger(x),$$

with θ real and constant.

With this we can prove that

$$\mathcal{L}' - \mathcal{L} = 0.$$

The infinitesimal transformation is

$$\delta\phi(x) = \phi'(x) - \phi(x) = -i\epsilon\phi(x),$$

$$\delta\phi^\dagger(x) = \phi^{\dagger\dagger}(x) - \phi^\dagger(x) = i\epsilon\phi^\dagger(x),$$

where ϵ is the infinitesimal parameter.

The Noether current is

$$J_\epsilon^\mu = \delta\phi \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} + \delta\phi^\dagger \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^\dagger)} - K^\mu,$$

with

$$K^\mu = 0$$

for internal symmetries.

Therefore,

$$J_\epsilon^\mu = i\epsilon \left(\phi^\dagger(x) \partial^\mu \phi(x) - \phi(x) \partial^\mu \phi^\dagger(x) \right).$$

The parameter-independent current is

$$J^\mu = i \left(\phi^\dagger \partial^\mu \phi - \phi \partial^\mu \phi^\dagger \right).$$

The corresponding conserved charge is

$$Q = \int d^3x J^0(x) \tag{5.150}$$

$$= i \int d^3x \left(\phi^\dagger \dot{\phi} - \dot{\phi}^\dagger \phi \right) \tag{5.151}$$

$$= \int d^3p \left(a^\dagger(\vec{p}) a(\vec{p}) - b(\vec{p}) b^\dagger(\vec{p}) \right). \tag{5.152}$$

After normal ordering,

$$:Q:= \int d^3p \left(a^\dagger(\vec{p}) a(\vec{p}) - b^\dagger(\vec{p}) b(\vec{p}) \right).$$

Thus,

$$:Q:= N_a - N_b,$$

the difference between the number of particles created by $a^\dagger$ and $b^\dagger$, with

$$N_a = \int d^3p a^\dagger(\vec{p}) a(\vec{p}), \quad N_b = \int d^3p b^\dagger(\vec{p}) b(\vec{p}).$$

For the one-particle states,

$$Q|\vec{p}\rangle = Q a^\dagger(\vec{p})|0\rangle = +a^\dagger(\vec{p})|0\rangle = +|\vec{p}\rangle,$$

$$Q|\tilde{\vec{p}}\rangle = Q b^\dagger(\vec{p})|0\rangle = -b^\dagger(\vec{p})|0\rangle = -|\tilde{\vec{p}}\rangle.$$

Therefore, Q is the total charge of the system.

The operator $a^\dagger$ generates particles with charge +1, while $b^\dagger$ generates particles with charge -1.

The Hamiltonian is

$$H = \int d^3p E \left(a^\dagger a + b^\dagger b \right).$$

There is another symmetry,

$$a \longleftrightarrow b,$$

which corresponds to charge conjugation.

A complex Klein-Gordon field describes particle-antiparticle systems of spin 0, such as the charged pions

$$\pi^+, \pi^-.$$

The particles

$$\pi^0, \quad h^0$$

are described by real Klein-Gordon fields.

5.7 Problems

Problem 5.1 Consider the quantized spinless Schrödinger field for fermions. Show that

$$\hat{H} |\vec{p}_1, \dots, \vec{p}_n\rangle = (E_1 + \dots + E_n) |\vec{p}_1, \dots, \vec{p}_n\rangle, \quad (5.153)$$

where the Hamiltonian is defined as

$$\hat{H} = \int d^3\vec{p} \frac{\vec{p}^2}{2m} \hat{a}^\dagger(\vec{p}) \hat{a}(\vec{p}), \quad (5.154)$$

and

$$E_i = \frac{\vec{p}_i^2}{2m}, \quad i = 1, \dots, n. \quad (5.155)$$

Solution.

The state of a particle with momentum $\vec{p}$ is

$$a^\dagger(\vec{p})|0\rangle = |\vec{p}\rangle.$$

The state of N particles with momenta $\vec{p}_i$ is

$$a^\dagger(\vec{p}_1) \cdots a^\dagger(\vec{p}_n)|0\rangle = |\vec{p}_1, \dots, \vec{p}_n\rangle.$$

We use

$$[a(\vec{p}), a^\dagger(\vec{p}')] = \delta^3(\vec{p} - \vec{p}'),$$

and therefore

$$\{a(\vec{p}), a^\dagger(\vec{p}')\} = \delta^3(\vec{p} - \vec{p}').$$

Then

$$\begin{aligned} H|\vec{p}_1, \dots, \vec{p}_n\rangle &= \int d^3p \frac{\vec{p}^2}{2m} a^\dagger(\vec{p}) a(\vec{p}) a^\dagger(\vec{p}_1) \cdots a^\dagger(\vec{p}_n)|0\rangle \\ &= \int d^3p \frac{\vec{p}^2}{2m} a^\dagger(\vec{p}) \delta^3(\vec{p} - \vec{p}_1) a^\dagger(\vec{p}_2) \cdots a^\dagger(\vec{p}_n)|0\rangle \\ &\quad - \int d^3p \frac{\vec{p}^2}{2m} a^\dagger(\vec{p}) a^\dagger(\vec{p}_1) a(\vec{p}) a^\dagger(\vec{p}_2) \cdots a^\dagger(\vec{p}_n)|0\rangle \\ &= \frac{\vec{p}_1^2}{2m} a^\dagger(\vec{p}_1) a^\dagger(\vec{p}_2) \cdots a^\dagger(\vec{p}_n)|0\rangle \\ &\quad - \int d^3p \frac{\vec{p}^2}{2m} a^\dagger(\vec{p}) a^\dagger(\vec{p}_1) a(\vec{p}) a^\dagger(\vec{p}_2) \cdots a^\dagger(\vec{p}_n)|0\rangle. \end{aligned}$$

Proceeding in an analogous fashion for all the pairs,

$$H|\vec{p}_1, \vec{p}_2, \dots, \vec{p}_n\rangle = (E_1 + E_2 + \dots + E_n) |\vec{p}_1, \vec{p}_2, \dots, \vec{p}_n\rangle.$$

□

Problem 5.2 Consider the quantized real Klein–Gordon field.

Verify that the commutation relations between the creation and annihilation operators,

$$\hat{a}(\vec{p}), \quad \hat{a}^\dagger(\vec{p}'), \quad (5.156)$$

imply the canonical equal-time commutation relations between the field operator $\hat{\phi}(\vec{x}, t)$ and its canonical momentum $\hat{\Pi}(\vec{y}, t)$.

Solution.

We write

$$\phi(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (e^{-ipx} a(\vec{p}) + e^{ipx} a^\dagger(\vec{p})),$$

and

$$\Pi(x) = \dot{\phi}(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (-iE) (e^{-ipx} a(\vec{p}) - e^{ipx} a^\dagger(\vec{p})).$$

The commutator is

$$[\phi(x), \Pi(y)] = \phi(x)\Pi(y) - \Pi(y)\phi(x).$$

First,

$$\begin{aligned} \phi(x)\Pi(y) &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} (-iE') \\ &\times (e^{-ipx} a(\vec{p}) + e^{ipx} a^\dagger(\vec{p})) (e^{-ip'y} a(\vec{p}') - e^{ip'y} a^\dagger(\vec{p}')). \end{aligned}$$

Similarly,

$$\begin{aligned} \Pi(y)\phi(x) &= \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (-iE') \\ &\times (e^{-ip'y} a(\vec{p}') - e^{ip'y} a^\dagger(\vec{p}')) (e^{-ipx} a(\vec{p}) + e^{ipx} a^\dagger(\vec{p})). \end{aligned}$$

Therefore,

$$\begin{aligned} [\phi(x), \Pi(y)] &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} (-iE') \\ &\times \left[e^{-i(px+p'y)} a(\vec{p}) a(\vec{p}') - e^{-i(px-p'y)} a(\vec{p}) a^\dagger(\vec{p}') \right. \\ &\quad + e^{i(px-p'y)} a^\dagger(\vec{p}) a(\vec{p}') - e^{i(px+p'y)} a^\dagger(\vec{p}) a^\dagger(\vec{p}') \\ &\quad - e^{-i(p'y+px)} a(\vec{p}') a(\vec{p}) - e^{-i(p'y-px)} a(\vec{p}') a^\dagger(\vec{p}) \\ &\quad \left. + e^{i(p'y-px)} a^\dagger(\vec{p}') a(\vec{p}) + e^{i(p'y+px)} a^\dagger(\vec{p}') a^\dagger(\vec{p}) \right]. \end{aligned}$$

Using the commutation relations,

$$\begin{aligned} [\phi(x), \Pi(y)] &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} (-iE') \\ &\times \left[-e^{-i(px-p'y)} [a(\vec{p}), a^\dagger(\vec{p}')] \right. \\ &\quad \left. + e^{i(px-p'y)} [a^\dagger(\vec{p}), a(\vec{p}')] \right] \\ &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} (-iE') \\ &\times \left\{ -e^{-i(px-p'y)} \delta^3(\vec{p} - \vec{p}') - e^{i(px-p'y)} \delta^3(\vec{p} - \vec{p}') \right\} \\ &= \int \frac{d^3p}{(2\pi)^3 2E} iE \left\{ e^{-ip(x-y)} + e^{ip(x-y)} \right\}. \end{aligned}$$

At equal times, $x^0 = y^0$, and therefore $e^0 = 1$:

$$\begin{aligned} [\phi(x), \Pi(y)]_{x^0=y^0} &= \int \frac{d^3p}{(2\pi)^3 2} i \left\{ e^{-i\vec{p} \cdot (\vec{x} - \vec{y})} + e^{i\vec{p} \cdot (\vec{x} - \vec{y})} \right\} \\ &= \frac{i}{2} [\delta^3(\vec{x} - \vec{y}) + \delta^3(\vec{y} - \vec{x})] \\ &= i\delta^3(\vec{x} - \vec{y}), \end{aligned}$$

where

$$\delta^3(\vec{x}) = \delta^3(-\vec{x}).$$

□

Problem 5.3 Show that the Hamiltonian of the quantized real Klein–Gordon field can be written as

$$\hat{H} = \int d^3\vec{p} \frac{E_{\vec{p}}}{2} [\hat{a}^\dagger(\vec{p}) \hat{a}(\vec{p}) + \hat{a}(\vec{p}) \hat{a}^\dagger(\vec{p})], \quad (5.157)$$

where

$$E_{\vec{p}} = \sqrt{\vec{p}^2 + m^2}. \quad (5.158)$$

Solution.

We start from

$$H = \int d^3x \left[\frac{1}{2} \Pi(\vec{x}) \Pi(\vec{x}) + \frac{1}{2} \vec{\nabla} \phi(\vec{x}) \cdot \vec{\nabla} \phi(\vec{x}) + \frac{1}{2} m^2 \phi^2(\vec{x}) \right].$$

The field and its conjugate momentum are

$$\phi(\vec{x}) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (e^{-ipx} a(\vec{p}) + e^{ipx} a^\dagger(\vec{p})),$$

$$\Pi(\vec{x}) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (-iE) (e^{-ipx} a(\vec{p}) - e^{ipx} a^\dagger(\vec{p})),$$

and

$$\vec{\nabla} \phi(\vec{x}) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (i\vec{p}) (-e^{-ipx} a(\vec{p}) + e^{ipx} a^\dagger(\vec{p})).$$

For the momentum contribution,

$$\begin{aligned} \Pi(\vec{x}) \Pi(\vec{x}) &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} (-iE) (-iE') \\ &\quad \times (e^{-ipx} a(\vec{p}) - e^{ipx} a^\dagger(\vec{p})) (e^{-ip'x} a(\vec{p}') - e^{ip'x} a^\dagger(\vec{p}')) \\ &= - \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} EE' \\ &\quad \times \left[e^{-ix(p+p')} a(\vec{p}) a(\vec{p}') - e^{-ix(p-p')} a(\vec{p}) a^\dagger(\vec{p}') \right. \\ &\quad \left. - e^{ix(p-p')} a^\dagger(\vec{p}) a(\vec{p}') + e^{ix(p+p')} a^\dagger(\vec{p}) a^\dagger(\vec{p}') \right]. \end{aligned}$$

After integrating over $\vec{x}$,

$$\begin{aligned} \frac{1}{2} \int d^3x \Pi(\vec{x}) \Pi(\vec{x}) &= -\frac{1}{2} \iint \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} EE' (2\pi)^3 \\ &\quad \times \left[e^{-it(E+E')} a(\vec{p}) a(\vec{p}') \delta^3(\vec{p} + \vec{p}') \right. \\ &\quad - e^{-it(E-E')} a(\vec{p}) a^\dagger(\vec{p}') \delta^3(\vec{p} - \vec{p}') \\ &\quad - e^{it(E-E')} a^\dagger(\vec{p}) a(\vec{p}') \delta^3(\vec{p} - \vec{p}') \\ &\quad \left. + e^{it(E+E')} a^\dagger(\vec{p}) a^\dagger(\vec{p}') \delta^3(\vec{p} + \vec{p}') \right] \\ &= -\frac{E}{4} \int d^3p \left[e^{-2iEt} a(\vec{p}) a(-\vec{p}) - a(\vec{p}) a^\dagger(\vec{p}) \right. \\ &\quad \left. - a^\dagger(\vec{p}) a(\vec{p}) + e^{2iEt} a^\dagger(\vec{p}) a^\dagger(-\vec{p}) \right]. \end{aligned}$$

Proceeding in an analogous fashion for $\vec{\nabla} \phi(\vec{x}) \cdot \vec{\nabla} \phi(\vec{x})$,

$$\begin{aligned} \frac{1}{2} \int d^3x \vec{\nabla} \phi(\vec{x}) \cdot \vec{\nabla} \phi(\vec{x}) &= -\frac{1}{4} \int d^3p \frac{\vec{p}^2}{E} \left[e^{-2iEt} a(\vec{p}) a(-\vec{p}) \right. \\ &\quad - a(\vec{p}) a^\dagger(\vec{p}) - a^\dagger(\vec{p}) a(\vec{p}) \\ &\quad \left. + e^{2iEt} a^\dagger(\vec{p}) a^\dagger(-\vec{p}) \right]. \end{aligned}$$

For $\phi^2(\vec{x})$,

$$\begin{aligned} \frac{1}{2} m^2 \phi^2(\vec{x}) d^3x &= \frac{1}{4} \int d^3p \frac{m^2}{E} \left[e^{-2iEt} a(\vec{p}) a(-\vec{p}) \right. \\ &\quad + a(\vec{p}) a^\dagger(\vec{p}) + a^\dagger(\vec{p}) a(\vec{p}) \\ &\quad \left. + e^{2iEt} a^\dagger(\vec{p}) a^\dagger(-\vec{p}) \right]. \end{aligned}$$

Therefore,

$$\begin{aligned}
H = \int d^3p \frac{1}{4E} & \left[\frac{m^2 + \vec{p}^2 - E^2}{4E} e^{-2iEt} a(\vec{p}) a(-\vec{p}) \right. \\
& + (E^2 + \vec{p}^2 + m^2) a(\vec{p}) a^\dagger(\vec{p}) \\
& + (E^2 + \vec{p}^2 + m^2) a^\dagger(\vec{p}) a(\vec{p}) \\
& \left. + (m^2 + \vec{p}^2 - E^2) e^{2iEt} a^\dagger(\vec{p}) a^\dagger(-\vec{p}) \right].
\end{aligned}$$

Using

$$E^2 = \vec{p}^2 + m^2,$$

the terms proportional to $a(\vec{p})a(-\vec{p})$ and $a^\dagger(\vec{p})a^\dagger(-\vec{p})$ vanish, and hence

$$H = \int d^3p \frac{E}{2} (a^\dagger(\vec{p})a(\vec{p}) + a(\vec{p})a^\dagger(\vec{p})).$$

□

Problem 5.4 Consider the complex scalar-field Lagrangian density

$$\mathcal{L} = (\partial_\mu \phi^\dagger)(\partial^\mu \phi) - m^2 \phi^\dagger \phi. \quad (5.159)$$

Verify that the commutation relations between the creation and annihilation operators

$$\hat{a}(\vec{p}), \quad \hat{a}^\dagger(\vec{p}), \quad \hat{b}(\vec{p}), \quad \hat{b}^\dagger(\vec{p}), \quad (5.160)$$

imply the canonical equal-time commutation relations between $\hat{\phi}(x)$, $\hat{\phi}^\dagger(x)$ and their corresponding canonical momenta $\hat{\Pi}(x)$ and $\hat{\Pi}^\dagger(x)$.

Solution.

We use

$$\begin{aligned}
\phi(x) &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (e^{-ipx} a(\vec{p}) + e^{ipx} b^\dagger(\vec{p})), \\
\phi^\dagger(x) &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (e^{-ipx} b(\vec{p}) + e^{ipx} a^\dagger(\vec{p})), \\
\Pi(x) = \dot{\phi}(x) &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (-iE) (e^{-ipx} a(\vec{p}) - e^{ipx} b^\dagger(\vec{p})),
\end{aligned}$$

and

$$\Pi^\dagger(x) = \dot{\phi}^\dagger(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (-iE) (e^{-ipx} b(\vec{p}) - e^{ipx} a^\dagger(\vec{p})).$$

We will use

$$[a_1 + a_2, b_1 + b_2] = [a_1, b_1] + [a_1, b_2] + [a_2, b_1] + [a_2, b_2].$$

For the fields,

$$\begin{aligned}
[\phi(x), \phi^\dagger(y)] &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} \\
&\times \left(e^{-i(px+p'y)} [a(\vec{p}), b(\vec{p}')] \right. \\
&\quad + e^{-i(px-p'y)} [a(\vec{p}), a^\dagger(\vec{p}')] \\
&\quad + e^{i(px-p'y)} [b^\dagger(\vec{p}), b(\vec{p}')] \\
&\quad \left. + e^{i(px+p'y)} [b^\dagger(\vec{p}), a^\dagger(\vec{p}')] \right) \\
&= \int \frac{d^3p}{(2\pi)^3 2E} \left(e^{-i\vec{p} \cdot (\vec{x}-\vec{y})} - e^{i\vec{p} \cdot (\vec{x}-\vec{y})} \right) \\
&= 0.
\end{aligned}$$

For the conjugate momenta,

$$\begin{aligned}
[\Pi(x), \Pi^\dagger(y)] &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} (-iE)(-iE') \\
&\quad \times \left(-e^{-i(px-p'y)} [a(\vec{p}), a^\dagger(\vec{p}')] \right. \\
&\quad \left. - e^{i(px-p'y)} [b^\dagger(\vec{p}), b(\vec{p}')] \right) \\
&= \int \frac{d^3p}{(2\pi)^3 2E} (-iE)^2 \left(-e^{-i\vec{p} \cdot (\vec{x} - \vec{y})} + e^{i\vec{p} \cdot (\vec{x} - \vec{y})} \right) \\
&= 0.
\end{aligned}$$

For the mixed commutator,

$$\begin{aligned}
[\phi(x), \Pi^\dagger(y)] &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} (-iE') \\
&\quad \times \left(-e^{-i(px-p'y)} [a(\vec{p}), a^\dagger(\vec{p}')] \right. \\
&\quad \left. + e^{i(px-p'y)} [b^\dagger(\vec{p}), b(\vec{p}')] \right) \\
&= \int \frac{d^3p}{(2\pi)^3 2E} (iE) \left(e^{-i\vec{p} \cdot (\vec{x} - \vec{y})} + e^{i\vec{p} \cdot (\vec{x} - \vec{y})} \right) \\
&= \frac{i}{2} [\delta^3(\vec{x} - \vec{y}) + \delta^3(\vec{x} - \vec{y})] \\
&= i\delta^3(\vec{x} - \vec{y}).
\end{aligned}$$

Similarly,

$$\begin{aligned}
[\phi^\dagger(x), \Pi(y)] &= \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} (-iE') \\
&\quad \times \left(-e^{-i(px-p'y)} [b(\vec{p}), b^\dagger(\vec{p}')] \right. \\
&\quad \left. + e^{i(px-p'y)} [a^\dagger(\vec{p}), a(\vec{p}')] \right) \\
&= \int \frac{d^3p}{(2\pi)^3 2E} (iE) \left(e^{-i\vec{p} \cdot (\vec{x} - \vec{y})} + e^{i\vec{p} \cdot (\vec{x} - \vec{y})} \right) \\
&= i\delta^3(\vec{x} - \vec{y}).
\end{aligned}$$

□

Problem 5.5 Prove that the Lagrangian density

$$\mathcal{L} = (\partial_\mu \phi^\dagger)(\partial^\mu \phi) - m^2 \phi^\dagger \phi \quad (5.161)$$

is invariant under the global phase transformation

$$\phi \longrightarrow \phi' = e^{i\theta} \phi, \quad (5.162)$$

where θ is a constant.

Find the associated Noether current.

Solution.

Under the transformation,

$$\phi'(x) = e^{i\theta} \phi(x), \quad \phi'^\dagger(x) = e^{-i\theta} \phi^\dagger(x).$$

Therefore,

$$\begin{aligned}
\mathcal{L}' &= (\partial_\mu \phi'^\dagger) (\partial^\mu \phi') - m^2 \phi'^\dagger \phi' \\
&= e^{-i\theta} (\partial_\mu \phi^\dagger) e^{i\theta} (\partial^\mu \phi) - m^2 e^{-i\theta} e^{i\theta} \phi^\dagger \phi \\
&= e^{-i\theta} e^{i\theta} (\partial_\mu \phi^\dagger) (\partial^\mu \phi) - m^2 \phi^\dagger \phi \\
&= (\partial_\mu \phi^\dagger) (\partial^\mu \phi) - m^2 \phi^\dagger \phi \\
&= \mathcal{L}.
\end{aligned}$$

The infinitesimal variations are

$$\delta\phi(x) = \phi'(x) - \phi(x) = i\theta\phi(x),$$

and

$$\delta\phi^\dagger(x) = \phi'^\dagger(x) - \phi^\dagger(x) = -i\theta\phi^\dagger(x).$$

The Noether current is

$$\begin{aligned}
J^\mu &= \delta\phi \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} + \delta\phi^\dagger \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^\dagger)} \\
&= i\theta\phi(x) \partial^\mu \phi^\dagger(x) - i\theta\phi^\dagger(x) \partial^\mu \phi(x) \\
&= i\theta (\phi \partial^\mu \phi^\dagger - \phi^\dagger \partial^\mu \phi).
\end{aligned}$$

□

6 Quantization of the Dirac Field

The Dirac equation describes spin-1/2 particles:

$$(-i\gamma^\mu \partial_\mu - m)\psi = 0.$$

The field

$$\psi_\alpha(x), \quad \alpha = 1, 2, 3, 4,$$

is a four-component complex spinor.

We also have the Dirac equation for the adjoint field:

$$i\partial_\mu \bar{\psi} \gamma^\mu + m\bar{\psi} = 0.$$

The adjoint spinor is

$$\bar{\psi} = \psi^\dagger \gamma^0.$$

The fields ψ and $\bar{\psi}$ are two different independent fields.

The Lagrangian density for the Dirac theory is

$$\mathcal{L} = \bar{\psi} (i\gamma^\mu \partial_\mu - m) \psi,$$

or equivalently,

$$\mathcal{L} = i\bar{\psi} \gamma^\mu \partial_\mu \psi - m\bar{\psi} \psi.$$

The Lagrangian density is non-Hermitian:

$$\begin{aligned} \mathcal{L}^\dagger &= [\bar{\psi} (i\gamma^\mu \partial_\mu - m) \psi]^\dagger \\ &= -i\partial_\mu \bar{\psi} \gamma^\mu \psi - m\bar{\psi} \psi \neq i\bar{\psi} \gamma^\mu \partial_\mu \psi - m\bar{\psi} \psi = \mathcal{L}. \end{aligned}$$

The difference is a total divergence:

$$\begin{aligned} \mathcal{L}^\dagger - \mathcal{L} &= -i\partial_\mu \bar{\psi} \gamma^\mu \psi - m\bar{\psi} \psi - i\bar{\psi} \gamma^\mu \partial_\mu \psi + m\bar{\psi} \psi \\ &= -i\partial_\mu \bar{\psi} \gamma^\mu \psi - i\bar{\psi} \gamma^\mu \partial_\mu \psi \\ &= -i\partial_\mu (\bar{\psi} \gamma^\mu \psi). \end{aligned}$$

A total divergence does not contribute to the action, so the physics is unchanged.

A Hermitian Lagrangian that also gives rise to the Dirac equations is

$$\begin{aligned} \mathcal{L} &= -\frac{1}{2} \bar{\psi} (-i\gamma^\mu \partial_\mu + m) \psi - \frac{1}{2} (i\partial_\mu \bar{\psi} \gamma^\mu + m\bar{\psi}) \psi \\ &= \bar{\psi} \left(\frac{i}{2} \overleftrightarrow{\partial} - m \right) \psi, \end{aligned}$$

where

$$\overleftrightarrow{\partial}_\mu = \overrightarrow{\partial}_\mu - \overleftarrow{\partial}_\mu.$$

6.1 Field Decomposition

We know that we deal with particle/antiparticle fields, so ψ and $\bar{\psi}$ are not Hermitian.

We will expand the fields in the basis of plane waves, with

$$E = \sqrt{\vec{p}^2 + m^2} > 0,$$

as for the Klein-Gordon field:

$$\psi_\alpha(x) = \sum_{s=\pm 1/2} \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} (e^{-ipx} c(\vec{p}, s) u_\alpha(\vec{p}, s) + e^{ipx} d^\dagger(\vec{p}, s) v_\alpha(\vec{p}, s)).$$

Here,

$$u_\alpha(\vec{p}, s) = \sqrt{E+m} \begin{pmatrix} \chi_s \\ \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \chi_s \end{pmatrix}, \quad v_\alpha(\vec{p}, s) = \sqrt{E+m} \begin{pmatrix} \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \hat{\chi}_s \\ \hat{\chi}_s \end{pmatrix}.$$

The two independent spinors are

$$\chi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \hat{\chi}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \hat{\chi}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The Hermitian conjugate field is

$$\psi_\alpha^\dagger(x) = \sum_{s=\pm 1/2} \int \frac{d^3 p}{\sqrt{(2\pi)^3 2E}} (e^{ipx} c^\dagger(\vec{p}, s) u_\alpha^\dagger(\vec{p}, s) + e^{-ipx} d(\vec{p}, s) v_\alpha^\dagger(\vec{p}, s)).$$

Therefore,

$$\bar{\psi}(x) = \psi^\dagger(x) \gamma^0 = \sum_{s=\pm 1/2} \int \frac{d^3 p}{\sqrt{(2\pi)^3 2E}} (e^{ipx} c^\dagger(\vec{p}, s) \bar{u}_\alpha(\vec{p}, s) + e^{-ipx} d(\vec{p}, s) \bar{v}_\alpha(\vec{p}, s)).$$

We have now seen four different sets of creation and annihilation operators:

$$\begin{aligned} c(\vec{p}, s) & \text{ annihilates a particle with } \vec{p}, s, \\ c^\dagger(\vec{p}, s) & \text{ creates a particle with } \vec{p}, s, \\ d(\vec{p}, s) & \text{ annihilates an antiparticle with } \vec{p}, s, \\ d^\dagger(\vec{p}, s) & \text{ creates an antiparticle with } \vec{p}, s. \end{aligned}$$

6.2 Hamiltonian of the Dirac Field

$$\mathcal{L} = \bar{\psi} (i\gamma^\mu \partial_\mu - m) \psi.$$

Defining the conjugate momenta,

$$\Pi_\psi = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = i\bar{\psi} \gamma^0 = i\psi^\dagger,$$

and

$$\Pi_{\bar{\psi}} = \frac{\partial \mathcal{L}}{\partial \dot{\bar{\psi}}} = 0.$$

We can now construct the Hamiltonian density:

$$\begin{aligned} \mathcal{H} &= \Pi_\psi \dot{\psi} - \mathcal{L} \\ &= -i\bar{\psi} \vec{\gamma} \cdot \vec{\nabla} \psi + m\bar{\psi} \psi. \end{aligned} \tag{6.1}$$

Therefore,

$$H = \int d^3 x \mathcal{H} = \int d^3 x (-i\bar{\psi} \vec{\gamma} \cdot \vec{\nabla} \psi + m\bar{\psi} \psi).$$

After a very lengthy computation, we can rewrite the Hamiltonian in terms of the creation and annihilation operators:

$$H = \sum_{s=\pm 1/2} \int d^3 p E [c^\dagger(\vec{p}, s) c(\vec{p}, s) - d(\vec{p}, s) d^\dagger(\vec{p}, s)].$$

Using bosonic normal ordering would give

$$: H := \sum_{s=\pm 1/2} \int d^3 p E [c^\dagger(\vec{p}, s) c(\vec{p}, s) - d^\dagger(\vec{p}, s) d(\vec{p}, s)].$$

Having done the normal ordering for bosons, we find that E is not bounded below by zero. Dirac theory does not describe bosons, but fermions.

Therefore, we are forced to use anticommutation relations for fermionic fields.

Normal ordering for fermions generates a factor (-1) . Hence, by virtue of the spin-statistics connection,

$$: H := \sum_{s=\pm 1/2} \int d^3 p E \left[c^\dagger(\vec{p}, s) c(\vec{p}, s) + d^\dagger(\vec{p}, s) d(\vec{p}, s) \right].$$

For the Dirac field,

$$H = \sum_{s=\pm 1/2} \int d^3 p E (c^\dagger c - d d^\dagger).$$

Normal ordering for fermions gives

$$: d d^\dagger := -d^\dagger d, \quad : d^\dagger d := d^\dagger d,$$

and

$$: c c^\dagger := -c^\dagger c, \quad : c^\dagger c := c^\dagger c.$$

Therefore,

$$: H := \sum_{s=\pm 1/2} \int d^3 p E (c^\dagger c + d^\dagger d).$$

Since the fields are fermionic,

$$(c^\dagger(\vec{p}, s))^2 = c^\dagger(\vec{p}, s) c^\dagger(\vec{p}, s) = 0.$$

We define the number operators

$$N(\vec{p}, s) = c^\dagger(\vec{p}, s) c(\vec{p}, s),$$

and

$$\tilde{N}(\vec{p}, s) = d^\dagger(\vec{p}, s) d(\vec{p}, s),$$

which are the number operators for

$$|p\rangle, \quad |\tilde{p}\rangle.$$

Putting all this together,

$$\begin{aligned} N^2(\vec{p}, s) &= c^\dagger(\vec{p}, s) c(\vec{p}, s) c^\dagger(\vec{p}, s) c(\vec{p}, s) \\ &= c^\dagger(\vec{p}, s) [1 - c^\dagger(\vec{p}, s) c(\vec{p}, s)] c(\vec{p}, s) \\ &= c^\dagger(\vec{p}, s) c(\vec{p}, s) - c^\dagger(\vec{p}, s) c^\dagger(\vec{p}, s) c(\vec{p}, s) c(\vec{p}, s) \\ &= N(\vec{p}, s). \end{aligned} \tag{6.2}$$

Thus,

$$N^2 - N = 0,$$

and consequently

$$N(\vec{p}, s) = 0, 1.$$

The equal-time canonical anticommutation rules are

$$\{\psi_\alpha(x), \psi_\beta(y)\} = \{(\Pi_\psi)_\alpha(x), (\Pi_\psi)_\beta(y)\} = 0,$$

$$\{\psi_\alpha(x), (\Pi_\psi)_\beta(y)\} = i\delta_{\alpha\beta} \delta^3(\vec{x} - \vec{y}),$$

$$\left\{ \psi_\alpha(x), \psi_\beta^\dagger(y) \right\} = \delta_{\alpha\beta} \delta^3(\vec{x} - \vec{y}),$$

and

$$\left\{ \psi_\alpha^\dagger(x), \psi_\beta^\dagger(y) \right\} = 0.$$

The anticommutation relations for the operators are

$$\{c(\vec{p}, s), c^\dagger(\vec{p}', s')\} = \delta_{ss'} \delta^3(\vec{p} - \vec{p}'),$$

$$\{d(\vec{p}, s), d^\dagger(\vec{p}', s')\} = \delta_{ss'} \delta^3(\vec{p} - \vec{p}'),$$

$$\{c^\dagger(\vec{p}, s), c^\dagger(\vec{p}', s')\} = 0 = \{d^\dagger(\vec{p}, s), d^\dagger(\vec{p}', s')\},$$

and

$$\{c(\vec{p}, s), c(\vec{p}', s')\} = 0 = \{d(\vec{p}, s), d(\vec{p}', s')\}.$$

Now let us perform the very lengthy computation:

$$H = \int d^3x \left(-i\bar{\psi} \vec{\gamma} \cdot \vec{\nabla} \psi + m\bar{\psi}\psi \right) = \int d^3x \bar{\psi} \left(-i\vec{\gamma} \cdot \vec{\nabla} + m \right) \psi.$$

Substituting the plane-wave expansions,

$$\begin{aligned} H &= \int \frac{d^3x}{(2\pi)^3} \sum_{s,s'=\pm 1/2} \int d^3p' \int d^3p \frac{1}{\sqrt{2E}} \frac{1}{\sqrt{2E'}} \\ &\times \left[\bar{u}(\vec{p}', s') c^\dagger(\vec{p}', s') e^{ip'x} + \bar{v}(\vec{p}', s') d(\vec{p}', s') e^{-ip'x} \right] \\ &\times \left(-i\vec{\gamma} \cdot \vec{\nabla} + m \right) \left[e^{-ipx} c(\vec{p}, s) u(\vec{p}, s) + e^{ipx} d^\dagger(\vec{p}, s) v(\vec{p}, s) \right]. \end{aligned} \quad (6.3)$$

$$\begin{aligned} H &= \sum_{s,s'} \int \frac{d^3x}{(2\pi)^3} \int \frac{d^3p}{\sqrt{2E}} \int \frac{d^3p'}{\sqrt{2E'}} \\ &\times \left[\bar{u}(\vec{p}', s') c^\dagger(\vec{p}', s') e^{ip'x} + \bar{v}(\vec{p}', s') d(\vec{p}', s') e^{-ip'x} \right] \\ &\times \left[e^{-ipx} c(\vec{p}, s) (-i\vec{\gamma} \cdot \vec{p} + m) u(\vec{p}, s) \right. \\ &\quad \left. + e^{ipx} d^\dagger(\vec{p}, s) (i\vec{\gamma} \cdot \vec{p} + m) v(\vec{p}, s) \right]. \end{aligned} \quad (6.4)$$

Using the Dirac equation in momentum space,

$$(\not{p} - m)u(\vec{p}, s) = (\gamma^0 p^0 - \vec{\gamma} \cdot \vec{p} - m) u(\vec{p}, s) = 0,$$

and

$$(\not{p} + m)v(\vec{p}, s) = (\gamma^0 p^0 - \vec{\gamma} \cdot \vec{p} + m) v(\vec{p}, s) = 0,$$

we obtain

$$\begin{aligned} H &= \sum_{s,s'} \int \frac{d^3x}{(2\pi)^3} \int \frac{d^3p}{\sqrt{2E}} \int \frac{d^3p'}{\sqrt{2E'}} p^0 \\ &\times \left[\bar{u}(\vec{p}', s') c^\dagger(\vec{p}', s') e^{ip'x} + \bar{v}(\vec{p}', s') d(\vec{p}', s') e^{-ip'x} \right] \\ &\times \gamma^0 \left[e^{-ipx} c(\vec{p}, s) u(\vec{p}, s) + e^{ipx} d^\dagger(\vec{p}, s) v(\vec{p}, s) \right]. \end{aligned} \quad (6.5)$$

Integrating over x ,

$$\begin{aligned} H &= \sum_{s,s'} \int d^3p \int d^3p' \frac{1}{\sqrt{2E}} \frac{1}{\sqrt{2E'}} p^0 \left[\delta^3(\vec{p} - \vec{p}') \bar{u}(\vec{p}', s') \gamma^0 u(\vec{p}, s) c^\dagger(\vec{p}', s') c(\vec{p}, s) \right. \\ &\quad - \delta^3(\vec{p} - \vec{p}') \bar{v}(\vec{p}', s') \gamma^0 v(\vec{p}, s) d(\vec{p}', s') d^\dagger(\vec{p}, s) \\ &\quad - \delta^3(\vec{p} + \vec{p}') e^{-i(p'^0 + p^0)x^0} \bar{u}(\vec{p}', s') \gamma^0 v(\vec{p}, s) c^\dagger(\vec{p}', s') d^\dagger(\vec{p}, s) \\ &\quad \left. + \delta^3(\vec{p} + \vec{p}') e^{i(p'^0 + p^0)x^0} \bar{v}(\vec{p}', s') \gamma^0 u(\vec{p}, s) d(\vec{p}', s') c(\vec{p}, s) \right]. \end{aligned} \quad (6.6)$$

Applying $\delta^3(\vec{p} - \vec{p}')$,

$$\begin{aligned} H &= \sum_{s,s'} \frac{1}{2} \int d^3p \left[\bar{u}(\vec{p}, s') \gamma^0 u(\vec{p}, s) c^\dagger(\vec{p}, s') c(\vec{p}, s) \right. \\ &\quad - \bar{v}(\vec{p}, s') \gamma^0 v(\vec{p}, s) d(\vec{p}, s') d^\dagger(\vec{p}, s) \\ &\quad - e^{2ip^0 x^0} \bar{u}(-\vec{p}, s') \gamma^0 v(\vec{p}, s) c^\dagger(-\vec{p}, s') d^\dagger(\vec{p}, s) \\ &\quad \left. + e^{-2ip^0 x^0} \bar{v}(-\vec{p}, s') \gamma^0 u(\vec{p}, s) d(-\vec{p}, s') c(\vec{p}, s) \right]. \end{aligned} \quad (6.7)$$

The spinors are orthogonal:

$$\begin{aligned} u_{s'}^\dagger u_s &= 2E \delta_{ss'} = v_{s'}^\dagger v_s, \\ u^\dagger v &= 0, \quad v^\dagger u = 0. \end{aligned}$$

Therefore,

$$\begin{aligned} H &= \sum_{s,s'} \frac{1}{2} \int d^3p \, 2E \delta_{ss'} [c^\dagger(\vec{p}, s') c(\vec{p}, s) - d(\vec{p}, s') d^\dagger(\vec{p}, s)] \\ &= \sum_{s=\pm 1/2} \int d^3p \, E [c^\dagger(\vec{p}, s) c(\vec{p}, s) - d(\vec{p}, s) d^\dagger(\vec{p}, s)]. \end{aligned} \quad (6.8)$$

After fermionic normal ordering,

$$: H := \sum_{s=\pm 1/2} \int d^3p \, E [c^\dagger(\vec{p}, s) c(\vec{p}, s) + d^\dagger(\vec{p}, s) d(\vec{p}, s)].$$

The vacuum of the theory is the state simultaneously annihilated by $c(\vec{p}, s)$ and $d(\vec{p}, s)$:

$$c(\vec{p}, s)|0\rangle = 0 = \langle 0|c^\dagger(\vec{p}, s),$$

$$d(\vec{p}, s)|0\rangle = 0 = \langle 0|d^\dagger(\vec{p}, s),$$

and

$$H|0\rangle = 0.$$

We have two one-particle states:

$$|p, s\rangle = c^\dagger(\vec{p}, s)|0\rangle,$$

and

$$|\widetilde{p}, s\rangle = d^\dagger(\vec{p}, s)|0\rangle.$$

In both cases,

$$E > 0,$$

and the states are degenerate in mass and energy.

6.3 Charge Operator and Conserved Charges

Analogously to the charged Klein–Gordon field, the Dirac Lagrangian density is invariant under a global phase transformation:

$$\psi(x) \longrightarrow \psi'(x) = e^{-i\Theta} \psi(x), \quad (6.9)$$

$$\bar{\psi}(x) \longrightarrow \bar{\psi}'(x) = \bar{\psi}(x) e^{i\Theta}, \quad (6.10)$$

with Θ constant.

The infinitesimal transformations are

$$\delta\psi = \psi'(x) - \psi(x) = -i\varepsilon\psi(x), \quad (6.11)$$

$$\delta\bar{\psi} = \bar{\psi}'(x) - \bar{\psi}(x) = i\varepsilon\bar{\psi}(x), \quad (6.12)$$

ε being an infinitesimal parameter.

The transformation of the Lagrangian density is then

$$\mathcal{L}(\psi, \bar{\psi}) \longrightarrow \mathcal{L}(\psi', \bar{\psi}') = \bar{\psi}' (i\gamma^\mu \partial_\mu - m) \psi' \quad (6.13)$$

$$= \bar{\psi} e^{i\Theta} (i\gamma^\mu \partial_\mu - m) e^{-i\Theta} \psi \quad (6.14)$$

$$= \bar{\psi} (i\gamma^\mu \partial_\mu - m) \psi \quad (6.15)$$

$$= \mathcal{L}(\psi, \bar{\psi}). \quad (6.16)$$

Therefore, the Lagrangian density remains unchanged.

Now let us find the conserved charge. The Noether current is

$$J_\varepsilon^\mu = \delta\bar{\psi}_\alpha \frac{\partial \mathcal{L}}{\partial(\partial_\mu \bar{\psi}_\alpha)} + \delta\psi_\alpha \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_\alpha)}, \quad (6.17)$$

since, as for the charged Klein–Gordon field, for an internal infinitesimal transformation,

$$K^\mu = 0. \quad (6.18)$$

Using the infinitesimal transformations,

$$J_\varepsilon^\mu = (-i\varepsilon\psi)_\alpha (-i\bar{\psi}\gamma^\mu)_\alpha \quad (6.19)$$

$$= \varepsilon\bar{\psi}(x)\gamma^\mu\psi(x), \quad (6.20)$$

where the ordering and the change of sign are due to anticommutation.

The Noether current is therefore

$$J^\mu = \bar{\psi}(x)\gamma^\mu\psi(x), \quad (6.21)$$

which is a vector current.

The conserved charge is

$$Q = \int d^3x J^0(x) \quad (6.22)$$

$$= \int d^3x \psi^\dagger(x)\psi(x). \quad (6.23)$$

We associate this current with electric charge, although it can also be associated with baryon number or lepton number.

The charge operator in terms of creation and annihilation operators is

$$: Q := \sum_{s=\pm 1/2} \int d^3p \left[c^\dagger(\vec{p}, s) c(\vec{p}, s) - d^\dagger(\vec{p}, s) d(\vec{p}, s) \right]. \quad (6.24)$$

It operates in the following way:

$$Q |\vec{p}, s\rangle = + |\vec{p}, s\rangle, \quad (6.25)$$

$$Q |\vec{p}, s\rangle = - |\vec{p}, s\rangle. \quad (6.26)$$

6.4 Green Function and Dirac Propagator

The Dirac Green function satisfies

$$(i\cancel{\partial} - m) S(x - y) = \delta^4(x - y), \quad (6.27)$$

where the left-hand side is a 4×4 matrix in spinor space, whereas the right-hand side is the 4×4 identity.

When we first found the Dirac equation from Klein–Gordon, we did something like

$$\sqrt{\square + m^2} \psi(x) = 0. \quad (6.28)$$

Can we do something of the same sort with the Klein–Gordon propagator?

$$\sqrt{\square + m^2} G(x - y) = \sqrt{\delta^4(x - y)}. \quad (6.29)$$

Let us check:

$$(i\cancel{\partial} - m)(i\cancel{\partial} + m) = (i\cancel{\partial})^2 - m^2 \quad (6.30)$$

$$= -\gamma^\mu \partial_\mu \gamma^\nu \partial_\nu - m^2 \quad (6.31)$$

$$= -\gamma^\mu \gamma^\nu \partial_\mu \partial_\nu - m^2 \quad (6.32)$$

$$= -\left(\frac{\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu}{2}\right) \partial_\mu \partial_\nu - m^2 \quad (6.33)$$

$$= -\frac{1}{2} \{\gamma^\mu, \gamma^\nu\} \partial_\mu \partial_\nu - m^2 \quad (6.34)$$

$$= -\frac{1}{2} 2\eta^{\mu\nu} \partial_\mu \partial_\nu - m^2 \quad (6.35)$$

$$= -\eta^{\mu\nu} \partial_\mu \partial_\nu - m^2 \quad (6.36)$$

$$= -\partial_\mu \partial^\mu - m^2 \quad (6.37)$$

$$= -(\partial_\mu \partial^\mu + m^2) \quad (6.38)$$

$$= -(\square + m^2). \quad (6.39)$$

Therefore, for Klein–Gordon,

$$(\partial_\mu \partial^\mu + m^2) G(x - y) = -\delta^4(x - y), \quad (6.40)$$

which can be rewritten as

$$(i\cancel{\partial} - m)(i\cancel{\partial} + m) G(x - y) = \delta^4(x - y). \quad (6.41)$$

Comparing this with the Dirac equation

$$(i\cancel{\partial} - m) S(x - y) = \delta^4(x - y), \quad (6.42)$$

we can identify

$$S(x - y) = (i\cancel{\partial} + m) G(x - y). \quad (6.43)$$

For the Klein–Gordon propagator,

$$i\Delta_+(x - y) = \langle 0 | [\phi^{(+)}(x), \phi^{(-)}(y)] | 0 \rangle \quad (6.44)$$

$$= \langle 0 | \phi^{(+)}(x) \phi^{(-)}(y) | 0 \rangle \quad (6.45)$$

$$= \langle 0 | \phi(x) \phi(y) | 0 \rangle, \quad (6.46)$$

where

$$\phi^{(+)}(x) | 0 \rangle = 0. \quad (6.47)$$

Similarly,

$$i\Delta_-(x - y) = \langle 0 | [\phi^{(-)}(x), \phi^{(+)}(y)] | 0 \rangle \quad (6.48)$$

$$= \langle 0 | \phi^{(+)}(y) \phi^{(-)}(x) | 0 \rangle \quad (6.49)$$

$$= \langle 0 | \phi(y) \phi(x) | 0 \rangle. \quad (6.50)$$

Therefore,

$$i\Delta_F(x - y) = i [\Theta(x^0 - y^0) \Delta_+(x - y) - \Theta(y^0 - x^0) \Delta_-(x - y)] \quad (6.51)$$

$$= \langle 0 | T \phi(x) \phi(y) | 0 \rangle, \quad (6.52)$$

since for bosons time ordering has a positive sign.

In momentum space,

$$\Delta_F(p) = \frac{i}{p^2 - m^2 + i\varepsilon}, \quad (6.53)$$

or equivalently

$$G_F(p) = \frac{1}{p^2 - m^2 + i\varepsilon}. \quad (6.54)$$

Focusing on the Feynman prescription,

$$S_F(p) = (\not{p} + m) G_F(p) \quad (6.55)$$

$$= \lim_{\varepsilon \rightarrow 0^+} \frac{\not{p} + m}{p^2 - m^2 + i\varepsilon} \quad (6.56)$$

$$= \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\not{p} - m + i\varepsilon} \quad (6.57)$$

$$= S_F(p). \quad (6.58)$$

Thus,

$$S(x - y) = (i\not{\partial} + m) G(x - y), \quad (6.59)$$

where the first object is the Dirac propagator and the second is the Klein–Gordon propagator.

For Klein–Gordon,

$$G_F(p) \propto \frac{1}{p^2}, \quad (6.60)$$

whereas for Dirac,

$$S_F(p) \propto \frac{1}{p}. \quad (6.61)$$

We now evaluate the vacuum expectation values of

$$\{\psi_+(x), \bar{\psi}_-(y)\} \quad \text{and} \quad \{\psi_-(x), \bar{\psi}_+(y)\}. \quad (6.62)$$

Using

$$\psi_+(x) |0\rangle = 0, \quad (6.63)$$

we obtain

$$iS_+(x - y) = \langle 0 | \{\psi_+(x), \bar{\psi}_-(y)\} | 0 \rangle \quad (6.64)$$

$$= \langle 0 | \psi_+(x) \bar{\psi}_-(y) | 0 \rangle \quad (6.65)$$

$$= \langle 0 | \psi(x) \bar{\psi}(y) | 0 \rangle, \quad (6.66)$$

and

$$iS_-(x - y) = \langle 0 | \{\psi_-(x), \bar{\psi}_+(y)\} | 0 \rangle \quad (6.67)$$

$$= \langle 0 | \bar{\psi}_+(y) \psi_-(x) | 0 \rangle \quad (6.68)$$

$$= \langle 0 | \bar{\psi}(y) \psi(x) | 0 \rangle. \quad (6.69)$$

Hence,

$$S_F(x - y) = \Theta(x^0 - y^0) S_+(x - y) - \Theta(y^0 - x^0) S_-(y - x) \quad (6.70)$$

$$= \langle 0 | T \psi(x) \bar{\psi}(y) | 0 \rangle, \quad (6.71)$$

since for fermions time ordering introduces an additional negative sign.

For two fermionic fields,

$$T \psi_1(x_1) \psi_2(x_2) = \Theta(t_1 - t_2) \psi_1(x_1) \psi_2(x_2) - \Theta(t_2 - t_1) \psi_2(x_2) \psi_1(x_1). \quad (6.72)$$

6.5 Recapitulation on Propagators

For the neutral Klein–Gordon field,

$$i\Delta_F(x - y) = \langle 0 | T \phi(x) \phi(y) | 0 \rangle. \quad (6.73)$$

The time ordering for bosons is

$$T \phi(x) \phi(y) = \Theta(x^0 - y^0) \phi(x) \phi(y) + \Theta(y^0 - x^0) \phi(y) \phi(x). \quad (6.74)$$

For the charged Klein–Gordon field,

$$i\Delta_F(x - y) = \langle 0 | T \phi(x) \phi^\dagger(y) | 0 \rangle. \quad (6.75)$$

For the Dirac spin- $\frac{1}{2}$ field,

$$iS_{F,\alpha\beta}(x-y) = \langle 0 | T \psi_\alpha(x) \bar{\psi}_\beta(y) | 0 \rangle. \quad (6.76)$$

Time ordering for fermions has a minus sign:

$$T \psi_\alpha(x) \bar{\psi}_\beta(y) = \Theta(x^0 - y^0) \psi_\alpha(x) \bar{\psi}_\beta(y) - \Theta(y^0 - x^0) \bar{\psi}_\beta(y) \psi_\alpha(x). \quad (6.77)$$

In momentum space,

$$G_F(p) \propto \frac{1}{p^2}, \quad \text{Klein-Gordon}, \quad (6.78)$$

$$S_F(p) \propto \frac{1}{p}, \quad \text{Dirac}. \quad (6.79)$$

6.6 Majorana Fermions

Consider that

$$\psi = \psi^\dagger. \quad (6.80)$$

Recall that for the Hermitian Klein-Gordon field,

$$\phi \propto a e^{-ipx} + a^\dagger e^{ipx}, \quad \pi^0 = \bar{\pi}^0. \quad (6.81)$$

For the non-Hermitian Klein-Gordon field,

$$\phi \propto a e^{-ipx} + b^\dagger e^{ipx}. \quad (6.82)$$

For a non-Hermitian Dirac field,

$$\psi \propto c e^{-ipx} + d^\dagger e^{ipx}. \quad (6.83)$$

The question is: what would a Hermitian Dirac field look like?

For a Hermitian Dirac field,

$$\psi \propto c e^{-ipx} + c^\dagger e^{ipx}. \quad (6.84)$$

Do we have any candidate for a fermion whose antiparticle is itself?

$$\nu_e = \bar{\nu}_e? \quad (6.85)$$

A Majorana particle is its own antiparticle.

We define a Majorana field as

$$\psi_M = \mathcal{C} \psi_M \mathcal{C}^{-1}, \quad (6.86)$$

where $\mathcal{C}$ is the charge-conjugation operator.

Its action on one-particle states is

$$\mathcal{C} |\psi(\vec{p}, s)\rangle = \eta_C |\bar{\psi}(\vec{p}, s)\rangle, \quad (6.87)$$

with

$$\eta_C = e^{i\theta}, \quad |\eta_C| = 1, \quad (6.88)$$

where η_C is an arbitrary non-observable phase.

The vacuum is invariant:

$$\mathcal{C} |0\rangle = |0\rangle. \quad (6.89)$$

The particle and antiparticle states are

$$|\psi(\vec{p}, s)\rangle = c^\dagger(\vec{p}, s) |0\rangle, \quad |\bar{\psi}(\vec{p}, s)\rangle = d^\dagger(\vec{p}, s) |0\rangle. \quad (6.90)$$

Applying charge conjugation to the particle state,

$$\mathcal{C} |\psi(\vec{p}, s)\rangle = \eta_C |\bar{\psi}(\vec{p}, s)\rangle \quad (6.91)$$

$$= \eta_C d^\dagger(\vec{p}, s) |0\rangle. \quad (6.92)$$

On the other hand,

$$\mathcal{C}c^\dagger(\vec{p}, s)|0\rangle = \mathcal{C}c^\dagger(\vec{p}, s)\mathcal{C}^{-1}\mathcal{C}|0\rangle \quad (6.93)$$

$$= \mathcal{C}c^\dagger(\vec{p}, s)\mathcal{C}^{-1}|0\rangle. \quad (6.94)$$

Therefore,

$$\mathcal{C}c^\dagger(\vec{p}, s)\mathcal{C}^{-1} = \eta_C d^\dagger(\vec{p}, s). \quad (1)$$

Analogously,

$$\mathcal{C}d^\dagger(\vec{p}, s)\mathcal{C}^{-1} = \eta_C c^\dagger(\vec{p}, s). \quad (2)$$

Taking the Hermitian conjugate of Eq. (1),

$$\mathcal{C}^{-1}c(\vec{p}, s)\mathcal{C} = \eta_C^* d(\vec{p}, s). \quad (6.95)$$

Equivalently,

$$\mathcal{C}c(\vec{p}, s)\mathcal{C}^{-1} = \eta_C^* d(\vec{p}, s). \quad (6.96)$$

Using the expansion of the Dirac field,

$$\psi'(x) = \mathcal{C}\psi(x)\mathcal{C}^{-1} \quad (6.97)$$

$$= \sum_{s=\pm 1/2} \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \left[\mathcal{C}c(\vec{p}, s)\mathcal{C}^{-1}u(\vec{p}, s)e^{-ipx} + \mathcal{C}d^\dagger(\vec{p}, s)\mathcal{C}^{-1}v(\vec{p}, s)e^{ipx} \right] \quad (6.98)$$

$$= \sum_{s=\pm 1/2} \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \left[\eta_C^* d(\vec{p}, s)u(\vec{p}, s)e^{-ipx} + \eta_C c^\dagger(\vec{p}, s)v(\vec{p}, s)e^{ipx} \right]. \quad (6.99)$$

Using the definition of the Majorana field,

$$\psi_M = \mathcal{C}\psi_M\mathcal{C}^{-1}, \quad (6.100)$$

we obtain the relations

$$c(\vec{p}, s) = \eta_C^* d(\vec{p}, s), \quad d^\dagger(\vec{p}, s) = \eta_C c^\dagger(\vec{p}, s). \quad (6.101)$$

Thus, the creation and annihilation operators for particles and antiparticles differ only by a non-observable arbitrary phase.

Therefore,

$$\psi_M(x) = \sum_{s=\pm 1/2} \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \left[c(\vec{p}, s)u(\vec{p}, s)e^{-ipx} + \lambda c^\dagger(\vec{p}, s)v(\vec{p}, s)e^{ipx} \right], \quad (6.102)$$

where

$$\lambda = \eta_C. \quad (6.103)$$

6.7 Problems

Problem 6.1 Show that the canonical anticommutation relations for the quantized Dirac field imply the usual anticommutation relations for the creation and annihilation operators.

Hint: Start by proving that

$$c(\vec{p}, s) = \int d^3x \sqrt{\frac{m}{(2\pi)^3 E}} e^{ipx} u_\alpha^\dagger(\vec{p}, s) \psi_\alpha(x), \quad (6.104)$$

$$d^\dagger(\vec{p}, s) = \int d^3x \sqrt{\frac{m}{(2\pi)^3 E}} e^{-ipx} v_\alpha^\dagger(\vec{p}, s) \psi_\alpha(x), \quad (6.105)$$

$$c^\dagger(\vec{p}, s) = \int d^3x \sqrt{\frac{m}{(2\pi)^3 E}} e^{-ipx} \psi_\alpha^\dagger(x) u_\alpha(\vec{p}, s), \quad (6.106)$$

$$d(\vec{p}, s) = \int d^3x \sqrt{\frac{m}{(2\pi)^3 E}} e^{ipx} \psi_\alpha^\dagger(x) v_\alpha(\vec{p}, s). \quad (6.107)$$

Solution.

We start by writing the field in the plane-wave basis:

$$\psi_\alpha(x) = \sum_s \int d^3p \sqrt{\frac{m}{(2\pi)^3 E}} \left[e^{-ipx} c(\vec{p}, s) u_\alpha(\vec{p}, s) + e^{ipx} d^\dagger(\vec{p}, s) v_\alpha(\vec{p}, s) \right]. \quad (6.108)$$

The spinors satisfy the orthogonality relations

$$u^\dagger(\vec{p}, s) u(\vec{p}, s') = \frac{E}{m} \delta_{ss'}, \quad (6.109)$$

$$v^\dagger(\vec{p}, s) v(\vec{p}, s') = \frac{E}{m} \delta_{ss'}, \quad (6.110)$$

$$u^\dagger(\vec{p}, s) v(-\vec{p}, s') = v^\dagger(\vec{p}, s) u(-\vec{p}, s') = 0. \quad (6.111)$$

The equal-time canonical anticommutation relations are

$$\left\{ \psi_\alpha(t, \vec{x}), \psi_\beta^\dagger(t, \vec{y}) \right\} = \delta_{\alpha\beta} \delta^3(\vec{x} - \vec{y}), \quad (6.112)$$

$$\left\{ \psi_\alpha(t, \vec{x}), \psi_\beta(t, \vec{y}) \right\} = \left\{ \psi_\alpha^\dagger(t, \vec{x}), \psi_\beta^\dagger(t, \vec{y}) \right\} = 0. \quad (6.113)$$

Let us start by substituting the field expansion into $c(\vec{p}, s)$:

$$\begin{aligned} c(\vec{p}, s) &= \sum_{s'} \int d^3p' \sqrt{\frac{m}{(2\pi)^3 E}} \sqrt{\frac{m}{(2\pi)^3 E'}} \int d^3x e^{ipx} \\ &\quad \times \left[e^{-ip'x} c(\vec{p}', s') u_\alpha(\vec{p}', s') u_\alpha^\dagger(\vec{p}, s) + e^{ip'x} d^\dagger(\vec{p}', s') v_\alpha(\vec{p}', s') u_\alpha^\dagger(\vec{p}, s) \right]. \end{aligned} \quad (6.114)$$

Taking $t = 0$, the spatial integrals are

$$\int d^3x e^{i(\vec{p}-\vec{p}') \cdot \vec{x}} = (2\pi)^3 \delta^3(\vec{p} - \vec{p}'), \quad (6.115)$$

$$\int d^3x e^{i(\vec{p}+\vec{p}') \cdot \vec{x}} = (2\pi)^3 \delta^3(\vec{p} + \vec{p}'). \quad (6.116)$$

Therefore,

$$\begin{aligned} c(\vec{p}, s) &= \sum_{s'} \frac{m}{(2\pi)^3 \sqrt{EE'}} (2\pi)^3 \left[c(\vec{p}, s') u_\alpha(\vec{p}, s') u_\alpha^\dagger(\vec{p}, s) \right. \\ &\quad \left. + d^\dagger(-\vec{p}, s') v_\alpha(-\vec{p}, s') u_\alpha^\dagger(\vec{p}, s) \right] \end{aligned} \quad (6.117)$$

$$= \sum_{s'} \frac{m}{E} \left[c(\vec{p}, s') \frac{E}{m} \delta_{ss'} + d^\dagger(-\vec{p}, s') 0 \right] \quad (6.118)$$

$$= c(\vec{p}, s). \quad (6.119)$$

Hence,

$$c(\vec{p}, s) = \int d^3x \sqrt{\frac{m}{(2\pi)^3 E}} e^{ipx} u_\alpha^\dagger(\vec{p}, s) \psi_\alpha(x). \quad (6.120)$$

Proceeding analogously, we recover the remaining inverse relations:

$$d^\dagger(\vec{p}, s) = \int d^3x \sqrt{\frac{m}{(2\pi)^3 E}} e^{-ipx} v_\alpha^\dagger(\vec{p}, s) \psi_\alpha(x), \quad (6.121)$$

$$c^\dagger(\vec{p}, s) = \int d^3x \sqrt{\frac{m}{(2\pi)^3 E}} e^{-ipx} \psi_\alpha^\dagger(x) u_\alpha(\vec{p}, s), \quad (6.122)$$

$$d(\vec{p}, s) = \int d^3x \sqrt{\frac{m}{(2\pi)^3 E}} e^{ipx} \psi_\alpha^\dagger(x) v_\alpha(\vec{p}, s). \quad (6.123)$$

Now let us compute the anticommutator:

$$\begin{aligned} \{c(\vec{p}, s), c^\dagger(\vec{p}', s')\} &= \sqrt{\frac{m}{(2\pi)^3 E}} \sqrt{\frac{m}{(2\pi)^3 E'}} \int d^3x d^3x' e^{ipx} e^{-ip'x'} \\ &\quad \times u_\alpha^\dagger(\vec{p}, s) u_\beta(\vec{p}', s') \left\{ \psi_\alpha(x), \psi_\beta^\dagger(x') \right\}. \end{aligned} \quad (6.124)$$

Using

$$\left\{ \psi_\alpha(x), \psi_\beta^\dagger(x') \right\} = \delta_{\alpha\beta} \delta^3(\vec{x} - \vec{x}'), \quad (6.125)$$

we obtain

$$\{c(\vec{p}, s), c^\dagger(\vec{p}', s')\} = NN' \int d^3x e^{i(\vec{p}-\vec{p}')\cdot\vec{x}} u^\dagger(\vec{p}, s) u(\vec{p}', s') \quad (6.126)$$

$$= NN' (2\pi)^3 \delta^3(\vec{p} - \vec{p}') \frac{E}{m} \delta_{ss'} \quad (6.127)$$

$$= \delta_{ss'} \delta^3(\vec{p} - \vec{p}'), \quad (6.128)$$

where

$$N = \sqrt{\frac{m}{(2\pi)^3 E}}, \quad N' = \sqrt{\frac{m}{(2\pi)^3 E'}}. \quad (6.129)$$

Analogously,

$$\{d(\vec{p}, s), d^\dagger(\vec{p}', s')\} = \delta_{ss'} \delta^3(\vec{p} - \vec{p}'). \quad (6.130)$$

From

$$\{\psi, \psi\} = \{\psi^\dagger, \psi^\dagger\} = 0, \quad (6.131)$$

it follows immediately that all the remaining anticommutators vanish:

$$\{c(\vec{p}, s), c(\vec{p}', s')\} = 0, \quad \{c^\dagger(\vec{p}, s), c^\dagger(\vec{p}', s')\} = 0, \quad (6.132)$$

$$\{d(\vec{p}, s), d(\vec{p}', s')\} = 0, \quad \{d^\dagger(\vec{p}, s), d^\dagger(\vec{p}', s')\} = 0. \quad (6.133)$$

All mixed anticommutators between c and d operators also vanish.

□

Problem 6.2 Show that the number operator $\hat{N}$ of the quantized Dirac field can be written as

$$\hat{N} = \int d^3x \psi^\dagger \psi = \int \frac{d^3p}{(2\pi)^3} \sum_{\lambda=+,-} \left[\hat{c}_\lambda^\dagger(\vec{p}) \hat{c}_\lambda(\vec{p}) + \hat{d}_\lambda(\vec{p}) \hat{d}_\lambda^\dagger(\vec{p}) \right]. \quad (6.134)$$

Solution.

We begin with

$$\hat{N} = \int d^3x \psi^\dagger(x) \psi(x). \quad (6.135)$$

Using the plane-wave expansions,

$$\psi(x) = \sum_s \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \left[e^{-ipx} c(\vec{p}, s) u(\vec{p}, s) + e^{ipx} d^\dagger(\vec{p}, s) v(\vec{p}, s) \right], \quad (6.136)$$

$$\psi^\dagger(x) = \sum_{s'} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} \left[e^{ip'x} c^\dagger(\vec{p}', s') u^\dagger(\vec{p}', s') + e^{-ip'x} d(\vec{p}', s') v^\dagger(\vec{p}', s') \right]. \quad (6.137)$$

Therefore,

$$\begin{aligned}
N = \int d^3x \int \frac{d^3p}{\sqrt{(2\pi)^3} 2E} \int \frac{d^3p'}{\sqrt{(2\pi)^3} 2E'} \sum_{s,s'} & \left[e^{-i(p-p')x} c^\dagger(\vec{p}', s') c(\vec{p}, s) u^\dagger(\vec{p}', s') u(\vec{p}, s) \right. \\
& + e^{i(p+p')x} c^\dagger(\vec{p}', s') d^\dagger(\vec{p}, s) u^\dagger(\vec{p}', s') v(\vec{p}, s) \\
& + e^{-i(p+p')x} d(\vec{p}', s') c(\vec{p}, s) v^\dagger(\vec{p}', s') u(\vec{p}, s) \\
& \left. + e^{-i(p'-p)x} d(\vec{p}', s') d^\dagger(\vec{p}, s) v^\dagger(\vec{p}', s') v(\vec{p}, s) \right]. \tag{6.138}
\end{aligned}$$

Using

$$u^\dagger v = 0, \quad v^\dagger u = 0, \tag{6.139}$$

$$u^\dagger u = 2E \delta_{ss'}, \quad v^\dagger v = 2E \delta_{ss'}, \tag{6.140}$$

the mixed terms vanish, and hence

$$\begin{aligned}
N = \int d^3x \int \frac{d^3p}{\sqrt{(2\pi)^3} 2E} \int \frac{d^3p'}{\sqrt{(2\pi)^3} 2E'} \sum_s & \left[e^{-i(p-p')x} c^\dagger(\vec{p}', s) c(\vec{p}, s) 2E \right. \\
& \left. + e^{-i(p'-p)x} d(\vec{p}', s) d^\dagger(\vec{p}, s) 2E \right]. \tag{6.141}
\end{aligned}$$

Performing the spatial integral,

$$\int d^3x e^{i(\vec{p}-\vec{p}') \cdot \vec{x}} = (2\pi)^3 \delta^3(\vec{p} - \vec{p}'), \tag{6.142}$$

we obtain

$$\begin{aligned}
N = \int d^3p d^3p' \frac{1}{\sqrt{(2\pi)^3} 2E} \frac{1}{\sqrt{(2\pi)^3} 2E'} \sum_s & (2\pi)^3 2E \\
& \times [c^\dagger(\vec{p}', s) c(\vec{p}, s) + d(\vec{p}', s) d^\dagger(\vec{p}, s)] \delta^3(\vec{p} - \vec{p}') \tag{6.143}
\end{aligned}$$

$$= \int d^3p \sum_s [c^\dagger(\vec{p}, s) c(\vec{p}, s) + d(\vec{p}, s) d^\dagger(\vec{p}, s)]. \tag{6.144}$$

Thus,

$$\hat{N} = \int \frac{d^3p}{(2\pi)^3} \sum_s [c^\dagger(\vec{p}, s) c(\vec{p}, s) + d(\vec{p}, s) d^\dagger(\vec{p}, s)]. \tag{6.145}$$

Using the fermionic anticommutation relation,

$$d(\vec{p}, s) d^\dagger(\vec{p}, s) = 1 - d^\dagger(\vec{p}, s) d(\vec{p}, s), \tag{6.146}$$

normal ordering gives

$$: \hat{N} := \int \frac{d^3p}{(2\pi)^3} \sum_s [c^\dagger(\vec{p}, s) c(\vec{p}, s) - d^\dagger(\vec{p}, s) d(\vec{p}, s)]. \tag{6.147}$$

□

Problem 6.3 Consider the spin operator

$$\hat{S}^i = \int d^3\vec{x} \psi^\dagger \frac{\Sigma_i}{2} \psi, \tag{6.148}$$

where Σ_i are the generalized Pauli matrices. Show that

$$[\hat{S}^i, \hat{S}^j] = i\epsilon^{ijk} \hat{S}^k. \tag{6.149}$$

Hint: Use both

$$[AB, C] = A[B, C] + [A, C]B \tag{6.150}$$

and

$$[AB, C] = A\{B, C\} - \{A, C\}B, \tag{6.151}$$

where A , B and C are arbitrary operators.

Solution.

The spin operator is

$$\hat{S}_i = \int d^3x \psi^\dagger(x) \frac{\Sigma_i}{2} \psi(x). \quad (6.152)$$

Therefore,

$$\begin{aligned} [\hat{S}_i, \hat{S}_j] &= \int d^3x d^3x' \left(\frac{\Sigma_i}{2} \right)_{\alpha\beta} \left(\frac{\Sigma_j}{2} \right)_{\gamma\delta} \\ &\quad \times [\psi_\alpha^\dagger(x) \psi_\beta(x), \psi_\gamma^\dagger(x') \psi_\delta(x')] . \end{aligned} \quad (6.153)$$

We use

$$[AB, C] = A[B, C] + [A, C]B, \quad (6.154)$$

$$[AB, C] = A\{B, C\} - \{A, C\}B. \quad (6.155)$$

First,

$$\begin{aligned} [\psi_\alpha^\dagger(x) \psi_\beta(x), \psi_\gamma^\dagger(x') \psi_\delta(x')] &= \psi_\alpha^\dagger(x) [\psi_\beta(x), \psi_\gamma^\dagger(x') \psi_\delta(x')] \\ &\quad + [\psi_\alpha^\dagger(x), \psi_\gamma^\dagger(x') \psi_\delta(x')] \psi_\beta(x). \end{aligned} \quad (6.156)$$

For the first commutator,

$$[\psi_\beta(x), \psi_\gamma^\dagger(x') \psi_\delta(x')] = - [\psi_\gamma^\dagger(x') \psi_\delta(x'), \psi_\beta(x)] \quad (6.157)$$

$$= - [\psi_\gamma^\dagger(x') \{ \psi_\delta(x'), \psi_\beta(x) \} - \{ \psi_\gamma^\dagger(x'), \psi_\beta(x) \} \psi_\delta(x')] \quad (6.158)$$

$$= \delta_{\gamma\beta} \delta^3(\vec{x} - \vec{x}') \psi_\delta(x'). \quad (6.159)$$

For the second commutator,

$$[\psi_\alpha^\dagger(x), \psi_\gamma^\dagger(x') \psi_\delta(x')] = - [\psi_\gamma^\dagger(x') \psi_\delta(x'), \psi_\alpha^\dagger(x)] \quad (6.160)$$

$$= - [\psi_\gamma^\dagger(x') \{ \psi_\delta(x'), \psi_\alpha^\dagger(x) \} - \{ \psi_\gamma^\dagger(x'), \psi_\alpha^\dagger(x) \} \psi_\delta(x')] \quad (6.161)$$

$$= -\delta_{\delta\alpha} \delta^3(\vec{x} - \vec{x}') \psi_\gamma^\dagger(x'). \quad (6.162)$$

Hence,

$$\begin{aligned} [\psi_\alpha^\dagger(x) \psi_\beta(x), \psi_\gamma^\dagger(x') \psi_\delta(x')] &= \psi_\alpha^\dagger(x) \delta_{\gamma\beta} \delta^3(\vec{x} - \vec{x}') \psi_\delta(x') \\ &\quad - \delta_{\delta\alpha} \delta^3(\vec{x} - \vec{x}') \psi_\gamma^\dagger(x') \psi_\beta(x). \end{aligned} \quad (6.163)$$

Substituting this result,

$$\begin{aligned} [\hat{S}_i, \hat{S}_j] &= \int d^3x d^3x' \left(\frac{\Sigma_i}{2} \right)_{\alpha\beta} \left(\frac{\Sigma_j}{2} \right)_{\gamma\delta} \\ &\quad \times [\psi_\alpha^\dagger(x) \delta_{\gamma\beta} \delta^3(\vec{x} - \vec{x}') \psi_\delta(x') - \delta_{\delta\alpha} \delta^3(\vec{x} - \vec{x}') \psi_\gamma^\dagger(x') \psi_\beta(x)] \end{aligned} \quad (6.164)$$

$$\begin{aligned} &= \int d^3x d^3x' \delta^3(\vec{x} - \vec{x}') \left(\frac{\Sigma_i}{2} \right)_{\alpha\beta} \left(\frac{\Sigma_j}{2} \right)_{\gamma\delta} \\ &\quad \times [\psi_\alpha^\dagger \delta_{\gamma\beta} \psi_\delta - \psi_\gamma^\dagger \delta_{\delta\alpha} \psi_\beta] \end{aligned} \quad (6.165)$$

$$= \int d^3x \left[\psi_\alpha^\dagger \left(\frac{\Sigma_i \Sigma_j}{4} \right)_{\alpha\delta} \psi_\delta - \psi_\gamma^\dagger \left(\frac{\Sigma_j \Sigma_i}{4} \right)_{\gamma\beta} \psi_\beta \right] \quad (6.166)$$

$$= \int d^3x \psi^\dagger \left[\frac{\Sigma_i}{2}, \frac{\Sigma_j}{2} \right] \psi. \quad (6.167)$$

Using

$$\left[\frac{\Sigma_i}{2}, \frac{\Sigma_j}{2} \right] = i \varepsilon_{ijk} \frac{\Sigma_k}{2}, \quad (6.168)$$

we obtain

$$[\hat{S}_i, \hat{S}_j] = \int d^3x \psi^\dagger \left(i\varepsilon_{ijk} \frac{\Sigma_k}{2} \right) \psi \quad (6.169)$$

$$= i\varepsilon_{ijk} \int d^3x \psi^\dagger \frac{\Sigma_k}{2} \psi \quad (6.170)$$

$$= i\varepsilon_{ijk} \hat{S}_k. \quad (6.171)$$

□

Problem 6.4 Show that a Majorana mass term represented by

$$m (\bar{\psi} \psi^c + \text{h.c.}), \quad (6.172)$$

where

$$\psi^c = i\gamma^2 \psi^*, \quad (6.173)$$

is a Lorentz scalar.

Solution.

The charge-conjugated spinor is

$$\psi^c = i\gamma^2 \psi^*. \quad (6.174)$$

Under a Lorentz transformation,

$$\psi(x) \longrightarrow \psi'(x') = S(\Lambda) \psi(x), \quad (6.175)$$

$$\bar{\psi}(x) \longrightarrow \bar{\psi}'(x') = \bar{\psi}(x) S^{-1}(\Lambda). \quad (6.176)$$

For $\bar{\psi} \psi^c$ to transform as a scalar, we need ψ^c to transform in the same way as ψ . We have

$$\psi^{c'}(x') = i\gamma^2 (S(\Lambda) \psi(x))^* \quad (6.177)$$

$$= i\gamma^2 S^*(\Lambda) \psi^*(x). \quad (6.178)$$

The spinorial Lorentz transformation is

$$S(\Lambda) = \exp \left[-\frac{i}{4} \omega_{\mu\nu} \sigma^{\mu\nu} \right], \quad \sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu]. \quad (6.179)$$

Using

$$i\gamma^2 \gamma^{\mu*} (i\gamma^2)^{-1} = -\gamma^\mu, \quad (6.180)$$

we find

$$i\gamma^2 \sigma^{\mu\nu*} (i\gamma^2)^{-1} = -\frac{i}{2} [-\gamma^\mu, -\gamma^\nu] \quad (6.181)$$

$$= -\frac{i}{2} [\gamma^\mu, \gamma^\nu] \quad (6.182)$$

$$= -\sigma^{\mu\nu}. \quad (6.183)$$

Therefore,

$$i\gamma^2 S^*(\Lambda) (i\gamma^2)^{-1} = \exp \left[\frac{i}{4} \omega_{\mu\nu} (i\gamma^2 \sigma^{\mu\nu*} (i\gamma^2)^{-1}) \right] \quad (6.184)$$

$$= \exp \left[-\frac{i}{4} \omega_{\mu\nu} \sigma^{\mu\nu} \right] \quad (6.185)$$

$$= S(\Lambda). \quad (6.186)$$

Hence,

$$i\gamma^2 S^*(\Lambda) = S(\Lambda) i\gamma^2. \quad (6.187)$$

It follows that

$$\psi^{c'}(x') = i\gamma^2 S^*(\Lambda) \psi^*(x) \quad (6.188)$$

$$= S(\Lambda) i\gamma^2 \psi^*(x) \quad (6.189)$$

$$= S(\Lambda) \psi^c(x). \quad (6.190)$$

Thus, ψ^c transforms as a Lorentz spinor. Consequently,

$$\bar{\psi} \psi^c \longrightarrow (\bar{\psi} \psi^c)' \quad (6.191)$$

$$= \bar{\psi}(x) S^{-1}(\Lambda) S(\Lambda) \psi^c(x) \quad (6.192)$$

$$= \bar{\psi}(x) \psi^c(x). \quad (6.193)$$

Therefore, $\bar{\psi} \psi^c$ is a Lorentz scalar. Its Hermitian conjugate is also a Lorentz scalar, and hence the complete Majorana mass term

$$m (\bar{\psi} \psi^c + \text{h.c.}) \quad (6.194)$$

is Lorentz invariant.

□

7 Quantization of the Photon Field

We will quantize spin-1 particles, i.e. Maxwell theory.

A massive spin-1 particle has three degrees of freedom: three eigenvalues of spin in the direction of its momentum.

A massless spin-1 particle has two degrees of freedom.

We begin by writing Maxwell's equations:

$$\vec{B} = \vec{\nabla} \times \vec{A}, \quad (7.1)$$

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} - \vec{\nabla} \Phi, \quad (7.2)$$

where $\vec{A}(x)$ and $\Phi(x)$ are the vector potential and the scalar potential, respectively.

The four-vector potential is

$$A^\mu = (\Phi, \vec{A}), \quad (7.3)$$

$$A_\mu = (\Phi, -\vec{A}). \quad (7.4)$$

We define the four-dimensional curl of the four-vector potential:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -F_{\nu\mu}, \quad \mu, \nu = 0, 1, 2, 3. \quad (7.5)$$

In matrix form,

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix}. \quad (7.6)$$

Maxwell's equations can be written as

$$\partial_\mu F^{\mu\nu} = 0. \quad (7.7)$$

The field that generates the photons is $A^\mu(x)$.

However, $A^\mu(x)$ has four components, and therefore apparently four degrees of freedom, whereas there are only two photon states: left-handed and right-handed. Thus, we need to remove two degrees of freedom.

If $A_\mu(x)$ is a solution, then

$$A_\mu(x) + \partial_\mu \Theta(x) \quad (7.8)$$

is also a solution.

This is called a gauge transformation:

$$A_\mu(x) \longrightarrow A'_\mu(x) = A_\mu(x) + \partial_\mu \Theta(x). \quad (7.9)$$

Under this transformation,

$$F_{\mu\nu} \longrightarrow F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu \quad (7.10)$$

$$= \partial_\mu (A_\nu + \partial_\nu \Theta) - \partial_\nu (A_\mu + \partial_\mu \Theta) \quad (7.11)$$

$$= \partial_\mu A_\nu + \partial_\mu \partial_\nu \Theta - \partial_\nu A_\mu - \partial_\nu \partial_\mu \Theta \quad (7.12)$$

$$= \partial_\mu A_\nu - \partial_\nu A_\mu \quad (7.13)$$

$$= F_{\mu\nu}. \quad (7.14)$$

We have found a special symmetry called gauge symmetry.

We can define $A_\mu(x)$ uniquely by imposing additional conditions on $A_\mu(x)$.

- Imposing

$$\partial_\mu A^\mu = 0 \quad (7.15)$$

gives the Lorenz gauge.

- Imposing

$$\vec{\nabla} \cdot \vec{A} = 0 \quad (7.16)$$

gives the Coulomb gauge, together with

$$A_0(x) = 0. \quad (7.17)$$

7.1 Quantization of the Maxwell Field

The field $A_\mu(x)$ will now be identified as an operator.

The Lagrangian density that gives rise to Maxwell's equations is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = \frac{1}{2}(\vec{E}^2 - \vec{B}^2). \quad (7.18)$$

Since $F^{\mu\nu}$ is gauge invariant, $\mathcal{L}$ is also gauge invariant.

A theory invariant under gauge transformations is called a gauge theory. The field $A_\mu(x)$ is known as a gauge field.

The conjugate momenta are

$$\Pi^{\mu\nu}(x) = \frac{\partial \mathcal{L}}{\partial_\mu A_\nu(x)} = -F^{\mu\nu}. \quad (7.19)$$

In particular,

$$\Pi^{00}(x) = -F^{00} = 0, \quad (7.20)$$

which is a constraint of the theory.

Therefore,

$$\Pi^0(x) = 0, \quad (7.21)$$

so the conjugate momentum of $A^0(x)$ vanishes. This implies that A^0 is not an independent variable.

The field $A_0(x)$ has to commute with the other fields of the theory. Therefore, we can choose any value for $A_0(x)$. We choose

$$A_0(x) = 0, \quad (7.22)$$

which fixes the gauge.

For the spatial components,

$$\Pi^i(x) = -F^{0i} = \dot{A}^i(x) = E^i. \quad (7.23)$$

We can now compute the Hamiltonian density:

$$\mathcal{H} = \Pi^i \dot{A}_i - \mathcal{L} \quad (7.24)$$

$$= \frac{1}{2}(\vec{\Pi}^2 + \vec{B}^2) \quad (7.25)$$

$$= \frac{1}{2}(\vec{E}^2 + \vec{B}^2). \quad (7.26)$$

In the Coulomb gauge,

$$A_0 = 0, \quad \vec{\nabla} \cdot \vec{A} = 0, \quad (7.27)$$

which implies

$$\partial^\mu A_\mu = 0. \quad (7.28)$$

Maxwell's equations become

$$0 = \partial_\mu F^{\mu\nu} = \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) \quad (7.29)$$

$$= \partial_\mu \partial^\mu A^\nu - \partial^\nu \partial_\mu A^\mu \quad (7.30)$$

$$= \square A^\nu \quad (7.31)$$

$$= 0, \quad (7.32)$$

where the second term vanishes because of the gauge condition.

For $\nu = 0$,

$$\square A^0 = 0, \quad (7.33)$$

which is automatically satisfied because

$$A^0 = 0. \quad (7.34)$$

For $\nu = i$,

$$\square A^i = 0. \quad (7.35)$$

Thus, the three spatial components of the vector potential satisfy the massless Klein-Gordon equation. Therefore, the solutions are plane waves:

$$\vec{A}(x) \propto \vec{\epsilon}(\vec{p}) e^{\pm i p x}. \quad (7.36)$$

The vector $\vec{\varepsilon}(\vec{p})$ represents the polarization vector of a plane-wave solution travelling along $\hat{\vec{p}}$. Since the photon is massless,

$$p^0 = E = \sqrt{\vec{p}^2} = |\vec{p}|, \quad (7.37)$$

and therefore

$$p^\mu p_\mu = p^2 = 0. \quad (7.38)$$

The vector potential is transverse. From the gauge condition,

$$\vec{\nabla} \cdot \vec{A} = 0, \quad (7.39)$$

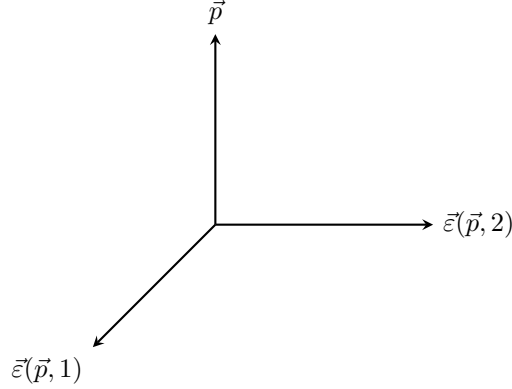
and using

$$\vec{\nabla} \longrightarrow i\vec{p}, \quad \vec{A} \propto \vec{\varepsilon}, \quad (7.40)$$

we obtain

$$\vec{p} \cdot \vec{\varepsilon}(\vec{p}) = 0. \quad (7.41)$$

Therefore, the polarization vector must be transverse to the direction of propagation of the plane wave. There are only two independent transverse directions.



We choose the polarization vectors to be orthonormal and complex:

$$\vec{p} \cdot \vec{\varepsilon}(\vec{p}, \lambda) = 0, \quad \lambda = 1, 2, \quad (7.42)$$

$$\vec{\varepsilon}^*(\vec{p}, \lambda') \cdot \vec{\varepsilon}(\vec{p}, \lambda) = \delta_{\lambda\lambda'}. \quad (7.43)$$

We can decompose $\vec{A}(x)$ as

$$\vec{A}(x) = \sum_{\lambda=1}^2 \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \left[\vec{\varepsilon}(\vec{p}, \lambda) e^{-ipx} a(\vec{p}, \lambda) + \vec{\varepsilon}^*(\vec{p}, \lambda) e^{ipx} a^\dagger(\vec{p}, \lambda) \right], \quad (7.44)$$

where

$$E = |\vec{p}|, \quad (7.45)$$

and $a(\vec{p}, \lambda)$ and $a^\dagger(\vec{p}, \lambda)$ are operators.

The quantization rules are

$$[a(\vec{p}, \lambda), a(\vec{p}', \lambda')] = 0, \quad (7.46)$$

$$[a^\dagger(\vec{p}, \lambda), a^\dagger(\vec{p}', \lambda')] = 0, \quad (7.47)$$

$$[a(\vec{p}, \lambda), a^\dagger(\vec{p}', \lambda')] = \delta^3(\vec{p} - \vec{p}') \delta_{\lambda\lambda'}. \quad (7.48)$$

In terms of creation and annihilation operators,

$$H = \sum_{\lambda=1}^2 \int d^3p E a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda), \quad (7.49)$$

and

$$\vec{P} = \sum_{\lambda=1}^2 \int d^3p \vec{p} a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda). \quad (7.50)$$

Normal ordering has already been applied.
The vacuum of the theory satisfies

$$a(\vec{p}, \lambda) |0\rangle = 0, \quad \langle 0| a^\dagger(\vec{p}, \lambda) = 0. \quad (7.51)$$

Therefore,

$$H |0\rangle = 0, \quad \vec{P} |0\rangle = 0. \quad (7.52)$$

The one-particle state is

$$|\vec{p}, \lambda\rangle = a^\dagger(\vec{p}, \lambda) |0\rangle. \quad (7.53)$$

This state satisfies

$$H |\vec{p}, \lambda\rangle = E |\vec{p}, \lambda\rangle, \quad (7.54)$$

$$\vec{P} |\vec{p}, \lambda\rangle = \vec{p} |\vec{p}, \lambda\rangle, \quad (7.55)$$

and

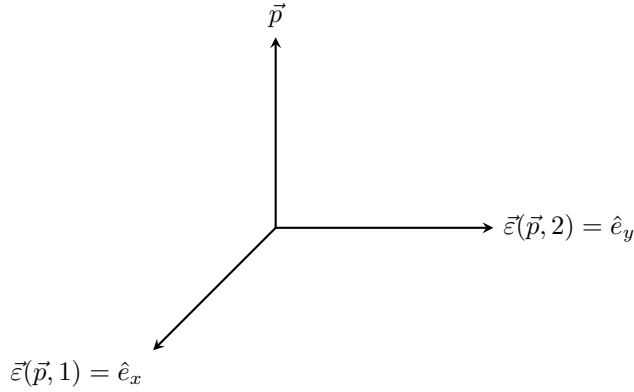
$$(H^2 - \vec{P}^2) |\vec{p}, \lambda\rangle = (E^2 - \vec{p}^2) |\vec{p}, \lambda\rangle = 0. \quad (7.56)$$

Looking at H or $A_\mu(x)$, we see that $A_\mu(x)$ is Hermitian. Therefore, the states do not carry electric charge.

For a convenient choice of axes, we take

$$\vec{\varepsilon}(\vec{p}, 1) = \hat{e}_x, \quad \vec{\varepsilon}(\vec{p}, 2) = \hat{e}_y, \quad (7.57)$$

with $\vec{p}$ along the remaining orthogonal direction.



We identify $a^\dagger(\vec{p}, 1)$ as creating a photon polarized along the x -axis, and $a^\dagger(\vec{p}, 2)$ as creating a photon polarized along the y -axis.

We define

$$\frac{1}{\sqrt{2}} [a^\dagger(\vec{p}, 1) - i a^\dagger(\vec{p}, 2)] = a^\dagger(\vec{p}, -) = a_L^\dagger(\vec{p}), \quad (7.58)$$

$$\frac{1}{\sqrt{2}} [a^\dagger(\vec{p}, 1) + i a^\dagger(\vec{p}, 2)] = a^\dagger(\vec{p}, +) = a_R^\dagger(\vec{p}). \quad (7.59)$$

The operators $a_L^\dagger(\vec{p})$ and $a_R^\dagger(\vec{p})$ create left- and right-circularly polarized photons, respectively. These states are eigenstates of the helicity operator with eigenvalues

$$-1 \quad \text{and} \quad +1, \quad (7.60)$$

respectively.

7.2 Spin of the Photons

The spin comes from angular momentum:

$$\vec{J} = \int d^3x \left(\vec{x} \times \vec{P} \right) = \int d^3x \left[\vec{x} \times \left(\vec{E} \times \left(\vec{\nabla} \times \vec{A} \right) \right) \right], \quad (7.61)$$

where

$$\vec{P} = \vec{E} \times \vec{B}, \quad \vec{B} = \vec{\nabla} \times \vec{A}. \quad (7.62)$$

We rewrite this in component form:

$$J^i = \epsilon^{ijk} \epsilon^{klm} \epsilon^{mnq} \int d^3x x^j E^l \partial_n A^q. \quad (7.63)$$

Using

$$\epsilon^{klm} \epsilon^{mnq} = \delta^{kn} \delta^{lq} - \delta^{kq} \delta^{ln}, \quad (7.64)$$

we obtain

$$J^i = \epsilon^{ijk} \int d^3x \left(x^j E^q \partial_k A^q - x^j E^n \partial_n A^k \right). \quad (7.65)$$

Next, we inject a total derivative:

$$J^i = \epsilon^{ijk} \int d^3x \left[x^j E^q \partial_k A^q - \partial_n \left(x^j E^n A^k \right) + x^j \left(\partial_n E^n \right) A^k + E^j A^k \right]. \quad (7.66)$$

The first term is the angular-momentum contribution,

$$L^i = \epsilon^{ijk} \int d^3x x^j E^q \partial_k A^q. \quad (7.67)$$

The second term is a total divergence and does not contribute. Moreover,

$$\vec{\nabla} \cdot \vec{E} = 0. \quad (7.68)$$

The remaining term does not depend explicitly on the position and can therefore be identified with the spin density:

$$S^i = \epsilon^{ijk} \int d^3x E^j A^k. \quad (7.69)$$

Thus,

$$\vec{S} = \int d^3x \left(\vec{E} \times \vec{A} \right). \quad (7.70)$$

In terms of the creation and annihilation operators,

$$\vec{S} = i \int d^3p \hat{p} \left[a^\dagger(\vec{p}, 2) a(\vec{p}, 1) - a^\dagger(\vec{p}, 1) a(\vec{p}, 2) \right]. \quad (7.71)$$

This expression is not written in terms of number operators. Substituting $a^\dagger(\vec{p}, 1)$ and $a^\dagger(\vec{p}, 2)$ by operators with definite helicity, we obtain

$$\vec{S} = \int d^3p \hat{p} \left[a^\dagger(\vec{p}, +) a(\vec{p}, +) - a^\dagger(\vec{p}, -) a(\vec{p}, -) \right]. \quad (7.72)$$

Equivalently,

$$\vec{S} = \int d^3p \hat{p} (N_R(\vec{p}) - N_L(\vec{p})), \quad (7.73)$$

where

$$N_R(\vec{p}) = a_R^\dagger(\vec{p}) a_R(\vec{p}), \quad (7.74)$$

$$N_L(\vec{p}) = a_L^\dagger(\vec{p}) a_L(\vec{p}). \quad (7.75)$$

The right-handed one-photon state satisfies

$$\vec{S} a_R^\dagger(\vec{p}) |0\rangle = +\hat{p} a_R^\dagger(\vec{p}) |0\rangle, \quad (7.76)$$

whereas the left-handed one-photon state satisfies

$$\vec{S} a_L^\dagger(\vec{p}) |0\rangle = -\hat{p} a_L^\dagger(\vec{p}) |0\rangle. \quad (7.77)$$

Therefore, photons are spin-1 particles.

7.3 Photon Propagator

We have found

$$\Delta(p) \propto \frac{1}{p^2 - m^2}, \quad S(p) \propto \frac{1}{p - m}, \quad G(p) \propto \frac{1}{p^2}. \quad (7.78)$$

In order to find the propagator for the photon field, we need to solve the inhomogeneous Maxwell equations:

$$\partial_\nu F^{\mu\nu} = j^\mu = \partial_\nu (\partial^\mu A^\nu - \partial^\nu A^\mu). \quad (7.79)$$

Therefore,

$$\square A^\mu - \partial^\mu (\partial_\nu A^\nu) = j^\mu. \quad (7.80)$$

We can rewrite this as

$$(\eta^{\nu\lambda} \square - \partial^\nu \partial^\lambda) A_\lambda = j^\nu. \quad (7.81)$$

We want to isolate A_λ , so we would need to invert

$$\eta^{\nu\lambda} \square - \partial^\nu \partial^\lambda, \quad (7.82)$$

but this operator does not have an inverse.

We choose the Lorenz gauge,

$$\partial_\lambda A^\lambda = 0. \quad (7.83)$$

Then,

$$\eta^{\nu\lambda} \square A_\lambda = j^\nu. \quad (7.84)$$

This is the inhomogeneous equation. Applying Green-function techniques,

$$\eta^{\nu\lambda} \square D_{\lambda\mu}(x' - x) = \delta_\mu^\nu \delta^4(x' - x). \quad (7.85)$$

Due to the presence of $\eta^{\nu\lambda}$, the Green function is a tensor.

The most general form for $D_{\mu\nu}(k)$ is

$$D_{\mu\nu}(k) = D(k^2) \eta_{\mu\nu} + D^{(1)}(k^2) k_\mu k_\nu, \quad (7.86)$$

where $D(k^2)$ and $D^{(1)}(k^2)$ are scalar functions of k^2 .

Substituting, we obtain

$$-k^2 \left[D(k^2) \eta^{\nu\lambda} \eta_{\lambda\mu} + D^{(1)}(k^2) \eta^{\nu\lambda} k_\lambda k_\mu \right] = \delta_\mu^\nu. \quad (7.87)$$

Therefore,

$$D(k^2) \delta_\mu^\nu + D^{(1)}(k^2) k^\nu k_\mu = -\frac{1}{k^2} \delta_\mu^\nu. \quad (7.88)$$

Comparing the left- and right-hand sides,

$$D(k^2) = -\frac{1}{k^2}, \quad D^{(1)}(k^2) = 0. \quad (7.89)$$

Hence, the photon propagator in momentum space in the Lorenz gauge is

$$D_F^{\mu\nu}(k^2) = -\frac{\eta^{\mu\nu}}{k^2}. \quad (7.90)$$

7.4 Problems

Problem 7.1 Show that the commutation relations for the creation and annihilation operators of the quantized electromagnetic field imply the canonical commutation relations

$$\left[\hat{A}_T^i(\vec{x}, t), \hat{A}_T^j(\vec{y}, t) \right] = 0, \quad (7.91)$$

$$\left[\hat{\Pi}_T^i(\vec{x}, t), \hat{\Pi}_T^j(\vec{y}, t) \right] = 0, \quad (7.92)$$

$$\left[\hat{A}_T^i(\vec{x}, t), \hat{\Pi}_T^j(\vec{y}, t) \right] = i \int \frac{d^3 \vec{k}}{(2\pi)^3} \left(\delta^{ij} - \frac{k^i k^j}{\vec{k}^2} \right) e^{i\vec{k} \cdot (\vec{x} - \vec{y})}. \quad (7.93)$$

Recall that

$$\hat{\Pi}_T^i(\vec{x}, t) = \partial_0 \hat{A}_T^i(\vec{x}, t). \quad (7.94)$$

Solution.

We can expand the transverse electromagnetic field as

$$A_T^i(x) = \sum_{\lambda=1}^2 \int \frac{d^3 k}{\sqrt{(2\pi)^3 2E}} \left[\varepsilon^i(\vec{k}, \lambda) e^{-ikx} a(\vec{k}, \lambda) + \varepsilon^{*i}(\vec{k}, \lambda) e^{ikx} a^\dagger(\vec{k}, \lambda) \right]. \quad (7.95)$$

The polarization vectors satisfy

$$\vec{k} \cdot \vec{\varepsilon}(\vec{k}, \lambda) = 0, \quad (7.96)$$

and

$$\vec{\varepsilon}^*(\vec{k}, \lambda) \cdot \vec{\varepsilon}(\vec{k}, \lambda') = \delta_{\lambda\lambda'}. \quad (7.97)$$

The commutation relations are

$$\left[a(\vec{k}, \lambda), a(\vec{k}', \lambda') \right] = 0, \quad (7.98)$$

$$\left[a^\dagger(\vec{k}, \lambda), a^\dagger(\vec{k}', \lambda') \right] = 0, \quad (7.99)$$

$$\left[a(\vec{k}, \lambda), a^\dagger(\vec{k}', \lambda') \right] = \delta^3(\vec{k} - \vec{k}') \delta_{\lambda\lambda'}. \quad (7.100)$$

We can easily compute the conjugate momentum:

$$\Pi_T^i(x) = \partial_0 A_T^i(x) \quad (7.101)$$

$$= \sum_{\lambda=1}^2 \int \frac{d^3 k}{\sqrt{(2\pi)^3 2E}} \left[-iE \varepsilon^i(\vec{k}, \lambda) e^{-ikx} a(\vec{k}, \lambda) + iE \varepsilon^{*i}(\vec{k}, \lambda) e^{ikx} a^\dagger(\vec{k}, \lambda) \right]. \quad (7.102)$$

Let us now compute the commutators. First,

$$\begin{aligned} \left[A_T^i(x), A_T^j(y) \right] &= \sum_{\lambda, \lambda'} \int \frac{d^3 k}{\sqrt{(2\pi)^3 2E}} \frac{d^3 k'}{\sqrt{(2\pi)^3 2E'}} \\ &\quad \times \left[\varepsilon^i(\vec{k}, \lambda) e^{-ikx} a(\vec{k}, \lambda) + \varepsilon^{*i}(\vec{k}, \lambda) e^{ikx} a^\dagger(\vec{k}, \lambda), \right. \\ &\quad \left. \varepsilon^j(\vec{k}', \lambda') e^{-ik'y} a(\vec{k}', \lambda') + \varepsilon^{*j}(\vec{k}', \lambda') e^{ik'y} a^\dagger(\vec{k}', \lambda') \right] \end{aligned} \quad (7.103)$$

$$\begin{aligned} &= \sum_{\lambda, \lambda'} \int \frac{d^3 k}{\sqrt{(2\pi)^3 2E}} \frac{d^3 k'}{\sqrt{(2\pi)^3 2E'}} \\ &\quad \times \left[\varepsilon^i(\vec{k}, \lambda) \varepsilon^{*j}(\vec{k}', \lambda') e^{-ikx} e^{ik'y} \left[a(\vec{k}, \lambda), a^\dagger(\vec{k}', \lambda') \right] \right. \\ &\quad \left. + \varepsilon^{*i}(\vec{k}, \lambda) \varepsilon^j(\vec{k}', \lambda') e^{ikx} e^{-ik'y} \left[a^\dagger(\vec{k}, \lambda), a(\vec{k}', \lambda') \right] \right] \end{aligned} \quad (7.104)$$

$$= 0, \quad x^0 = y^0. \quad (7.105)$$

Analogously,

$$\left[\Pi_T^i(x), \Pi_T^j(y) \right] = 0, \quad x^0 = y^0. \quad (7.106)$$

Finally,

$$\begin{aligned} [A_T^i(x), \Pi_T^j(y)] &= \sum_{\lambda, \lambda'} \int \frac{d^3k}{\sqrt{(2\pi)^3} 2E} \frac{d^3k'}{\sqrt{(2\pi)^3} 2E'} \\ &\times \left[\varepsilon^i(\vec{k}, \lambda) e^{-ikx} a(\vec{k}, \lambda) + \varepsilon^{*i}(\vec{k}, \lambda) e^{ikx} a^\dagger(\vec{k}, \lambda), \right. \\ &\quad \left. - iE' \varepsilon^j(\vec{k}', \lambda') e^{-ik'y} a(\vec{k}', \lambda') + iE' \varepsilon^{*j}(\vec{k}', \lambda') e^{ik'y} a^\dagger(\vec{k}', \lambda') \right] \end{aligned} \quad (7.107)$$

$$\begin{aligned} &= \sum_{\lambda, \lambda'} \int \frac{d^3k}{\sqrt{(2\pi)^3} 2E} \frac{d^3k'}{\sqrt{(2\pi)^3} 2E'} \\ &\times \left[iE' \varepsilon^i(\vec{k}, \lambda) \varepsilon^{*j}(\vec{k}', \lambda') e^{-ikx} e^{ik'y} \left[a(\vec{k}, \lambda), a^\dagger(\vec{k}', \lambda') \right] \right. \\ &\quad \left. - iE' \varepsilon^{*i}(\vec{k}, \lambda) \varepsilon^j(\vec{k}', \lambda') e^{ikx} e^{-ik'y} \left[a^\dagger(\vec{k}, \lambda), a(\vec{k}', \lambda') \right] \right] \end{aligned} \quad (7.108)$$

$$\begin{aligned} &= \frac{i}{2} \sum_{\lambda} \int \frac{d^3k}{(2\pi)^3} \left[\varepsilon^i(\vec{k}, \lambda) \varepsilon^{*j}(\vec{k}, \lambda) e^{-ik(x-y)} \right. \\ &\quad \left. + \varepsilon^{*i}(\vec{k}, \lambda) \varepsilon^j(\vec{k}, \lambda) e^{ik(x-y)} \right]. \end{aligned} \quad (7.109)$$

At equal times, $x^0 = y^0$, this becomes

$$[A_T^i(x), \Pi_T^j(y)] = i\delta_T^{ij}(\vec{x} - \vec{y}) \quad (7.110)$$

$$= i \int \frac{d^3k}{(2\pi)^3} \left(\delta^{ij} - \frac{k^i k^j}{\vec{k}^2} \right) e^{i\vec{k} \cdot (\vec{x} - \vec{y})}. \quad (7.111)$$

Therefore,

$$[A_T^i(x), A_T^j(y)]_{x^0=y^0} = 0, \quad (7.112)$$

$$[\Pi_T^i(x), \Pi_T^j(y)]_{x^0=y^0} = 0, \quad (7.113)$$

$$[A_T^i(x), \Pi_T^j(y)]_{x^0=y^0} = i \int \frac{d^3k}{(2\pi)^3} \left(\delta^{ij} - \frac{k^i k^j}{\vec{k}^2} \right) e^{i\vec{k} \cdot (\vec{x} - \vec{y})}. \quad (7.114)$$

□

Problem 7.2 Obtain the four-momentum operator

$$\hat{P}^\mu \quad (7.115)$$

using Noether's theorem.

Write the resulting four-momentum operator in terms of the creation and annihilation operators of the quantized electromagnetic field.

Solution.

Consider the Lagrangian density for the free electromagnetic field

$$\mathcal{L} = \frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \quad (7.116)$$

We obtain the momentum operator P^μ in terms of creation and annihilation operators from

$$P^\mu = \int d^3x T^{0\mu}, \quad (7.117)$$

where

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\beta)} \partial^\nu A^\beta - \eta^{\mu\nu} \mathcal{L}. \quad (7.118)$$

Since

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu A^\beta)} = -F^\mu{}_\beta, \quad (7.119)$$

we have

$$T^{\mu\nu} = -F^\mu{}_\beta \partial^\nu A^\beta + \eta^{\mu\nu} \mathcal{L}. \quad (7.120)$$

Therefore,

$$P^\mu = \int d^3x T^{0\mu} \quad (7.121)$$

$$= \int d^3x (-F^0{}_\beta \partial^\mu A^\beta + \eta^{0\mu} \mathcal{L}) \quad (7.122)$$

$$= \int d^3x \left(\Pi_\beta \partial^\mu A^\beta + \eta^{0\mu} \frac{1}{2} (-\Pi_\beta \Pi^\beta + B^\beta B_\beta) \right), \quad (7.123)$$

where

$$F_{\mu\nu} F^{\mu\nu} = -F_{0i} F^{0i} + 2F_{0i} F^{0i} + F_{ij} F^{ij} = -2\Pi_i \Pi_i + \frac{\varepsilon^{ijk} \varepsilon^{ilm} B^k B^m}{2}. \quad (7.124)$$

For $\mu = 0$,

$$P^0 = \int d^3x \left[\Pi_\beta \partial^0 A^\beta + \eta^{00} \frac{1}{2} (-\Pi_\beta \Pi^\beta + B^\beta B_\beta) \right] \quad (7.125)$$

$$= \int d^3x \frac{1}{2} (\Pi_\beta \Pi^\beta + B_\beta B^\beta) \quad (7.126)$$

$$= \int d^3x \frac{1}{2} (\vec{E}^2 + \vec{B}^2). \quad (7.127)$$

For $\mu = i$,

$$P^i = \int d^3x \Pi_j \partial^i A^j \quad (7.128)$$

$$= \int d^3x (\Pi_j \partial^j A^i + A_j \partial^j \Pi^i) \quad (7.129)$$

$$= \int d^3x (\Pi_j \partial^j A^i - \partial^i A^j \Pi_j) \quad (7.130)$$

$$= - \int d^3x \Pi_j F^{ij} \quad (7.131)$$

$$= \int d^3x \Pi_j \varepsilon^{ijk} B^k \quad (7.132)$$

$$= \int d^3x (\vec{E} \times \vec{B})^i. \quad (7.133)$$

We now need to write the electric and magnetic fields in terms of creation and annihilation operators. In the Coulomb gauge, $A^0 = 0$, and hence

$$E^\delta = -\partial_0 A^\delta - \partial^\delta A^0 = -\partial_0 A^\delta. \quad (7.134)$$

Using

$$A^\delta(x) = \sum_\lambda \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} [\varepsilon^\delta(\vec{p}, \lambda) a(\vec{p}, \lambda) e^{-ipx} + \varepsilon^{*\delta}(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda) e^{ipx}], \quad (7.135)$$

we obtain

$$E^\delta(x) = -\partial_0 \sum_\lambda \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} [\varepsilon^\delta(\vec{p}, \lambda) a(\vec{p}, \lambda) e^{-ipx} + \varepsilon^{*\delta}(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda) e^{ipx}] \quad (7.136)$$

$$= i \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} E \sum_\lambda [\varepsilon^\delta(\vec{p}, \lambda) a(\vec{p}, \lambda) e^{-ipx} - \varepsilon^{*\delta}(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda) e^{ipx}]. \quad (7.137)$$

Similarly,

$$B^\delta(x) = \varepsilon^{\delta hk} \partial_h A^k \quad (7.138)$$

$$= i \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \sum_\lambda \varepsilon^{\delta hk} p^h [\varepsilon^k(\vec{p}, \lambda) a(\vec{p}, \lambda) e^{-ipx} - \varepsilon^{*k}(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda) e^{ipx}]. \quad (7.139)$$

We first compute

$$\begin{aligned}
E^\delta E^\delta &= - \int \frac{d^3 p}{\sqrt{(2\pi)^3} 2E} \frac{d^3 p'}{\sqrt{(2\pi)^3} 2E'} E E' \sum_{\lambda, \lambda'} \\
&\quad \times \left[\varepsilon^\delta(\vec{p}, \lambda) a(\vec{p}, \lambda) e^{-ipx} - \varepsilon^{*\delta}(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda) e^{ipx} \right] \\
&\quad \times \left[\varepsilon^\delta(\vec{p}', \lambda') a(\vec{p}', \lambda') e^{-ip'x} - \varepsilon^{*\delta}(\vec{p}', \lambda') a^\dagger(\vec{p}', \lambda') e^{ip'x} \right] \quad (7.140)
\end{aligned}$$

$$\begin{aligned}
&= - \int \frac{d^3 p}{\sqrt{(2\pi)^3} 2E} \frac{d^3 p'}{\sqrt{(2\pi)^3} 2E'} E E' \sum_{\lambda, \lambda'} \left[\right. \\
&\quad \varepsilon^\delta(\vec{p}, \lambda) \varepsilon^\delta(\vec{p}', \lambda') a(\vec{p}, \lambda) a(\vec{p}', \lambda') e^{-i(p+p')x} \\
&\quad - \varepsilon^\delta(\vec{p}, \lambda) \varepsilon^{*\delta}(\vec{p}', \lambda') a(\vec{p}, \lambda) a^\dagger(\vec{p}', \lambda') e^{-i(p-p')x} \\
&\quad - \varepsilon^{*\delta}(\vec{p}, \lambda) \varepsilon^\delta(\vec{p}', \lambda') a^\dagger(\vec{p}, \lambda) a(\vec{p}', \lambda') e^{i(p-p')x} \\
&\quad \left. + \varepsilon^{*\delta}(\vec{p}, \lambda) \varepsilon^{*\delta}(\vec{p}', \lambda') a^\dagger(\vec{p}, \lambda) a^\dagger(\vec{p}', \lambda') e^{i(p+p')x} \right]. \quad (7.141)
\end{aligned}$$

Using

$$\delta^3(\vec{k}) = \int \frac{d^3 x}{(2\pi)^3} e^{i\vec{k} \cdot \vec{x}}, \quad \vec{k} = \vec{p} \pm \vec{p}', \quad (7.142)$$

we find

$$\begin{aligned}
\int d^3 x E^\delta E^\delta &= - \int \frac{d^3 p d^3 p'}{2} \frac{E E'}{\sqrt{E E'}} \sum_{\lambda, \lambda'} \left[\right. \\
&\quad \delta^3(\vec{p} + \vec{p}') \varepsilon^\delta(\vec{p}, \lambda) \varepsilon^\delta(\vec{p}', \lambda') a(\vec{p}, \lambda) a(\vec{p}', \lambda') e^{-i(E+E')t} \\
&\quad + \delta^3(\vec{p} + \vec{p}') \varepsilon^{*\delta}(\vec{p}, \lambda) \varepsilon^{*\delta}(\vec{p}', \lambda') a^\dagger(\vec{p}, \lambda) a^\dagger(\vec{p}', \lambda') e^{i(E+E')t} \\
&\quad - \delta^3(\vec{p} - \vec{p}') \varepsilon^{*\delta}(\vec{p}, \lambda) \varepsilon^\delta(\vec{p}', \lambda') a^\dagger(\vec{p}, \lambda) a(\vec{p}', \lambda') e^{i(E-E')t} \\
&\quad \left. - \delta^3(\vec{p} - \vec{p}') \varepsilon^\delta(\vec{p}, \lambda) \varepsilon^{*\delta}(\vec{p}', \lambda') a(\vec{p}, \lambda) a^\dagger(\vec{p}', \lambda') e^{-i(E-E')t} \right] \quad (7.143)
\end{aligned}$$

$$\begin{aligned}
&= - \int \frac{d^3 p}{2} E \sum_{\lambda, \lambda'} \left[\right. \\
&\quad \varepsilon^\delta(\vec{p}, \lambda) \varepsilon^\delta(-\vec{p}, \lambda') a(\vec{p}, \lambda) a(-\vec{p}, \lambda') e^{-2iEt} \\
&\quad + \varepsilon^{*\delta}(\vec{p}, \lambda) \varepsilon^{*\delta}(-\vec{p}, \lambda') a^\dagger(\vec{p}, \lambda) a^\dagger(-\vec{p}, \lambda') e^{2iEt} \\
&\quad - \varepsilon^{*\delta}(\vec{p}, \lambda) \varepsilon^\delta(\vec{p}, \lambda') a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda') \\
&\quad \left. - \varepsilon^\delta(\vec{p}, \lambda) \varepsilon^{*\delta}(\vec{p}, \lambda') a(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda') \right]. \quad (7.144)
\end{aligned}$$

For the magnetic field,

$$\begin{aligned}
B^\delta B^\delta &= - \int \frac{d^3 p}{\sqrt{(2\pi)^3} 2E} \frac{d^3 p'}{\sqrt{(2\pi)^3} 2E'} \sum_{\lambda, \lambda'} \varepsilon^{\delta h k} \varepsilon^{\delta m n} p^h p'^m \\
&\quad \times \left[\varepsilon^k(\vec{p}, \lambda) a(\vec{p}, \lambda) e^{-ipx} - \varepsilon^{*k}(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda) e^{ipx} \right] \\
&\quad \times \left[\varepsilon^n(\vec{p}', \lambda') a(\vec{p}', \lambda') e^{-ip'x} - \varepsilon^{*n}(\vec{p}', \lambda') a^\dagger(\vec{p}', \lambda') e^{ip'x} \right] \quad (7.145)
\end{aligned}$$

$$\begin{aligned}
&= - \int \frac{d^3 p}{\sqrt{(2\pi)^3} 2E} \frac{d^3 p'}{\sqrt{(2\pi)^3} 2E'} \sum_{\lambda, \lambda'} p^h p'^m \left[\right. \\
&\quad \varepsilon^k(\vec{p}, \lambda) \varepsilon^n(\vec{p}', \lambda') a(\vec{p}, \lambda) a(\vec{p}', \lambda') e^{-i(p+p')x} \\
&\quad + \varepsilon^{*k}(\vec{p}, \lambda) \varepsilon^{*n}(\vec{p}', \lambda') a^\dagger(\vec{p}, \lambda) a^\dagger(\vec{p}', \lambda') e^{i(p+p')x} \\
&\quad - \varepsilon^k(\vec{p}, \lambda) \varepsilon^{*n}(\vec{p}', \lambda') a(\vec{p}, \lambda) a^\dagger(\vec{p}', \lambda') e^{-i(p-p')x} \\
&\quad - \varepsilon^{*k}(\vec{p}, \lambda) \varepsilon^n(\vec{p}', \lambda') a^\dagger(\vec{p}, \lambda) a(\vec{p}', \lambda') e^{i(p-p')x} \left. \right] \\
&\quad \times (\delta^{hm} \delta^{kn} - \delta^{hn} \delta^{km}). \quad (7.146)
\end{aligned}$$

Since

$$\varepsilon^{\delta h k} \varepsilon^{\delta m n} = \delta^{h m} \delta^{k n} - \delta^{h n} \delta^{k m}, \quad (7.147)$$

and

$$\vec{p} \cdot \vec{\varepsilon}(\vec{p}, \lambda) = 0, \quad (7.148)$$

the terms proportional to $\delta^{h n} \delta^{k m}$ vanish. Therefore,

$$\begin{aligned} \int d^3 x B^\delta B^\delta &= - \int \frac{d^3 p}{2} E \sum_{\lambda, \lambda'} \left[\right. \\ &\quad - \varepsilon^k(\vec{p}, \lambda) \varepsilon^k(-\vec{p}, \lambda') a(\vec{p}, \lambda) a(-\vec{p}, \lambda') e^{-2iEt} \\ &\quad - \varepsilon^{*k}(\vec{p}, \lambda) \varepsilon^{*k}(-\vec{p}, \lambda') a^\dagger(\vec{p}, \lambda) a^\dagger(-\vec{p}, \lambda') e^{2iEt} \\ &\quad - \varepsilon^{*k}(\vec{p}, \lambda) \varepsilon^k(\vec{p}, \lambda') a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda') \\ &\quad \left. - \varepsilon^k(\vec{p}, \lambda) \varepsilon^{*k}(\vec{p}, \lambda') a(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda') \right]. \end{aligned} \quad (7.149)$$

Using the orthonormality relation

$$\sum_k \varepsilon^{*k}(\vec{p}, \lambda) \varepsilon^k(\vec{p}, \lambda') = \delta_{\lambda \lambda'}, \quad (7.150)$$

the terms containing two creation or two annihilation operators cancel between the electric and magnetic contributions. Hence,

$$P^0 = \frac{1}{2} \int d^3 x \left(\vec{E}^2 + \vec{B}^2 \right) \quad (7.151)$$

$$\begin{aligned} &= \frac{1}{2} \int d^3 p \frac{E}{2} \sum_{\lambda} \left[2 \varepsilon^{*\delta}(\vec{p}, \lambda) \varepsilon^\delta(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda) \right. \\ &\quad \left. + 2 \varepsilon^\delta(\vec{p}, \lambda) \varepsilon^{*\delta}(\vec{p}, \lambda) a(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda) \right] \end{aligned} \quad (7.152)$$

$$= \frac{1}{2} \sum_{\lambda} \int d^3 p E \left[a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda) + a(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda) \right]. \quad (7.153)$$

After normal ordering,

$$: P^0 := \sum_{\lambda} \int d^3 p E a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda). \quad (7.154)$$

For the spatial components,

$$\int d^3 x \left(\vec{E} \times \vec{B} \right)^i = \int d^3 p \frac{p^i}{2} \sum_{\lambda} \left[a(\vec{p}, \lambda) a^\dagger(\vec{p}, \lambda) + a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda) \right]. \quad (7.155)$$

Therefore, after normal ordering,

$$: P^i := \sum_{\lambda} \int d^3 p p^i a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda). \quad (7.156)$$

Thus, the four-momentum operator of the quantized electromagnetic field is

$$\boxed{: P^\mu := \sum_{\lambda=1}^2 \int d^3 p p^\mu a^\dagger(\vec{p}, \lambda) a(\vec{p}, \lambda) :}. \quad (7.157)$$

□

Problem 7.3 Actions defined as

$$S = \int d^4x \mathcal{L} \quad (7.158)$$

have the same dimensions as $\hbar$. Therefore, in natural units $\hbar = 1$, the action is dimensionless.

Find the mass dimensions of the following fields in four spacetime dimensions:

1. A scalar field ϕ .
2. A spinor field ψ .
3. A gauge field A^μ .

Solution.

The action is defined as

$$S = \int d^4x \mathcal{L}. \quad (7.159)$$

Since the action has the same dimensions as $\hbar$, it is dimensionless in our $\hbar = 1$ units. Therefore,

$$[S] = [d^4x][\mathcal{L}] \quad (7.160)$$

$$= [\text{mass}]^{-4}[\mathcal{L}] = 1, \quad (7.161)$$

and hence the Lagrangian density has units of mass to the fourth power:

$$[\mathcal{L}] = [M]^4. \quad (7.162)$$

For the scalar Klein–Gordon field,

$$\mathcal{L}_{\text{KG}} = m^2 \phi^2, \quad (7.163)$$

$$[\mathcal{L}] = [M]^4 = [m^2 \phi^2]. \quad (7.164)$$

Therefore,

$$[\phi] = [M]. \quad (7.165)$$

For the Dirac field,

$$\mathcal{L}_{\text{Dirac}} = m \bar{\psi} \psi, \quad (7.166)$$

$$[\bar{\psi} \psi] = [M]^3. \quad (7.167)$$

Thus,

$$[\psi] = [\bar{\psi}] = [M]^{3/2}. \quad (7.168)$$

For a gauge field,

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad (7.169)$$

$$[F^{\mu\nu}] = [M]^2. \quad (7.170)$$

Since

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu, \quad (7.171)$$

it follows that

$$[A^\mu] = [M]. \quad (7.172)$$

□

8 Interactions

In quantum electrodynamics, we need to solve Maxwell's equations using a Dirac current as a source, this is extremely difficult to do, so we will use perturbation theory.

For quantum electrodynamics the coupling constant that measures the scale of the interactions is

$$\alpha = \frac{1}{137}. \quad (8.1)$$

We will separate the lagrangian into the free lagrangian and the interaction lagrangian

$$\mathcal{L} = \sum_i \mathcal{L}_i + \mathcal{L}_{int} = \mathcal{L}_0 + \mathcal{L}_I \quad (8.2)$$

$\mathcal{L}_{int}$ has to be

1. hermitian
2. Lorentz invariant
3. local
4. do not contain time derivatives(if possible)

If there are no time derivatives, then the conjugate momenta are not affected by the interaction

$$\rightarrow \mathcal{H} = \sum_i \mathcal{H}_i + \mathcal{H}_{int} = \mathcal{H}_0 + \mathcal{H}_I \quad (8.3)$$

where $\mathcal{H}_I = -\mathcal{L}_I$.

In practice $\mathcal{L}_{int}$ are polynomials of the fields, with one parameter defining the strength of the interaction, the coupling constant.

8.1 Interaction Representation

States and operators are not observables.

$\Rightarrow$ observables are expectation values of operators applied to states.

There are different pictures for describing the same state.

For $\phi(x, t)$, the field operator carries the time dependence in the Heisenberg representation.

In the Heisenberg picture, the state vector $|\psi\rangle$ does not carry time dependence, whereas the operators carry the time dependence.

In the Schrödinger picture, the operators are time independent, whereas the state vector carries the time dependence.

$$|\psi(t)\rangle^{(S)} = e^{-iH_H t} |\psi\rangle^{(H)}, \quad (8.4)$$

$$O^{(S)} = e^{-iH_H t} O^{(H)}(t) e^{iH_H t}, \quad (8.5)$$

where H_H is the Hamiltonian in the Heisenberg representation.

The expectation value is unchanged:

$${}^{(H)}\langle\psi| O^{(H)}(t) |\psi\rangle^{(H)} = {}^{(S)}\langle\psi(t)| O^{(S)} |\psi(t)\rangle^{(S)}. \quad (8.6)$$

For interactions, we introduce the interaction representation, in which the time dependence is carried by both states and operators.

Defining

$$|\psi(t)\rangle^{(I)} = e^{iH_0^{(S)}t} |\psi(t)\rangle^{(S)} \quad (8.7)$$

$$= e^{iH_0^{(S)}t} e^{-iHt} |\psi\rangle^{(H)}, \quad (8.8)$$

$$O^{(I)}(t) = e^{iH_0^{(S)}t} O^{(S)} e^{-iH_0^{(S)}t} \quad (8.9)$$

$$= e^{iH_0^{(S)}t} e^{-iHt} O^{(H)}(t) e^{iHt} e^{-iH_0^{(S)}t}. \quad (8.10)$$

We can identify

$$|\psi(0)\rangle^{(I)} = |\psi(0)\rangle^{(S)} = |\psi(0)\rangle^{(H)}, \quad (8.11)$$

$$O^{(I)}(0) = O^{(S)} = O^{(H)}(0), \quad (8.12)$$

$$H_0^{(I)}(t) = e^{iH_0^{(S)}t} H_0^{(S)} e^{-iH_0^{(S)}t} = H_0^{(S)}. \quad (8.13)$$

What is the time evolution of $|\psi(t)\rangle^{(I)}$ and $O^{(I)}(t)$?

The states evolve as

$$i \frac{\partial}{\partial t} |\psi(t)\rangle^{(I)} = i \frac{\partial}{\partial t} \left(e^{iH_0^{(S)}t} e^{-iHt} |\psi\rangle^{(H)} \right) \quad (8.14)$$

$$= -H_0^{(S)} e^{iH_0^{(S)}t} e^{-iHt} |\psi\rangle^{(H)} + e^{iH_0^{(S)}t} H e^{-iHt} |\psi\rangle^{(H)} \quad (8.15)$$

$$= -H_0^{(I)} |\psi(t)\rangle^{(I)} + e^{iH_0^{(S)}t} H e^{-iH_0^{(S)}t} |\psi(t)\rangle^{(I)} \quad (8.16)$$

$$= -H_0^{(I)} |\psi(t)\rangle^{(I)} + H^{(I)} |\psi(t)\rangle^{(I)} \quad (8.17)$$

$$= H_I^{(I)}(t) |\psi(t)\rangle^{(I)}. \quad (8.18)$$

Similarly, for operators,

$$\frac{\partial O^{(I)}}{\partial t} = \frac{\partial}{\partial t} \left(e^{iH_0^{(S)}t} O^{(S)} e^{-iH_0^{(S)}t} \right) \quad (8.19)$$

$$= \frac{1}{i} [O^{(I)}(t), H_0^{(I)}]. \quad (8.20)$$

States evolve with the interaction Hamiltonian, while operators evolve with the free Hamiltonian.

In summary:

- Schrödinger representation:

$$i \frac{\partial}{\partial t} |\psi\rangle_S = H |\psi\rangle_S, \quad H = H_0 + H_I, \quad (8.21)$$

$$i \frac{\partial O_S}{\partial t} = 0. \quad (8.22)$$

- Heisenberg representation:

$$i \frac{\partial}{\partial t} |\psi\rangle_H = 0, \quad (8.23)$$

$$i \frac{\partial O_H}{\partial t} = [O_H, H], \quad H = H_0 + H_I. \quad (8.24)$$

- Interaction representation:

$$i \frac{\partial}{\partial t} |\psi\rangle_I = H_I |\psi\rangle_I, \quad (8.25)$$

$$i \frac{\partial O_I}{\partial t} = [O_I, H_0]. \quad (8.26)$$

The commutation relations for operators are unchanged in the presence of interactions.

We define the time-evolution operator as

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle, \quad (8.27)$$

which transforms a state at t_0 into its evolved state at t .

The evolution operator is

$$U(t, t_0) = e^{iH_0^{(S)}t} e^{-iH(t-t_0)} e^{-iH_0^{(S)}t_0}. \quad (8.28)$$

It satisfies

$$i \frac{\partial}{\partial t} U(t, t_0) = H_I(t) U(t, t_0), \quad (8.29)$$

together with

$$U(t, t) = \mathbb{I}, \quad (8.30)$$

$$U(t_2, t_1) U(t_1, t_0) = U(t_2, t_0), \quad (8.31)$$

$$U^\dagger(t, t_0) = U^{-1}(t, t_0) = U(t_0, t). \quad (8.32)$$

The time-evolution operator can be expressed as

$$U(t, t_0) = T \left[\exp \left(-i \int_{t_0}^t dt' H_I(t') \right) \right], \quad (8.33)$$

where T denotes the time-ordered product.

We understand this operator through its expansion.

8.2 S-Matrix

Assume that we have incoming particles at $t = -\infty$ and outgoing particles at $t = +\infty$, which are free-particle states.

$$|\psi_i(-\infty)\rangle = |i\rangle. \quad (8.34)$$

The final state is

$$|\psi_i(+\infty)\rangle = U(+\infty, -\infty) |\psi_i(-\infty)\rangle \quad (8.35)$$

$$= U(+\infty, -\infty) |i\rangle \quad (8.36)$$

$$= S |i\rangle, \quad (8.37)$$

where we have defined

$$S \equiv U(+\infty, -\infty). \quad (8.38)$$

The probability amplitude for an initial state $|i\rangle$ to be in a final state $|f\rangle$ at $t = +\infty$ is

$$S_{fi} = \langle f | \psi_i(+\infty) \rangle \quad (8.39)$$

$$= \langle f | U(+\infty, -\infty) | i \rangle \quad (8.40)$$

$$= \langle f | S | i \rangle. \quad (8.41)$$

We identify

$$S = \lim_{\substack{t \rightarrow +\infty \\ t_0 \rightarrow -\infty}} U(t, t_0) = T \left[\exp \left(-i \int_{-\infty}^{+\infty} dt H_I(t) \right) \right]. \quad (8.42)$$

If the coupling constant is sufficiently small,

$$\begin{aligned} S &\simeq 1 - i \int_{-\infty}^{+\infty} dt T(H_I(t)) \\ &\quad + \frac{(-i)^2}{2!} \int_{-\infty}^{+\infty} dt_1 \int_{-\infty}^{+\infty} dt_2 T(H_I(t_1) H_I(t_2)) + \dots \end{aligned} \quad (8.43)$$

This is perturbation theory.

In our case,

$$H_I = -\mathcal{L}_{\text{int}} \quad (8.44)$$

at first order. Therefore,

$$S \simeq 1 + i \int d^4x \mathcal{L}_{\text{int}}. \quad (8.45)$$

Hence,

$$\langle f | S | i \rangle = \langle f | i \rangle + i \langle f | \int d^4x \mathcal{L}_{\text{int}} | i \rangle. \quad (8.46)$$

Let us compute the following example:

$$\pi^0 \longrightarrow e^+ e^-. \quad (8.47)$$

Consider the interaction Lagrangian

$$\mathcal{L}_{\text{int}} = ig\phi \bar{\psi} \gamma^5 \psi, \quad (8.48)$$

where ϕ is the π^0 field and the bilinear $\bar{\psi} \gamma^5 \psi$ changes parity for the π^0 .

We want to compute the transition amplitude for the process

$$\pi^0 \longrightarrow e^+ e^-. \quad (8.49)$$

The four-momentum of the π^0 is P , while the four-momenta and spins of the electron and positron are

$$(p_1, s_1) \quad \text{and} \quad (p_2, s_2). \quad (8.50)$$

Our initial state is

$$|i\rangle = |P\rangle = a^\dagger(\vec{P}) |0\rangle, \quad (8.51)$$

and our final state is

$$|f\rangle = |p_1, s_1; p_2, s_2\rangle = c^\dagger(\vec{p}_1, s_1) d^\dagger(\vec{p}_2, s_2) |0\rangle. \quad (8.52)$$

Therefore,

$$\langle f | = \langle 0 | c(\vec{p}_1, s_1) d(\vec{p}_2, s_2). \quad (8.53)$$

At first order in perturbation theory,

$$S \simeq 1 + i \int d^4x \mathcal{L}_{\text{int}} \quad (8.54)$$

$$= 1 + i \int d^4x (ig\phi \bar{\psi} \gamma^5 \psi). \quad (8.55)$$

The transition amplitude is

$$S_{fi} = \langle f | \left[- \int d^4x g\phi \bar{\psi} \gamma^5 \psi \right] | i \rangle. \quad (8.56)$$

After a lengthy computation, we find

$$\begin{aligned} S_{fi} = & -g \int d^3k \int d^3p \int d^3p' d^4x \sum_{s,s'} \frac{1}{\sqrt{(2\pi)^3 2E_k}} \frac{1}{\sqrt{(2\pi)^3 2E}} \frac{1}{\sqrt{(2\pi)^3 2E'}} \\ & \times e^{-i(k-p-p')x} [\bar{u}(p, s) \gamma^5 v(p', s')] \langle 0 | \dots | 0 \rangle. \end{aligned} \quad (8.57)$$

The only piece that depends on the interaction is

$$\mathcal{M} = -g \bar{u}(p, s) \gamma^5 v(p', s'), \quad (8.58)$$

which is the invariant matrix element.

Integrating over x ,

$$\begin{aligned} S_{fi} = & -g \int d^3k d^3p d^3p' \frac{1}{\sqrt{(2\pi)^3 2E_k}} \frac{1}{\sqrt{(2\pi)^3 2E}} \frac{1}{\sqrt{(2\pi)^3 2E'}} \\ & \times (2\pi)^4 \delta^4(k - p - p') \bar{u}(p, s) \gamma^5 v(p', s') \delta_{ss_1} \delta_{s's_2} \\ & \times \delta^3(\vec{p} - \vec{p}_1) \delta^3(\vec{p}' - \vec{p}_2) \delta^3(\vec{P} - \vec{k}). \end{aligned} \quad (8.59)$$

Thus,

$$S_{fi} = \frac{(2\pi)^4 \delta^4(P - p_1 - p_2)}{\sqrt{(2\pi)^3 2E_P} \sqrt{(2\pi)^3 2E_1} \sqrt{(2\pi)^3 2E_2}} \mathcal{M}. \quad (8.60)$$

The probability for the transition is

$$\text{Prob}_{i \rightarrow f} = |\langle f | S | i \rangle|^2. \quad (8.61)$$

What we can measure is the decay rate,

$$d\Gamma = \frac{\text{Prob}_{i \rightarrow f}}{T} = \frac{|\langle f | S | i \rangle|^2}{T}. \quad (8.62)$$

The differential decay rate is

$$d\Gamma_{fi} = \frac{(2\pi)^4}{2E_P} d\Phi_2 \overline{|\mathcal{M}|^2}, \quad (8.63)$$

where E_P is the energy of the incoming particle and

$$d\Phi_2 = \delta^4(P - p_1 - p_2) \frac{d^3 p_1}{(2\pi)^3 2E_1} \frac{d^3 p_2}{(2\pi)^3 2E_2} \quad (8.64)$$

is the differential phase-space term.

Let us compute $|\mathcal{M}|^2$. We have

$$\mathcal{M} = -g \bar{u}(p_1, s_1) \gamma^5 v(p_2, s_2). \quad (8.65)$$

Moreover,

$$|\mathcal{M}|^2 = \mathcal{M} \mathcal{M}^* = \mathcal{M} \mathcal{M}^\dagger, \quad (8.66)$$

$$\mathcal{M}^* = (-g \bar{u}(p_1, s_1) \gamma^5 v(p_2, s_2))^* \quad (8.67)$$

$$= -g \left[v^\dagger(p_2, s_2) (\gamma^5)^\dagger (u^\dagger(p_1, s_1) \gamma^0)^\dagger \right] \quad (8.68)$$

$$= -g v^\dagger(p_2, s_2) \gamma^5 \gamma^0 u(p_1, s_1) \quad (8.69)$$

$$= -g v^\dagger(p_2, s_2) \gamma^0 \gamma^0 \gamma^5 \gamma^0 u(p_1, s_1) \quad (8.70)$$

$$= g \bar{v}(p_2, s_2) \gamma^5 u(p_1, s_1). \quad (8.71)$$

We need to account for spin:

$$|\mathcal{M}|^2 \longrightarrow \sum_{s_1, s_2} |\mathcal{M}|^2. \quad (8.72)$$

Therefore,

$$\overline{|\mathcal{M}|^2} = -g^2 \sum_{s_1, s_2} [\bar{u}(p_1, s_1) \gamma^5 v(p_2, s_2)] [\bar{v}(p_2, s_2) \gamma^5 u(p_1, s_1)]. \quad (8.73)$$

Using the completeness relations

$$\sum_{s_2} v_\beta(p_2, s_2) \bar{v}_\gamma(p_2, s_2) = (\not{p}_2 - m)_{\beta\gamma}, \quad (8.74)$$

$$\sum_{s_1} u_\delta(p_1, s_1) \bar{u}_\alpha(p_1, s_1) = (\not{p}_1 + m)_{\delta\alpha}, \quad (8.75)$$

we obtain

$$\overline{|\mathcal{M}|^2} = -g^2 (\not{p}_1 + m)_{\delta\alpha} (\gamma^5)_{\alpha\beta} (\not{p}_2 - m)_{\beta\gamma} (\gamma^5)_{\gamma\delta} \quad (8.76)$$

$$= -g^2 \text{Tr} \left[(\not{p}_1 + m) \gamma^5 (\not{p}_2 - m) \gamma^5 \right]. \quad (8.77)$$

For

$$\pi^0 \longrightarrow e^+ e^-, \quad (8.78)$$

we have

$$m_{\pi^0} = 135 \text{ MeV}, \quad m_e = 0.5 \text{ MeV}, \quad (8.79)$$

so we can neglect the masses of the electron and positron. Then

$$\sum_{s_1, s_2} |\mathcal{M}|^2 \simeq -g^2 \text{Tr} \left(\not{p}_1 \gamma^5 \not{p}_2 \gamma^5 \right). \quad (8.80)$$

Using

$$-\text{Tr}(\not{p}_1 \gamma^5 \not{p}_2 \gamma^5) = \text{Tr}(\not{p}_1 \gamma^5 \gamma^5 \not{p}_2) \quad (8.81)$$

$$= \text{Tr}(\not{p}_1 \not{p}_2) \quad (8.82)$$

$$= 4p_1 \cdot p_2, \quad (8.83)$$

we find

$$\sum_{s_1, s_2} |\mathcal{M}|^2 = 4g^2 p_1 \cdot p_2. \quad (8.84)$$

The scalar product $p_1 \cdot p_2$ can be easily computed in the rest frame of the decaying particle:

$$P = (M, \vec{0}), \quad (8.85)$$

$$p_1 = (p, \vec{p}), \quad (8.86)$$

$$p_2 = (p, -\vec{p}), \quad (8.87)$$

with

$$p = \frac{M}{2}. \quad (8.88)$$

Therefore,

$$\sum_{s_1, s_2} |\mathcal{M}|^2 = 4g^2 p_1 \cdot p_2 \quad (8.89)$$

$$= 4g^2 (2p^2) \quad (8.90)$$

$$= 2g^2 M^2. \quad (8.91)$$

The differential decay rate is

$$d\Gamma = \frac{(2\pi)^4}{2M} d\Phi_2 2g^2 M^2. \quad (8.92)$$

Using

$$\int d\Phi_2 = \frac{1}{(2\pi)^5} \frac{|\vec{p}|}{2M}, \quad (8.93)$$

where $|\vec{p}|$ is the momentum of the electron or positron in the centre-of-mass frame of the π^0 , the total decay rate is

$$\Gamma = \int d\Gamma \quad (8.94)$$

$$= \frac{(2\pi)^4}{2M} 2g^2 M^2 \int d\Phi_2 \quad (8.95)$$

$$= \frac{(2\pi)^4}{2M} \frac{|\vec{p}|}{(2\pi)^5 2M} 2g^2 M^2 \quad (8.96)$$

$$= \frac{g^2 M}{8\pi}. \quad (8.97)$$

We can extrapolate this result to a particle decaying into n particles:

$$S_{fi} = \frac{(2\pi)^4 \delta^4(p_a - \sum_{i=1}^n p_i)}{\sqrt{(2\pi)^3 2E_a} \sqrt{\prod_{i=1}^n (2\pi)^3 2E_i}} \mathcal{M}, \quad (8.98)$$

and

$$d\Gamma = \frac{(2\pi)^4}{2E_a} d\Phi_n |\mathcal{M}|^2, \quad (8.99)$$

where

$$d\Phi_n = \delta^4\left(p_a - \sum_{i=1}^n p_i\right) \prod_{i=1}^n \frac{d^3 p_i}{(2\pi)^3 2E_i}. \quad (8.100)$$

Let us check the units of our computation:

$$\Gamma(\pi^0 \rightarrow e^+ e^-) = \frac{g^2}{8\pi} M. \quad (8.101)$$

Since

$$\Gamma = \frac{1}{\tau}, \quad (8.102)$$

we have

$$[\Gamma] = \left[\frac{1}{\text{time}} \right]. \quad (8.103)$$

In natural units,

$$\text{time} = \text{space} = \frac{1}{\text{energy}} = \frac{1}{\text{mass}}, \quad (8.104)$$

and therefore

$$[\Gamma] = \left[\frac{1}{1/\text{mass}} \right] = [\text{mass}]. \quad (8.105)$$

From this we can infer that g is dimensionless.

How do we know the units of the coupling constant? Since

$$S = \int d^4x \mathcal{L} \quad (8.106)$$

is dimensionless, d^4x has dimensions $[\text{space}]^4$, and the Lagrangian density has dimensions

$$[\mathcal{L}] = [\text{space}]^{-4} = [E]^4. \quad (8.107)$$

Looking at the mass terms of the Lagrangian,

$$\mathcal{L}_{\text{KG}} = m^2 \phi^2, \quad [\mathcal{L}] = [E]^4 = [m^2 \phi^2], \quad (8.108)$$

$$[\phi^2] = [E]^2, \quad [\phi] = [E], \quad (8.109)$$

while for the Dirac field,

$$\mathcal{L}_{\text{Dirac}} = m \bar{\psi} \psi, \quad [\bar{\psi} \psi] = [E]^3, \quad (8.110)$$

$$[\psi] = [E]^{3/2}, \quad [\bar{\psi}] = [E]^{3/2}. \quad (8.111)$$

For the interaction term,

$$\mathcal{L}_{\text{int}} = ig \phi \bar{\psi} \gamma^5 \psi, \quad (8.112)$$

and therefore

$$[\mathcal{L}_{\text{int}}] = [E]^4, \quad (8.113)$$

so g is dimensionless.

We have obtained

$$\Gamma(\pi^0 \rightarrow e^+ e^-) = \frac{g^2}{8\pi} M. \quad (8.114)$$

We define the branching ratio as

$$\text{Br}(a \rightarrow x) = \frac{\Gamma_{a \rightarrow x}}{\Gamma_{a \rightarrow \text{all}}} = \frac{\Gamma_{a \rightarrow x}}{\Gamma_{\text{total}}}. \quad (8.115)$$

Experimentally,

$$\Gamma_{\text{total}}^{-1}(\pi^0) = \tau_{\pi^0} = 8 \times 10^{-17} \text{ s}, \quad (8.116)$$

and

$$\text{Br}(\pi^0 \rightarrow e^+ e^-) = 6 \times 10^{-8}. \quad (8.117)$$

Since

$$\Gamma(\pi^0 \rightarrow e^+ e^-) = \frac{g^2}{8\pi} M = \frac{1}{\tau_{\text{total}}} \text{Br}(\pi^0 \rightarrow e^+ e^-), \quad (8.118)$$

we obtain

$$g = 3 \times 10^{-7}. \quad (8.119)$$

It was justified to use perturbation theory.

8.3 Feynman Rules for Yukawa Coupling

$$\mathcal{L}_{\text{int}} = ig\bar{\psi}\gamma^5\psi, \quad (8.120)$$

which is the Yukawa interaction for

$$\pi^0 \longrightarrow e^+e^-. \quad (8.121)$$

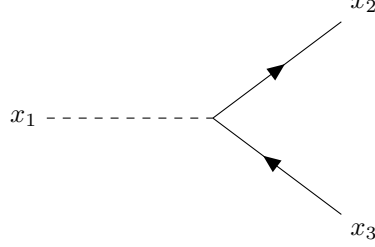


Figure 2: Yukawa interaction vertex, where x_1 is a scalar and x_2, x_3 are fermions.

Here,

$$x_1 = \text{scalar}, \quad x_2, x_3 = \text{fermions}. \quad (8.122)$$

In order to obtain S_{fi} :

- Include a factor $(-ig)$ for each vertex.
- Impose momentum conservation through a delta function.
- Each incoming or outgoing scalar contributes a factor 1.
- Include a factor $\bar{u}(p, s)$ for an outgoing fermion and $v(p, s)$ for an outgoing antifermion.
- Include a global factor $(2\pi)^4$.
- For each external boson or fermion, include the normalization factor

$$\frac{1}{\sqrt{(2\pi)^3 2E}}. \quad (8.123)$$

Therefore,

$$S_{fi} = (2\pi)^4 \frac{1}{\sqrt{(2\pi)^3 2E_P}} \frac{1}{\sqrt{(2\pi)^3 2E_{p_1}}} \frac{1}{\sqrt{(2\pi)^3 2E_{p_2}}} \\ \times (-ig) \bar{u}(p_1, s_1) \gamma^5 v(p_2, s_2) \delta^4(P - p_1 - p_2). \quad (8.124)$$

In order to obtain $d\Gamma$:

- Define $\mathcal{M}$ using the same rules for the vertex and compute $|\mathcal{M}|^2$.
- Include a global factor $(2\pi)^4$ and the momentum-conservation delta function.
- For every fermion in the final state, include a factor

$$\int \frac{d^3p}{(2\pi)^3 2E}. \quad (8.125)$$

- For the boson in the initial state, include a factor

$$\frac{1}{2E}. \quad (8.126)$$

Thus,

$$d\Gamma = (2\pi)^4 \frac{1}{2E_P} \int \frac{d^3p_1}{(2\pi)^3 2E_1} \int \frac{d^3p_2}{(2\pi)^3 2E_2} \\ \times [-ig\bar{u}(p_1, s_1) \gamma^5 v(p_2, s_2)] [-ig\bar{u}(p_1, s_1) \gamma^5 v(p_2, s_2)]^\dagger \\ \times \delta^4(P - p_1 - p_2). \quad (8.127)$$

8.4 Problems

Problem 8.1 The interaction picture is defined by

$$|\psi(t)\rangle_I \equiv e^{i\hat{H}_0 t} |\psi(t)\rangle_S, \quad (8.128)$$

and

$$\hat{O}_I(t) \equiv e^{i\hat{H}_0 t} \hat{O}_S e^{-i\hat{H}_0 t}, \quad (8.129)$$

where the subscript S denotes the Schrödinger picture. The full Hamiltonian is

$$\hat{H} = \hat{H}_0 + \hat{H}_1, \quad (8.130)$$

where $\hat{H}_0$ is the free Hamiltonian. Show that

1.

$$i \frac{d\hat{O}_I}{dt} = [\hat{O}_I, \hat{H}_0], \quad (8.131)$$

2.

$$i \frac{d}{dt} |\psi(t)\rangle_I = \hat{H}_I |\psi(t)\rangle_I, \quad (8.132)$$

where $\hat{H}_I$ is $\hat{H}_1$ in the interaction picture.

Solution. For the operator,

$$i \frac{d\hat{O}_I}{dt} = i \frac{d}{dt} \left(e^{i\hat{H}_0 t} \hat{O}_S e^{-i\hat{H}_0 t} \right) \quad (8.133)$$

$$= -\hat{H}_0 e^{i\hat{H}_0 t} \hat{O}_S e^{-i\hat{H}_0 t} + e^{i\hat{H}_0 t} \hat{O}_S \hat{H}_0 e^{-i\hat{H}_0 t} \quad (8.134)$$

$$= -\hat{H}_0 \hat{O}_I + \hat{O}_I \hat{H}_0 \quad (8.135)$$

$$= [\hat{O}_I, \hat{H}_0]. \quad (8.136)$$

For the state,

$$i \frac{d}{dt} |\psi(t)\rangle_I = i \frac{d}{dt} \left(e^{i\hat{H}_0 t} |\psi(t)\rangle_S \right) \quad (8.137)$$

$$= -\hat{H}_0 e^{i\hat{H}_0 t} |\psi(t)\rangle_S + e^{i\hat{H}_0 t} i \frac{d}{dt} |\psi(t)\rangle_S \quad (8.138)$$

$$= -\hat{H}_0 e^{i\hat{H}_0 t} |\psi(t)\rangle_S + e^{i\hat{H}_0 t} \hat{H} |\psi(t)\rangle_S \quad (8.139)$$

$$= \left(-\hat{H}_0 + \hat{H}_0 + \hat{H}_I \right) e^{i\hat{H}_0 t} |\psi(t)\rangle_S \quad (8.140)$$

$$= \hat{H}_I |\psi(t)\rangle_I. \quad (8.141)$$

□

Problem 8.2 The scattering of a low-energy photon by a charged particle of spin s at rest, known as Thomson scattering, can be described by a non-relativistic Schrödinger field coupled to the electromagnetic field.

1. Obtain the interaction Lagrangian density $\mathcal{L}_I$.
2. At low energies, the photon energy satisfies $E_\gamma \ll m$, where m is the mass of the charged particle. Show that the only relevant term in the calculation of the scattering amplitude at leading order in the expansion of $\mathcal{L}_I$ is

$$\mathcal{L}_I \simeq -\frac{q^2}{2m} \vec{A} \cdot \vec{A} \psi^\dagger \psi. \quad (8.142)$$

3. Calculate the scattering amplitude in the laboratory frame using the Coulomb gauge.

Solution. The Lagrangian density for a non-relativistic field with minimal coupling may be written as

$$\mathcal{L} = \psi^\dagger (i\partial_t - qA_0)\psi - \frac{1}{2m} \left[(-i\vec{\nabla} - q\vec{A})\psi \right]^\dagger \left[(-i\vec{\nabla} - q\vec{A})\psi \right]. \quad (8.143)$$

Therefore the interaction part is

$$\mathcal{L}_I = -qA_0\psi^\dagger\psi - \frac{iq}{2m}\vec{A} \cdot \left(\psi^\dagger\vec{\nabla}\psi - (\vec{\nabla}\psi^\dagger)\psi \right) - \frac{q^2}{2m}\vec{A}^2\psi^\dagger\psi. \quad (8.144)$$

If spin is included one may add

$$\mathcal{L}_{\text{spin}} \simeq \frac{gq}{2m}\psi^\dagger \vec{S} \cdot \vec{B} \psi, \quad (8.145)$$

but this is not required at leading order in the low-energy limit.

The process

$$\gamma + \psi \longrightarrow \gamma + \psi \quad (8.146)$$

receives a contribution of order q^2 in two different ways. At leading order in $\mathcal{L}_I$, the seagull term contributes,

$$-\frac{q^2}{2m}\vec{A}^2\psi^\dagger\psi. \quad (8.147)$$

At the next order there are two insertions of

$$-\frac{iq}{2m}\vec{A} \cdot \left(\psi^\dagger\vec{\nabla}\psi - (\vec{\nabla}\psi^\dagger)\psi \right). \quad (8.148)$$

These vertices carry momenta of order

$$|\vec{k}| = \omega = E_\gamma \ll m, \quad (8.149)$$

so their contribution is suppressed. Hence, at leading order,

$$\mathcal{L}_I \simeq -\frac{q^2}{2m}\vec{A}^2\psi^\dagger\psi. \quad (8.150)$$

In the Coulomb gauge,

$$A_0 = 0, \quad \vec{\nabla} \cdot \vec{A} = 0, \quad (8.151)$$

and

$$\vec{A}(x) = \sum_{\lambda} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E}} \left[\vec{\epsilon}_{\lambda}(\vec{k}) a_{\lambda}(\vec{k}) e^{-ikx} + \vec{\epsilon}_{\lambda}^*(\vec{k}) a_{\lambda}^{\dagger}(\vec{k}) e^{ikx} \right]. \quad (8.152)$$

The part of $\vec{A}^2$ that annihilates the incoming photon and creates the outgoing photon gives

$$\mathcal{M}_{\text{Th}} = -\frac{q^2}{2m} \vec{\epsilon}_{\lambda'}^*(\vec{k}') \cdot \vec{\epsilon}_{\lambda}(\vec{k}). \quad (8.153)$$

□

Problem 8.3 Show that, for a two-body decay

$$A \longrightarrow 1 + 2, \quad (8.154)$$

the decay width is

$$\Gamma_{A \rightarrow 12} = \frac{|\mathcal{M}|^2}{8\pi} \frac{p}{m_A^2}, \quad (8.155)$$

where

$$p = |\vec{p}_1| = |-\vec{p}_2|. \quad (8.156)$$

If $m_1 = m_2 = m$, show that

$$\Gamma_{A \rightarrow 12} = \frac{|\mathcal{M}|^2}{16\pi m_A} \sqrt{1 - \frac{4m^2}{m_A^2}}. \quad (8.157)$$

Solution. For the decay $A \rightarrow 1 + 2$ in the rest frame of the parent particle,

$$p_A = (m_A, \vec{0}), \quad p_1 = (E_1, \vec{p}), \quad p_2 = (E_2, -\vec{p}). \quad (8.158)$$

The differential decay rate is

$$d\Gamma = \frac{1}{2m_A} |\mathcal{M}|^2 d\Phi_2, \quad (8.159)$$

with

$$d\Phi_2 = (2\pi)^4 \delta^{(4)}(p_A - p_1 - p_2) \frac{d^3 p_1}{(2\pi)^3 2E_1} \frac{d^3 p_2}{(2\pi)^3 2E_2}. \quad (8.160)$$

The three-momentum delta function fixes

$$\vec{p}_1 = -\vec{p}_2. \quad (8.161)$$

The remaining delta function is

$$\delta(m_A - E_1 - E_2). \quad (8.162)$$

Using

$$\frac{d}{dp}(E_1 + E_2) = p \left(\frac{1}{E_1} + \frac{1}{E_2} \right) = p \frac{E_1 + E_2}{E_1 E_2} = p \frac{m_A}{E_1 E_2}, \quad (8.163)$$

the two-body phase space becomes

$$d\Phi_2 = \frac{p}{16\pi^2 m_A} d\Omega. \quad (8.164)$$

Hence

$$d\Gamma = \frac{|\mathcal{M}|^2 p}{32\pi^2 m_A^2} d\Omega. \quad (8.165)$$

Integrating over the solid angle,

$$\Gamma_{A \rightarrow 12} = \frac{|\mathcal{M}|^2}{8\pi} \frac{p}{m_A^2}. \quad (8.166)$$

In general,

$$p = \frac{\sqrt{[m_A^2 - (m_1 + m_2)^2][m_A^2 - (m_1 - m_2)^2]}}{2m_A}. \quad (8.167)$$

For $m_1 = m_2 = m$,

$$p = \frac{m_A}{2} \sqrt{1 - \frac{4m^2}{m_A^2}}, \quad (8.168)$$

and therefore

$$\Gamma_{A \rightarrow 12} = \frac{|\mathcal{M}|^2}{16\pi m_A} \sqrt{1 - \frac{4m^2}{m_A^2}}. \quad (8.169)$$

□

Problem 8.4 Show that the differential cross section for a process

$$A + B \longrightarrow 1 + 2 \quad (8.170)$$

in the centre-of-mass frame is

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CoM}} = \frac{|\mathcal{M}|^2}{64\pi^2 s} \frac{p_f}{p_i}, \quad (8.171)$$

where

$$p_f = |\vec{p}_1| = |-\vec{p}_2|, \quad p_i = |\vec{p}_A| = |-\vec{p}_B|, \quad (8.172)$$

and

$$s = (p_A + p_B)^2 = (p_1 + p_2)^2. \quad (8.173)$$

Solution. For the process $A + B \rightarrow 1 + 2$, the invariant cross section is

$$d\sigma = \frac{1}{4\sqrt{(p_A \cdot p_B)^2 - m_A^2 m_B^2}} |\mathcal{M}|^2 d\Phi_2, \quad (8.174)$$

where

$$d\Phi_2 = (2\pi)^4 \delta^{(4)}(p_A + p_B - p_1 - p_2) \frac{d^3 p_1}{(2\pi)^3 2E_1} \frac{d^3 p_2}{(2\pi)^3 2E_2}. \quad (8.175)$$

In the centre-of-mass frame,

$$p_A = (E_A, \vec{p}_i), \quad p_B = (E_B, -\vec{p}_i), \quad (8.176)$$

and

$$p_1 = (E_1, \vec{p}_f), \quad p_2 = (E_2, -\vec{p}_f). \quad (8.177)$$

Moreover,

$$s = (p_A + p_B)^2 = (E_A + E_B)^2. \quad (8.178)$$

The flux factor becomes

$$4\sqrt{(p_A \cdot p_B)^2 - m_A^2 m_B^2} = 4p_i \sqrt{s}. \quad (8.179)$$

Integrating the two-body phase space over the momentum magnitude gives

$$d\Phi_2 = \frac{p_f}{16\pi^2 \sqrt{s}} d\Omega. \quad (8.180)$$

Therefore

$$d\sigma = \frac{|\mathcal{M}|^2}{4p_i \sqrt{s}} \frac{p_f}{16\pi^2 \sqrt{s}} d\Omega, \quad (8.181)$$

and hence

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CoM}} = \frac{|\mathcal{M}|^2}{64\pi^2 s} \frac{p_f}{p_i}. \quad (8.182)$$

□

Problem 8.5 The interactions between charged and neutral pions at low energies are described by

$$\mathcal{L}_I = -\frac{1}{3f_\pi^2} [\phi_{\pi^0} \phi_{\pi^0} \partial_\mu \phi_{\pi^+} \partial^\mu \phi_{\pi^+}^* + \phi_{\pi^+} \phi_{\pi^+}^* \partial_\mu \phi_{\pi^0} \partial^\mu \phi_{\pi^0}]. \quad (8.183)$$

Calculate the amplitude, differential cross section, and total cross section for $\pi^+ \pi^- \rightarrow \pi^0 \pi^0$, taking equal pion masses.

Solution. We study

$$\pi^+(p_1) + \pi^-(p_2) \longrightarrow \pi^0(k_1) + \pi^0(k_2), \quad (8.184)$$

with

$$p_1^2 = p_2^2 = k_1^2 = k_2^2 = m_\pi^2. \quad (8.185)$$

$$S_{fi} \simeq i \int d^4 x \langle \pi^0(k_1) \pi^0(k_2) | \mathcal{L}_I | \pi^+(p_1) \pi^-(p_2) \rangle \quad (8.186)$$

$$= i(2\pi)^4 \delta^{(4)}(p_1 + p_2 - k_1 - k_2) \mathcal{M}. \quad (8.187)$$

For the first term,

$$\mathcal{L}_I \simeq -\frac{1}{3f_\pi^2} \phi_{\pi^0} \phi_{\pi^0} \partial_\mu \phi_{\pi^+} \partial^\mu \phi_{\pi^+}^*, \quad (8.188)$$

we have

$$\partial_\mu \phi_{\pi^+}(x) | \pi^+(p_1) \rangle \simeq -ip_{1\mu} e^{-ip_1 x}, \quad (8.189)$$

$$\partial^\mu \phi_{\pi^+}^*(x) | \pi^-(p_2) \rangle \simeq -ip_2^\mu e^{-ip_2 x}, \quad (8.190)$$

and

$$\langle \pi^0(k) | \phi_{\pi^0}(x) \simeq e^{ikx}. \quad (8.191)$$

Therefore

$$(-ip_{1\mu})(-ip_2^\mu) = -p_1 \cdot p_2. \quad (8.192)$$

There are two ways of contracting the two identical neutral-pion fields with the final state, so the first term gives

$$\frac{2}{3f_\pi^2} p_1 \cdot p_2. \quad (8.193)$$

By the same argument, the second term gives

$$\frac{2}{3f_\pi^2} k_1 \cdot k_2. \quad (8.194)$$

Thus

$$\mathcal{M} = \frac{2}{3f_\pi^2} (p_1 \cdot p_2 + k_1 \cdot k_2). \quad (8.195)$$

Using

$$s = (p_1 + p_2)^2 = 2m_\pi^2 + 2p_1 \cdot p_2 \quad (8.196)$$

and

$$s = (k_1 + k_2)^2 = 2m_\pi^2 + 2k_1 \cdot k_2, \quad (8.197)$$

we obtain

$$\mathcal{M} = \frac{2}{3f_\pi^2} (s - 2m_\pi^2). \quad (8.198)$$

For the differential cross section,

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CoM}} = \frac{|\mathcal{M}|^2}{64\pi^2 s} \frac{p_f}{p_i}. \quad (8.199)$$

Since all pion masses are equal,

$$\frac{p_f}{p_i} = 1. \quad (8.200)$$

Because the final state contains two identical particles, a factor $1/2!$ must be included:

$$\frac{d\sigma}{d\Omega} = \frac{|\mathcal{M}|^2}{128\pi^2 s} = \frac{(s - 2m_\pi^2)^2}{288\pi^2 f_\pi^4 s}. \quad (8.201)$$

Since

$$d\Omega = d\varphi d(\cos \theta) = 2\pi \sin \theta d\theta, \quad (8.202)$$

we obtain

$$\frac{d\sigma}{d\theta} = \frac{(s - 2m_\pi^2)^2}{144\pi f_\pi^4 s} \sin \theta. \quad (8.203)$$

Integrating from $\theta = 0$ to $\theta = \pi$,

$$\sigma = \frac{(s - 2m_\pi^2)^2}{72\pi f_\pi^4 s}. \quad (8.204)$$

□

Problem 8.6 The charged-pion decay into leptons is described by

$$\mathcal{L}_I = -G_F f_\pi \sum_{\ell=e,\mu} [\bar{\psi}_{\nu_\ell} \gamma^\mu (1 - \gamma^5) \psi_\ell \partial_\mu \phi_{\pi^+} + \text{h.c.}]. \quad (8.205)$$

Calculate the amplitude, decay width, and the relative fractions of π^+ decays into electron–neutrino and muon–neutrino pairs.

Solution. The term describing

$$\pi^+ \longrightarrow e^+ + \nu_e \quad (8.206)$$

is

$$-G_F f_\pi \bar{\psi}_{\nu_e} \gamma^\mu (1 - \gamma^5) \psi_e \partial_\mu \phi_{\pi^+}. \quad (8.207)$$

For each vertex,

$$-ig\Gamma, \quad \Gamma = \gamma^\mu(1 - \gamma^5), \quad (8.208)$$

the derivative acting on the pion field gives

$$-ip_\mu, \quad (8.209)$$

and the outgoing fermion and antifermion give respectively

$$\bar{u}(k, s), \quad v(q, s'). \quad (8.210)$$

Thus

$$\mathcal{M} = -iG_F f_\pi \bar{u}(k, s) \gamma^\mu (1 - \gamma^5) v(q, s') p_\mu. \quad (8.211)$$

Using momentum conservation,

$$p_\mu = k_\mu + q_\mu, \quad (8.212)$$

where p , k and q are respectively the pion, neutrino and lepton momenta,

$$\mathcal{M} = iG_F f_\pi \bar{u}(k, s) \not{p} (1 - \gamma^5) v(q, s') \quad (8.213)$$

$$= iG_F f_\pi [\bar{u}(k, s) \not{k} (1 - \gamma^5) v(q, s') + \bar{u}(k, s) (1 + \gamma^5) \not{q} v(q, s')]. \quad (8.214)$$

Using

$$\bar{u}(k, s) (\not{k} - m_\nu) = 0 \quad \longrightarrow \quad \bar{u}(k, s) \not{k} = 0 \quad (8.215)$$

and

$$(-\not{q} - m_e) v(q, s') = 0 \quad \longrightarrow \quad \not{q} v(q, s') = -m_e v(q, s'), \quad (8.216)$$

neglecting the neutrino mass,

$$\mathcal{M} = -iG_F f_\pi m_e \bar{u}(k, s) (1 + \gamma^5) v(q, s'). \quad (8.217)$$

The two-body decay rate is

$$\Gamma(\pi^+ \rightarrow e^+ \nu_e) = \frac{|\vec{k}|_{\text{CoM}}}{8\pi m_{\pi^+}^2} |\mathcal{M}|^2. \quad (8.218)$$

Summing over spins,

$$|\mathcal{M}|^2 = G_F^2 f_\pi^2 m_e^2 \text{Tr}[(\not{q} - m_e)(1 - \gamma^5) \not{k} (1 + \gamma^5)] \quad (8.219)$$

$$= G_F^2 f_\pi^2 m_e^2 \text{Tr}(2\not{q} \not{k}) \quad (8.220)$$

$$= 8G_F^2 f_\pi^2 m_e^2 E_\nu (E_e + E_\nu). \quad (8.221)$$

Here

$$q \cdot k = E_e E_\nu + E_\nu^2 \quad (8.222)$$

in the pion rest frame. □

Problem 8.7 The interaction of the Higgs boson h with leptons is described by

$$\mathcal{L}_I = \sum_{\ell=e,\mu,\tau} \lambda_\ell \phi_h \bar{\psi}_\ell \psi_\ell. \quad (8.223)$$

Calculate the matrix element, the decay width, and the relative fractions for $h \rightarrow e^+ e^-$, $h \rightarrow \mu^+ \mu^-$ and $h \rightarrow \tau^+ \tau^-$.

Solution. We start by writing the field expansions

$$\phi_h(x) = \int \frac{d^3 k}{\sqrt{(2\pi)^3} 2E} \left(e^{-ikx} a(\vec{k}) + e^{ikx} a^\dagger(\vec{k}) \right), \quad (8.224)$$

and

$$\psi_e(y) = \int \frac{d^3 p_e}{\sqrt{(2\pi)^3} 2E_p} \sum_s \left(e^{-ip_e y} u(p_e, s) c(p_e, s) + e^{ip_e y} v(p_e, s) d^\dagger(p_e, s) \right). \quad (8.225)$$

The initial and final states are

$$|i\rangle = a^\dagger(\vec{k})|0\rangle, \quad (8.226)$$

and

$$|f\rangle = |p_+, s_+; p_-, s_-\rangle = c^\dagger(\vec{p}_+, s_+)d^\dagger(\vec{p}_-, s_-)|0\rangle. \quad (8.227)$$

We can now calculate

$$S_{fi} = \langle f|S|i\rangle \quad (8.228)$$

$$= \left\langle p_+, s_+; p_-, s_- \left| \int d^4x \mathcal{L}_I \right| h \right\rangle \quad (8.229)$$

$$= \lambda_e \langle 0|c(p_+, s_+)d(p_-, s_-)\bar{\psi}\psi\phi a^\dagger(h')|0\rangle. \quad (8.230)$$

Substituting the field expansions,

$$\begin{aligned} S_{fi} &= \lambda_e \langle 0|c(p_+, s_+)d(p_-, s_-) \int \frac{d^3k}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p}{\sqrt{(2\pi)^3 2E}} \int \frac{d^3p'}{\sqrt{(2\pi)^3 2E'}} \\ &\quad \times \sum_{s, s'} e^{-i(k-p-p')x} \bar{u}(p, s)v(p', s')a(h)c^\dagger(p, s)d^\dagger(p', s')a^\dagger(h')|0\rangle. \end{aligned} \quad (8.231)$$

Using the commutation relations,

$$\langle 0|c(p_+, s_+)d(p_-, s_-)a(h)c^\dagger(p, s)d^\dagger(p', s')a^\dagger(h')|0\rangle \quad (8.232)$$

$$= \delta_{ss_+}\delta_{s's_-}\delta^3(p_+ - p)\delta^3(p_- - p')\delta^3(h - h'). \quad (8.233)$$

Therefore,

$$S_{fi} = \frac{(2\pi)^4 \delta^4(k - p - p')}{\sqrt{(2\pi)^3 2E_k} \sqrt{(2\pi)^3 2E} \sqrt{(2\pi)^3 2E'}} \lambda_e \bar{u}(p, s)v(p', s'). \quad (8.234)$$

Defining

$$\mathcal{M} = \lambda_e \bar{u}(p, s)v(p', s'), \quad (8.235)$$

the decay width is

$$\Gamma(h \rightarrow e^+e^-) = \frac{|\vec{p}|_{\text{CoM}}}{8\pi m_h^2} |\mathcal{M}|^2. \quad (8.236)$$

Summing over spins,

$$\sum_{s, s'} |\mathcal{M}|^2 = \lambda_e^2 \sum_{s, s'} \bar{u}(p, s)v(p', s')\bar{v}(p', s')u(p, s) \quad (8.237)$$

$$= \lambda_e^2 \text{Tr}[(\not{p}' - m_e)(\not{p} + m_e)] \quad (8.238)$$

$$= \lambda_e^2 (4p \cdot p' - 4m_e^2) \quad (8.239)$$

$$= \lambda_e^2 (2m_h^2 - 8m_e^2). \quad (8.240)$$

The momentum in the centre-of-mass frame is

$$|\vec{p}| = \frac{\sqrt{m_h^4 + m_e^4 + m_e^4 - 2m_h^2 m_e^2 - 2m_h^2 m_e^2 - 2m_e^4}}{2m_h} \quad (8.241)$$

$$= \frac{1}{2}m_h \sqrt{1 - \frac{4m_e^2}{m_h^2}}. \quad (8.242)$$

Hence,

$$\Gamma(h \rightarrow \ell^+\ell^-) = \sum_{\ell=e, \mu, \tau} \lambda_\ell^2 \frac{1}{8\pi m_h^2} (m_h^2 - 4m_\ell^2)^{3/2}. \quad (8.243)$$

Putting numbers,

$$\frac{\Gamma(h \rightarrow e^+e^-)}{\Gamma(h \rightarrow \ell^+\ell^-)} = 8.25 \times 10^{-8}, \quad (8.244)$$

$$\frac{\Gamma(h \rightarrow \mu^+\mu^-)}{\Gamma(h \rightarrow \ell^+\ell^-)} = 3.5 \times 10^{-3}, \quad (8.245)$$

$$\frac{\Gamma(h \rightarrow \tau^+\tau^-)}{\Gamma(h \rightarrow \ell^+\ell^-)} = 0.9965. \quad (8.246)$$

□

Problem 8.8 The decay of the top quark into a bottom quark is described by

$$\mathcal{L}_I = -\frac{g}{2\sqrt{2}} V_{tb} \bar{\psi}_b \gamma^\mu (1 - \gamma^5) \psi_t W_\mu^\dagger + \text{h.c.} \quad (8.247)$$

Calculate $\mathcal{M}(t \rightarrow bW^+)$, neglecting colour, and calculate the total top-quark decay width. Neglect m_b .

Solution. For a massive vector boson,

$$\sum_{\lambda=1}^3 \epsilon^{*\mu}(p, \lambda) \epsilon^\nu(p, \lambda) = -\eta^{\mu\nu} + \frac{p^\mu p^\nu}{m_W^2}. \quad (8.248)$$

The interaction Lagrangian is

$$\mathcal{L}_I = -\frac{g}{2\sqrt{2}} V_{tb} [\bar{\psi}_b \gamma^\mu (1 - \gamma^5) \psi_t W_\mu^\dagger + W_\mu \bar{\psi}_t (1 - \gamma^5)^\dagger \gamma^{\mu\dagger} \gamma^0 \psi_b] \quad (8.249)$$

$$= -\frac{g}{2\sqrt{2}} V_{tb} [\bar{\psi}_b \gamma^\mu (1 - \gamma^5) \psi_t W_\mu^\dagger + W_\mu \bar{\psi}_t (1 + \gamma^5) \gamma^\mu \psi_b]. \quad (8.250)$$

The fields are expanded as

$$\psi_Q(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3} 2E_p} \sum_s [e^{-ipx} u(p, s) c(p, s) + e^{ipx} v(p, s) d^\dagger(p, s)], \quad (8.251)$$

$$\bar{\psi}_Q(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3} 2E_p} \sum_s [e^{ipx} \bar{u}(p, s) c^\dagger(p, s) + e^{-ipx} \bar{v}(p, s) d(p, s)], \quad (8.252)$$

and

$$W^\mu(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3} 2E_p} \sum_\lambda [e^{-ipx} \epsilon^\mu(p, \lambda) a(p, \lambda) + e^{ipx} \epsilon^{*\mu}(p, \lambda) a^\dagger(p, \lambda)]. \quad (8.253)$$

The initial and final states are

$$|i\rangle = |p_t, s_t\rangle = c^\dagger(p_t, s_t) |0\rangle, \quad (8.254)$$

and

$$|f\rangle = |p_b, s_b; p_W, \lambda_W\rangle = c^\dagger(p_b, s_b) a^\dagger(p_W, \lambda_W) |0\rangle. \quad (8.255)$$

The S -matrix element is

$$S_{fi} = i(2\pi)^4 \delta^{(4)}(p_t - p_b - p_W) \mathcal{M} \quad (8.256)$$

apart from the normalization factors, with

$$\mathcal{M} = -\frac{g}{2\sqrt{2}} V_{tb} \bar{u}(p_b, s_b) \gamma^\mu (1 - \gamma^5) u(p_t, s_t) \epsilon_\mu^*(p_W, \lambda_W). \quad (8.257)$$

The decay width is

$$\Gamma = \overline{|\mathcal{M}|^2} \frac{|\vec{p}|_{\text{CoM}}}{8\pi m_t^2}. \quad (8.258)$$

Averaging over the initial spin,

$$\overline{|\mathcal{M}|^2} = \frac{1}{2} \sum_{s_b, s_t, \lambda_W} |\mathcal{M}|^2. \quad (8.259)$$

Thus

$$\overline{|\mathcal{M}|^2} = \frac{g^2}{16} |V_{tb}|^2 \text{Tr} \left[\not{p}_b (1 + \gamma^5) \gamma^\nu (\not{p}_t + m_t) \gamma^\mu (1 - \gamma^5) \right] \left(-\eta_{\mu\nu} + \frac{p_{W\mu} p_{W\nu}}{m_W^2} \right) \quad (8.260)$$

$$= \frac{g^2}{16} |V_{tb}|^2 \left[2 \cdot 4 (\eta^{\mu\alpha} \eta^{\nu\beta} - \eta^{\mu\nu} \eta^{\alpha\beta} + \eta^{\mu\beta} \eta^{\alpha\nu} + i \epsilon^{\mu\alpha\nu\beta}) p_{t\alpha} p_{b\beta} \right] \left(-\eta_{\mu\nu} + \frac{p_{W\mu} p_{W\nu}}{m_W^2} \right) \quad (8.261)$$

$$= \frac{g^2}{2} |V_{tb}|^2 \left[\frac{2}{m_W^2} (p_W \cdot p_t)(p_W \cdot p_b) + p_t \cdot p_b \right]. \quad (8.262)$$

In the centre-of-mass frame, neglecting m_b ,

$$|\vec{p}|_{\text{CoM}} = \frac{m_t^2 - m_W^2}{2m_t} = 68 \text{ GeV}. \quad (8.263)$$

Using the numerical values,

$$\Gamma(t \rightarrow bW) = 1.5 \text{ GeV}. \quad (8.264)$$

The branching fraction is written as

$$\frac{\Gamma(t \rightarrow bW)}{\Gamma_{\text{tot}}} = 99.7\%. \quad (8.265)$$

Therefore, in the photograph,

$$\Gamma_{\text{tot}} = 1.42_{-0.15}^{+0.19} \text{ GeV}. \quad (8.266)$$

□

Problem 8.9 The decay $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$ is described by

$$\mathcal{L} = -\frac{G_F}{\sqrt{2}} (J_\mu^\dagger(\mu) J^\mu(e) + \text{h.c.}). \quad (8.267)$$

Determine the amplitude and evaluate the spin-averaged squared amplitude, neglecting the electron and neutrino masses.

Solution. The interaction Lagrangian can be written as

$$\begin{aligned} \mathcal{L}_{\text{int}} = & -\frac{G_F}{\sqrt{2}} \left[(\bar{\psi}_{\nu_\mu} (1 + \gamma^5) \gamma_\mu \psi_\mu) (\bar{\psi}_e \gamma^\mu (1 - \gamma^5) \psi_{\nu_e}) \right. \\ & \left. + (\bar{\psi}_{\nu_e} (1 + \gamma^5) \gamma_\mu \psi_e) (\bar{\psi}_\mu \gamma^\mu (1 - \gamma^5) \psi_{\nu_\mu}) \right]. \end{aligned} \quad (8.268)$$

Only the first term contributes. The S -matrix element is

$$S_{fi} = -i \frac{G_F}{\sqrt{2}} (2\pi)^4 \delta^{(4)}(p_\mu - p_{\nu_\mu} - p_e - p_{\nu_e}) \prod_{\lambda=1}^4 \frac{1}{\sqrt{(2\pi)^3 2E_\lambda}} \mathcal{M}, \quad (8.269)$$

where

$$\mathcal{M} = [\bar{u}(p_{\nu_\mu}, s_{\nu_\mu}) (1 + \gamma^5) \gamma_\alpha u(p_\mu, s_\mu)] [\bar{u}(p_e, s_e) \gamma^\alpha (1 - \gamma^5) v(p_{\nu_e}, s_{\nu_e})]. \quad (8.270)$$

The complex conjugate is

$$\mathcal{M}^* = [\bar{v}(p_{\nu_e}, s_{\nu_e}) (1 + \gamma^5) \gamma_\beta u(p_e, s_e)] [\bar{u}(p_\mu, s_\mu) \gamma^\beta (1 - \gamma^5) u(p_{\nu_\mu}, s_{\nu_\mu})]. \quad (8.271)$$

Averaging over the initial muon spin and summing over the final spins,

$$\begin{aligned} |\overline{\mathcal{M}_{fi}}|^2 = & \frac{G_F^2}{4} \sum_{\text{spins}} [\bar{u}_{\nu_\mu} (1 + \gamma^5) \gamma_\alpha u_\mu] [\bar{u}_e \gamma^\alpha (1 - \gamma^5) v_{\nu_e}] \\ & \times [\bar{v}_{\nu_e} (1 + \gamma^5) \gamma_\beta u_e] [\bar{u}_\mu \gamma^\beta (1 - \gamma^5) u_{\nu_\mu}]. \end{aligned} \quad (8.272)$$

Using the completeness relations, this becomes a product of traces. The result written in the photograph is

$$|\overline{\mathcal{M}_{fi}}|^2 = 32 G_F^2 [(p_{\nu_\mu} \cdot p_e)(p_\mu \cdot p_{\nu_e}) + (p_{\nu_\mu} \cdot p_{\nu_e})(p_\mu \cdot p_e)]. \quad (8.273)$$

□

Problem 8.10 The decay $W^+ \rightarrow e^+ \nu_e$ is described by

$$\mathcal{L}_{\text{int}} = \frac{ig}{2\sqrt{2}} [\bar{\psi}_\nu \gamma^\mu (1 - \gamma^5) \psi_e W_\mu + \text{h.c.}]. \quad (8.274)$$

Determine the amplitude, evaluate the spin-averaged squared amplitude, and calculate the decay width, neglecting the electron and neutrino masses.

Solution. For a massive vector boson,

$$\sum_{\lambda=1}^3 \epsilon^{*\mu}(p, \lambda) \epsilon^\nu(p, \lambda) = -\eta^{\mu\nu} + \frac{p^\mu p^\nu}{m_W^2}. \quad (8.275)$$

The hermitian-conjugate term is

$$W_\mu^+ \bar{\psi}_e (1 + \gamma^5) \gamma^\mu \psi_\nu. \quad (8.276)$$

The initial and final states are

$$|i\rangle = a^\dagger(h, \lambda_W) |0\rangle, \quad (8.277)$$

and

$$|f\rangle = c^\dagger(p_\nu, s_\nu) d^\dagger(p_e, s_e) |0\rangle. \quad (8.278)$$

The S -matrix element is

$$S_{fi} = i(2\pi)^4 \delta^{(4)}(h - p_e - p_\nu) \prod_{\lambda=1}^3 \frac{1}{\sqrt{(2\pi)^3 2E_\lambda}} \mathcal{M}, \quad (8.279)$$

with

$$\mathcal{M} = \frac{ig}{2} \bar{u}(p_\nu, s_\nu) \gamma^\mu (1 - \gamma^5) v(p_e, s_e) \epsilon_\mu(h, \lambda_W). \quad (8.280)$$

The squared amplitude, averaged over the three initial polarizations and summed over final spins, is

$$\begin{aligned} \overline{|\mathcal{M}|^2} &= \frac{g^2}{12} \sum_{s_\nu, s_e, \lambda_W} [\bar{u}(p_\nu, s_\nu) \gamma^\mu (1 - \gamma^5) v(p_e, s_e) \epsilon_\mu] \\ &\quad \times [\epsilon_\nu^* \bar{v}(p_e, s_e) (1 + \gamma^5) \gamma^\nu u(p_\nu, s_\nu)]. \end{aligned} \quad (8.281)$$

Using the polarization and spin sums,

$$\overline{|\mathcal{M}|^2} = \frac{g^2}{3} \left[p_\nu \cdot p_e + \frac{2(h \cdot p_\nu)(h \cdot p_e)}{m_W^2} \right]. \quad (8.282)$$

For massless final particles in the W rest frame,

$$|\vec{p}|_{\text{CoM}} = \frac{m_W}{2}, \quad (8.283)$$

and

$$p_\nu \cdot p_e = \frac{m_W^2}{2}, \quad h \cdot p_e = \frac{m_W^2}{2}, \quad h \cdot p_\nu = \frac{m_W^2}{2}. \quad (8.284)$$

Thus

$$\overline{|\mathcal{M}|^2} = \frac{g^2}{3} m_W^2. \quad (8.285)$$

Using the two-body formula,

$$\Gamma(W^+ \rightarrow e^+ \nu_e) = \frac{|\vec{p}|_{\text{CoM}}}{8\pi m_W^2} \overline{|\mathcal{M}|^2} = \frac{g^2}{48\pi} m_W. \quad (8.286)$$

□

Problem 8.11 Consider the Lagrangian

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi + \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2 \phi^2 + \frac{1}{4}\lambda \phi^4 - ig\bar{\psi}\gamma^5\psi\phi. \quad (8.287)$$

Find the equations of motion for all fields.

Solution. For $\bar{\psi}$,

$$\frac{\partial \mathcal{L}}{\partial \bar{\psi}} = 0, \quad (8.288)$$

and therefore

$$(i\gamma^\mu \partial_\mu - m)\psi - ig\gamma^5\psi\phi = 0. \quad (8.289)$$

For ψ ,

$$\frac{\partial \mathcal{L}}{\partial \psi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \psi)} \right) = 0. \quad (8.290)$$

We have

$$\frac{\partial \mathcal{L}}{\partial \psi} = -m\bar{\psi} - ig\bar{\psi}\gamma^5\phi, \quad (8.291)$$

and

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} = i\bar{\psi}\gamma^\mu. \quad (8.292)$$

Hence

$$i(\partial_\mu \bar{\psi})\gamma^\mu + m\bar{\psi} + ig\bar{\psi}\gamma^5\phi = 0. \quad (8.293)$$

For ϕ ,

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = 0. \quad (8.294)$$

We have

$$\frac{\partial \mathcal{L}}{\partial \phi} = -m^2\phi + \lambda\phi^3 - ig\bar{\psi}\gamma^5\psi, \quad (8.295)$$

and

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi. \quad (8.296)$$

Therefore

$$(\square + m^2)\phi - \lambda\phi^3 + ig\bar{\psi}\gamma^5\psi = 0. \quad (8.297)$$

□

9 Quantum Electrodynamics

Consider first the scattering process

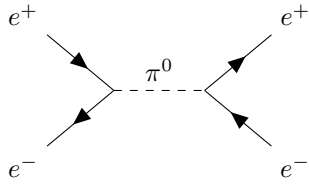
$$e^+e^- \longrightarrow e^+e^-. \quad (9.1)$$

If the interaction is mediated by a neutral pion, the effective coupling is very small,

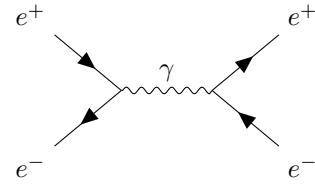
$$g \sim 10^{-7}, \quad (9.2)$$

whereas in QED the interaction is mediated by a photon and the relevant coupling is

$$\alpha = \frac{1}{137}. \quad (9.3)$$



(a) Exchange of a π^0



(b) Exchange of a photon

Figure 3: Comparison between a Yukawa-type interaction mediated by a π^0 and the QED interaction mediated by a photon.

We start by writing the QED Lagrangian:

$$\begin{aligned} \mathcal{L}_{\text{QED}} &= \mathcal{L}_0 + \mathcal{L}_{\text{int}} \\ &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi - e\bar{\psi}\gamma^\mu\psi A_\mu. \end{aligned} \quad (9.4)$$

Equivalently,

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - m\bar{\psi}\psi + i\bar{\psi}\gamma^\mu(\partial_\mu + ieA_\mu)\psi, \quad (9.5)$$

where

$$D_\mu = \partial_\mu + ieA_\mu \quad (9.6)$$

is the covariant derivative, corresponding to minimal coupling.

A physical observable in scattering theory is the cross section

$$\sigma(1+2 \rightarrow 3+4). \quad (9.7)$$

We define the differential cross section as

$$d\sigma = \frac{\text{rate}}{\text{incident flux}} = \frac{1}{4E_1E_2|\vec{v}_1 - \vec{v}_2|}(2\pi)^4\delta^4\left(\sum_i p_i - \sum_f p_f\right)|\mathcal{M}|^2 d\Phi_2. \quad (9.8)$$

Using the Mandelstam variable

$$s = (p_1 + p_2)^2, \quad (9.9)$$

this becomes

$$d\sigma = \frac{1}{4|\vec{p}_1|_{\text{cm}}\sqrt{s}}|\mathcal{M}|^2 d\Phi_2. \quad (9.10)$$

The two-body phase-space element is

$$d\Phi_2 = \frac{|\vec{p}_f|}{16\pi^2\sqrt{s}} d\Omega, \quad (9.11)$$

where $d\Omega$ is the solid angle. Therefore, in the center-of-mass frame,

$$\left.\frac{d\sigma}{d\Omega}\right|_{\text{cm}} = \frac{|\mathcal{M}|^2}{64\pi^2s} \frac{|\vec{p}_f|_{\text{cm}}}{|\vec{p}_1|_{\text{cm}}}. \quad (9.12)$$

The interaction part of the QED Lagrangian is

$$\mathcal{L}_{\text{int}}^{\text{QED}} = -e\bar{\psi}\gamma^\mu\psi A_\mu. \quad (9.13)$$

As an example, consider the process

$$e^+e^- \longrightarrow \mu^+\mu^-, \quad (9.14)$$

namely

$$e^-(p_1, s_1) e^+(p_2, s_2) \longrightarrow \mu^-(q_1, s'_1) \mu^+(q_2, s'_2). \quad (9.15)$$

The initial state is

$$|i\rangle = c^\dagger(p_1, s_1) d^\dagger(p_2, s_2) |0\rangle, \quad (9.16)$$

and the final state is

$$|f\rangle = c^\dagger(q_1, s'_1) d^\dagger(q_2, s'_2) |0\rangle. \quad (9.17)$$

The transition amplitude is

$$\langle f|S|i\rangle = \left\langle f \left| 1 + \sum_{n=1}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{+\infty} dt_1 \cdots dt_n T(H_I(t_1) \cdots H_I(t_n)) \right| i \right\rangle. \quad (9.18)$$

Expanding the first terms,

$$\langle f|S|i\rangle = \left\langle f \left| 1 - i \int_{-\infty}^{+\infty} dt H_I(t) + \cdots \right| i \right\rangle. \quad (9.19)$$

Since

$$H_I(t) = \int d^3x (-\mathcal{L}_{\text{int}}) = e \int d^3x \bar{\psi}\gamma^\mu\psi A_\mu, \quad (9.20)$$

the first-order term does not contribute to the process $e^+e^- \rightarrow \mu^+\mu^-$, so we must go to second order in perturbation theory:

$$S_{fi} \simeq \left\langle f \left| -\frac{1}{2} \int_{-\infty}^{+\infty} dx^0 \int_{-\infty}^{+\infty} dy^0 T(h(x^0)h(y^0)) \right| i \right\rangle. \quad (9.21)$$

Hence,

$$S_{fi} = \left\langle f \left| -\frac{e^2}{2} \int_{-\infty}^{+\infty} dx^0 \int_{-\infty}^{+\infty} dy^0 T \left\{ \left(\int d^3x \bar{\psi}_e \gamma^\mu \psi_e A_\mu \right) (x) \left(\int d^3y \bar{\psi}_\mu \gamma^\nu \psi_\mu A_\nu \right) (y) \right\} \right| i \right\rangle. \quad (9.22)$$

Through Wick's theorem, this gives schematically

$$(\bar{u}\gamma^\mu v)_e \langle 0|T(A_\mu(x)A_\nu(y))|0\rangle (\bar{u}\gamma^\nu v)_\mu, \quad (9.23)$$

where the middle factor is the photon propagator.

All this is difficult to compute directly, but it admits a diagrammatic representation in perturbation theory.

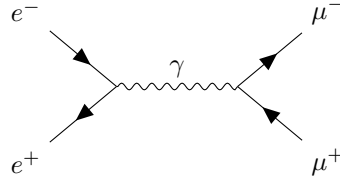


Figure 4: Tree-level Feynman diagram for the process $e^+e^- \rightarrow \mu^+\mu^-$. The internal photon is virtual.

This diagram should not be interpreted as a literal classical trajectory of particles in space-time. Rather, it is a compact graphical representation of a term in the perturbative expansion of the S -matrix.

The photon in the internal line cannot be directly measured; it is a *virtual particle*.

As long as the interaction vertices are respected, we may also consider higher-order contributions in perturbation theory, for example:

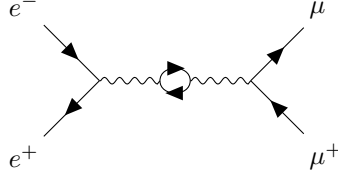


Figure 5: Example of a higher-order correction to the tree-level process.

9.1 Feynman Rules for QED

Draw all possible diagrams connecting the initial and final states with the allowed vertex

$$\gamma \bar{f} f. \quad (9.24)$$

To construct $\mathcal{M}$:

- For an outgoing fermion:

$$\bar{u}(p). \quad (9.25)$$

- For an outgoing antifermion:

$$v(p). \quad (9.26)$$

- For an incoming fermion:

$$u(p). \quad (9.27)$$

- For an incoming antifermion:

$$\bar{v}(p). \quad (9.28)$$

- For an outgoing photon:

$$\varepsilon^{\mu*}(p). \quad (9.29)$$

- For an incoming photon:

$$\varepsilon^\mu(p). \quad (9.30)$$

- For each vertex:

$$-ie\gamma^\mu. \quad (9.31)$$

- For an internal fermion line:

$$S_F(p) = \frac{i}{\not{p} - m + i\varepsilon} = \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon}. \quad (9.32)$$

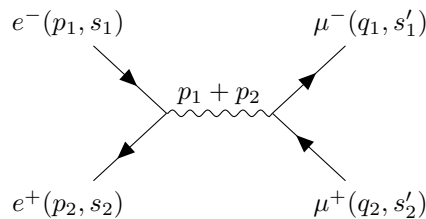
- For an internal photon line:

$$D_{\mu\nu}(k) = \frac{-i\eta_{\mu\nu}}{k^2 + i\varepsilon}. \quad (9.33)$$

For the process

$$e^+ e^- \longrightarrow \mu^+ \mu^-, \quad (9.34)$$

we have the diagram



The invariant matrix element is

$$\mathcal{M} = (-ie) \bar{v}(p_2) \gamma_\mu u(p_1) \frac{-i\eta^{\mu\nu}}{(p_1 + p_2)^2 + i\varepsilon} \bar{u}(q_1) \gamma_\nu v(q_2) (-ie) \quad (9.35)$$

$$= ie^2 \bar{v}(p_2) \gamma_\mu u(p_1) \frac{\eta^{\mu\nu}}{(p_1 + p_2)^2} \bar{u}(q_1) \gamma_\nu v(q_2). \quad (9.36)$$

We can now compute $|\overline{\mathcal{M}}|^2$:

$$|\overline{\mathcal{M}}|^2 = \frac{1}{2s_1 + 1} \frac{1}{2s_2 + 1} \sum_{\text{initial spins}} \sum_{\text{final spins}} |\mathcal{M}|^2 = \frac{1}{4} \sum_i \sum_f |\mathcal{M}|^2. \quad (9.37)$$

We write

$$\mathcal{M} = ie^2 \bar{v}_k(p_2, s_2) (\gamma_\mu)_{k\ell} u_\ell(p_1, s_1) \frac{1}{(p_1 + p_2)^2} \bar{u}_i(q_1, s'_1) (\gamma^\mu)_{ij} v_j(q_2, s'_2), \quad (9.38)$$

and

$$\mathcal{M}^* = -ie^2 \bar{v}_j(q_2, s'_2) (\gamma^\nu)_{ji} u_i(q_1, s'_1) \frac{1}{(p_1 + p_2)^2} \bar{u}_\ell(p_1, s_1) (\gamma_\nu)_{\ell k} v_k(p_2, s_2). \quad (9.39)$$

Therefore,

$$|\overline{\mathcal{M}}|^2 = \frac{1}{4} \sum_i \sum_f |\mathcal{M}|^2 \quad (9.40)$$

$$\begin{aligned} &= \frac{1}{4} \sum_i \sum_f \frac{e^4}{(p_1 + p_2)^4} \left(\bar{v}_k(p_2, s_2) (\gamma_\mu)_{k\ell} u_\ell(p_1, s_1) \right) \\ &\quad \times \left(\bar{u}_i(q_1, s'_1) (\gamma^\mu)_{ij} v_j(q_2, s'_2) \right) \left(\bar{v}_{j'}(q_2, s'_2) (\gamma^\nu)_{j'i'} u_{i'}(q_1, s'_1) \right) \\ &\quad \times \left(\bar{u}_{\ell'}(p_1, s_1) (\gamma_\nu)_{\ell'k'} v_{k'}(p_2, s_2) \right). \end{aligned} \quad (9.41)$$

Using the spinor completeness relations, this becomes

$$|\overline{\mathcal{M}}|^2 = \frac{e^4}{4(p_1 + p_2)^4} \text{Tr} \left[(\not{p}_1 + m_e) \gamma_\nu (\not{p}_2 - m_e) \gamma_\mu \right] \text{Tr} \left[(\not{q}_1 + m_\mu) \gamma^\mu (\not{q}_2 - m_\mu) \gamma^\nu \right]. \quad (9.42)$$

The first trace is

$$\text{Tr} \left[(\not{p}_1 + m_e) \gamma_\nu (\not{p}_2 - m_e) \gamma_\mu \right] = 4 \left[p_{1\nu} p_{2\mu} - (p_1 \cdot p_2) \eta_{\mu\nu} + p_{1\mu} p_{2\nu} - m_e^2 \eta_{\mu\nu} \right]. \quad (9.43)$$

The second trace is

$$\text{Tr} \left[(\not{q}_1 + m_\mu) \gamma^\mu (\not{q}_2 - m_\mu) \gamma^\nu \right] = 4 \left[q_1^\mu q_2^\nu - (q_1 \cdot q_2) \eta^{\mu\nu} + q_1^\nu q_2^\mu - m_\mu^2 \eta^{\mu\nu} \right]. \quad (9.44)$$

Multiplying the two traces,

$$\begin{aligned} \text{Tr}_1 \text{Tr}_2 &= 32 \left[(p_1 \cdot q_2)(p_2 \cdot q_1) + (p_1 \cdot q_1)(p_2 \cdot q_2) + m_e^2 (q_1 \cdot q_2) \right. \\ &\quad \left. + m_\mu^2 (p_1 \cdot p_2) + 2m_e^2 m_\mu^2 \right]. \end{aligned} \quad (9.45)$$

Hence,

$$\begin{aligned} |\overline{\mathcal{M}}|^2 &= \frac{8e^4}{(p_1 + p_2)^4} \left[(p_1 \cdot q_2)(p_2 \cdot q_1) + (p_1 \cdot q_1)(p_2 \cdot q_2) + m_e^2 (q_1 \cdot q_2) \right. \\ &\quad \left. + m_\mu^2 (p_1 \cdot p_2) + 2m_e^2 m_\mu^2 \right]. \end{aligned} \quad (9.46)$$

To express the scalar products, we go to the center-of-mass frame.

The four-momenta are

$$p_1^\mu = (E, \vec{p}), \quad p_2^\mu = (E, -\vec{p}), \quad (9.47)$$

$$q_1^\mu = (E, \vec{q}), \quad q_2^\mu = (E, -\vec{q}). \quad (9.48)$$

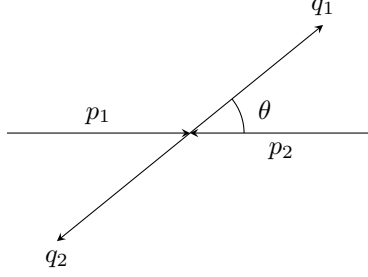


Figure 6: Center-of-mass kinematics and scattering angle.

Therefore,

$$p_1 \cdot p_2 = E^2 + p^2, \quad (9.49)$$

$$q_1 \cdot q_2 = E^2 + q^2, \quad (9.50)$$

$$p_1 \cdot q_1 = E^2 - pq \cos \theta = p_2 \cdot q_2, \quad (9.51)$$

$$p_1 \cdot q_2 = E^2 + pq \cos \theta = p_2 \cdot q_1. \quad (9.52)$$

Substituting,

$$|\overline{\mathcal{M}}|^2 = \frac{8e^4}{16E^4} \left[2E^4 + 2p^2q^2 \cos^2 \theta + (m_e^2 + m_\mu^2)E^2 + m_e^2q^2 + m_\mu^2p^2 + 2m_e^2m_\mu^2 \right]. \quad (9.53)$$

Since

$$m_e \ll m_\mu, \quad m_e^2 \ll E^2, \quad m_\mu^2 \ll E^2, \quad (9.54)$$

we set

$$m_e \simeq m_\mu \simeq 0, \quad q \simeq E, \quad p \simeq E. \quad (9.55)$$

Thus,

$$|\overline{\mathcal{M}}|^2 \simeq e^4(1 + \cos^2 \theta). \quad (9.56)$$

The differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} |\overline{\mathcal{M}}|^2 \frac{q}{p} \quad (9.57)$$

$$= \frac{e^4}{64\pi^2 s} (1 + \cos^2 \theta) \quad (9.58)$$

$$= \frac{\alpha^2}{4s} (1 + \cos^2 \theta), \quad (9.59)$$

where

$$\alpha = \frac{e^2}{4\pi}. \quad (9.60)$$

Integrating over the solid angle,

$$\sigma = \int \frac{d\sigma}{d\Omega} d\Omega \quad (9.61)$$

$$= \frac{\alpha^2 2\pi}{4s} \int_{-1}^1 (1 + \cos^2 \theta) d(\cos \theta) \quad (9.62)$$

$$= \frac{4\pi\alpha^2}{3s}. \quad (9.63)$$

Therefore,

$$\boxed{\sigma(e^+e^- \longrightarrow \mu^+\mu^-) = \frac{4\pi\alpha^2}{3s}}. \quad (9.64)$$

Let us now consider another process,

$$e^-(p_1)\mu^+(p_2) \longrightarrow e^-(q_1)\mu^+(q_2). \quad (9.65)$$

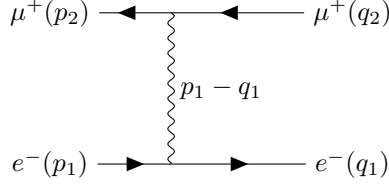


Figure 7: Elastic $e^-\mu^+$ scattering by photon exchange.

Using the Feynman rules,

$$\mathcal{M} = ie^2 [\bar{v}(q_2, s'_2) \gamma^\nu v(p_2, s_2)] \frac{\eta_{\mu\nu}}{(p_1 - q_1)^2} [\bar{u}(q_1, s'_1) \gamma^\mu u(p_1, s_1)]. \quad (9.66)$$

The Mandelstam variables are

$$s = (p_1 + p_2)^2, \quad (9.67)$$

$$t = (p_1 - q_1)^2, \quad (9.68)$$

$$u = (p_1 - q_2)^2. \quad (9.69)$$

For a general process

$$a + b \longrightarrow 1 + 2, \quad (9.70)$$

they satisfy

$$s + t + u = m_a^2 + m_b^2 + m_1^2 + m_2^2. \quad (9.71)$$

For

$$e^+ e^- \longrightarrow \mu^+ \mu^-, \quad (9.72)$$

in the massless limit,

$$|\overline{\mathcal{M}}|^2 = 2e^4 \frac{t^2 + u^2}{s^2}. \quad (9.73)$$

The relation between

$$e^+ e^- \longrightarrow \mu^+ \mu^- \quad (9.74)$$

and

$$e^- \mu^+ \longrightarrow e^- \mu^+ \quad (9.75)$$

is obtained through crossing:

$$\vec{p}_2 \longleftrightarrow -\vec{q}'_1, \quad (9.76)$$

$$\vec{q}_1 \longleftrightarrow -\vec{p}'_2. \quad (9.77)$$

Then,

$$t = (p_1 - q_1)^2 \longleftrightarrow (p'_1 + p'_2)^2 = s', \quad (9.78)$$

$$s = (p_1 + p_2)^2 \longleftrightarrow (p'_1 - q'_1)^2 = t', \quad (9.79)$$

$$u = (p_1 - q_2)^2 \longleftrightarrow (p'_1 - q'_2)^2 = u'. \quad (9.80)$$

Therefore,

$$\boxed{|\overline{\mathcal{M}}|^2 = 2e^4 \frac{s^2 + u^2}{t^2}}. \quad (9.81)$$

9.2 Problems

Problem 9.1 Consider the Lagrangian density

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 A_\mu A^\mu, \quad (9.82)$$

with

$$m \neq 0. \quad (9.83)$$

1. Verify that this theory is not gauge invariant.
2. Derive the equations of motion and show that they can be written as

$$(\square + m^2) A^\mu = 0, \quad (9.84)$$

together with the constraint

$$\partial_\mu A^\mu = 0. \quad (9.85)$$

Solution.

Solution. Under a gauge transformation,

$$A_\mu \longrightarrow A'_\mu = A_\mu + \partial_\mu \Lambda. \quad (9.86)$$

The field-strength tensor is gauge invariant:

$$F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu \quad (9.87)$$

$$= \partial_\mu (A_\nu + \partial_\nu \Lambda) - \partial_\nu (A_\mu + \partial_\mu \Lambda) \quad (9.88)$$

$$= F_{\mu\nu}. \quad (9.89)$$

Therefore,

$$-\frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \quad (9.90)$$

However, the mass term transforms as

$$\frac{1}{2}m^2 A'_\mu A'^\mu = \frac{1}{2}m^2 (A_\mu + \partial_\mu \Lambda) (A^\mu + \partial^\mu \Lambda) \quad (9.91)$$

$$= \frac{1}{2}m^2 A_\mu A^\mu + m^2 A^\mu \partial_\mu \Lambda + \frac{1}{2}m^2 \partial_\mu \Lambda \partial^\mu \Lambda. \quad (9.92)$$

The last two terms are in general nonzero. Hence,

$$\mathcal{L}' \neq \mathcal{L} \quad \text{for} \quad m \neq 0. \quad (9.93)$$

Thus, the theory is not gauge invariant.

Applying the Euler–Lagrange equations gives

$$\partial_\nu F^{\nu\mu} + m^2 A^\mu = 0. \quad (9.94)$$

Using

$$F^{\nu\mu} = \partial^\nu A^\mu - \partial^\mu A^\nu, \quad (9.95)$$

we obtain

$$\partial_\nu \partial^\nu A^\mu - \partial_\nu \partial^\mu A^\nu + m^2 A^\mu = 0, \quad (9.96)$$

$$\square A^\mu - \partial^\mu (\partial_\nu A^\nu) + m^2 A^\mu = 0. \quad (9.97)$$

Taking the divergence,

$$\partial_\mu \partial_\nu F^{\nu\mu} + m^2 \partial_\mu A^\mu = 0. \quad (9.98)$$

Since $F^{\nu\mu}$ is antisymmetric while $\partial_\mu \partial_\nu$ is symmetric,

$$\partial_\mu \partial_\nu F^{\nu\mu} = 0. \quad (9.99)$$

Therefore,

$$m^2 \partial_\mu A^\mu = 0. \quad (9.100)$$

For $m \neq 0$,

$$\partial_\mu A^\mu = 0. \quad (9.101)$$

Substituting this result into the equation of motion gives

$$\boxed{(\square + m^2) A^\mu = 0}, \quad (9.102)$$

together with

$$\boxed{\partial_\mu A^\mu = 0}. \quad (9.103)$$

□
□

Problem 9.2 Consider elastic electron–proton scattering at low energies,

$$e^- + p \longrightarrow e^- + p. \quad (9.104)$$

1. Calculate the scattering amplitude

$$i\mathcal{M} \quad (9.105)$$

in quantum electrodynamics.

2. Assuming unpolarized initial and final states, calculate

$$|\overline{\mathcal{M}}|^2. \quad (9.106)$$

Express the result in terms of the Mandelstam variables.

3. Obtain the dominant contribution in the non-relativistic limit, assuming that all three-momenta in the centre-of-mass frame are much smaller than the electron mass.
4. In the non-relativistic limit, obtain the corresponding scattering amplitude using the conventional normalization of non-relativistic states.

Warning: Recall that the conventional normalization of states is different in relativistic and non-relativistic theories.

Solution.

Solution. For low-energy elastic electron–proton scattering,

$$e^-(p_1, s_1) + p(p_2, s_2) \longrightarrow e^-(p_3, s_3) + p(p_4, s_4). \quad (9.107)$$

We define

$$q = p_3 - p_1 = p_2 - p_4, \quad t = q^2. \quad (9.108)$$

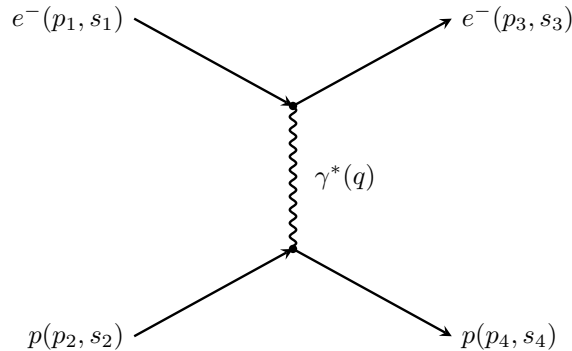


Figure 8: Tree-level Feynman diagram for elastic electron–proton scattering through the exchange of a virtual photon.

Using the QED Feynman rules,

$$i\mathcal{M} = \bar{u}_e(p_3) (-ie\gamma^\mu) u_e(p_1) \frac{-i\eta_{\mu\nu}}{q^2} \bar{u}_p(p_4) (+ie\gamma^\nu) u_p(p_2). \quad (9.109)$$

Therefore, up to an overall sign,

$$\mathcal{M} = \frac{e^2}{t} [\bar{u}_e(p_3)\gamma^\mu u_e(p_1)] [\bar{u}_p(p_4)\gamma_\mu u_p(p_2)]. \quad (9.110)$$

For unpolarized scattering, we average over the two initial electron and proton spin configurations and sum over the final spins:

$$\overline{|\mathcal{M}|^2} = \frac{1}{4} \sum_{s_1, s_2} \sum_{s_3, s_4} |\mathcal{M}|^2. \quad (9.111)$$

Using the completeness relations

$$\sum_s u_e(p, s) \bar{u}_e(p, s) = \not{p} + m_e, \quad (9.112)$$

$$\sum_s u_p(p, s) \bar{u}_p(p, s) = \not{p} + m_p, \quad (9.113)$$

we obtain

$$\begin{aligned} \overline{|\mathcal{M}|^2} &= \frac{e^4}{4t^2} \text{Tr} \left[(\not{p}_1 + m_e) \gamma^\nu (\not{p}_3 + m_e) \gamma^\mu \right] \\ &\quad \times \text{Tr} \left[(\not{p}_2 + m_p) \gamma_\nu (\not{p}_4 + m_p) \gamma_\mu \right]. \end{aligned} \quad (9.114)$$

Using

$$\text{Tr} \left[(\not{a} + m) \gamma^\mu (\not{b} + m) \gamma^\nu \right] = 4 (a^\mu b^\nu + a^\nu b^\mu - (a \cdot b - m^2) \eta^{\mu\nu}), \quad (9.115)$$

and the Mandelstam variables

$$s = (p_1 + p_2)^2, \quad (9.116)$$

$$t = (p_3 - p_1)^2, \quad (9.117)$$

$$u = (p_1 - p_4)^2, \quad (9.118)$$

with

$$s + t + u = 2m_e^2 + 2m_p^2, \quad (9.119)$$

we find

$$\boxed{\overline{|\mathcal{M}|^2} = \frac{2e^4}{t^2} \left[(s - m_e^2 - m_p^2)^2 + (u - m_e^2 - m_p^2)^2 + 2t (m_e^2 + m_p^2) \right]}. \quad (9.120)$$

In the nonrelativistic limit, all three-momenta in the centre-of-mass frame are much smaller than the fermion masses. The dominant current components are temporal:

$$\bar{u}_e(p_3) \gamma^0 u_e(p_1) \simeq 2m_e, \quad (9.121)$$

$$\bar{u}_p(p_4) \gamma^0 u_p(p_2) \simeq 2m_p. \quad (9.122)$$

Also,

$$t = q^2 \simeq -|\vec{q}|^2. \quad (9.123)$$

Thus,

$$\mathcal{M}_{\text{rel}} \simeq -\frac{4e^2 m_e m_p}{|\vec{q}|^2}. \quad (9.124)$$

Relativistic one-particle states have the normalization

$$\langle p | p' \rangle = (2\pi)^3 2E \delta^3(\vec{p} - \vec{p}'), \quad (9.125)$$

whereas nonrelativistic states do not contain the factor $2E$.

Removing the external-state normalization factors gives

$$\boxed{\mathcal{M}_{\text{NR}} \simeq -\frac{e^2}{|\vec{q}|^2}}, \quad (9.126)$$

which is the familiar electrostatic Coulomb interaction. □

□

10 Experimental Particle Physics

In a scattering experiment, we shoot particles at a target and observe where they end up.

We study the angular distribution of the scattered particles to obtain an idea of the target and the interaction.

This is measured through the differential cross section:

$$\frac{dN}{d\Omega} = \mathcal{L} \frac{d\sigma}{d\Omega}. \quad (10.1)$$

We will consider two kinds of processes:

- Scattering processes:

measure cross sections.

- Decays:

measure decay rates,

for instance, beta decay.

We will use Fermi's golden rule:

$$\Gamma_{fi} = 2\pi |\mathcal{M}_{fi}|^2 \rho(E_f), \quad (10.2)$$

where $\rho(E_f)$ is the density of states.

The total decay rate is

$$\Gamma_{\text{tot}} = \sum_f \Gamma_{fi} = \frac{1}{\tau}. \quad (10.3)$$

Resonances appear as peaks in the invariant mass, and the width of the peak gives the decay rate.

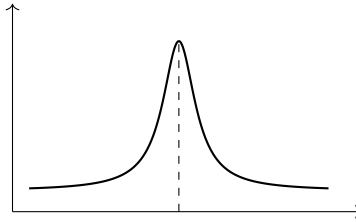


Figure 9: Schematic resonance peak in the invariant-mass distribution.

There are four fundamental particle interactions:

- **Gravitation:** negligible and not included in the Standard Model of particle physics.
- **Electromagnetism:** for example, Rutherford scattering.
- **Strong force:** for example, alpha decay.
- **Weak force:** for example, beta decay.

10.1 Particle Accelerators

We typically want to accelerate stable charged particles.

We can make collisions of the following types:

- beam against a fixed target,
- beam against beam.

There are two main types of accelerators:

- linear accelerators,

- circular accelerators.

Two important parameters are the instantaneous luminosity $\mathcal{L}$ and the centre-of-mass energy E_{cm} . The number of events of a given process is

$$N = \sigma \int dt \mathcal{L}(t), \quad (10.4)$$

where σ is the total cross section.

The centre-of-mass energy is defined by

$$E_{\text{cm}}^2 = (p_1 + p_2)^\mu (p_1 + p_2)_\mu. \quad (10.5)$$

For a fixed-target experiment,

$$E_{\text{cm}}^2 = m_1^2 + m_2^2 + 2E_1 m_2, \quad (10.6)$$

where particle 2 is at rest.

For a collider,

$$E_{\text{cm}}^2 = (E_1 + E_2)^2 - (\vec{p}_1 + \vec{p}_2)^2. \quad (10.7)$$

In particle accelerators, particles are packed into bunches. If

$$f = \frac{1}{\Delta t}, \quad (10.8)$$

then f is the bunch crossing frequency.

The luminosity can be written as

$$\mathcal{L} = \frac{f N_1 N_2}{A}, \quad (10.9)$$

where N_1 and N_2 are the number of particles per bunch and A is the transverse overlap area of the beams.

For Gaussian beams,

$$A = 2\pi \sqrt{\sigma_{x1}^2 + \sigma_{x2}^2} \sqrt{\sigma_{y1}^2 + \sigma_{y2}^2}. \quad (10.10)$$

If both beams have the same transverse sizes,

$$\boxed{\mathcal{L} = \frac{f N_1 N_2}{4\pi \sigma_x \sigma_y}}. \quad (10.11)$$

An accelerator has three main tasks:

- acceleration,
- steering,
- focusing.

For a charged particle in electric and magnetic fields,

$$\frac{d\vec{p}}{dt} = \vec{F} = q\vec{E} + q\vec{v} \times \vec{B}. \quad (10.12)$$

The variation of the kinetic energy is

$$\frac{dT}{ds} = \vec{v} \cdot \vec{F} = q\vec{v} \cdot \vec{E}. \quad (10.13)$$

Therefore electric fields are responsible for the acceleration, while magnetic fields are especially useful for steering and focusing.

We first consider radio-frequency linear accelerators. In a linac, particles gain energy by crossing a sequence of drift tubes or cavities.

After passing through n tubes, the energy gain is

$$E_n = nqV_0 \sin(\psi_s), \quad (10.14)$$

where V_0 is the peak voltage and ψ_s is the synchronous phase.

The length of the n -th tube is chosen as

$$L_n = \beta_n \frac{c}{2f}, \quad (10.15)$$

with

$$\beta_n = \sqrt{1 - \frac{1}{\left(1 + \frac{E_n}{mc^2}\right)^2}}. \quad (10.16)$$

The beam gets naturally grouped into bunches.

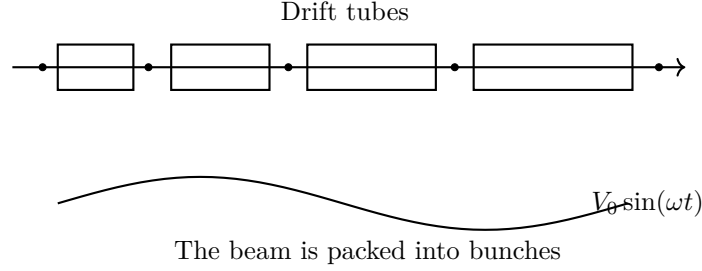


Figure 10: Schematic operation of a linear radio-frequency accelerator.

A useful improvement is to place the drift tubes inside resonance cavities. This gives rise to the *Alvarez structure*, allowing higher frequencies and larger energies over shorter distances.

Generally, the cavities have the geometry of a pillbox. Inside the cavity, the electromagnetic field satisfies

$$\nabla^2 \vec{E} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2}. \quad (10.17)$$

Using a harmonic ansatz,

$$\vec{E}(\vec{r}, t) = \vec{E}_0(\vec{r}) e^{i\omega t}, \quad (10.18)$$

we obtain the Helmholtz equation

$$\nabla^2 \vec{E} + k^2 \vec{E} = 0, \quad k = \frac{\omega}{c} = \frac{2\pi f}{c}. \quad (10.19)$$

Assuming cylindrical symmetry and no dependence on z and ϕ , one gets

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dR}{dr} \right) + h^2 R = 0, \quad (10.20)$$

which is Bessel's equation. Its regular solution is

$$R(r) = J_0(hr). \quad (10.21)$$

Imposing the boundary condition $E_z(r = a) = 0$ gives

$$J_0(ha) = 0 \quad \implies \quad ha = 2.405. \quad (10.22)$$

Therefore the fundamental mode is the TM_{010} mode, with frequency

$$f_{010} = \frac{2.405 c}{2\pi a}. \quad (10.23)$$

The corresponding fields can be written schematically as

$$E_z = E_0 f(r) e^{i(\omega t - kz)}, \quad B_\phi = B_0 g(r) e^{i(\omega t - kz)}, \quad (10.24)$$

with

$$f(r) = J_0\left(\frac{2.405 r}{R}\right). \quad (10.25)$$

In the beam pipe, the magnetic field can be expanded as a multipole series:

$$\frac{e}{p} B_y = \frac{1}{\rho} + kx + \frac{1}{2} m x^2 + \dots \quad (10.26)$$

The first terms correspond to

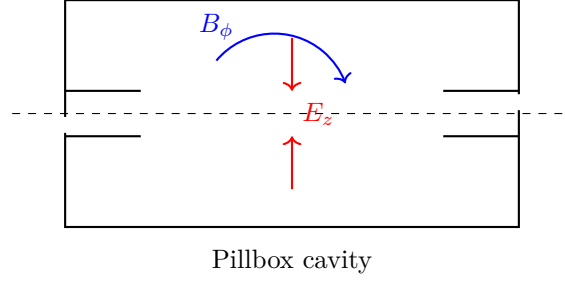


Figure 11: Schematic pillbox cavity supporting the TM_{010} mode.

- dipole fields $\rightarrow$ steering,
- quadrupole fields $\rightarrow$ focusing,
- sextupole fields $\rightarrow$ chromaticity correction.

Dipole magnets are used to bend the beam. The Lorentz force provides the centripetal force:

$$evB = F_L = F_{\text{cent}} = \frac{\gamma m v^2}{\rho}. \quad (10.27)$$

This implies

$$\boxed{\frac{p}{e} = B\rho}. \quad (10.28)$$

Hence the bending radius ρ of a particle with momentum p is controlled by the magnetic field B . Equivalently,

$$\frac{1}{\rho} \quad (10.29)$$

is the bending strength of the accelerator.

As the particle momentum increases, the magnetic field must also increase. A limitation then comes from the power dissipated in conventional magnets,

$$B \propto I, \quad P = RI^2, \quad (10.30)$$

which is why superconducting magnets are so important.

A moving charge radiates according to the Liénard–Wiechert fields:

$$\vec{E}(\vec{r}, t) = \frac{e}{4\pi\epsilon_0} \left[\frac{\hat{n} - \vec{\beta}}{\gamma^2(1 - \hat{n} \cdot \vec{\beta})^3 R^2} \right]_{t_r} + \frac{e}{4\pi\epsilon_0 c} \left[\frac{\hat{n} \times ((\hat{n} - \vec{\beta}) \times \dot{\vec{\beta}})}{(1 - \hat{n} \cdot \vec{\beta})^3 R} \right]_{t_r}. \quad (10.31)$$

The radiated power is given by the relativistic Larmor formula:

$$P = \frac{e^2}{6\pi\epsilon_0 c} \gamma^6 \left[\dot{\vec{\beta}}^2 - (\vec{\beta} \times \dot{\vec{\beta}})^2 \right]. \quad (10.32)$$

In a circular accelerator, $\vec{\beta} \perp \dot{\vec{\beta}}$, so synchrotron-radiation losses become important. In particular, the energy loss per turn scales as

$$\Delta E_{\text{loss}} \propto \frac{E^4}{m^4 \rho}. \quad (10.33)$$

This explains why radiation losses are much more severe for electrons than for protons. For instance,

$$\frac{\Delta E_{\text{loss}}^{\text{LEP}}}{\Delta E_{\text{loss}}^{\text{LHC}}} \sim \left(\frac{E_{\text{LEP}}}{E_{\text{LHC}}} \right)^4 \left(\frac{m_p}{m_e} \right)^4 \sim 10^6. \quad (10.34)$$

Quadrupole magnets are used to focus the beam in the transverse plane. They act as lenses with focal length f .

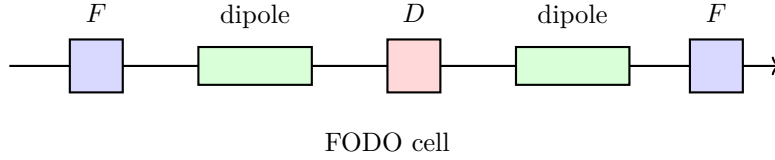


Figure 12: Typical FODO cell in a circular accelerator.

By alternating focusing and defocusing magnets one can obtain an overall focusing effect. If $f_1 = -f_2$, then

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} = \frac{d}{f_1^2} > 0. \quad (10.35)$$

In circular accelerators the magnets are often arranged in *FODO cells*.

The transverse motion is described by an equation of the form

$$x'' + \left(h(s) + \frac{1}{\ell(s)^2} \right) x = \frac{1}{\rho(s)} \frac{\Delta p}{p}, \quad (10.36)$$

where the different terms are associated with strong focusing, weak focusing, and dispersion.

The solutions describe quasi-harmonic oscillations along the reference orbit:

$$y(s) = \sqrt{\epsilon_y \beta_y(s)} \cos \phi_y(s), \quad (10.37)$$

and

$$x(s) = \sqrt{\epsilon_x \beta_x(s)} \cos \phi_x(s) + D(s) \frac{\Delta p}{p}. \quad (10.38)$$

Here $\beta(s)$ determines the beam size (the beam envelope), and the number of oscillations per turn is the tune,

$$Q_{x,y} = \frac{1}{2\pi} \oint \frac{ds}{\beta_{x,y}(s)}. \quad (10.39)$$

10.2 Particle Detectors

We want to detect different kinds of particles, including both stable and decaying particles.

Examples of stable or sufficiently long-lived particles include

- electrons and positrons,
- muons,
- neutral particles,
- charged hadrons.

These particles either do not decay or have a sufficiently long lifetime to be detected.

The mean decay length in the laboratory frame is

$$\ell = \gamma \ell_{\text{lab}} = \beta \gamma c \tau = \frac{p}{m} c \tau. \quad (10.40)$$

Decaying particles are detected through their decay products.

To understand particle detection, we need to consider the electromagnetic interaction of charged particles with matter.

Charged particles interact mainly with

- atomic electrons, producing excitation and mostly ionisation,
- atomic nuclei, producing Coulomb scattering.

The combined effect of multiple Coulomb scatterings is described approximately by the Highland formula,

$$\theta_0 \simeq \frac{13.6 \text{ MeV}}{\beta pc} z \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \left(\frac{x}{X_0} \right) \right]. \quad (10.41)$$

For charged particles other than electrons, the mean energy loss through ionisation is described by the Bethe–Bloch formula:

$$-\frac{1}{\rho} \left\langle \frac{dE}{dx} \right\rangle = 4\pi N_A r_e^2 m_e c^2 \frac{Z}{A} \frac{z^2}{\beta^2} \left[\frac{1}{2} \ln \left(\frac{2m_e c^2 \beta^2 \gamma^2 T_{\max}}{I^2} \right) - \beta^2 - \frac{\delta(\beta\gamma)}{2} - \frac{C(\beta\gamma, I)}{Z} \right]. \quad (10.42)$$

Here,

- the term proportional to $1/\beta^2$ dominates at low velocity,
- $\delta(\beta\gamma)$ is the density-effect correction,
- $C(\beta\gamma, I)$ is the shell correction.

By measuring the momentum p and the energy loss dE/dx , one can identify the type of particle. For electrons and positrons, radiative energy losses due to bremsstrahlung must also be considered. Defining the radiation length X_0 , the energy decreases approximately as

$$E(x) = E_0 e^{-x/X_0}. \quad (10.43)$$

The main photon interactions with matter are the photoelectric effect, Compton scattering, and pair production.

10.2.1 Photoelectric effect

In the photoelectric effect, the photon is completely absorbed by an atom, which becomes ionised by emitting an electron.

The process dominates when the photon energy is comparable to the atomic binding energy:

$$E_\gamma \gtrsim E_B, \quad E_B \sim \text{eV} - \text{keV}. \quad (10.44)$$

The total cross section approximately scales as

$$\sigma_{\text{tot}} \propto \frac{Z^5}{E_\gamma^{3.5}}. \quad (10.45)$$

10.2.2 Compton scattering

Compton scattering is the quasi-elastic scattering of a photon by an atomic-shell electron. It dominates when

$$E_\gamma \gg E_B. \quad (10.46)$$

The energy of the scattered photon is

$$E'_\gamma = \frac{E_\gamma}{1 + \frac{E_\gamma}{m_e c^2} (1 - \cos \theta_\gamma)}. \quad (10.47)$$

10.2.3 Pair production

At sufficiently high energy, a photon can convert into an electron–positron pair:

$$\gamma \longrightarrow e^+ e^-. \quad (10.48)$$

The threshold condition is

$$E_\gamma \gtrsim 2m_e c^2. \quad (10.49)$$

Electromagnetic showers consist of alternating bremsstrahlung emission and pair production.

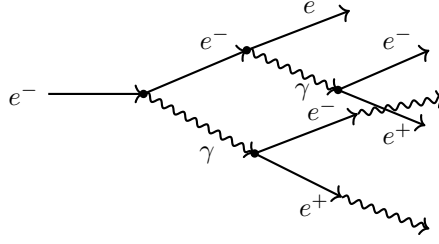


Figure 13: Schematic development of an electromagnetic shower through repeated bremsstrahlung and pair production.

Both electrons, positrons, and photons can continue the cascade.

In a simple model, the number of particles after a time t , measured in radiation lengths, is

$$N(t) = 2^t = e^{t \ln 2}. \quad (10.50)$$

The average energy per particle is

$$E(t) = \frac{E_0}{2^t}. \quad (10.51)$$

The shower continues until the energy per particle falls below the critical energy E_c :

$$E(t_{\max}) \simeq E_c. \quad (10.52)$$

Therefore,

$$t_{\max} = \frac{\ln(E_0/E_c)}{\ln 2}. \quad (10.53)$$

Charged hadrons can interact with matter through the strong interaction.

These interactions produce secondary hadrons, which can interact again, giving rise to a hadronic shower.

Hadronic showers also contain electromagnetic subshowers, mainly because of processes such as

$$\pi^0 \longrightarrow \gamma\gamma. \quad (10.54)$$

Hadronic showers typically

- contain many electromagnetic subshowers,
- exhibit large event-by-event fluctuations,
- are longer and wider than electromagnetic showers because of neutrons and nuclear fragments.

The shower energy decreases approximately as

$$E(x) = E_0 e^{-x/\lambda_a}, \quad (10.55)$$

where λ_a is the nuclear interaction length:

$$\lambda_a = \frac{A}{N_A \rho \sigma_{\text{inelastic}}}. \quad (10.56)$$

A commonly used approximation is

$$\lambda_a \simeq \frac{25 \text{ g cm}^{-2}}{\rho} A^{1/3}. \quad (10.57)$$

The depth of the shower maximum increases approximately logarithmically with the initial energy:

$$t_{\max} = \frac{x_{\max}}{\lambda_a} \simeq 0.2 \ln \left(\frac{E_0}{\text{GeV}} \right) + 0.7. \quad (10.58)$$

Cherenkov radiation is emitted when a charged particle moves through a medium faster than the phase velocity of light in that medium.

For a medium with refractive index n ,

$$v > \frac{c}{n}, \quad (10.59)$$

or equivalently

$$\beta n > 1. \quad (10.60)$$

The emitted light forms a cone with Cherenkov angle θ_C :

$$\boxed{\cos \theta_C = \frac{1}{n\beta}}. \quad (10.61)$$

Transition radiation is emitted when a relativistic charged particle crosses the boundary between two media with different refractive indices.

It is particularly relevant for highly relativistic particles, with a characteristic photon energy scaling approximately as

$$E_{\text{transition}} \sim \gamma. \quad (10.62)$$

Neutrinos interact only through the weak interaction.

They can interact with nucleons through charged-current or neutral-current processes.

The mean free path is

$$\lambda = \frac{A}{N_A \rho \sigma_{\nu N}}. \quad (10.63)$$

Because the neutrino cross section is extremely small, neutrino detection requires either very large detector volumes or very intense neutrino fluxes.

Photons interact with a photocathode through the photoelectric effect, releasing electrons.

The electrons are accelerated from one dynode to the next, each dynode being held at a different electric potential.

After n dynodes, the total multiplication factor is approximately

$$G = \delta^n, \quad (10.64)$$

where δ is the average secondary-emission factor per dynode.

Typical gains are

$$G \sim 10^6 - 10^7. \quad (10.65)$$

The photon-detection efficiency is

$$\text{PDE} = \text{QE} \times \text{CE}, \quad (10.66)$$

where QE is the quantum efficiency and CE is the collection efficiency.

For example,

$$\text{PDE} \simeq 25\% \times 80\% \simeq 20\%. \quad (10.67)$$

In a scintillator, atoms or molecules are excited by the passage of a particle.

The subsequent recombination or de-excitation produces luminescence.

Scintillators can be divided into two main classes.

Organic scintillators include plastics and aromatic compounds.

Their main properties are

- fast response,
- low cost,
- mechanical flexibility,
- relatively low stopping power,
- possible sensitivity to radiation damage.

They are commonly used in tracking and timing detectors.

Inorganic scintillators include crystals and liquid noble gases.

Their main properties are

- slower response,
- high light yield,

- high stopping power,
- higher cost,
- possible sensitivity to heat and moisture.

They are commonly used in calorimeters.

In organic scintillators, the Stokes shift must be considered: the emitted wavelength is longer than the absorbed wavelength.

This reduces the self-absorption of scintillation photons.

Semiconductor detectors can be used to construct tracking devices based on *p-n* junction diodes.

A charged particle crossing the depletion region produces electron–hole pairs.

The motion of these charges in the electric field induces a measurable electrical signal.

Calorimeters measure the energy of a particle through its total absorption.

During the shower development, most of the particle energy is eventually converted into heat or detectable signals.

Electrons, positrons, and photons produce electromagnetic showers and are measured with electromagnetic calorimeters.

Hadrons produce hadronic showers and are measured with hadronic calorimeters.

There are two main types of calorimeters:

- homogeneous calorimeters,
- sampling calorimeters.

A sampling calorimeter consists of alternating layers of

- passive absorber material, usually dense and with high atomic number,
- active detector material that measures the deposited energy.

The sampling fraction is defined as

$$f = \frac{\text{energy deposited in the active material}}{\text{total deposited energy}}. \quad (10.68)$$

If the total shower energy is shared among N charged particles,

$$E \propto N. \quad (10.69)$$

Because the number of particles approximately follows Poisson statistics,

$$\sigma_N \propto \sqrt{N}. \quad (10.70)$$

Therefore, for an ideal calorimeter,

$$\frac{\sigma_E}{E} \propto \frac{1}{\sqrt{E}}. \quad (10.71)$$

For a realistic calorimeter, the energy resolution is commonly parametrised as

$$\boxed{\frac{\sigma_E}{E} = \sqrt{\frac{a^2}{E} + \frac{b^2}{E^2} + c^2}}. \quad (10.72)$$

Here,

- $a/\sqrt{E}$ is the stochastic term,
- b/E is the electronic-noise term,
- c is the constant term due to systematic effects.

10.3 Particle Identification

There are several particle-identification techniques. Information from different detector subsystems can be combined.

The average ionisation energy loss can be studied as a function of the momentum:

$$\left\langle \frac{dE}{dx} \right\rangle \quad \text{vs.} \quad p. \quad (10.73)$$

Different particle species follow different curves in the $\langle dE/dx \rangle$ -versus- p plane.

For a detector located at a distance L , the time of flight is

$$\Delta t = \frac{L}{\beta c}. \quad (10.74)$$

Using

$$\beta = \frac{p}{\sqrt{p^2 + m^2}}, \quad (10.75)$$

we obtain

$$\Delta t = \frac{L}{c} \sqrt{\frac{p^2 + m^2}{p^2}}. \quad (10.76)$$

Thus, particles with the same momentum but different masses have different flight times.

Cherenkov detectors separate particles by using the momentum threshold required to produce Cherenkov light in a radiator with refractive index n .

The Cherenkov condition is

$$\beta n > 1. \quad (10.77)$$

At threshold,

$$\beta_{\text{th}} = \frac{1}{n}. \quad (10.78)$$

Since

$$p = \gamma m \beta c, \quad (10.79)$$

the threshold momentum is

$$p_{\text{th}} = \frac{mc}{\sqrt{n^2 - 1}}. \quad (10.80)$$

Cherenkov light is emitted at the Cherenkov angle θ_C , defined by

$$\cos \theta_C = \frac{1}{n\beta}. \quad (10.81)$$

For a particle entering perpendicularly to a planar detector surface, the internal reflection angle θ satisfies

$$\sin \theta = \sin \left(\frac{\pi}{2} - \theta_C \right) \quad (10.82)$$

$$= \cos \theta_C \quad (10.83)$$

$$= \frac{1}{n\beta}. \quad (10.84)$$

For an ultrarelativistic particle, $\beta \simeq 1$, so

$$\sin \theta \simeq \frac{1}{n}. \quad (10.85)$$

Total internal reflection requires

$$\sin \theta > \frac{1}{n}, \quad (10.86)$$

and in the geometry considered this implies approximately

$$n > \sqrt{2}. \quad (10.87)$$

The Cherenkov angle depends on the particle velocity:

$$\theta_C = \arccos\left(\frac{1}{n\beta}\right). \quad (10.88)$$

These particle-identification methods must be combined with a momentum measurement obtained from the curvature of the track in a magnetic field.

For a particle of charge q moving in a uniform magnetic field,

$$p = |q|Br. \quad (10.89)$$

Therefore,

$$m = \frac{p}{\gamma\beta c}. \quad (10.90)$$

10.4 Problems

Problem 10.1 A cosmic ray shower strikes Earth, which we assume to be uniform quartz rock (SiO_2) of density

$$\rho = 3 \text{ g/cm}^3.$$

The shower has a core of 1000 GeV muons.

Assuming that the muons ionize minimally until they stop, determine how deeply into the Earth the muons penetrate.

Solution. Under the minimum-ionizing-particle approximation,

$$\frac{dE}{dx} \simeq (2 \text{ MeV cm}^2 \text{ g}^{-1}) (3 \text{ g cm}^{-3}) = 6 \text{ MeV/cm}. \quad (10.91)$$

The initial energy is

$$E_0 = 1000 \text{ GeV} = 10^6 \text{ MeV}. \quad (10.92)$$

Neglecting radiative energy losses, the range is

$$R \simeq \frac{E_0}{dE/dx} \quad (10.93)$$

$$= \frac{10^6 \text{ MeV}}{6 \text{ MeV/cm}} \quad (10.94)$$

$$= 1.67 \times 10^5 \text{ cm} \quad (10.95)$$

$$= 1.67 \text{ km}. \quad (10.96)$$

Therefore, the muons penetrate approximately

$$\boxed{R \simeq 1.7 \text{ km}} \quad (10.97)$$

into the quartz rock.

At TeV energies, radiative losses of muons can in fact be relevant, but the problem explicitly asks us to neglect them. $\square$

Problem 10.2 Assume an unstable charged particle with mean lifetime τ circulating in a synchrotron whose dipoles have a magnetic field B and occupy half its circumference (dipole fill factor equal to 0.5).

1. Obtain an expression for the number of turns that the particle completes during its mean lifetime in the laboratory frame as a function of the dipole magnetic field B .
2. Apply the result to the case of a muon.

Solution. Inside the dipole magnets, the bending radius satisfies

$$\rho = \frac{p}{|q|B} = \frac{\gamma m \beta c}{|q|B}. \quad (10.98)$$

The dipoles must provide a total bending angle of 2π per revolution, so the total length occupied by the dipoles is

$$L_{\text{dip}} = 2\pi\rho. \quad (10.99)$$

Since the dipoles occupy one half of the circumference,

$$L_{\text{dip}} = \frac{C}{2}. \quad (10.100)$$

Therefore,

$$C = 4\pi\rho. \quad (10.101)$$

The revolution period is

$$T_{\text{rev}} = \frac{C}{\beta c} \quad (10.102)$$

$$= \frac{4\pi\rho}{\beta c}. \quad (10.103)$$

The mean lifetime in the laboratory frame is

$$\tau_{\text{lab}} = \gamma\tau. \quad (10.104)$$

Hence, the number of turns completed during one mean lifetime is

$$N_{\text{turns}} = \frac{\gamma\tau}{T_{\text{rev}}} \quad (10.105)$$

$$= \frac{\gamma\tau\beta c}{4\pi\rho}. \quad (10.106)$$

Using

$$\rho = \frac{\gamma m\beta c}{|q|B}, \quad (10.107)$$

we obtain

$$N_{\text{turns}} = \frac{\gamma\tau\beta c}{4\pi} \frac{|q|B}{\gamma m\beta c} \quad (10.108)$$

$$= \boxed{\frac{|q|B\tau}{4\pi m}}. \quad (10.109)$$

All relativistic factors cancel. Therefore, for a fixed dipole fill factor, the number of turns does not depend on the particle momentum.

For a muon,

$$N_{\mu} = \frac{eB\tau_{\mu}}{4\pi m_{\mu}}. \quad (10.110)$$

Using the muon mass and lifetime,

$$N_{\mu} \simeq 148.7 B[\text{T}]. \quad (10.111)$$

Thus, a muon completes approximately

$$\boxed{N_{\mu} \simeq 1.5 \times 10^2 \text{ turns per tesla}} \quad (10.112)$$

during one mean lifetime.

For example,

$$B = 1 \text{ T} \quad \Rightarrow \quad N_{\mu} \simeq 149, \quad (10.113)$$

$$B = 8.3 \text{ T} \quad \Rightarrow \quad N_{\mu} \simeq 1.23 \times 10^3. \quad (10.114)$$

□

Problem 10.3 Multiple scattering often limits the ability to measure the direction of motion of charged particles.

To what accuracy can the initial direction of a 100 GeV muon be estimated if its direction is measured after traversing 30 cm of steel?

Solution. We use the Highland formula for a singly charged particle, $z = 1$, with $\beta \simeq 1$:

$$\theta_{\text{ms}} \simeq \frac{13.6 \text{ MeV}}{pc} \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \left(\frac{x}{X_0} \right) \right]. \quad (10.115)$$

For steel, which we approximate by iron,

$$X_0 \simeq 1.76 \text{ cm}. \quad (10.116)$$

With

$$x = 30 \text{ cm}, \quad (10.117)$$

we obtain

$$\frac{x}{X_0} = \frac{30}{1.76} \quad (10.118)$$

$$\simeq 17.0. \quad (10.119)$$

The muon momentum is

$$p = 100 \text{ GeV}/c = 10^5 \text{ MeV}/c. \quad (10.120)$$

Therefore,

$$\theta_{\text{ms}} \simeq \frac{13.6}{10^5} \sqrt{17.0} [1 + 0.038 \ln(17.0)] \quad (10.121)$$

$$\simeq 6.2 \times 10^{-4} \text{ rad}. \quad (10.122)$$

Thus, the initial direction can only be reconstructed with an accuracy of approximately

$$\boxed{\theta_{\text{ms}} \simeq 0.62 \text{ mrad}} \quad (10.123)$$

or, equivalently,

$$\boxed{\theta_{\text{ms}} \simeq 0.036^\circ}. \quad (10.124)$$

□